100 DAYS OF
QUALITY ENGINEERING

From Industrial Insights to Engineering Excellence

JAGRITI SINGH

Independent Learning • Industry Case Studies • Quality Methods

FIRST EDITION | 2026

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Copyright and Source Note

Copyright © 2026 Jagriti Singh. All rights reserved.

This manuscript is an independent learning and research compilation developed from the original PDF documents supplied in the 100 Days of Quality Engineering Challenge repository ZIP. The source documents are the primary basis for the day-wise material reproduced and organised in this book.

The organisation-specific reports and case studies are educational research documents. Unless a source independently establishes otherwise, they should not be interpreted as official reports issued by the named organisations, evidence of employment or collaboration, or confirmation of access to confidential operational data. Hypothetical scenarios, illustrative calculations, proposed frameworks, and analytical assumptions retain the status assigned to them in their original documents.

This edition preserves source terminology and selected source text. It is an organised working manuscript and should undergo final technical, factual, citation, and editorial review before commercial publication.

Acknowledgements

This book represents one hundred days of curiosity, independent learning, research, and exploration of Quality Engineering across diverse industries. To everyone reading this book, thank you for being a part of this journey. Your interest in understanding quality, engineering excellence, reliability, and continuous improvement gives this work a meaningful purpose. May these pages encourage you to question processes, explore new perspectives, understand industrial practices, and develop a deeper appreciation for the role of quality in engineering. This journey began with a simple intention: to learn something new every day and connect engineering knowledge with real-world industrial applications. I hope it inspires you to continue learning, observing, questioning, and improving. With warm regards, Jagriti Singh, Quality Engineer, Creator, 100 Days of Quality Engineering Challenge.

Preface

Quality Engineering is not confined to final inspection or the detection of defects after production. It brings together process understanding, measurement, risk assessment, prevention, reliability, customer requirements, and continuous improvement. Across industries, the form of quality work changes, but the underlying responsibility remains: to make processes and outcomes dependable, consistent, safe, and fit for purpose.

The 100 Days of Quality Engineering Challenge was designed as an independent learning journey across manufacturing, technology, aerospace, energy, infrastructure, transportation, services, and other fields represented in the original research documents. This book organises that material into a reference-style volume while retaining the identity of each day and the limits of the original evidence.

How to Use This Book

Part I introduces the broad foundations and recurring methods that appear across the source collection. Part II follows the day-wise sequence and records the original document titles, source themes, and extracted technical material. Part III draws careful comparisons across industries using the supplied documents. Part IV reflects on the learning journey, and the final sections provide a source inventory and editorial audit notes.

PART I — FOUNDATIONS OF QUALITY ENGINEERING

1. Quality Engineering as a System of Prevention and Improvement

Quality Engineering can be understood as the systematic application of engineering methods, measurement, analysis, risk management, and process control to achieve requirements consistently. It connects product and service expectations with the processes that create them. Its focus is not only on identifying nonconforming output, but also on understanding variation, preventing failure, improving process capability, and learning from performance evidence.

The source collection demonstrates this breadth through reports on 8D problem solving, process capability, cost of quality, supplier evaluation, first article inspection, overall equipment effectiveness, Taguchi quality loss, Lean and TQM frameworks, service quality, reliability, and industry-specific quality systems. These methods address different questions: What failed? How capable is the process? Where is time or output being lost? Which supplier risks require attention? How can a process be made more robust? How can a service become more consistent?

2. The Quality Management Cycle

A practical quality management cycle begins with requirements and planning, proceeds through controlled execution and measurement, and uses evidence to drive corrective and preventive action. Quality planning translates customer, regulatory, technical, and operational needs into acceptance criteria and controls. Quality assurance builds confidence that the system can meet those requirements. Quality control measures outputs and identifies deviations. Improvement uses analysis and learning to strengthen the process.

The PDCA cycle provides a repeatable structure: plan the change and define the measure; do the work under controlled conditions; check the results against evidence; and act by standardising an effective change or revising the approach. The source documents use related approaches such as 8D, DMAIC, Kaizen, Lean, FMEA, SPC, audits, and root-cause analysis.

3. Problem Solving, Risk and Process Control

Structured problem solving prevents teams from jumping directly from a symptom to a corrective action. The 8D approach commonly moves from team formation and problem description through containment, root-cause analysis, corrective-action selection, validation, prevention of recurrence, and closure. The exact sequence and report content should follow the relevant source document.

Failure Mode and Effects Analysis (FMEA) is a forward-looking risk analysis method. It identifies potential failure modes, their effects and causes, existing controls, and actions to reduce risk. Where the source uses severity, occurrence, detection, or a Risk Priority Number, those values must be understood in the context of the source's stated scoring system and assumptions.

Statistical Process Control (SPC) uses process data over time to distinguish common-cause variation from signals that may indicate special causes. Process capability indices such as Cp and Cpk compare process spread and centring with specification limits, subject to assumptions about process stability and data distribution. Control charts are not substitutes for engineering judgement; they are evidence tools that help determine when investigation is warranted.

4. Lean, TQM and Continuous Improvement

Lean thinking seeks to improve flow and remove activities that do not create value from the customer's perspective. The source collection includes material on TIMWOOD waste, inventory, ABC analysis, JIT, Kanban, 5S, and service-process improvement. Total Quality Management (TQM) treats quality as an organisation-wide responsibility supported by leadership, customer focus, employee involvement, process thinking, and continuous improvement.

The source documents also cover quality thinkers and strategic frameworks. Their historical contributions should be explained in context rather than reduced to slogans. Deming's systems perspective, Taguchi's emphasis on robust design and quality loss, and other TQM frameworks connect design decisions, process behaviour, and customer outcomes.

5. Supplier, Inspection and Measurement Systems

Supplier quality is an extension of the organisation's own process. Supplier rating systems, supplier quality procedures, incoming inspection, First Article Inspection (FAI), traceability, and audit dashboards provide structured ways to assess conformity and control supply-chain risk. A rating or dashboard is only as reliable as its criteria, evidence, sampling plan, and data integrity.

Inspection and testing provide evidence that specified requirements have been met. Their effectiveness depends on calibrated equipment, competent personnel, suitable methods, representative sampling, documented acceptance criteria, and traceable records. Inspection alone cannot guarantee quality if process design and prevention controls are weak.

6. Reliability and Quality in Different Sectors

In aerospace, defence, energy, transport, and infrastructure, quality is closely linked with reliability, safety, traceability, lifecycle performance, and configuration control. In software and digital services, quality also includes availability, performance, usability, security, data integrity, and resilience. In hospitality, retail, healthcare-related service processes, and logistics, consistency and customer experience are prominent. These are broad sector perspectives; the day-wise chapters retain the exact focus and qualifications of each source.

PART II — THE 100-DAY LEARNING JOURNEY

The following chapters are organised by the day numbers identified in the supplied ZIP. Each chapter retains the original source document name and uses its extracted content as the evidence base. Where multiple documents map to one day, each is identified. A missing day is not filled with an invented case study.

DAY 01

Original source file(s): Day 01 _8D_Report_Automotive_Engine_Shaft (1) (1).pdf

Source focus: The Day 1 source material focuses on 8D PROBLEM SOLVING REPORT. The source text explicitly discusses the following quality themes and methods: 8D, Taguchi, Inspection, ISOLATION, ISO 9001:2015, isolate, ROOT CAUSE, RCA, FMEA, RPN, Six Sigma, DMAIC.

Section headings identified in the source: 1. HEADER DETAILS & TARGET SPECIFICATIONS OVERVIEW & SPECS • Specification Philosophy Comparison: • 1. Quality Inspection Department Quality Inspector, QA/QC Lead Personnel • 2. Production Department Line Supervisors & Machining Technicians • 3. Manufacturing Department Process Engineers & Tooling Specialists • 4. Management Department Overall Cross-Functional Team Coordination & Handling

Technical terms detected in the source: 8D, Taguchi, Inspection, ISOLATION, ISO 9001:2015, isolate, ROOT CAUSE, RCA, FMEA, RPN, Six Sigma, DMAIC, PDCA, Kaizen

Source extract 1: Day 01 _8D_Report_Automotive_Engine_Shaft (1) (1).pdf

8D PROBLEM SOLVING REPORT

Quality Engineering & Technical Methodology — Automotive Engine Shaft Component

1. HEADER DETAILS & TARGET SPECIFICATIONS OVERVIEW & SPECS

Report Title 8D Report: Automotive Engine Shaft — Oversized Diameter

Part Name & Number Engine Shaft (Part No: ES-2026-X)

Actual: Diameter > 25.03 mm

Target Specification Standard / Ideal Target: 25.00 ± 0.02 mm

Max = 25.02 mm | Min = 24.98 mm

Specification Philosophy Comparison:

• Taguchi Philosophy: Specification must be as near to the target (25.00 mm) as possible to minimize quality loss.

• Traditional Philosophy: Any value within limits (24.98 mm < X < 25.02 mm) is accepted as pass.

• Most Ideal Case: 25.00 mm (Zero deviation from target center).

D1 — FORM CROSS-FUNCTIONAL TEAM (CFT) TEAM STRUCTURE

1. Quality Inspection Department Quality Inspector, QA/QC Lead Personnel

2. Production Department Line Supervisors & Machining Technicians

3. Manufacturing Department Process Engineers & Tooling Specialists

4. Management Department Overall Cross-Functional Team Coordination & Handling

5. Operational Department Shopfloor Logistics & Workflow Supervisors

D2 — DEFINE PROBLEM (5W2H FRAMEWORK) PROBLEM DESCRIPTION

Diameter of automotive engine shaft exceeds specifications, measuring

1. What

more than 25.03 mm (Standard: 25.00 ± 0.02 mm).

2. Where Detected at Production Line / Quality Inspection Bench.

3. When Detected when parts were about to be dispatched to customer.

1st Level: Quality Inspector / QC Department | 2nd Level: Customer

4. Who

feedback/escalation.

Standard limit is 24.98 mm < X < 25.02 mm. Deviation observed: +0.01 mm

5. Why (Deviation)

above maximum allowable specification limit.

Rejection Rate increased from 2% → 8.5% (15% of manufactured parts are

6. How Many

currently out of tolerance).

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D3 — IMMEDIATE CONTAINMENT ACTION (ICA) CONTAINMENT & ISOLATION

Standard Protocol (ISO 9001:2015 Clause 8.7 — Control of Non-Conforming Outputs):

• Red Flag Tagging: Immediately attach Red Flags / Tags to all defective and non-conforming parts to isolate them.

• Line Halt: Stop the production line immediately to prevent further manufacturing of out-of-tolerance parts.

D4 — IDENTIFY, VERIFY & VALIDATE ROOT CAUSE ANALYSIS ANALYSIS &

DIAGNOSTICS

(RCA)

Step 1: Ishikawa Fishbone Diagram & 5 Why Analysis (6M Breakdown)

• Man: Unskilled operator or lack of training with respect to metrology, inspection procedures, and precision accuracy.

• Material: Excessively tough / hard material properties affecting tool cutting dynamics.

• Measurement: Metrological accuracy and precision incorrect → Uncalibrated measurement instruments

used; defects present in precision measurement tools.

• Machine: Increased operational speed causing reduced precision & accuracy; excessive machine vibrations.

• Environment: Natural and ambient environmental errors affecting measurement thermal expansion.

Step 2: Applying Pareto Chart (80/20 Rule)

Prioritize root c

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Taguchi Philosophy and Zero Defect System.

2. Overall System Auditing (ISO 19011 Guidelines):

• Internal Audit: Conducted periodically by internal company audit team.

• External Audit: Conducted by third-party certification bodies / ISO agencies.

3. Continuous Improvement: Always embed Kaizen philosophy for ongoing incremental process refinements.

4. System Benchmarking: Implement External, Functional, and Cross-Functional Benchmarking across all operational lines.

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D8 — TEAM RECOGNITION & PROJECT CLOSURE SIGN-OFF & CLOSURE

• Sign-off & Official Documentation: Formally sign off 8D documentation with authorized signatures from QA/QC Lead and Quality Inspection (QI) Department.

• Recognition & Motivation: Empower team members for their contribution, dedication, and hard work using Deming Award Methodology for team motivation.

• Leadership Philosophy: Follow Deming's principle of providing authority, empowering team members, and striving to "be a leader, not a boss."

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DAY 02

Original source file(s): Day 02_8D_Official_Quality_Report_v2 (1).pdf

Source focus: The Day 2 source material focuses on 8D PROBLEM-SOLVING REPORT REPORT REF: 8D-ECB-2026-01. The source text explicitly discusses the following quality themes and methods: 8D, ROOT CAUSE, Inspection, Taguchi, Isolate, ISO 9001:2015, RCA, FMEA, RPN, 5S, Kaizen, Six Sigma.

Section headings identified in the source: LEAD ENGINEER: Jagriti Singh • Component Details: Engine Cylinder Block / Brake Disc • 1. Automobile Department • 2. Operations Department • 4. Manufacturing Department • 5. Industrial Department

Technical terms detected in the source: 8D, ROOT CAUSE, Inspection, Taguchi, Isolate, ISO 9001:2015, RCA, FMEA, RPN, 5S, Kaizen, Six Sigma, DMAIC, PDCA

Source extract 1: Day 02_8D_Official_Quality_Report_v2 (1).pdf

8D PROBLEM-SOLVING REPORT REPORT REF: 8D-ECB-2026-01

LEAD ENGINEER: Jagriti Singh

AUTOMOTIVE TIER-1 QUALITY ENGINEERING & ROOT CAUSE

ANALYSIS DATE: July 29, 2026

Company Context: Automotive Tier-1 Casting & Precision Machining (Bosch, Bharat Forge, Sundram Fasteners,

Endress+Hauser, etc.)

Component Details: Engine Cylinder Block / Brake Disc

Target Specification: Surface Roughness Ra ≤ 1.6 µm | Observed Defect: Surface Finish Failure (Ra = 3.2 µm – 4.5 µm) | Rejection Rate Spike: 1.2% → 6.8%

D1: ESTABLISH TEAM (CROSS-FUNCTIONAL TEAM)

Formed an inter-departmental Cross-Functional Team (CFT) leveraging Agile methodology and departmental balance:

1. Automobile Department

2. Operations Department

3. Core Management Department (For proper implementation of Agile methodology & cross-functional team balance)

4. Manufacturing Department

5. Industrial Department

6. Supply Chain Department (For material check with respect to parts)

Toyota Philosophy Integration:

If we take suggestions from vendors for good material, it will also help to find the best material for automotive parts. Involving material suppliers directly ensures defect prevention at the raw material stage.

7. Quality Inspection / QA / QC Department

8. Production Department

D2: PROBLEM DEFINITION & DESCRIPTION

Problem Statement: In an Automotive Tier-1 casting and precision machining setup (Bosch, Bharat Forge, Sundram Fasteners, Endress+Hauser), facing a major problem of surface finish with respect to Automotive Brake Disc / Cylinder Block face which exceeds more than 1.6 µm than its actual limit, which causes rejection rate increment from 1.2% to 6.8%, preventing customer retention as well as maximizing customer escalation.

Describing Problem (5W2H Framework):

1. What: Surface finish standard target R ≤ 1.6 µm (Ideal / Standard). Actual measured finish: R = 3.2 µm to 4.5 µm.

a a

Taguchi Quality Loss Philosophy:

According to Taguchi's Quality Loss Function ("Smaller-is-Better" characteristic), driving surface roughness Ra as far below the upper specification limit of 1.6 µm as possible minimizes variation and prevents customer quality loss.

2. Where: Detected during casting inspection & at manufacturing line / production point.

3. When: When the parts were getting rejected at the final stage inspection.

4. Who:

■ 1st Level: Producer (Who is responsible + Caster for matching parts) + Quality Inspection (QI).

■ 2nd Level: Customer escalations.

5. Why:

■ Surface finish exceeds its actual limit due to defects: porosity, holes, gaps, or micro-cracks.

■ Caused if the material inside the parts is not watertight (detected during CAD modelling).

■ Caused if the material does not settle easily during parts making.

6. How Many: Rejection rate increased from 1.2% → 6.8%, which causes customer escalation due to vibration, noise, etc.

8D Problem-Solving Report | Quality Engineering & Manufacturing Page 1 of 3

D3: INTERIM CONTAINMENT ACTIONS (ICA)

1. Stop the production immediately at the manufacturing line

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based on SPC data (Reformation of SOPs based on overall analysis, customer feedback, and

specification limit).

2. ISO 19011 Compliance: Follow auditing principle standard given under ISO 19011 .

3. Benchmarking: Benchmarking with top companies in Function / Machine / Method / System.

4. Zero-Defect Philosophy & Juran's Trilogy: Follow Zero-Defect System Philosophy of Philip Crosby together with

implementing Juran's Trilogy (Quality Planning, Quality Control, Quality Improvement).

5. Poka-Yoke: Integrate Poka-Yoke (Mistake-proofing) on the shop floor.

D8: RECOGNIZE TEAM & SYSTEM CLOSURE

1. Official signature of all Department Heads for formal closure.

2. Recognize, empower, give authority, and appreciate team members for their outstanding approach & hard work.

3. Follow Dr. W. Edwards Deming management philosophy for long-term continuous transformation.

Prepared By: Reviewed By: Approved By:

Jagriti Singh Head of QA / QC Plant Operations Manager

Lead Quality Engineer Quality Assurance Dept. Executive Sign-off

8D Problem-Solving Report | Quality Engineering & Manufacturing Page 3 of 3

DAY 03

Original source file(s): Day 03_Full Analysis 8D Report(1) (1).pdf

Source focus: The Day 3 source material focuses on 8D PROBLEM SOLVING REPORT (QUALITY + SAFETY. The source text explicitly discusses the following quality themes and methods: 8D, ISO 9001, ISO 14000, ISO 45001, ISO 19011, Inspection, Taguchi, Cpk, Isolation, ISO 9001:2015, NDT, Root Cause.

Section headings identified in the source: Target Industry: Aerospace Precision Manufacturing • Inspection Criteria: Fluorescent Penetrant Inspection (FPI) & CMM Measurement • Defect Identified: Thermal Micro-cracks (Grit Burn) & Dimensional Thickness Drift • 1. Manufacturing Department • 2. Production Department • 3. Operational Department

Technical terms detected in the source: 8D, ISO 9001, ISO 14000, ISO 45001, ISO 19011, Inspection, Taguchi, Cpk, Isolation, ISO 9001:2015, NDT, Root Cause, RCA, FMEA

Source extract 1: Day 03_Full Analysis 8D Report(1) (1).pdf

8D PROBLEM SOLVING REPORT (QUALITY + SAFETY

+ EMS)

Day - 3

Project Reference Code: AER-8D2026-TB03

Target Industry: Aerospace Precision Manufacturing

Component Name: High Pressure Gas Turbine Blade (Inconel 718)

Manufacturing Process: 5-Axis CNC Precision Grinding

Applicable Standards: AS9100 Rev D, ISO 9001, ISO 14000 (series), ISO 45001, ISO 19011

Inspection Criteria: Fluorescent Penetrant Inspection (FPI) & CMM Measurement

Target Specification 12.500 ± 0.015 mm (Max: 12.515 mm | Min: 12.485 mm)

(Thickness): Acc. to Taguchi: 12.500 mm (Better) — (Taguchi Philosophy used)

Defect Identified: Thermal Micro-cracks (Grit Burn) & Dimensional Thickness Drift

Baseline Rejection Rate: 6.8% → 5.4%

Date of Report: 30 July, 2026

D1: Team Formation (CFT Formation)

1. Manufacturing Department

2. Production Department

3. Operational Department

4. Quality Management Department & (QA / QC / QI)

5. Supply Chain & Procurement Department

6. Management Department (For overall balance & management of all departments)

7. Research & Development Department (For making methodologies based on overall research)

8. Finance & Accounting Department (For management of overall data & metrics system and its analysis — Responsible for financial budget for overall project analysis.)

9. Human Resources Department (For communication & maintaining connection b/w team — Responsibility of maintaining ethical & strategic workspace planning.)

10. Legal and Compliance Department (For maintaining government regulations)

11. Maintenance / Facility Management: (For servicing of tools & machining)

12. IT Department: (For maintaining data security)

D2: Problem Description (5W2H Framework)

What:

1. Found thermal micro-cracks on FPI.

2. Dimensional variation in CMM Inspections — Blade root thickness exceeds limit (12.500 ± 0.015 mm).

Note: Rejection rate 6.8% → 5.4%

During 5-Axis CNC grinding, due to more vibration, sound level reaches above 95 dB (which is harmful for worker, violates ISO 45001 Worker Safety).

Where:

Aerospace precision machining cell & 5-Axis CNC grinding line.

When:

Batch Inspection During Batch inspection for (TB-2026.07) (30 July 2026).

Who:

• Overall system

• Delay in delivery

• Effect on customer retention / requirements

• Assembly Plant

• Effect on worker

• Effect on Environment

• Aerospace Engine Quality (AS9100)

Why?

1. Because in Aerospace industry Zero Defect philosophy of Philip Crosby is followed with integration of ISO 9001 + AS9100.

2. As per Taguchi specification, must be near to target to prevent defect.

3. Effecting overall Quality & violating standard norms for (QMS, EMS & Worker Safety).

4. Increasing Internal and External failure cost and minimizing prevention cost.

How Much?

Rejection rate reaches to 5.4%. Due to which rework + scrap both cost increases (COPQ), and Cpk index is decreasing.

D3: Containment Actions & Measurement Validation

1. Immediate Containment

• Isolation of stock for Batch (TB-2026-07).

• Putting red tag on all parts (ISO 9001:2015, Cl

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ption:

■ Viscoelastic Damping pads

■ Acoustic Enclosure (Noise reduction) → (Due to which noise level intensity decreased)

6. For Environment:

■ Apply EMS Standard → ISO 14000

■ ISO 14001, ISO 14004, ISO 140011, ISO 140012, ISO 140013

■ For PES: ISO 14021, ISO 14022, ISO 14023

D6: Validation & Verification Process

1. SPC (Statistical Process Control):

Control Chart: S-chart (Deviation), R-chart (Range / Variation)

2. Cpk index measured which indicates control limit specification.

3. Re-inspection for FPI & CMM.

4. Environmental Labelling & Auditing Performed.

D7: Systematic Prevention

1. Use of (PDCA + Kaizen) to prevent failure.

2. Zero-defect system implementation (Philip Crosby).

3. Use of AI Integration + In-situ sensors.

4. Use of high-end NDT technology for micro-cracks and micro-pores.

5. Implementation of SQC.

D8: Closure

1. Signature of all departments head & manager.

2. Employee Empowerment (Edward Deming):

■ Employee appreciation (Edward Deming)

■ Providing of incentives & achievement trophies to employees for their hard work.

■ Provide scope of future or upcoming project.

DAY 04

Original source file(s): Day04_spc_capability_report_final (1).pdf

Source focus: The Day 4 source material focuses on TECHNICAL REPORT: SPC & PROCESS CAPABILITY (Cp & Cpk). The source text explicitly discusses the following quality themes and methods: SPC, PROCESS CAPABILITY, Cp, Cpk.

Section headings identified in the source: TECHNICAL REPORT: SPC & PROCESS CAPABILITY (Cp & Cpk) • Parameter Checked: Blade Root Thickness Lead Inspector: Jagriti Singh (Quality Engineer) • 1. Document Details & Objective • Objectives of the Study: • 2. Sample Data Collection Table • 3. Process Averages & Standard Deviation

Technical terms detected in the source: SPC, PROCESS CAPABILITY, Cp, Cpk

Source extract 1: Day04_spc_capability_report_final (1).pdf

TECHNICAL REPORT: SPC & PROCESS CAPABILITY (Cp & Cpk)

Day-4 Quality Engineering Challenge — 5-Axis CNC Blade Grinding

Component Name: High Pressure Gas Turbine Blade Material: Inconel 718 (Superalloy)

Parameter Checked: Blade Root Thickness Lead Inspector: Jagriti Singh (Quality Engineer)

Target Industry: Aerospace Manufacturing Target Specification: USL = 12.515 mm | LSL = 12.485 mm

1. Document Details & Objective

Target Industry & Application: Aerospace Industry (Precision Superalloy Component Machining).

Objectives of the Study:

• To check whether the 5-Axis CNC machine is manufacturing components to the correct dimension/size or not.

• To evaluate whether our precision grinding process is operating strictly within the customer-provided engineering specifications.

• To perform Statistical Process Control (SPC) charting (X-bar & R chart) and assess Process Capability (Cp, Cpk) against aerospace benchmark thresholds (Cpk ≥ 1.33).

2. Sample Data Collection Table

Subgroup Sample 1 (mm) Sample 2 (mm) Sample 3 (mm) Sample 4 (mm) Mean (X-bar) (mm) Range (R) (mm)

SG-1 12.502 12.505 12.498 12.501 12.5015 0.007

SG-2 12.499 12.503 12.502 12.500 12.5010 0.004

SG-3 12.506 12.501 12.497 12.504 12.5020 0.009

SG-4 12.496 12.500 12.501 12.499 12.4995 0.003

SG-5 12.503 12.502 12.504 12.501 12.5025 0.003

3. Process Averages & Standard Deviation

1. Process Grand Mean (X-double-bar):

X-double-bar = (12.5015 + 12.5010 + 12.5020 + 12.4995 + 12.5025) / 5 = 12.5013 mm

2. Average Range (R-bar):

R-bar = (0.007 + 0.004 + 0.009 + 0.003 + 0.003) / 5 = 0.0052 mm

3. Standard Constants (for subgroup size n = 4):

A2 = 0.729 | d2 = 2.059 | D3 = 0 | D4 = 2.282

4. Standard Deviation Estimation (σ):

σ = R-bar / d2 = 0.0052 / 2.059 = 0.00252 mm

4. Process Capability Calculations (Cp & Cpk)

1. Process Capability Index (Cp):

Cp = (USL - LSL) / (6 × σ) = (12.515 - 12.485) / (6 × 0.00252) = 0.030 / 0.01512 = 1.98

2. Upper Capability Index (Cpu):

Cpu = (USL - X-double-bar) / (3 × σ) = (12.515 - 12.5013) / (3 × 0.00252) = 0.0137 / 0.00756 =

1.81

Day-4 SPC & Capability Study | Quality Engineering Page 1 of 2

3. Lower Capability Index (Cpl):

Cpl = (X-double-bar - LSL) / (3 × σ) = (12.5013 - 12.485) / (3 × 0.00252) = 0.0163 / 0.00756 =

2.15

4. Critical Capability Index (Cpk):

Cpk = min(Cpu, Cpl) = min(1.81, 2.15) = 1.81

Cp Index Cpu Index Cpl Index Final Cpk

1.98 1.81 2.15 1.81

5. Control Limits for X-bar & R Charts

A) X-bar Chart Control Limits:

• Center Line (CL): 12.5013 mm

• Upper Control Limit (UCL): X-double-bar + (A2 × R-bar) = 12.5013 + 0.00379 = 12.5052 mm

• Lower Control Limit (LCL): X-double-bar - (A2 × R-bar) = 12.5013 - 0.00379 = 12.4945 mm

B) R Chart (Range Chart) Control Limits:

• Center Line (CL): 0.0052 mm

• Upper Control Limit (UCL): D4 × R-bar = 2.282 × 0.0052 = 0.0118 mm

• Lower Control Limit (LCL): D3 × R-bar = 0 × 0.0052 = 0 mm

6. Final Observations & Engineering Conclusions

1. Process Capability Status: Cpk = 1.81 (> 1.33 benchmark). The CNC grinding process is highly capable of meeting aerospace quality standards.

2. Process Centering: Process Mean (12.5013 mm) is virtually centered on the target dimension (12.500 mm).

3. Statistical Control Status: PROCESS IN STATISTICAL CONTROL. All subgroup means and ranges remain within control limits with no out-of-control signals.

Day-4 SPC & Capability Study | Quality Engineering Page 2 of 2

DAY 05

Original source file(s): Day 05_ ISRO Process Audit Report.pdf

Source focus: The Day 5 source material focuses on PROCESS AUDIT REPORT. The source text explicitly discusses the following quality themes and methods: AUDIT, inspection, Kaizen, PDCA, FMEA, Process capability, Cpk, Cp, ISO 14000, ISO 14001, ISO 14012, ISO limit.

Section headings identified in the source: PROCESS AUDIT REPORT • Department: Space & Satellite Audit Standard: ISRO Space Quality Standard • Auditor Name & Designation: Jagriti Singh (Quality Component Name: Titanium Mounting Bracket • Objective: Our objective is to combine the aspects of daily routine inspection through • Kaizen (Japanese philosophy) with Deming Wheel Cycle (PDCA) for proper implementation • so that Satellite Titanium Mounting Bracket will achieve zero defect system by eliminating

Technical terms detected in the source: AUDIT, inspection, Kaizen, PDCA, FMEA, Process capability, Cpk, Cp, ISO 14000, ISO 14001, ISO 14012, ISO limit, Six Sigma, 5S

Source extract 1: Day 05_ ISRO Process Audit Report.pdf

PROCESS AUDIT REPORT

SECTION 1: AUDIT OVERVIEW & HEADER DETAILS

Organization: Indian Space Research Organisation Audit Date: 01 / 08 / 2026

(ISRO)

Department: Space & Satellite Audit Standard: ISRO Space Quality Standard

Auditor Name & Designation: Jagriti Singh (Quality Component Name: Titanium Mounting Bracket

Engineer) (Ti-6Al-4V)

Auditee Name & Designation: Senior Quality Part No. & Batch ID: 3S3-ISRO-TMB

Engineer, ISRO

SECTION 2: AUDIT OBJECTIVE & SCOPE

Objective: Our objective is to combine the aspects of daily routine inspection through

Kaizen (Japanese philosophy) with Deming Wheel Cycle (PDCA) for proper implementation

so that Satellite Titanium Mounting Bracket will achieve zero defect system by eliminating

potential system failure before it arises.

Tools Used / Philosophy: PDCA (Deming Wheel Cycle), Kaizen (Japanese Philosophy),

Zero Defect System (Philip Crosby), FMEA (System, Process).

SECTION 3: AUDIT CHECKLIST

Chec Clause / Phase Requirements Observations Statu

klist s

ID

01 PLAN ●■ Operator Competency ●■ Checked operator training & PASS

(Competency & ●■ SOP Technological Standard stamps.

SOP) ●■ Titanium processing certified.

●■ Verification checked.

●■ Process capability checked

[Trend: Cpk + Cp checked]

Chec Clause / Phase Requirements Observations Statu

klist s

ID

02 DO ●■ Tool Calibration ●■ Dimensional Conformance PASS

(Environment & ●■ Environment Management (Bracket length, thickness, bolt

Equipment) System Standards (ISO 14000, hole found in true specification

ISO 14001, ISO 14012, ISO limit set under SOPs)

14013) compliance ●■ MSA performed [Gage R&R

●■ Material risk & Process validation Study performed which

& deviation confirms measurement system

acceptable & reproducible]

03 CHECK ●■ Measurement capability and ●■ Inspection performed for both PASS

(Measurement inspection of two stages (PLAN & stages for their requirements &

Capability & DO) observations using Six Sigma,

Inspection) ●■ MSA (Measurement System System, FMEA.

Analysis)

●■ Digital Micrometer & Vernier

Caliper Inspection

●■ Environmental degradation

detected: Zero surface oxidation

inspection

04 ACT ●■ Sort (Seiri) & Set in order ●■ 5S Methodology properly PASS

(CAPA & 5S (Seiton): Non-inspected and implemented.

Integration) inspected bracket has been ●■ Placement of un-inspected &

separated by red & green visual fully inspected Titanium

bins. brackets properly identified.

●■ Shine & Standardize: Inspection

table layout re-designed with

standard color code.

●■ Sustain: Daily 5S audit

SECTION 4: NC LOG & CAPA TRACKING

NC Reference ID: NC-TB-107 NC Severity Level: Minor NC

Description: Risk of mix-up of un-inspected & inspected Titanium brackets found during check & inspection

stage.

Root Cause Analysis (RCA): Due to unavailability & tagging of visual tags as per SOP standards.

CAPA: Automated Poka-Yoke / AI sensors are used for visual tagging.

Target Completion Date: 01 / 08 / 2026 Action Owner / Responsibility: Jagriti Singh

SECTION 5: AUDITOR CONCLUSION & FINAL DISPOSITION

Overall Audit Assessment Summary:

●■ Properly identified failure risk with System & Process FMEA.

●■ Six Sigma & SQC are applied for reducing variation of specification.

●■ PDCA cycle and 5S methodology are applied for Audit Checklist & Findings stage.

●■ Titanium Bracket manufacturing process satisfies ISRO Space Quality Standard.

Integrated Quality Management tools & philosophy to ensure high repeatability & zero

defect delivery.

Final Audit Status: [✓] APPROVED

Auditor Signature: Jagriti Singh Date: 01 / 08 / 2026

DAY 06

Original source file(s): Day 06 _COQ_Analysis_Report_Jagriti_Singh (1).pdf | Day_69_Editorial_Theme (1) (1).pdf | day_65_nasa_systems_engineering.pdf

Source focus: The Day 6 source material focuses on COST OF QUALITY (COQ) ANALYSIS REPORT. The source text explicitly discusses the following quality themes and methods: audit, Root Cause, Inspection, FMEA, RPN, Six Sigma, NDT, reliability, SPC, Cp, Cpk, RCA. More than one PDF is associated with this day in the ZIP; the files are listed together and have not been silently treated as identical.

Section headings identified in the source: COST OF QUALITY (COQ) ANALYSIS REPORT • Prepared By: Jagriti Singh (Quality Engineer) Status: Approved & Locked for Document Record • 1. Material Extrusion (FDM) • 2. Powder Bed Fusion (SLS / SLM) — [Primary Selected Process] • 3. Vat Photopolymerization (SLA) • 4. Material Jetting (PolyJet)

Technical terms detected in the source: audit, Root Cause, Inspection, FMEA, RPN, Six Sigma, NDT, reliability, SPC, Cp, Cpk, RCA, traceability

Source extract 1: Day 06 _COQ_Analysis_Report_Jagriti_Singh (1).pdf

COST OF QUALITY (COQ) ANALYSIS REPORT

(Additive Manufacturing Line)

Document ID: COQ/AM-3D/2026/08-06 Date: August 02, 2026

Prepared By: Jagriti Singh (Quality Engineer) Status: Approved & Locked for Document Record

SECTION 1: HEADER & PROCESS CLASSIFICATION

• Target Benchmark Enterprises: L&T Advanced Manufacturing , Siemens Additive Solutions , &

Wipro 3D

• Product Line: 3D Printed Precision Enclosures & Brackets

• Batch Size: 1,000 Units

• Standard Framework: PAF Model (Prevention, Appraisal, Internal Failure, External Failure)

Evaluated AM Categories (ISO/ASTM 52900 Standards):

1. Material Extrusion (FDM)

2. Powder Bed Fusion (SLS / SLM) — [Primary Selected Process]

3. Vat Photopolymerization (SLA)

4. Material Jetting (PolyJet)

5. Binder Jetting

6. Directed Energy Deposition (DED)

7. Sheet Lamination (LOM)

Chosen Process: Powder Bed Fusion (Selective Laser Sintering - SLS)

Reason: We selected SLS because it gives strong mechanical parts, needs zero support structure wastage, and gives good dimensional accuracy for industrial enclosures as seen in benchmarks of L&T and Wipro 3D .

SECTION 2: OBJECTIVE & ZERO-DEFECT STRATEGY

Main Objective: Our main goal is to use the PAF model in additive manufacturing to make sure our 3D printing process becomes super effective. We want to build a Zero Defect System so that no failure happens at client side, while saving money for the company at the same time. We are targeting both financial savings and quality improvement together. By spending more money on Prevention and Appraisal early on, we cut down internal scrap and bring external customer failures straight to 0%.

SECTION 3: PAF MODEL COST BREAKDOWN MATRIX

COQ Category Sl. Key Quality Activity Details & Practical Execution Logic Cost (■) %

1. Prevention 1.1 Workforce Training & • First give proper training to all workers and ■ 35,000 7.0%
Costs Skill Assessments employees on how to operate machines.
(Good Quality • Do regular skill assessment sessions to find weak
Cost) areas.
• Make performance and audit reports for each
worker to see where they lack and give targeted
training to save prevention cost.

1.2 CAD Design & FEA • Before printing, focus heavily on CAD modeling. ■ 40,000 8.0%
Analysis • Engineers can analyze stress and strain right
inside CAD to fix issues early.
• Unlike conventional machining where whole part
gets ruined, CAD lets us target and fix the exact
spot.

1.3 Software Updates & • Keep software updated according to new ■ 25,000 5.0%
File Formats standards.
• STL format has issues like not saving colors or
textures, so use better formats (like 3MF/AMF) that
fulfill all needs without drawbacks.

1.4 AI & ML Integration • Integrate AI and Machine Learning into slicers and ■ 30,000 6.0%
3D printing process to make things way more

effective and smart.

1.5 Environmental & • Check materials for environmental degradation ■ 20,000 4.0%

Material Safety and harmful gas emissions.

• Follow air pollution laws so company doesn't face

risk or get ba

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SECTION 4: FINANCIAL & QUALITY IMPACT ANALYSIS

• Good Quality Investment (Prevention + Appraisal): ■ 3,00,000 (60.0%)

• Bad Quality Cost (Internal Scrap/Rework): ■ 2,00,000 (40.0%)

• External Failure Cost: ■ 0 (0.0%)

Summary Analysis: By spending 60% on proper worker training, CAD analysis, AI integration, MSA, Control Charts, and Micro-CT scanning, we stopped defects from reaching the customer. All failure costs stayed inside factory as routine internal scrap (40%), completely eliminating external claims and brand damage.

SECTION 5: RECOMMENDATIONS & FINAL SIGN-OFF

1. AI Auto-Correction: Hook up the AI-based hybrid control charts directly with 3D printer controls so it catches warping and nozzle issues early and adjusts settings automatically.

2. File Format Shift: Move completely away from basic STL format to formats like 3MF/AMF so texture and color data are not lost during slicing.

3. Fume Control: Set up better exhaust and air filtration inside factory so escaping gases don't damage factory equipment or degrade raw materials.

Report Prepared By:

Jagriti Singh

Quality Engineer

The next file is retained as a separate source version; differences have not been reconciled without a version audit.

Source extract 2: Day_69_Editorial_Theme (1) (1).pdf

L’ O R É A L

P R O D U C T C O N S I S T E N C Y & Q U A L I T Y AT S C A L E

4 T H O C TO B E R | DAY 6 9

DISCLAIMER: This is an independent educational case study created as part of my “100 Days Quality Management Challenge.” L’Oréal is a

separate organisation. This is not an official L’Oréal project, report, collaboration, or representation. The analysis applies general Quality

Engineering principles using publicly available information.

Q UA L IT Y C HA L L E N G E

In large-scale cosmetic production, maintaining consistent viscosity, texture, appearance and performance across batches is an important quality challenge.

A batch may remain within specification while showing a gradual increase in variation or process drift. If this trend is not detected early, it can eventually affect product consistency and increase quality risk.

Quality Engineering Question:

How can process variation be detected and controlled before it becomes a consumer-facing quality issue?

Q UA L IT Y E N G I N E E R I N G TO O LS

MSA — Verify measurement reliability SPC — Monitor variation & trends Cp/Cpk — Evaluate capability & centering

Trend Analysis — Detect gradual drift RCA — Identify underlying causes FMEA — Assess & prioritize risks

PRO P O S E D Q UA L I T Y A PPROAC H

MEASURE → MONITOR → DETECT → ANALYSE → CORRECT → VERIF Y → PREVENT

• 1. Measurement Reliability: Validate the measurement system before making process decisions.

• 2. Process Monitoring: Use SPC and trend analysis to identify unusual variation or gradual drift.

• 3. Capability Assessment: Use Cp/Cpk to evaluate process spread and centering against specifications.

• 4. Root Cause & Corrective Action: Investigate significant variation and address the underlying process cause.

• 5. Verification & Prevention: Verify corrective-action effectiveness and continue monitoring to prevent recurrence.

E X P E C T E D Q U A L I T Y I M PA C T

• Improved batch-to-batch consistency • Fewer recurring quality issues

• Earlier detection of process drift • Better process predictability

• Reduced process variation • Stronger prevention-based quality

« “Quality at scale is not achieved by inspecting more. It is achieved by controlling variation before it becomes a defect.” »

I N D U S T R Y: B E A U T Y & C O S M E T I C S • Q U A L I T Y F O C U S : P R O C E S S Q U A L I T Y • P R O D U C T

C O N S I S T E N CY • VA R I AT I O N C O N T R O L

The next file is retained as a separate source version; differences have not been reconciled without a version audit.

Source extract 3: day_65_nasa_systems_engineering.pdf

NASA MISSION ASSURANCE DAY 6 5 • TEC H N I CA L CAS E STU DY

Systems Engineering, Risk Management & Mission Assurance

How Quality Engineering helps control risk, manage changes, verify systems and improve mission reliability.

Date: 30 September 2026 Author: Jagriti Singh Domain: Aerospace Quality & Systems

Educational Note: This is an educational, illustrative case study based on publicly available systems-engineering and quality principles. It does not describe NASA's confidential or proprietary internal processes.

1. THE QUALITY CHALLENGE

A space mission is not made of one component. It is a combination of many connected systems such as:

Structure + Power + Communication + Sensors + Software + Thermal Systems + Control Systems

A problem in one subsystem can affect other parts of the mission. Therefore, Quality Engineering must look at the complete system, not only

at individual components.

Component → Subsystem → Spacecraft → Mission

Systems Engineering: Means looking at the complete system and understanding how its different parts work together to achieve the

mission objective.

2. RISK IDENTIFICATION

Before a mission moves forward, engineers need to ask: “What can go wrong?”

POSSIBLE RISKS INCLUDE RISK ANALYSIS & APPROACH

• Component failure or sensor error Risk Analysis: Identifying possible problems early,

• Software error / bus interruption understanding impact, and deciding priority.

• Power interruption / Communication loss Identify → Analyse → Control → Verify

• Manufacturing variation

• Incorrect design or specification

• Uncontrolled engineering changes

3. FAULT TREE ANALYSIS — FTA

What is FTA? A method used to start with a major failure and work backwards to find its possible causes. Instead of asking “What

component failed?” we ask “What different conditions could have caused this major failure?”

TOP EVENT: Spacecraft loses required function

• Power System Problem (OR) | • Communication System Problem (OR) | • Control System Problem

AND vs OR Logic: OR: Any one condition causes failure. | AND: Two or more conditions occur together.

Why FTA is useful: Helps engineers understand how major failures occur, identify failure paths, and focus on critical causes.

4. CONFIGURATION MANAGEMENT

Making sure that the correct and approved version of a design, drawing, software, specification or process is being used.

Design v1 → Change → Version 2 → Approval → Testing → Release

Why important? Prevents confusion between design, manufacturing, and testing teams through traceability.

DAY 65 — Jagriti Singh | NASA Mission Assurance Page 1 of 2

5. VERIFICATION VS VALIDATION

These two words sound similar, but they answer different questions in aerospace quality engineering.

VERIFICATION VALIDATION

Question: “Did we build the product according to specified Question: “Does the product actually work for its intended

requirements?” purpose?”

Checks requirements via inspection, analysis, calculation, Looks at system suitability for its intended operating

test

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. Problems identified must be: Recorded → Investigated → Corrected →

Retested.

7. ANOMALY ANALYSIS

What is an anomaly? Something unexpected happens during testing or operation (behavior differs from prediction).

Quality Approach: Anomaly Detected → Record → Collect Evidence → Root Cause → Corrective Action → Retest → Effectiveness

Check.

Important principle: Fixing the symptom is not enough; understand why it happened to prevent recurrence.

8. FINAL QUALITY ENGINEERING FRAMEWORK

NASA Mission Quality Approach:

System Requirements → Risk ID → Systems Eng → FTA → Config Control → Verification → Validation → Anomaly Analysis → Corrective Action →

Retesting → Effectiveness → Mission Assurance

KEY TAKEAWAY

Quality Engineering in space is about understanding the complete system, identifying risks early, controlling changes, testing carefully,

investigating anomalies, and ensuring mission readiness.

“Build it correctly, control every important change, test it with evidence, learn from failures, and continuously improve reliability.”

DAY 65 — Jagriti Singh | NASA Mission Assurance Page 2 of 2

DAY 07

Original source file(s): Day_07_Taguchi_QLF_Quality_Report_Jagriti (1).pdf

Source focus: Day 7 source document titled 1 0 0 - DAY Q UA L I T Y E N G I N E E R I N G C H A L L E N G E • DAY 0 7.

1 0 0 - DAY Q UA L I T Y E N G I N E E R I N G C H A L L E N G E • DAY 0 7

Taguchi’s Quality Loss Function (QLF) & Robust Design

PREPARED BY TARGET INDUSTRY COMPONENT / FEATURE INDUSTRY BENCHMARKS

Jagriti Singh Consumer Electronics & Smartwatch Unibody Groove Apple Hardware / Samsung

Hardware Depth MX

1. Executive Summary & Context

In consumer electronics hardware manufacturing (for top brands like Apple Hardware Supply Division and

Samsung MX Operations), traditional quality checks follow a simple rule: if a dimension is inside the Upper

Specification Limit (USL) and Lower Specification Limit (LSL), it passes with zero loss.

In this study, I used Genichi Taguchi’s Quality Loss Function (QLF) to check what happens when dimensions

deviate from the exact target depth (2.00 mm). My goal was to see how small variations inside the tolerance band

still create real financial losses because of seal wear, loose fitment, and future warranty risks.

MATHEMATICAL BASELINE & FORMULA SETUP

Quality Loss Formula: $L(y) = k \times (y - m)^2$ | Loss Coefficient: $k = A / (\Delta)^2$

Where: m = Target Value (2.00 mm), Δ = Tolerance Limit (±0.05 mm), A = Full Rejection / Scrap Cost (■50,000)

$k = 50,000 / (0.05)^2 = 50,000 / 0.0025 = 20,000,000$ (2 Crore)

2. Data & Calculation Analysis (Traditional QA vs. Taguchi Model)

TAGUCHI

SAMPLE MEASURED DEVIATION (Y TRADITIONAL PRACTICAL

FINANCIAL LOSS

ID SIZE (Y) - M) QA OPERATIONAL IMPACT

L(Y)

Exact Target Match

Sample 1 2.00 mm 0.00 mm PASS (■0) ■0

(Zero Defect)

Hidden Loss (Minor

Sample 2 2.02 mm +0.02 mm PASS (■0) ■8,000

Sealing Degradation)

Borderline Limit (Max

Sample 3 2.05 mm (USL) +0.05 mm PASS (■0) ■50,000

Tolerance Loss)

Critical Defect (Assembly

Sample 4 2.07 mm (Out) +0.07 mm REJECT (■50k) ■98,000

Failure)

3. Engineering Insights & Key Takeaways

1. Sample 1 as the Perfect Target Standard

Sample 1 (2.00 mm) hits the exact mathematical target size. Since it does not drift toward USL or LSL limits, it generates zero loss. Staying right on target reduces internal failure costs (rework/scrap) and external failure costs (warranty returns). In fact, appraisal costs also go down because we don't need excessive end-of-line sorting.

2. Catching Hidden Financial Losses (Samples 2 & 3)

Traditional inspection calls Sample 2 (2.02 mm) and Sample 3 (2.05 mm) fully acceptable. But applying Taguchi's model reveals hidden financial losses of ■8,000 and ■50,000. For high-precision suppliers like Foxconn Technology Group or L&T Precision Engineering , these small dimensional shifts compromise water resistance (IP68) over time and cause field failures.

3. Building a Zero-Defect System with Kaizen & Deming Cycle (PDCA)

To keep parts consistently close to the 2.00 mm target size, quality engineering needs to combine Kaizen and the Deming Wheel (PDCA Cycle):

• Plan: Identify root causes making dimensions shift away from 2.00 mm (like CNC tool wear or thermal variation).

• Do & Check: Run micro-Kaizen fixes on fixture holding and coolant temperature control.

• Act: Standardize control limits around the exact target size to keep overall process variation low.

DISCLAIMER & CONTEXT OF WORK

• Self-Learning & Domain Knowledge Initiative: This report is created as part of my self-directed learning journey for the 100-Day Quality Engineering Challenge.

• Raw / Dummy Data Utilization: All dimensions, parameters, formulas, and financial values used here are raw, synthetic dummy data prepared strictly for learning and demonstration purposes. No proprietary company data was used.

• Target Application Alignment: The framework and analysis shown here are designed to address practical quality engineering problems in modern consumer electronics and precision hardware manufacturing.

DAY 08

Original source file(s): Day 08_Supplier_Quality_Rating_Report_Day08_Updated (1).pdf

Source focus: The Day 8 source material focuses on SUPPLIER QUALITY RATING SYSTEM (SQRS) AUDIT. The source text explicitly discusses the following quality themes and methods: SUPPLIER QUALITY, AUDIT, reliability, 8D, PDCA, Cp, Cpk.

Section headings identified in the source: SUPPLIER QUALITY RATING SYSTEM (SQRS) AUDIT • Prepared By: Jagriti Singh (Lead Quality Engineer) Supplier Name: Bharat Forge Limited • 1. AUDIT BACKGROUND & KEY PRIORITIES • Core Operational Focus Areas: • 2. PARAMETER CALCULATION & FORMULA BREAKDOWN • Evaluation Metric Formula Used (Mathematical Rule) Monthly Data & Calculation

Technical terms detected in the source: SUPPLIER QUALITY, AUDIT, reliability, 8D, PDCA, Cp, Cpk

Source extract 1: Day 08_Supplier_Quality_Rating_Report_Day08_Updated (1).pdf

SUPPLIER QUALITY RATING SYSTEM (SQRS) AUDIT

Monthly Vendor Evaluation Report — Day 08 | Quality Engineering Challenge

Report Ref: SQRS-BFL-2026-07 Audit Month: July 2026

Prepared By: Jagriti Singh (Lead Quality Engineer) Supplier Name: Bharat Forge Limited

Component: Precision Helical Gears Target Limit: < 1,000 PPM Defect Rate

1. AUDIT BACKGROUND & KEY PRIORITIES

In our transmission line assembly, overall reliability relies heavily on the quality of incoming gear components. During July 2026, we audited Bharat Forge Limited to assess their monthly supply of precision helical gears. This rating framework helps us catch quality gaps early, avoid line stoppage due to delivery delays, and build a practical zero-defect workflow with our tier-1 vendors.

Core Operational Focus Areas:

• Defect Minimization (PPM): Keeping rejected parts as low as possible during receiving checks.

• Delivery Timeliness (OTD): Ensuring shipments arrive strictly on schedule so the assembly line never sits idle.

• Drawing Specification Compliance: Making sure every gear strictly meets engineered micro-tolerances and surface finish requirements.

• Responsiveness: Quick turnaround on quality queries and submitting effective 8D root-cause reports when issues pop up.

2. PARAMETER CALCULATION & FORMULA BREAKDOWN

Below is the step-by-step scoring breakdown showing exact evaluation formulas alongside actual operational figures:

Score

Evaluation Metric Formula Used (Mathematical Rule) Monthly Data & Calculation

Earned

Quality Rating (PPM) PPM = (Rejections / Total Received) Total Received = 50,000 units 39.96 / 40

(40% Weightage) × 10^6 Total Rejections = 50 units

PPM = 1,000 PPM

Score = 40 × [1 - (Rejections / Score = 40 × [1 - (50 / 50,000)]

Total)]

Delivery Rating (OTD) Delivery Score = 30 × (On-Time Scheduled Batches = 35 30.00 / 30

(30% Weightage) Batches / Total Scheduled) On-Time Batches = 35

Score = 30 × (35 / 35)

System Audit Score Audit Score = 15 × (Audit Compliance Shopfloor Audit = 90% 13.50 / 15

(15% Weightage) % / 100) Score = 15 × (90 / 100)

8D Responsiveness Response Score = 15 × (Turnaround Requirement: < 48 Hours 15.00 / 15

(15% Weightage) Compliance Factor) Actual: Submitted in 24 Hours

Score = 15 × 1.00

3. FINAL SQRS AUDIT SCORECARD

Evaluation Parameter Weightage Target Benchmark Actual Result Final Earned Score

Quality Performance (PPM) 40% < 1,000 PPM 1,000 PPM (50 Rejections) 39.96 / 40

Delivery Performance (OTD) 30% 100% On-Time 35 / 35 Batches 30.00 / 30

Process System Audit 15% > 85% Compliance 90% Audit Rating 13.50 / 15

8D Issue Turnaround 15% < 48 Hours Submitted in 24 Hours 15.00 / 15

TOTAL OVERALL SCORE 100% Grade A (> 90%) GRADE A PREFERRED 98.46 / 100

4. NEXT STEPS & VENDOR IMPROVEMENT PLAN

Scoring 98.46% keeps Bharat Forge in our Grade A Preferred Vendor category. To push our quality standards further towards a zero-defect target, we are putting three main plans into action:

• Direct Vendor Integration (Early Involvement): We are involving Bharat Forge application engineers directly in our plant design reviews. Catching tooling and tolerance issues during design saves both teams time and prevents unexpected floor rejections.

• PDCA Action Plan:

■ Plan: Work together to reduce incoming defects down from 1,000 PPM to under 500 PPM by next quarter.

■ Do: Run regular spot-checks at receiving and verify critical pitch measurements.

■ Check: Review monthly defect Pareto charts and double-check shopfloor capability stats (Cp/Cpk).

■ Act: Standardize working steps across both teams once a corrective fix proves successful.

• Mistake-Proofing (Poka-Yoke): Bharat Forge has installed an optical pitch sensor on their gear hobbing machine to check tooth geometry before packing, stopping defective gears right at their plant.

Prepared By: Approved By:

Jagriti Singh Head of Supplier Quality Assurance (SQA)

Lead Quality Engineer Plant Operations & Management

DAY 09

Original source file(s): Day09_OEE_Report_Nestle_Targeted (1).pdf | Day_94_Quality_Engineering_NABL_Case_Study-v3 (1).pdf

Source focus: The Day 9 source material focuses on TARGET COMPANY: NESTLÉ S.A.. The source text explicitly discusses the following quality themes and methods: OEE, Audit, ISO 14001, ISO Compliance, ISO eco-sustainability, root cause, RCA, FMEA, PFMEA, RPN, PDCA, Kaizen. More than one PDF is associated with this day in the ZIP; the files are listed together and have not been silently treated as identical.

Section headings identified in the source: Lead Engineer & Quality Auditor: Jagriti Singh • Project Scope: OEE Audit & Quality Control Analysis • 1. TARGET INDUSTRY, PROBLEM STATEMENT & AUDIT OBJECTIVE • Target Industry: Fast-Moving Consumer Goods (FMCG) / Food & Beverage Packaging • Target Production Line: Automated VFFS High-Speed Eco-Friendly Packaging Line • The Core Operational Challenges Identified:

Technical terms detected in the source: OEE, Audit, ISO 14001, ISO Compliance, ISO eco-sustainability, root cause, RCA, FMEA, PFMEA, RPN, PDCA, Kaizen, ISO GOVERNANCE, ISO Standards

Source extract 1: Day09_OEE_Report_Nestle_Targeted (1).pdf

TARGET COMPANY: NESTLÉ S.A.

DAY 09: OEE & 6 BIG LOSSES ANALYSIS REPORT

Lead Engineer & Quality Auditor: Jagriti Singh

Project Scope: OEE Audit & Quality Control Analysis

Project Context: Part of 100-Day Self-Learning Challenge Series (Personal Knowledge & Skill Growth)

1. TARGET INDUSTRY, PROBLEM STATEMENT & AUDIT OBJECTIVE

Target Company: Nestlé S.A.

Target Industry: Fast-Moving Consumer Goods (FMCG) / Food & Beverage Packaging

Target Production Line: Automated VFFS High-Speed Eco-Friendly Packaging Line

During the transition of a high-speed Vertical Form-Fill-Seal (VFFS) snack packaging line from standard flexible plastic to eco-friendly biodegradable pouch material, key operational challenges were identified during shift execution.

The Core Operational Challenges Identified:

• Sealing Leakage & Efficiency Losses: Thermal sensitivity of biodegradable film causes seam line weakness upon minor temperature fluctuations, resulting in continuous pouch sealing leaks and line efficiency drop.

• Environmental & Regulatory Compliance: Non-biodegradable plastic usage poses environmental risks; fixing pouch integrity issues is mandatory to comply with ISO 14001 (Environmental Management System) standards.

Main Objectives of the OEE Audit:

1. Reduce unplanned machine downtime by replacing breakdown maintenance with proactive checks.

2. Eliminate product scrap and rework arising from defective heat sealing.

3. Optimize line throughput and overall shopfloor productivity.

4. Ensure strict adherence to ISO 14001 environmental sustainability guidelines.

2. SHIFT OPERATIONAL DATA & STEP-BY-STEP OEE CALCULATIONS

Operational numbers recorded from an 8-hour shift execution on the line:

Given Input Data (Collected from Shift Logs):

• Total Shift Duration: 8 Hours = 480 Minutes

• Planned Breaks (Lunch/Tea): 30 Minutes

• Planned Production Time: 480 - 30 = 450 Minutes

• Unplanned Downtime: 60 Minutes (approx 7.5 mins lost per hour due to sealing bar adjustments and film alignment)

• Actual Operating Time: 450 - 60 = 390 Minutes

• Target Machine Speed: 100 Pouches per Minute (PPM)

• Expected Output in 390 Mins: 390 × 100 = 39,000 Pouches

• Total Output Produced: 39,000 Pouches

• Rejection Rate: 5% (due to improper seal leakage and weight variation)

• Rejected Pouches (Scrap): 39,000 × 0.05 = 1,950 Pouches

• Good Pouches (First Time Right): 39,000 - 1,950 = 37,050 Pouches

Step-by-Step OEE Metric Calculations:

1. Availability Rate (%)

Formula: (Operating Time / Planned Production Time) × 100

Calculation: (390 / 450) × 100 = 86.67%

2. Performance Rate (%)

Formula: (Actual Output / Target Output) × 100

Calculation: (39,000 / 39,000) × 100 = 100.00%

3. Quality Rate (%)

Formula: (Good Pouches / Total Output) × 100

Calculation: (37,050 / 39,000) × 100 = 95.00%

Final Overall OEE Score:

OEE = Availability × Performance × Quality

OEE = 0.8667 × 1.0000 × 0.9500 × 100 = 82.33%

Audit Inference: World-class target benchmark is 85%. Current performance at 82.33% highlights the need

t

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r handling.

2. Daily Audit Report File: Logbook tracking shift OEE indices, 6 Big Losses breakdowns, scrap quantities, and sensor performance metrics.

3. ISO Standards Compliance Report: Mapping shopfloor execution directly to international ISO frameworks:

■ ISO 9001: Quality management framework for raw material integrity and First-Time-Right production.

■ ISO 14001 & ISO 14004: Operational guidelines for Environmental Management Systems (EMS) and sustainable packaging compliance.

■ ISO 14011 & ISO 14012: Governance procedures for internal quality/environmental audits and auditor competencies.

■ ISO 14020, 14021, 14022 & 14023: Eco-labeling and self-declared environmental claim validation for biodegradable polymers.

Self-Learning & Knowledge Ownership Declaration:

This OEE report and industrial audit framework has been prepared by Jagriti Singh through independent technical research and analytical study. Executed as part of the 100-Day Self-Learning Challenge Series to strengthen core competencies in Quality Engineering, Total Quality Management (TQM), and Sustainable Manufacturing.

The next file is retained as a separate source version; differences have not been reconciled without a version audit.

Source extract 2: Day_94_Quality_Engineering_NABL_Case_Study-v3 (1).pdf

1 0 0 DAYS Q UA L I T Y E N G I N E E R I N G C H A L L E N G E — DAY 9 4

Building Reliable Laboratory Results: An

Integrated Approach to Quality

Management and Measurement Assurance

NABL-Inspired Case Study on ISO/IEC 17025, Calibration, Measurement Uncertainty,

and Laboratory Quality Improvement

Document Objective: Technical Educational Case Study & Advanced Quality Engineering Guide

Target Audience: Quality Engineers, Metrologists, Laboratory Managers, and Technical Interview Candidates

Core Standards: ISO/IEC 17025:2017, NABL Specific Criteria, ISO 9001

Author: Quality Engineering Excellence Initiative

Notice: This document is an educational case study inspired by laboratory accreditation principles. It does not represent actual NABL events, official audits, or specific commercial laboratory data.

1. Document Objective & Context

This technical educational case study forms Day 94 of the 100 Days Quality Engineering Challenge. The central objective is to explain how a testing and calibration laboratory can transform variable, inconsistent testing data into highly reliable, metrologically sound results through an integrated Laboratory Quality Management System (LQMS).

Using the National Accreditation Board for Testing and Calibration Laboratories (NABL) as the organizational context, this guide illustrates the practical interplay between ISO/IEC 17025 requirements, calibration management, measurement uncertainty, Measurement System Analysis (MSA), personnel competence, and proficiency testing.

Educational Disclaimer: This case study is a hypothetical educational model created for professional training and interview preparation. It is inspired by established accreditation frameworks (ISO/IEC 17025 and NABL guidance). It does not present real-world

security incidents, confidential client data, or official NABL audit findings.

4. Introduction to NABL & Laboratory Accreditation

What is NABL?

The National Accreditation Board for Testing and Calibration Laboratories (NABL) is an autonomous body operating

under the guidance of the Quality Council of India (QCI). NABL provides formal third-party recognition of the technical competence of testing and calibration laboratories, medical laboratories, reference material producers (RMPs), and proficiency testing providers (PTPs) in accordance with international standards.

Understanding Laboratory Accreditation vs. ISO 9001 Certification

A common misconception among early-career engineers is confusing ISO 9001 certification with laboratory accreditation under ISO/IEC 17025. The distinction is critical:

Parameter ISO 9001 Certification NABL Accreditation (ISO/IEC 17025)

Core Focus Quality Management System (QMS) pro■ Technical competence, valid testing meth■

cesses and customer satisfaction. odologies, and measurement accuracy.

Evaluation Basis Conformity to broad management system Demonstrated technical capability evalu■

requirements across any organization. ated by qualified technical assessors in

specific te

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ry.

Intermediate Checks In-house periodic performance checks between external calibra■

tions using check-standards and control charts.

Decision Rules & Guardbanding Documented rules for declaring pass/fail near specification limits considering expanded measurement uncertainty.

Key Formula Reference

• Combined Standard Uncertainty: $u_c = \sqrt{(u_1^2 + u_2^2 + ... + u_n^2)}$

• Expanded Uncertainty: $U = k \cdot u_c$ (typically k = 2 for 95% coverage probability)

• PT Z-Score: $Z = (x - X) / \sigma$

22. References & Standards Framework

• ISO/IEC 17025:2017: General requirements for the competence of testing and calibration laboratories.

• NABL-100: General Information Brochure on NABL Accreditation Services.

• JCGM 100:2008 (GUM): Evaluation of measurement data — Guide to the expression of uncertainty in measurement.

• ILAC-G8:09/2019: Guidelines on Decision Rules and Statements of Conformity.

• ISO 5725: Accuracy (trueness and precision) of measurement methods and results.

• NABL-163: Policy for Participation in Proficiency Testing Activities.

100 Days Quality Engineering Challenge — Day 94 Page 15 of 15

DAY 10

Original source file(s): Day 10_Quality_Engineering_Master_PlaybookFinal_v3.pdf.pdf

Source focus: The Day 10 source material focuses on QUALITY ENGINEERING & TQM MASTER PLAYBOOK. The source text explicitly discusses the following quality themes and methods: ISO Standards, Audit, SPC, Process Capability, Cp, Cpk, Root Cause, ISO 9001, ISO 2859-1, OEE, ISO 14001, ISO 22000.

Section headings identified in the source: QUALITY ENGINEERING & TQM MASTER PLAYBOOK • 1. STRATEGIC RATIONALE: WHY DAY 10 IS A MASTER COMPILATION OVER A NEW CASE STUDY • 2. FULL DAY-BY-DAY AUDIT BREAKDOWN WITH ISO COMPLIANCE (DAYS 01–09) • 3. INTERNATIONAL ISO STANDARDS & INDUSTRIAL GOVERNANCE MATRIX • 4. TOTAL QUALITY MANAGEMENT GURUS PHILOSOPHICAL MATRIX • 5. COMPREHENSIVE QUALITY ENGINEERING & TQM TOOLKIT MATRIX

Technical terms detected in the source: ISO Standards, Audit, SPC, Process Capability, Cp, Cpk, Root Cause, ISO 9001, ISO 2859-1, OEE, ISO 14001, ISO 22000, isolated, Taguchi

Source extract 1: Day 10_Quality_Engineering_Master_PlaybookFinal_v3.pdf.pdf

QUALITY ENGINEERING & TQM MASTER PLAYBOOK

(COMPLETE SYNTHESIS)

Full Industrial Breakdown, ISO Standards, TQM Gurus & 13 Core Tools | Days 01–09 Master Audit

Author & Lead Quality Engineer: Jagriti Singh | Ref Code: QE-100D-MASTER-2026-10

TARGETED INDUSTRIAL ORGANIZATIONS & BENCHMARKS (DAYS 01–09)

Day 01–03 (Automotive & Precision Eng.): Tata Motors Limited [Hole Diameter Specification Limits (USL/LSL), SPC Control Charts & Process Capability Cp/Cpk]

Day 04 (Heavy Infrastructure & EPC): Larsen & Toubro (L&T;) Construction [Defect Pareto Analysis & Ishikawa Root Cause Analysis]

Day 05 (Aerospace & Defense): Indian Space Research Organisation (ISRO) [Titanium Mounting Bracket Audit, AS9100D & ISO 9001 Compliance]

Day 06 (Advanced Manufacturing): Additive Manufacturing / 3D Printing Operations [Cost of Quality (COQ) Ledger & ISO/ASTM 52900 Compliance]

Day 07 (Consumer Electronics): Apple Hardware & Samsung MX Benchmarks [Smartwatch Body Groove Depth & ISO 2859-1 Sampling]

Day 08 (Automotive Tier-1 Supply Chain): Bharat Forge Limited [Precision Helical Gears SQRS Audit & IATF 16949 Compliance]

Day 09 (FMCG High-Speed Packaging): Nestlé S.A. [Automated VFFS Packaging Line OEE, ISO 14001 & ISO 22000 Compliance]

1. STRATEGIC RATIONALE: WHY DAY 10 IS A MASTER COMPILATION OVER A NEW CASE STUDY

• Consolidation & Cross-Domain Synergy: Days 01–09 covered isolated domains (Machining, Civil Infrastructure, Aerospace, 3D Printing, Electronics,

Automotive Supply Chain, FMCG). Day 10 acts as the strategic bridge to connect these frameworks into an enterprise-level Quality Management Playbook.

• Transition from Operational Execution to Systemic Audit: Rather than solving a 10th standalone problem, Day 10 establishes a unified Quality

Governance Framework (QGF) that aligns ISO Standards (ISO 9001, IATF 16949, AS9100D, etc.) with classical TQM Philosophies.

• Executive-Ready Portfolio & Reference Architecture: Standardizing metrics like Cp/Cpk, Taguchi Loss Function L(y), COQ, SQRS, and OEE into a single

Master Index transforms isolated exercises into an industry-grade audit playbook for senior quality leadership roles.

• Knowledge Retention & Synthesis Milestone: Completing a 10-day sprint requires a milestone review to lock in fundamental learnings, verify compliance

completeness, and prepare the foundation for advanced Quality 4.0 implementations.

2. FULL DAY-BY-DAY AUDIT BREAKDOWN WITH ISO COMPLIANCE (DAYS 01–09)

• Day 01–03 | Precision Machining & Statistical Control (Tata Motors): Analyzed engine bore hole specs (USL/LSL), built X-bar/R control charts, and

calculated Cp/Cpk indices under ISO 9001 (Quality Management System) guidelines.

• Day 04 | Infrastructure Structural Audit (L&T; Construction): Applied Pareto 80/20 rule and Fishbone RCA on structural welding defects in compliance

with ISO 3834 (Quality Requirements for Fusion Welding).

• Day 05 | Aerospace Space-Grade Process Audit (ISRO): Audited titanium brackets using PDCA, 5S visual bins, and AI visual

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r L&T; structural flaws and Nestlé VFFS sealing bar

downtime (Days 04, 09).

PDCA (Deming Cycle) Process Audit & Governance Structured 4-step loop for ISRO space quality compliance and Nestlé packaging line

audits (Days 05, 09).

Poka-Yoke & AI Sensors Error Proofing Mechanisms Installed visual tagging at ISRO and sensor checks at Nestlé to prevent part mix-ups

(Days 05, 09).

Cost of Quality (COQ) Quality Financial Ledger Divided 3D printing budget into Prevention, Appraisal, and Failure to ensure ■0 external

returns (Day 06).

Taguchi Loss Function L(y) Robust Tolerancing Quantified hidden monetary loss inside spec limits for smartwatch sealing groove

depths (Day 07).

SQRS & PPM / OTD Vendor Rating System Rated Bharat Forge helical gear shipments on PPM defects, OTD, and 8D response

(98.46% score) (Day 08).

OEE & 6 Big Losses Equipment Efficiency Measured Availability (86.67%), Performance (100%), and Quality (95%) on Nestlé

packaging line (Day 09).

ISO Governance Framework Standards Compliance Aligned operations with ISO 9001, IATF 16949, AS9100D, ISO 14001, ISO 22000,

ISO/ASTM 52900.

DAY 11

Original source file(s): Day11_Universal_Supplier_Quality_SOP_Clean_White.pdf

Source focus: The Day 11 source material focuses on 1 0 0 - D AY Q U A L I T Y E N G I N E E R I N G C H A L L E N G E • D AY 1 1. The source text explicitly discusses the following quality themes and methods: SUPPLIER QUALITY, FAI, Inspection, ISO-BASED, ISO standard, Root Cause, process capability, ISO STANDARDS, ISO 9001:2015, 8D, ISO Clause.

Section headings identified in the source: UNIVERSAL SUPPLIER QUALITY & FAI COMPLIANCE • 1. OBJECTIVE & UNIVERSAL SCOPE • 2. AUTOMATED & ISO-BASED DUAL SENSOR GRADING SYSTEM • based monitoring system is integrated into the inspection framework: • 3. FAI REJECTION & MANDATORY 30-DAY SCAR RECOVERY PROTOCOL • Daily Documentation Requirements:

Technical terms detected in the source: SUPPLIER QUALITY, FAI, Inspection, ISO-BASED, ISO standard, Root Cause, process capability, ISO STANDARDS, ISO 9001:2015, 8D, ISO Clause

Source extract 1: Day11_Universal_Supplier_Quality_SOP_Clean_White.pdf

1 0 0 - D AY Q U A L I T Y E N G I N E E R I N G C H A L L E N G E • D AY 1 1

UNIVERSAL SUPPLIER QUALITY & FAI COMPLIANCE

SOP

Document Reference: SOP-SQA-2026-011 | Standard Operating Procedure

1. OBJECTIVE & UNIVERSAL SCOPE

This Standard Operating Procedure defines a universal framework for onboarding new suppliers, evaluating First Article Inspection (FAI) submissions, and enforcing structured recovery protocols when non-conformances occur. Designed as an industry-agnostic standard, this document applies universally across manufacturing, automotive, aerospace, precision engineering, and tech hardware sectors.

2. AUTOMATED & ISO-BASED DUAL SENSOR GRADING SYSTEM

To ensure objective, real-time evaluation of incoming materials and supplier performance, a dual-layer sensor-based monitoring system is integrated into the inspection framework:

• Layer 1 (Automated Operational Sensors): Processes operational and dimensional data as components pass through initial inspection. The system automatically categorizes suppliers into three distinct risk tiers:

■ Approved Supplier: Granted upon 100% compliance during FAI with zero critical or major defects.

■ Medium Risk Supplier: Identified when minor non-critical deviations are detected that can be rectified rapidly without structural impact.

■ High Risk Supplier: Triggered immediately upon FAI rejection, major dimensional out-of-tolerance conditions, or material specification failures.

• Layer 2 (ISO-Based Smart Compliance Sensors): Performs automated scanning of physical material properties, surface finishes, and geometry directly against ISO standard specifications. This eliminates human inspection error and ensures absolute technical verification against international standards.

3. FAI REJECTION & MANDATORY 30-DAY SCAR RECOVERY PROTOCOL

When a new supplier fails the First Article Inspection (FAI), a mandatory Supplier Corrective Action Request (SCAR) protocol is initiated. The top three root causes identified via Pareto Analysis are subjected to a strict 30-day recovery timeline divided into three 10-day phases.

30-Day Execution Matrix (10 Days Per Root Cause):

• Phase 1 (Days 01–10): Corrective action implementation and daily verification for Root Cause 1.

• Phase 2 (Days 11–20): Corrective action implementation and daily verification for Root Cause 2.

• Phase 3 (Days 21–30): Corrective action implementation, daily verification for Root Cause 3, and final FAI resubmission.

Daily Documentation Requirements:

During each 10-day block, the supplier must submit systematic daily progress logs. The documentation must explicitly detail:

• Specific measures and corrective actions implemented on each day.

• Quality engineering tools, methodologies, and techniques utilized to rectify the root cause and improve the

supplier rating.

• Daily measurable outputs and learnings applied toward achieving FAI resubmission readiness.

4. FINAL SIGN-OFF & APPROVAL AUTHORITY

Designated Authority: Head of Operations

Technical Justification: The authority to approve or reject a First Article Inspection report is assigned

exclusively to the Head of Operations. Rejecting an FAI requires extensive industrial expertise and deep

technical insight to validate non-conformance claims.

Furthermore, following a Toyota-style supplier partnership philosophy, the Head of Operations does not merely

reject submissions, but provides detailed guidance, feedback, and actionable corrective measures to enable

the supplier to elevate their process capability and successfully pass resubmission.

TQM TOOLS APPLIED ISO STANDARDS APPLIED

• Pareto Analysis (Root Cause Prioritization) • ISO 9001:2015 (Quality Management Systems)

• 8D Problem-Solving Framework • AS9102 (First Article Inspection Requirement)

• Daily Action & Progress Log Matrix • ISO Clause 8.4 (Control of Externally Provided

• Supplier Corrective Action Request (SCAR) Processes)

• Risk Classification Matrix

Notice: This document is created as part of a self-learning practice initiative and professional portfolio development. No real company

or proprietary data was utilized. All protocols are designed to reflect high-level industrial standards.

DAY 12

Original source file(s): Day12_FAI_Decision_and_Audit_Dashboard.pdf

Source focus: The Day 12 source material focuses on 1 0 0 - D AY Q U A L I T Y E N G I N E E R I N G C H A L L E N G E • D AY 1 2. The source text explicitly discusses the following quality themes and methods: INSPECTION, FAI, AUDIT, ISO compliance, ISO Smart, ISO 9001:2015, ISO 14001:2015, ISO 19011:2018, ISO 45001:2018, ISO 10474, ROOT CAUSE, SPC.

Section headings identified in the source: FIRST ARTICLE INSPECTION (FAI) DECISION & AUDIT • DASHBOARD • 1. EXECUTIVE SUMMARY & AUDIT DECISION • 2. DUAL-LAYER AUTOMATED SENSOR SCAN EVALUATION • underwent two-tier automated sensor evaluation: • and technical standards:

Technical terms detected in the source: INSPECTION, FAI, AUDIT, ISO compliance, ISO Smart, ISO 9001:2015, ISO 14001:2015, ISO 19011:2018, ISO 45001:2018, ISO 10474, ROOT CAUSE, SPC, ISO
9001, ISO 14001

Source extract 1: Day12_FAI_Decision_and_Audit_Dashboard.pdf

1 0 0 - D AY Q U A L I T Y E N G I N E E R I N G C H A L L E N G E • D AY 1 2

FIRST ARTICLE INSPECTION (FAI) DECISION & AUDIT

DASHBOARD

Document Reference: FAI-DEC-2026-012 | Executive Quality Audit Clearance

Supplier Name: AeroMotion Dynamics Ltd. Part Name: Brushless Motor Housing

Part Number: AMD-EV-7042 Application: EV & Drone Drive System

Inspection Date: August 2026 Final Status: HIGH RISK (FAI REJECTED)

1. EXECUTIVE SUMMARY & AUDIT DECISION

This document serves as the official First Article Inspection (FAI) Decision Certificate issued under the authority of the Head of Operations. Following rigorous automated sensor scanning and ISO compliance auditing, the initial production batch of Brushless Motor Housing (Part # AMD-EV-7042) submitted by AeroMotion Dynamics Ltd. has been REJECTED due to critical quality non-conformances.

2. DUAL-LAYER AUTOMATED SENSOR SCAN EVALUATION

To maintain zero-defect standards across critical aerospace and electric vehicle drive applications, the component batch underwent two-tier automated sensor evaluation:

• Layer 1: Operational & Dimensional Sensor Scan — [FAILED]

Dimensional inspection sensors detected severe out-of-tolerance conditions exceeding Permissible Limits (0.05mm deviation on key bearing seat diameters), accompanied by localized surface roughness defects.

• Layer 2: ISO Smart Compliance Scanner — [FAILED]

Automated multi-standard scanning flagged critical non-conformances against five core international management and technical standards:

■ ISO 9001:2015: Failure in shop-floor process control and quality assurance protocols.

■ ISO 14001:2015: Non-compliance in environmental waste and coolant disposal management.

■ ISO 19011:2018: Audit trail gaps and missing operational verification documentation.

■ ISO 45001:2018: Breach of shop-floor operator safety protocols during batch production.

■ ISO 10474 / EN 10204 3.1: Raw material composition and testing certificate verification failure.

3. PARETO-DRIVEN ROOT CAUSE ANALYSIS & 30-DAY SCAR RECOVERY PLAN

In accordance with Day 11 SOP protocol, a mandatory 30-Day Supplier Corrective Action Request (SCAR) timeline is enforced. The top three root causes identified via Pareto Analysis must be systematically rectified in three 10-day phases:

30-Day Execution & Daily Logging Matrix:

• Phase 1 (Days 01–10) — Raw Material Quality Failure: Rectify raw material sourcing, replace non-compliant alloy stock, and secure certified ISO 10474 / EN 10204 3.1 material test reports.

• Phase 2 (Days 11–20) — Dimensional & Operational Errors: Recalibrate CNC machinery, eliminate geometric deviations, and enforce Statistical Process Control (SPC) for tight tolerance holding.

• Phase 3 (Days 21–30) — ISO & EHS Standard Non-Compliance: Re-align shop-floor operations with ISO 9001, ISO 14001, ISO 19011, and ISO 45001 standards; submit completed daily logs for final FAI resubmission.

4. HEAD OF OPERATIONS GUIDANCE & APPROVAL AUTHORITY (TOYOTA-STYLE MENTORSHIP)

Authority: Head of Operations

Technical Justification & Actionable Feedback:

Rejecting an FAI is a measure to safeguard product integrity and end-user safety in critical drive systems. In line

with Toyota-style supplier development philosophy, rejection is accompanied by direct engineering support rather

than punitive disconnection:

• Material Procurement: Source raw materials exclusively from ISO-certified tier-1 suppliers and attach 3.1

Inspection Certificates with every batch.

• Process Control: Implement automated in-line probing on CNC machines to prevent dimensional drift before

batch completion.

• EHS & Audit Alignment: Retain shop-floor technicians on ISO 45001 safety and ISO 14001 environmental

protocols. Complete a internal audit as per ISO 19011 guidelines prior to resubmission.

Authorized Sign-off: Head of Operations | Quality Engineering Division

Status: Conditional Rejection — Enforced 30-Day SCAR Active

Notice: This document is created as part of a self-learning practice initiative and professional portfolio development. No real company or

proprietary data was utilized. All protocols reflect executive industrial standards.

DAY 13

Original source file(s): Day13_Executive_TQM_Gurus_and_Strategic_Frameworks_v2.pdf

Source focus: The Day 13 source material focuses on EXECUTIVE TQM GURUS & STRATEGIC QUALITY FRAMEWORKS. The source text explicitly discusses the following quality themes and methods: Taguchi, PDCA, inspection, ROOT CAUSE, Kaizen, audit.

Section headings identified in the source: EXECUTIVE TQM GURUS & STRATEGIC QUALITY FRAMEWORKS • HISTORICAL TITLE & MANUFACTURING • TQM GURU CORE QUALITY PHILOSOPHY KEY TOOLS & FRAMEWORKS • DISTINCTION IMPACT • Control (SQC) management charts) in shop-floor • variation using empirical

Technical terms detected in the source: Taguchi, PDCA, inspection, ROOT CAUSE, Kaizen, audit

Source extract 1: Day13_Executive_TQM_Gurus_and_Strategic_Frameworks_v2.pdf

DAY 13 • 100-DAY QUALITY ENGINEERING CHALLENGE

EXECUTIVE TQM GURUS & STRATEGIC QUALITY FRAMEWORKS

Comprehensive Industry Reference Guide: Deming, Shewhart, Juran, Crosby, Feigenbaum, Ishikawa, Taguchi & Hoshin Kanri

A. STATISTICAL QUALITY CONTROL PIONEERS (SHEWHART & DEMING)

HISTORICAL TITLE & MANUFACTURING

TQM GURU CORE QUALITY PHILOSOPHY KEY TOOLS & FRAMEWORKS

DISTINCTION IMPACT

Walter A. Shewhart Father of Statistical Quality • Statistical process variation • Control Charts (X■-R Charts, p- Eliminated guesswork

Control (SQC) management charts) in shop-floor

• Common Causes vs. Special Causes of • 3σ Natural Process Limits machining tolerances

variation using empirical

• Specification vs. Control

• Foundational principles of Process statistical limits.

Boundaries

Control

W. Edwards Deming Father of Modern Quality • 14 Points for Top Management • PDCA / PDSA Cycle (Plan-Do- Spearheaded Japan's

Management Transformation Check/Study-Act) post-WWII industrial

• System of Profound Knowledge (SoPK) • Red Bead Experiment renaissance

• Shift from inspection to system design • Funnel Experiment (Foundation of Toyota

Production System).

• Elimination of numerical quotas & fear

B. STRATEGIC & FINANCIAL QUALITY LEADERSHIP (JURAN & CROSBY)

HISTORICAL TITLE & MANUFACTURING

TQM GURU CORE QUALITY PHILOSOPHY KEY TOOLS & FRAMEWORKS

DISTINCTION IMPACT

Joseph M. Juran Pioneer of Quality • Quality defined as "Fitness for Use" • Juran's Quality Trilogy: Quality Linked defect metrics

Management & Cost of • Cost of Quality (COQ) Framework Planning, Quality Control, Quality directly to board-level

Quality • Pareto Principle applied to Quality ("Vital Improvement financial performance

few vs. Useful many") • COQ Breakdown: Prevention, and operational P&L.

Appraisal, Internal & External

Failures

Philip B. Crosby Advocate of Zero Defects & • "Zero Defects" Mindset & Philosophy • 4 Absolutes of Quality Enforced zero-defect

Price of Non-Conformance • "Quality is Free" (Prevention saves more Management standards across

than it costs) • Price of Non-Conformance aerospace and high-

• Quality is Conformance to (PONC) vs. Price of Conformance precision defense

Requirements (POC) manufacturing.

• 14-Step Quality Improvement

Program

C. TOTAL QUALITY CONTROL, ROOT CAUSE & ROBUST DESIGN (FEIGENBAUM, ISHIKAWA, TAGUCHI)

HISTORICAL TITLE & MANUFACTURING

TQM GURU CORE QUALITY PHILOSOPHY KEY TOOLS & FRAMEWORKS

DISTINCTION IMPACT

Armand V. Feigenbaum Father of Total Quality • Total Quality Control (TQC) as • Total Quality System Architecture Structured the modern

Control (TQC) & TQM company-wide accountability • Cross-Departmental Quality enterprise-wide Total

Originator • "Hidden Factory" Concept (capacity Ownership Matrix (R&D to Quality Management

wasted on rework) Purchase/Sales) (TQM) architecture.

• Customer-defined total quality system

Kaoru Ishikawa Pioneer of Shop-Floor • Democratization of quality control to • Cause & Effect Diagram Empowered front-line

Quality & CWQC shop-floor

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gement

CORE PRINCIPLES & STRATEGIC IMPLEMENTATION MANUFACTURING

FRAMEWORK / TOOL ORIGIN & PURPOSE

ALIGNMENT MECHANICS IMPACT

Hoshin Kanri X-Matrix Visual Strategic Execution • Single-page visual alignment linking 4 • 4-Quadrant Intersecting Matrix Provides line-of-sight

Tool for TQM key quadrants: • Correlation Mapping (Strong / visibility for plant

1. Long-Term Strategies (3-5 Yrs) Secondary Dots) managers and ISO/

2. Annual Objectives (1 Yr) • Primary & Secondary Ownership IATF audit executives.

3. Top Improvement Projects Assignment

4. Operational KPIs & Metrics

EXECUTIVE TQM FRAMEWORK MATRIX • SYNTHESIS SUMMARY

1. STATISTICAL QUALITY 2. FINANCIAL QUALITY 3. ORGANIZATIONAL TQM 4. STRATEGIC DEPLOYMENT

• Shewhart: 3σ Limits & Control • Juran: Trilogy & Cost of Quality • Feigenbaum: TQC & Hidden • Taguchi: L(y) = k(y - m)2

Charts (COQ) Factory • Hoshin Kanri: X-Matrix Alignment

• Deming: PDCA & Profound • Crosby: Zero Defects & PONC • Ishikawa: 7 QC Tools & Circles

Knowledge

Document Code: TQM-EXEC-2026-V2 • Prepared for Executive Quality Engineering Portfolio & Industrial Reference.

DAY 14

Original source file(s): Day14_Lean_TQM_Framework_Jagriti_Singh.pdf

Source focus: The Day 14 source material focuses on Eliminating TIMWOOD Inventory Waste via ABC Analysis, JIT Kanban & 100-Day QE Challenge. The source text explicitly discusses the following quality themes and methods: Lean, PDCA, 5S, KAIZEN, isolate, Audit.

Section headings identified in the source: PDCA Deployment Author: Jagriti Singh • Automotive OEM • sections covered in this report: • Waste (TIMWOOD) Titan Motors Shop-Floor Problem Lean-TQM Countermeasure • Frequent unguided movement of raw Point-of-Use storage & fixed Kanban routing • T - Transportation

Technical terms detected in the source: Lean, PDCA, 5S, KAIZEN, isolate, Audit

Source extract 1: Day14_Lean_TQM_Framework_Jagriti_Singh.pdf

Day 14: Lean-TQM Integration Framework

Eliminating TIMWOOD Inventory Waste via ABC Analysis, JIT Kanban & 100-Day QE Challenge

PDCA Deployment Author: Jagriti Singh

Target Enterprise: Titan Motors & Precision Assemblies Ltd. | Industry: Heavy

Automotive OEM

■ Framework Index & Executive Overview

This engineering case study bridges Lean Manufacturing tools with Total Quality Management (TQM) principles to systematically

identify, categorize, and eliminate factory inventory waste at Titan Motors & Precision Assemblies Ltd. Below is the blueprint of sections covered in this report:

Section 1: Enterprise Profile & TIMWOOD Mapping Section 2: ABC Inventory Prioritization

Identifies 8 Lean wastes in conventional Push manufacturing Classifies shop-floor items into A, B, and C categories to apply systems with a special focus on Inventory Waste (I). maximum control over high-value capital items.

Section 3 & 4: JIT Kanban & PDCA Mechanics Section 5: Step-by-Step Engineering Analysis

Introduces Pull-based replenishment (2-Bin/3-Bin) and Features quantitative calculations (Kanban sizing, financial

Deming's PDCA cycle to stabilize Work-in-Progress (WIP). savings, Little's Law) with formula references.

Section 6 & 7: Quick-Reference Glossary, Executive Dashboard & Author Declaration

Includes a core concept revision guide, pre vs. post KPI impact table, and author ownership statement.

SECTION 1: TIMWOOD WASTE IDENTIFICATION IN PUSH SYSTEMS

In traditional Push Manufacturing at Titan Motors, components were produced based on forecasts and pushed to workstations regardless of immediate demand, causing severe operational friction across all TIMWOOD categories:

Waste (TIMWOOD) Titan Motors Shop-Floor Problem Lean-TQM Countermeasure

Frequent unguided movement of raw Point-of-Use storage & fixed Kanban routing

T - Transportation

materials across shop-floor. pathways.

High WIP buildup (500 units) clogging

I - Inventory ABC Analysis + JIT Kanban Pull System.

assembly lines.

Operators searching for misplaced 5S Workplace Organization & ergonomic kit

M - Motion

components and toolkits. bins.

Downstream stations idling due to batch Takt Time alignment and balanced single-

W - Waiting

production delays. piece flow.

Making components earlier or in larger Strict Pull Signal: Produce ONLY when Kanban

O - Overproduction

quantities than requested. card arrives.

Adding unnecessary extra processes not

O - Overprocessing Standard Operating Procedures (SOPs) & VSM.

valued by customer.

Batch defects discovered late, leading to Poka-Yoke (Mistake-proofing) at source

D - Defects

massive scrap/rework. station.

SECTION 2: ABC INVENTORY PRIORITIZATION MATRIX

Not all inventory items carry equal financial weight. To eliminate Inventory Waste efficiently, we apply ABC Analysis (Pareto

Principle) to focus control where capital risk is highest:

Class A (High Value) Class B (Moderate Value)

Class C (Low Value)

• 10-15% of total items. • 20-30% of total items.

• 60-70% of total items (e.g., screw

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mentation Post-Implementation

Operational Metric Quantifiable Benefit

(Push) (Pull)

76% Reduction in shop-floor

Work-In-Progress (WIP) 500 Units 120 Units

inventory

Manufacturing Lead Time 10.0 Hours 2.4 Hours 7.6 Hours Faster order fulfillment

Capital Locked in WIP ■125,000 ■30,000 ■95,000 Capital Freed Up

Annual Inventory Holding

■25,000 / year ■6,000 / year ■19,000 Direct Annual Savings

Cost

■ Author Declaration & Knowledge Integrity Statement

This Day 14 Challenge Dashboard & Implementation Report has been conceptualized, structured, mathematically solved, and

authored independently by Jagriti Singh as part of the self-driven 100-Day Quality Engineering & Industrial Excellence

Challenge.

"This work is non-copied, authentic, and constructed purely from first-principles engineering knowledge, combining Total Quality

Management (TQM) philosophy, Lean Manufacturing tools, and quantitative industrial calculations."

Target Enterprise: Titan Motors & Precision Assemblies Ltd. Specially Created by: Jagriti Singh

Author: Jagriti Singh | Day 14/100 Quality Engineering Challenge Page 3 of 3

DAY 15

Original source file(s): Day15_Lean_TQM_Master_Reference_Jagriti_Singh.pdf

Source focus: The Day 15 source material focuses on Specialist. The source text explicitly discusses the following quality themes and methods: Lean, OEE, Process Capability, Cp, Cpk, Yield, Taguchi.

Section headings identified in the source: Author: Jagriti Singh Role: Quality Engineer & Industrial Excellence • Specialist • Hints & Lean Concepts Engineering-Challenge- • 1. Lean Manufacturing & Line Balancing • Takt Time (Tt) • Cycle Time (Tc) & Minimum Workstations (Nmin)

Technical terms detected in the source: Lean, OEE, Process Capability, Cp, Cpk, Yield, Taguchi

Source extract 1: Day15_Lean_TQM_Master_Reference_Jagriti_Singh.pdf

Day 15: Lean, Inventory & TQM Master Reference

100-Day Quality Engineering & Industrial Excellence Challenge

Author: Jagriti Singh Role: Quality Engineer & Industrial Excellence

Specialist

Focus Area: Core Formulas, Practice Questions with Repository: github.com/Jagriti1907/100-Day-Quality-

Hints & Lean Concepts Engineering-Challenge-

SECTION 1: STANDARD INDUSTRIAL & COMPETITIVE FORMULA BANK

1. Lean Manufacturing & Line Balancing

Takt Time (Tt)

Tt = Total Available Work Time per Shift / Customer Demand per Shift

Sets the required speed of production to meet customer demand on time.

Cycle Time (Tc) & Minimum Workstations (Nmin)

Tc = Available Production Time / Total Units Made

Nmin = Ceiling [(Σ ti) / Tc]

ti is the time needed for each task. Ceiling means rounding up to the next whole number.

Line Efficiency (η) & Balance Delay (BD)

η = [(Σ ti) / (N × Tc)] × 100%

BD = 100% - η = [(N × Tc - Σ ti) / (N × Tc)] × 100%

Shows how well work is shared across workstations and the percentage of idle time.

Smoothness Index (SI)

SI = √ [Σ (Tc - Tsi)²]

Tsi is the total time spent at station i. A lower SI value means a better balanced assembly line.

Overall Equipment Effectiveness (OEE)

OEE = Availability (A) × Performance (P) × Quality (Q)

A = Run Time / Planned Production Time | P = (Ideal Cycle Time × Total Units) / Run Time | Q = Good Units / Total Units

2. Kanban, Pull Systems & Flow Metrics

Kanban Bins Calculation (K)

K = [Demand (D) × Lead Time (L) × (1 + Safety Factor S)] / Bin Size (C)

Calculates how many bins or cards you need to keep production moving smoothly without running out.

100-Day QE Challenge | Jagriti Singh Page 1 of 5

Little’s Law

Manufacturing Lead Time (MLT) = Work-in-Progress (WIP) / Production Rate (TH)

Explains how stock on the shop floor directly affects how long orders take to finish.

Process Flow Efficiency

Flow Efficiency (%) = [Value-Added Time / Total Lead Time] × 100%

Measures the percentage of total time actually spent working on the product.

3. Inventory Planning & Cost Control

Economic Order Quantity (EOQ)

EOQ = √ [(2 × Annual Demand D × Order Cost S) / Holding Cost H]

Finds the best order size to balance ordering costs and stock holding costs.

Economic Production Quantity (EPQ)

EPQ = √ [(2 × D × S) / (H × (1 - d/p))]

Used when making items in-house. p = daily production rate, d = daily usage rate.

Reorder Point (ROP) & Safety Stock (SS)

ROP = (Daily Demand d × Lead Time L) + Safety Stock (SS)

SS = Z × σd × √L

Z is the standard normal value for target service level. σd is daily demand variation.

4. Process Capability & Quality Metrics

Process Capability Indices (Cp, Cpk, Cpm)

Cp = (USL - LSL) / (6σ)

Cpk = Minimum of [(USL - Mean μ) / (3σ) , (μ - LSL) / (3σ)]

Cpm = (USL - LSL) / [6 × √(σ² + (μ - Target T)²)]

Cp measures potential capability, Cpk accounts for off-center process, Cpm considers target alignment.

First Pass Yield (FPY) & Rolled First Pass Yield (RFPY)

FPY = Good Units Output / Total Units

[Middle content condensed for page fit]

fore they can cause defects.

Andon Visual signal board: Light or display system that workers use to call for quick help when a problem occurs.

Gemba The real place: The actual factory floor where work happens and value is created.

Jidoka Smart automation: Building machines that automatically stop working as soon as an issue is detected.

Mizusumashi Water Spider: A dedicated operator who supplies parts and materials to workstations so builders never stop.

100-Day QE Challenge | Jagriti Singh Page 4 of 5

TERM SIMPLE OPERATIONAL MEANING

Shojinka Flexible staffing: Adjusting the number of line workers smoothly based on changing customer demand.

AUTHOR DECLARATION & PROJECT NOTE

This Day 15 Master Reference Document was independently compiled and created by Jagriti Singh for the 100-Day Quality Engineering

Challenge.

"Designed as a clean, practical reference guide combining fundamental quality engineering formulas, solved numerical hints, and core Lean

manufacturing terms."

Jagriti Singh • Quality Engineer & Industrial Excellence Specialist

100-Day QE Challenge | Jagriti Singh Page 5 of 5

DAY 16

Original source file(s): Day16_TQM_Zomato_Swiggy_Case_Study_v2.pdf

Source focus: The Day 16 source material focuses on Quality Engineering Solutions for ZOMATO and SWIGGY. The source text explicitly discusses the following quality themes and methods: Root Cause, Cp, Cpk, Process Capability, inspection, 5S.

Section headings identified in the source: Quality Engineering Solutions for ZOMATO and SWIGGY • 1. Operational Challenges & Objective • 2. Root Cause Diagnostic: Fishbone (6M) & Pareto 80/20 Rule • 3. Statistical Process Control (p-Chart, np-Chart, c-Chart, Cp & Cpk) • sizes: • 4. Visual Management & 3-Color Kanban Bins

Technical terms detected in the source: Root Cause, Cp, Cpk, Process Capability, inspection, 5S

Source extract 1: Day16_TQM_Zomato_Swiggy_Case_Study_v2.pdf

Day 16: TQM Framework for Hyperlocal Logistics

Quality Engineering Solutions for ZOMATO and SWIGGY

1. Operational Challenges & Objective

Fast-paced delivery platforms like ZOMATO and SWIGGY depend heavily on speed and order correctness. During rush hours, hubs face food spillage from loose container lids, delivery delays, and wrong order handovers at pickup counters. Rather than fixing these issues separately, this study applies an integrated Total Quality Management framework to control process variation and reduce customer refunds.

2. Root Cause Diagnostic: Fishbone (6M) & Pareto 80/20 Rule

Using the 6M Fishbone framework, operational failures were mapped across six operational inputs:

• Man (Riders and Kitchen Crew): Riders rushing in traffic causing container drops. Kitchen staff handing over bags without verifying order IDs on the screen.

• Material (Containers and Seals): Non-vented lids opening under steam pressure and thin plastic containers cracking during transport.

• Method (Dispatch SOPs): Disorganized pickup tables causing bag mix-ups and high rider waiting times.

• Machine / Tech: App synchronization delay between merchant POS terminals and rider app screens.

• Measurement: Static SLA calculations failing to adapt during sudden order volume spikes.

• Milieu (Environment): Extreme weather conditions and traffic bottlenecks during lunch or dinner hours.

Pareto 80/20 Analysis: Defect tracking data shows that 80% of total order complaints come from 20% vital causes, specifically Man, Material, and Method failures. Directing corrective actions toward these three areas delivers the highest operational impact.

3. Statistical Process Control (p-Chart, np-Chart, c-Chart, Cp & Cpk)

To track order defects and process variation across delivery hubs, four statistical control tools are used based on sample sizes:

• p-Chart (Fraction Defective): Tracks the percentage of defective orders when daily batch sizes vary across different merchant hubs.

• np-Chart (Number of Defectives): Monitors the total count of defective orders when inspecting a fixed batch size (such as checking 500 orders daily per cluster).

• c-Chart (Defects per Unit): Tracks the total number of specific defects within a single order bag (e.g., item missing, gravy leakage, and missing cutlery in one order).

• Process Capability (Cp & Cpk): Evaluates if hub delivery times remain inside the target SLA window of 30 minutes (plus or minus 6 minutes). Maintaining Cpk above 1.33 ensures stable service delivery under peak load.

4. Visual Management & 3-Color Kanban Bins

Merchant pickup counters use visual cues and a 3-Color Kanban Bin system to standardize dispatch flow:

Kanban Bin

Food Category / Order Type Visual Control & TQM Directive

Color

High-Risk Liquids (Soups, Gravies, Ice Mandatory double-sealing with anti-spill thermal wraps before QA

RED BIN

Cream) inspection.

YELLOW BIN In-Progress / Packing Check Items packed and placed here for receipt item-count verification.

QR tamper seal applied. Rider picks directly from Green Bin with

GREEN BIN QA Verified / Ready for Dispatch

zero bag mix-ups.

5. Execution Plan: Poka-Yoke, 5S & Muda Elimination

• Poka-Yoke (Mistake Proofing): Smart QR tamper seals on delivery bags. The rider app blocks the "Order Picked

Up" confirmation button until the rider scans the QR seal printed on the bag, preventing wrong handovers.

• 5S Workplace Setup: Sorting and organizing merchant dispatch areas (Sort, Set in Order, Shine, Standardize,

Sustain) to eliminate overcrowding at kitchen doors.

• Muda (Waste) Elimination: Reduces waiting waste for riders and overburden on kitchen crew, creating a smooth

order dispatch flow without encouraging dangerous driving.

Summary: Implementing this structured TQM approach allows ZOMATO and SWIGGY to control process

variation, prevent packaging defects, and maintain reliable delivery SLAs.

DAY 17

Original source file(s): Day 17_Nykaa_PFMEA_Case_Study.pdf

Source focus: The Day 17 source material focuses on Project Author: Jagriti Singh | Role: Lead Quality & Process Improvement Engineer. The source text explicitly discusses the following quality themes and methods: PFMEA, RPN, Root Cause, inspection, CAPA, Audit, PDCA, Kaizen, FMEA, Lean.

Section headings identified in the source: 1. Operational Context & Project Objectives • 2. PFMEA Risk Mapping & RPN Clustering • Cluster / Operational Failure Mode Root Cause Drivers S O D RPN • Leakage) • 3. Inventory Optimization: ABC Analysis & JIT Integration • methods:

Technical terms detected in the source: PFMEA, RPN, Root Cause, inspection, CAPA, Audit, PDCA, Kaizen, FMEA, Lean

Source extract 1: Day 17_Nykaa_PFMEA_Case_Study.pdf

Day 17: Supply Chain Risk Management & PFMEA Framework

Project Author: Jagriti Singh | Role: Lead Quality & Process Improvement Engineer

1. Operational Context & Project Objectives

In fast-moving e-commerce setups like NYKAA , beauty and personal care supply chains constantly face risks like leaking containers, melted

cosmetics, broken glass packaging, and dispatching wrong shades. These issues directly hurt profit margins through scrap costs and customer

returns. I designed this case study to map out potential failure points across NYKAA 's supply network, restructure inventory flows, and build a

practical quality control loop.

2. PFMEA Risk Mapping & RPN Clustering

I evaluated major supply chain breakdown points across NYKAA 's fulfillment centers using Severity (S), Occurrence (O), and Detection (D)

scales to calculate Risk Priority Numbers (RPN = S x O x D):

Cluster / Operational Failure Mode Root Cause Drivers S O D RPN

Cluster 1: Primary Inventory Loss

(Heat-Induced Spoilage & Inbound High warehouse temperatures and improper container sealing from suppliers during transit. 9 8 7 504

Leakage)

Cluster 2: Customer Delivery Experience

Manual picking mix-ups at packing stations and insufficient protective wrap for fragile items. 8 7 6 336

(Wrong SKU Dispatch & Glass Breakage)

Cluster 3: Last-Mile Delivery Delays Regional courier bottlenecks leading to missed delivery windows and customer order

6 5 4 120

(Transit Lags & Order Rejections) cancellations.

3. Inventory Optimization: ABC Analysis & JIT Integration

To cut down stock holding risks and keep high-demand inventory safe across NYKAA 's warehouses, I combined two essential inventory

methods:

• ABC Inventory Categorization: Fast-moving, trending beauty products on NYKAA are assigned to Class A for rigorous daily checks.

Baseline staples go into Class B, while slow-moving SKUs sit in Class C, ensuring our inspection efforts target high-risk inventory.

• Just-In-Time (JIT) Replenishment: Stocking high-volume items closer to real-time order signals reduces warehouse storage duration,

preventing heat damage to sensitive creams and lipsticks.

4. CAPA Execution & Daily Audit Control Loops

To directly address NYKAA 's high-RPN failure points, I structured targeted Action Plans alongside daily operational checks:

• Fixes for Cluster 1: Added incoming inspection checklists for supplier shipments and set up dedicated climate-controlled rack zones in

warehouses.

• Fixes for Cluster 2: Mandated barcode scanners at packing counters to stop wrong shade dispatches, and switched to shock-absorbent

molded pulp trays for fragile glass bottles.

• Daily Dispatch Audits: Conducting spot checks on daily outgoing batches helps catch wrong SKU dispatches immediately, stopping

repeating errors before items leave the facility.

5. Continuous Improvement Strategy: PDCA & Kaizen

• Plan: Pinpoint critical risk areas in NYKAA 's supply chain, assign functional ownership, gather operational error logs, and set clear RPN

target reductions.

• Do: Roll out barcode validation, temperature checks, and ABC-JIT controls in pilot warehouse centers.

• Check: Review daily dispatch audit findings and tracking logs to measure real reduction in RPN levels.

• Act: Turn working fixes into standard operating procedure (SOP) documents and keep refining processes through regular Kaizen reviews.

■■ Core Theories, Quality Tools & Methodologies Applied

Tools & Methods: Process FMEA (PFMEA), Risk Priority Number (RPN) Calculation, ABC Classification Analysis, Just-In-Time (JIT) Pull

System, Daily Dispatch Audits, Barcode Verification, and CAPA Frameworks.

Underlying Theories & Philosophies: Deming's PDCA Continuous Improvement Cycle (Plan-Do-Check-Act), Kaizen Philosophy (Ongoing

Incremental Upgrades), and Lean Waste Elimination (Muda of Scrap, Defect, and Rework).

DAY 18

Original source file(s): Day18_Paytm_FinTech_Quality_Framework.pdf

Source focus: The Day 18 source material focuses on Project Author: Jagriti Singh | Role: Lead Quality & Process Improvement Engineer. The source text explicitly discusses the following quality themes and methods: Reliability, ISO 9001-aligned, FMEA, RPN, ISO 9001, ISO 9001:2015, 5S, PDCA, audit, CAPA, Lean.

Section headings identified in the source: 1. OPERATIONAL CONTEXT & PROBLEM STATEMENT (PAYTM / PHONEPE GATEWAY SCALE) • 2. KEY FAILURE MODES IDENTIFIED • 3. SYSTEM FMEA (FAILURE MODE AND EFFECTS ANALYSIS) • Failure Mode S O D RPN Mitigation & Poka-Yoke Strategy • 4. COMPREHENSIVE QUALITY ENGINEERING & ISO 9001 EXECUTION STRATEGY

Technical terms detected in the source: Reliability, ISO 9001-aligned, FMEA, RPN, ISO 9001, ISO 9001:2015, 5S, PDCA, audit, CAPA, Lean

Source extract 1: Day18_Paytm_FinTech_Quality_Framework.pdf

Day 18: FinTech System Reliability & Quality Assurance Framework

Project Author: Jagriti Singh | Role: Lead Quality & Process Improvement Engineer

1. OPERATIONAL CONTEXT & PROBLEM STATEMENT (PAYTM / PHONEPE GATEWAY SCALE)

During peak-load events (e.g., Festival Sales, IPL rush), transaction volumes spike drastically across digital gateways. When ■100 gets stuck, a user might

tolerate it; but if ■20,000 of a middle-class family gets trapped in pending status, user trust and retention vanish instantly. Irrespective of system

transparency, such delays damage brand reputation, management credibility, and user confidence. I designed this case study to address root-cause

transaction bottlenecks, restructure feedback loops, and implement ISO 9001-aligned quality management systems.

2. KEY FAILURE MODES IDENTIFIED

• High-Value Pending Transactions: High-value funds (e.g., ■20,000+) stuck in network queues due to bank server latency or API timeout.

• Server & App Crashes: Central server throttling and memory overflow during peak transaction spikes.

• Gateway & Network Bottlenecks: Inter-operable NPCI gateway delays, bank switch outages, and status synchronization lags.

3. SYSTEM FMEA (FAILURE MODE AND EFFECTS ANALYSIS)

Failure Mode S O D RPN Mitigation & Poka-Yoke Strategy

High-Value Pending Transaction (■20,000+ 10 7 6 420 Poka-Yoke Pre-Check: Verify bank server health before payment. Auto-route to secondary

Stuck) gateway if primary bank API latency exceeds threshold.

Peak-Load Server & App Crash 8 8 5 320 Dynamic Queueing: Auto-scale cloud servers and implement rate-limiting throttle to

prevent app crashes.

Timeout & Status Sync Lag 7 6 4 168 Auto-Reconciliation Loop: 5-minute automated polling & status resolution with immediate

refund triggers.

4. COMPREHENSIVE QUALITY ENGINEERING & ISO 9001 EXECUTION STRATEGY

• Poka-Yoke (Mistake Proofing): Pre-payment gateway health checks block payment attempts on failing bank servers, redirecting users to active

payment modes.

• ISO 9001:2015 Customer Focus & Immediate Post-Payment Feedback Loop: Integrated a mandatory micro-feedback loop right after payment

completion to capture real-time customer experience, pain points, and requirements to continuously optimize app functionality.

• Selective Benchmarking (Suggestive Adaptation): Used benchmarking as a suggestive framework rather than a copy-paste mechanism. Analyzed

global payment architectures to build a custom solution on top of the original problem baseline.

• Digital Data DBMS 5S Integration: Applied 5S methodology (Sort, Set in Order, Shine, Standardize, Sustain) to Database Management Systems

(DBMS) to streamline high-volume logs, purge redundant metadata, and accelerate transaction querying.

• PDCA Cycle & System Audits: Plan risk priorities, Do routing/feedback implementations, Check audit metrics against control limits, and Act by

codifying working fixes into SOPs.

"We don't have to reinvent the wheel; we just have to make the best wheel with what we have." — W. Edwards Deming

■■ Core Theories, Quality Tools & Methodologies Applied

Tools & Methods: System FMEA (SFMEA), Risk Priority Number (RPN) Calculation, Poka-Yoke (Mistake Proofing), DBMS 5S Integration, Post-Payment

Customer Feedback Loops, Selective Benchmarking, System Auditing, and CAPA Frameworks.

Underlying Theories & Philosophies: ISO 9001:2015 Quality Management Principles (Customer Focus & Continuous Improvement), Deming's PDCA

Improvement Cycle, and Lean Digital Waste Elimination.

DAY 19

Original source file(s): Day19_TQM_Framework_Jagriti_Singh.pdf

Source focus: The Day 19 source material focuses on Doc Ref: QMS/CA-CMA/2026/D19. The source text explicitly discusses the following quality themes and methods: AUDIT, FMEA, PFMEA, RPN, 5S, Kaizen, PDCA, Risk-based, CAPA.

Section headings identified in the source: QUALITY MANAGEMENT & AUDIT ASSURANCE Author: Jagriti Singh • Target Scale: Mid-to-Large Advisory • PRACTICE Agencies • Scope: Peak Season Operations & Tax • OPERATIONAL EXCELLENCE FRAMEWORK FOR CA & CMA ADVISORY FIRMS • Audits

Technical terms detected in the source: AUDIT, FMEA, PFMEA, RPN, 5S, Kaizen, PDCA, Risk-based, CAPA

Source extract 1: Day19_TQM_Framework_Jagriti_Singh.pdf

Doc Ref: QMS/CA-CMA/2026/D19

QUALITY MANAGEMENT & AUDIT ASSURANCE Author: Jagriti Singh

Target Scale: Mid-to-Large Advisory

PRACTICE Agencies

Scope: Peak Season Operations & Tax

OPERATIONAL EXCELLENCE FRAMEWORK FOR CA & CMA ADVISORY FIRMS

Audits

1. OPERATIONAL CONTEXT & PROBLEM STATEMENT

When peak tax filing deadlines and statutory audit cycles arrive, mid to large CA and CMA firms face enormous workload pressures. Working against strict deadlines often causes operational bottlenecks, data entry mistakes, and paperwork discrepancies like missing vouchers, unverified bank statements, or mismatched GSTIN and PAN details. Handling heavy volumes of financial data under tight deadlines makes errors much more likely. These mistakes can lead to statutory show-cause notices, financial penalties, interest liabilities for clients, and severe harm to the firm's reputation. To tackle this, firms need a practical quality assurance framework that fixes operational delays and standardizes document intake without adding useless administrative work.

2. KEY OPERATIONAL FAILURE MODES IDENTIFIED

• Data Disarray & Audit Trail Gaps: Incomplete client files, missing payment vouchers, mismatched GSTIN/PAN entries, and raw entry errors during peak intake.

• Process Bottlenecks & Trainee Onboarding Issues: Execution delays and reporting mistakes caused by peak rush, lack of clear guidance for beginners and articles, and frequent turnaround time slippage.

• Compliance Exposure & Client Trust Risks: Higher risk of legal notices due to incorrect filings, weak risk screening during client onboarding, and loss of client retention due to poor communication.

3. STRATEGIC SWOT ANALYSIS FOR CA/CMA ADVISORY OPERATIONS

Strengths & Opportunities Weaknesses & Threats

Strengths: Strong subject expertise, deep regulatory knowledge, and Weaknesses: Over-reliance on manual double-checks, established client trust. unorganized client file collection, and steep learning

curves for trainees.

Opportunities: Standardizing internal audit procedures, organizing

digital folder structures, creating automated checks, and streamlining Threats: Heavy tax penalties, statutory notices from document workflows. authorities, and losing clients due to missed deadlines.

4. PROCESS FMEA (PFMEA) MATRIX & RISK EVALUATION

Process Step Potential Failure Effect S O D RPN Mitigation Strategy (TQM Tool)

1. Data Intake & Data mismatches, filing 9 7 8 504 Digital 5S & Poka-Yoke Checklist: Standardized client

Audit Trail errors, and work delays. intake templates and pre-validation logic to block

incomplete uploads.

2. Review & Missed deadlines, 9 7 7 441 Gemba Walk & Kaizen SOPs: Visual work tracking,

Execution Phase reviewer burnout, high beginner cheat-sheets, and cutting out unnecessary

TAT. steps.

3. Client Statutory notices, client 10 9 7 630 PDCA & Risk-based CAPA: Structured KYC risk profiling

Onboarding & penalties, loss of trust. and automated pre-submission sanity reviews.

Signing

5. QUALITY IMPLEMENTATION RO

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APA: Turn working fixes into permanent firm SOPs and implement corrective actions to address remaining

risks and screen out high-risk accounts.

6. INTEGRATED TQM & QUALITY ENGINEERING TOOLS VERIFICATION

✓ Gemba Walks (Ground-Level Observation) ✓ Low-Cost Kaizen (Continuous Micro-Improvement)

✓ Digital 5S Methodology (DBMS Hygiene) ✓ Automated Poka-Yoke (Mistake Proofing)

✓ Process FMEA (RPN Risk Scoring) ✓ PDCA Verification Engine (Plan-Do-Check-Act)

✓ Strategic SWOT Analysis ✓ CAPA (Corrective and Preventive Action)

Prepared By: Approved By:

Jagriti Singh Senior Managing Partner / QCR Head

Quality Engineering Lead Audit & Compliance Practice

DISCLAIMER & COMPLIANCE NOTICE: This document is an independent educational case study developed strictly to demonstrate the application of Quality

Engineering and Total Quality Management (TQM) methodologies within CA and CMA advisory practices. References to audit workflows, risk scores, or

practice structures are simulated for academic purposes and do not represent internal data from any specific professional firm or statutory oversight body.

DAY 20

Original source file(s): Day_20_Case_Study_SQC1_TQM_v3.pdf

Source focus: The Day 20 source material focuses on CASE STUDY: IMPLEMENTING TQM & QUALITY ENGINEERING IN ICAI SQC. The source text explicitly discusses the following quality themes and methods: Audit, inspection, PFMEA, PDCA, Kaizen, RPN, 5S, RCA, Root Cause.

Section headings identified in the source: CASE STUDY: IMPLEMENTING TQM & QUALITY ENGINEERING IN ICAI SQC • Standardizing Audit Quality Management Framework for CA & CMA Firms • Continuance without thorough background and to engagement and evaluating risk • inspections due to an absence of Ishikawa (Fishbone) RCA with external peer reviews and • Regulatory Terminology & Full Forms: • Architectural Note on Integration:

Technical terms detected in the source: Audit, inspection, PFMEA, PDCA, Kaizen, RPN, 5S, RCA, Root Cause

Source extract 1: Day_20_Case_Study_SQC1_TQM_v3.pdf

CASE STUDY: IMPLEMENTING TQM & QUALITY ENGINEERING IN ICAI SQC

1

Standardizing Audit Quality Management Framework for CA & CMA Firms

Author: Jagriti Singh | Quality Engineering Lead Series: Day 20 of #100DaysOfTQM

SECTION 1: EXECUTIVE SUMMARY & OBJECTIVE

The Standard on Quality Control 1 (SQC 1) issued by ICAI lays down fundamental quality control requirements for audit and assurance

firms. However, compliance often remains a periodic documentation exercise rather than a smooth daily operational workflow.

Primary Objective: Integrate core Industrial Engineering, TQM tools, and quality philosophies such as Crosby, Deming, and Hoshin Kanri

directly into the 6 elements of SQC 1. This converts legal compliance into an automated, zero-defect process for CA & CMA practices.

SECTION 2: HIGH-RISK AUDIT GAPS & SYSTEMIC BOTTLENECK ANALYSIS

Based on Quality Review Board (QRB) inspection trends, CA firms primarily face compliance bottlenecks in two critical operational areas:

1. Audit Documentation & Large Data Management: Managing, indexing, and retrieving large volumes of audit evidence under high

pressure, which creates major compliance risks during internal and external reviews.

2. Pre-Issuance Engagement Quality Review (EQR): A lack of automated preventive checks before final report signing, increasing the

risk of missing critical statutory disclosures.

SECTION 3: SYSTEMIC SQC 1 IMPLEMENTATION MATRIX

SQC 1 Core Element Primary Quality Gap Integrated TQM Tool Operational Implementation

1. Leadership Quality is viewed only as a Crosby’s 4 Absolutes & Quality Aligning top leadership goals down to

Responsibilities partner responsibility instead of a Vaccine + Hoshin Kanri article clerks and treating quality as daily

shared team culture. operational nourishment.

2. Ethical Requirements Manual tracking of annual Poka-Yoke + PFMEA + Deming’s Blocking engagement setup

independence forms, leading to 14 Principles + Crosby’s Zero automatically until independence forms

potential conflicts of interest. Defects are signed, leaving zero room for ethical

compromises.

3. Acceptance & Onboarding high-risk clients PFMEA + PDCA Cycle + Kaizen Scoring RPN for client risk factors prior

Continuance without thorough background and to engagement and evaluating risk

integrity verification. continuously throughout client retention.

4. Human Resources Skill gaps among audit staff, Standardized Work (SOP) + Skill Mapping staff competencies to

workload burnout, and a lack of Matrix + 7 New Management engagement complexity, ensuring

clear SOPs. Tools balanced workload distribution and

continuous training.

5. Engagement Mismanaged audit working Digital 5S + Gemba Walk 5S file indexing for 30-second document

Performance papers, last-minute review retrieval, with senior partners performing

pressure, and missed checks. ground-level verification of live audit

trails.

6. Monitoring Panic during QRB or NFRA Dual-Tier Quality Audit + Combining internal quality assurance

inspections due to a

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combining internal audits with external peer reviews.

14. Fishbone (Ishikawa) Diagram / RCA: Cause-and-effect tool used to trace non-compliances back to their fundamental root cause.

Architectural Note on Integration:

The regulatory standards of ICAI (SQC 1) and the domain of Quality Engineering and TQM are historically independent disciplines. However,

in this case study, their implementation is brought together through rigorous cross-domain research to demonstrate how industrial quality

control systems can practically support statutory regulatory frameworks.

Disclaimer:

This case study has been conceptualized and documented strictly for educational and analytical purposes as part of the #100DaysOfTQM (Quality

Engineering) series. It does not replace official ICAI guidelines or statutory provisions.

Quality Engineering is not confined to assembly lines or manufacturing units, it is a practical mindset. While domain terminologies

may vary across industries, the core objective remains universal: eliminating operational waste, preventing errors at source, and

elevating system quality.

DAY 21

Original source file(s): Chapter_21_Clean_Book.pdf

Source focus: The Day 21 source material focuses on # 1 0 0 D AY S O F T Q M : D AY 2 1 B O O K C H A P T E R. The source text explicitly discusses the following quality themes and methods: 5S, audit, isolated, inspection, Root Cause.

Section headings identified in the source: Overcoming Cultural Resistance and Implementing SOPs in CA & CMA Firms • 1. EXECUTIVE SUMMARY & PROBLEM CONTEXT • Core Bottlenecks Identified: • 2. QUALITY TOOLS & FRAMEWORK OVERVIEW • mindset: • 3. BREAKDOWN OF KOTTER 8 STEPS (CHANGE MODEL)

Technical terms detected in the source: 5S, audit, isolated, inspection, Root Cause

Source extract 1: Chapter_21_Clean_Book.pdf

1 0 0 D AY S O F T Q M : D AY 2 1 B O O K C H A P T E R

CHAPTER 21: THE HUMAN SIDE OF TQM

Overcoming Cultural Resistance and Implementing SOPs in CA & CMA Firms

Author: Jagriti Singh | Quality Engineer

Document Objective: Establish a structured Change Management model for 50-article CA/CMA firms. This chapter addresses cultural

resistance, eliminates the Scratch Syndrome, and integrates Philip Crosby Quality Absolutes with Kotter 8-Step Model to build shared

quality ownership.

1. EXECUTIVE SUMMARY & PROBLEM CONTEXT

Implementing new Standard Operating Procedures (SOPs) in accounting and auditing firms often meets internal resistance. In a mid-

sized firm with 50 article trainees and senior staff, introducing Digital 5S or Poka-Yoke intake SOPs creates operational friction, especially during peak audit seasons.

Core Bottlenecks Identified:

• The Scratch Syndrome: Staff members fear that adopting new SOPs forces them to discard ongoing work and restart documentation from zero.

• Stakeholder Opinion Clashes: Partners prioritize regulatory compliance, senior managers prioritize speed, and trainees face conflicting instructions due to a lack of uniform process standards.

• Top-Down Mandate Failure: Quality Management fails when treated as an isolated task pushed by a single partner instead of a collective organization-wide responsibility.

2. QUALITY TOOLS & FRAMEWORK OVERVIEW

This implementation relies on three complementary Quality Engineering frameworks to tackle both process control and organizational

mindset:

• Kotter 8-Step Change Model: A structured framework for driving organizational transformation, overcoming employee resistance,

and embedding new behaviors into firm culture.

• Philip Crosby Quality Principles: Focuses on defect prevention rather than post-audit inspection, advocating that quality means absolute conformance to requirements.

• Poka-Yoke & Digital 5S: Technical error-proofing mechanisms (digital checklists and intake templates) designed to make non-compliance impossible and eliminate file clutter.

3. BREAKDOWN OF KOTTER 8 STEPS (CHANGE MODEL)

The 8 steps of John Kotter Change Management Model applied directly to CA/CMA audit workflows:

• 1. Create Urgency: Present historical Price of Non-Conformance (PONC) data showing lost billable hours due to manual re-work.

• 2. Form a Guiding Coalition: Appoint a Quality Committee with 1 Partner, 1 Senior Manager, and 2 Article Leads to represent all

staff levels.

• 3. Develop Vision & Strategy: Set a clear operational goal: "Zero-Defect Audit Filing Without Staff Burnout."

• 4. Communicate Vision: Conduct weekly 15-minute operational alignment sessions to answer doubts and explain benefits.

• 5. Empower Action & Remove Obstacles: Deploy pre-filled digital templates and Poka-Yoke tools so staff do not work from scratch.

• 6. Generate Short-Term Wins: Recognize audit teams that complete 10 consecutive client filings with 100% SOP adherence.

• 7. Consolidate Improvements: Standardize verified SOP workflows across Tax, GST, and Advisory divisions.

• 8. Anchor in Culture: Integrate SOP execution into formal employee performance reviews and onboarding modules.

4. PHILIP CROSBY 4 ABSOLUTES & QUALITY VACCINE

To support Kotter model, the firm adopts Crosby core principles:

• Conformance to Requirements: SOPs define the mandatory baseline for client deliverables.

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• Prevention System: Preventive validation at intake replaces late-stage inspection stress.

• Zero Defects Standard: Rejects routine error tolerance in filing compliance.

• Price of Non-Conformance (PONC): Measures error costs in lost billable hours.

• Quality Vaccine: Combines Leadership Determination, Employee Education, and Seamless Workflow Implementation.

5. SOP RESISTANCE-TO-ADOPTION MATRIX

TQM Solution

Stakeholder & Friction Point Root Cause Analysis Standard Action Directive

Framework

Senior Staff: Claim manual process is Fear of filing delays & Crosby PONC Publish time-log data proving manual errors

faster and SOPs waste time. change anxiety Analysis cause longer rework cycles.

Trainees: Concern over restarting active Scratch Syndrome Error Cause Removal Issue pre-populated Poka-Yoke checklists to

files from zero. auto-migrate legacy data.

Cross-Teams: Managers giving conflicting Lack of single standard Crosby First Absolute Publish one firm-wide approved SOP manual

directives. on internal portal.

6. Epilogue: Shared Quality Ownership

Quality Management is an operational shield for professional firms, not a policing tool. By combining Crosby prevention mindset

with Kotter structured change framework, a 50-article CA/CMA firm eliminates the fear of restarting work. SOPs cease to be an

individual burden and become a shared organizational standard.

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DAY 22

Original source file(s): Day_22_Full_One_Page.pdf

Source focus: The Day 22 source material focuses on # 1 0 0 D AY S O F T Q M : D AY 2 2. The source text explicitly discusses the following quality themes and methods: AUDIT, 5S, Root Cause.

Section headings identified in the source: Standardizing File Archival and Workpaper Closure in CA and CMA Firms • 1. EXECUTIVE SUMMARY AND PROBLEM CONTEXT • Core Bottlenecks Faced by Firms: • 2. TQM TOOLS APPLIED IN THIS PROTOCOL • 3. THE 5 STEP WIND UP PROTOCOL (48 HOURS) • 4. AUDIT CLOSURE READINESS MATRIX

Technical terms detected in the source: AUDIT, 5S, Root Cause

Source extract 1: Day_22_Full_One_Page.pdf

1 0 0 D AY S O F T Q M : D AY 2 2

DAY 22: THE ZERO LOOSE END AUDIT WIND UP PROTOCOL

Standardizing File Archival and Workpaper Closure in CA and CMA Firms

Author: Jagriti Singh | Quality Engineer

Document Objective: Solve post audit chaos in CA and CMA firms through a structured 48 hour closure protocol. This quality engineering

framework prevents loose working papers, eliminates file clutter, and streamlines future tax seasons.

1. EXECUTIVE SUMMARY AND PROBLEM CONTEXT

In many accounting firms, completing an audit report doesn't mean work is truly done. Draft files, unverified spreadsheets, and

unresolved client queries often linger on computers for weeks after signing.

Core Bottlenecks Faced by Firms:

• Open File Fatigue: Teams move to new client projects immediately after report submission, leaving older files partially organized and

cluttering local drives.

• Loss of Critical Context: Months later when tax assessments or notices arrive, finding exact working paper calculations becomes

very hard and time consuming.

• Repeated Rework: Lessons learned during tricky tax calculations are forgotten, forcing new article trainees to solve identical issues

all over again.

2. TQM TOOLS APPLIED IN THIS PROTOCOL

• Digital 5S (Sort and Set in Order): Delete temporary files, organize final working papers into master folders, and permanently lock

shared drives against edits.

• Poka Yoke Exit Gate: A digital check rule preventing team members from archiving a file until every required checklist item is

marked complete.

• Crosby's Price of Non Conformance (PONC): Measuring and tracking lost billable hours caused by missing documents or

unorganized files.

3. THE 5 STEP WIND UP PROTOCOL (48 HOURS)

• 1. Digital Cleanup (0 to 12 hrs): Delete draft spreadsheets and raw client downloads. Move verified calculations into the designated

master folder.

• 2. Poka Yoke Gate (12 to 24 hrs): Ensure every summary sheet links directly to supporting vouchers and partner review notes.

• 3. PONC Log (24 to 30 hrs): Record extra billable hours spent tracking down missing client documents during the audit lifecycle.

• 4. Client Confirmation (30 to 36 hrs): Issue a formal closing email with pending query summaries and final record handovers.

• 5. Knowledge Base (36 to 48 hrs): Save unique tax rulings or complex audit solutions into the firm shared portal for future reference.

4. AUDIT CLOSURE READINESS MATRIX

Closure Friction Point Root Cause Analysis TQM Solution Action Directive

Messy Folders: Multiple file versions like Version 1 No file naming standards Digital 5S Keep only one master folder and delete

Final, Version 2 Final saved on desktop. followed. draft sheets within 24 hours.

Unresolved Queries: Client claims papers were sent, Informal tracking via Poka Yoke Gate Require a single signed query tracker

team disagrees. messaging apps. before closing the file.

Forgotten Solutions: Solving same complex tax issue No central knowledge Knowledge Write a 5 line summary of unique tax

every year. base. Migration treatments in firm portal.

Epilogue: A disciplined wind up process isn't extra paperwork, it is how efficient firms protect their time. By closing audits cleanly

within 48 hours, CA and CMA firms eliminate rework, save billable hours, and ensure seamless handovers for future tax seasons.

Disclaimer: This document has been created independently as part of the #100DaysOfTQM challenge, based purely on personal knowledge, quality engineering

concepts, and practical research.

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DAY 23

Original source file(s): Day_23_Healthcare_TQM.pdf

Source focus: The Day 23 source material focuses on # 1 0 0 D AY S O F T Q M : D AY 2 3. The source text explicitly discusses the following quality themes and methods: 5S, Kaizen, PDCA, Audit, Root Cause.

Section headings identified in the source: 1. EXECUTIVE SUMMARY AND PROBLEM CONTEXT • Major Bottlenecks in Healthcare Operations: • 2. KEY TQM FRAMEWORKS APPLIED • 3. THE PDCA CYCLE FOR ER WORKFLOW OPTIMIZATION • 1. PLAN (Identify & Map) 2. DO (Implement Changes) • 3. CHECK (Monitor Metrics) 4. ACT (Standardize Best Practice)

Technical terms detected in the source: 5S, Kaizen, PDCA, Audit, Root Cause

Source extract 1: Day_23_Healthcare_TQM.pdf

1 0 0 D AY S O F T Q M : D AY 2 3

DAY 23: REDUCING PATIENT WAIT TIME IN EMERGENCY ROOMS

Applying 5S, Kaizen, and PDCA Cycle for Operational Excellence in Healthcare

Author: Jagriti Singh | Quality Engineer

Document Objective: Optimize hospital Emergency Room (ER) workflows and outpatient department (OPD) processing time using core

Total Quality Management (TQM) tools. This operational framework aims to eliminate response delays, reduce patient waiting times, and

prevent medical errors.

1. EXECUTIVE SUMMARY AND PROBLEM CONTEXT

In hospital emergency departments, speed and precision directly impact patient outcomes. However, operational bottlenecks frequently cause

critical delays during peak admission hours.

Major Bottlenecks in Healthcare Operations:

• Extended Wait Times: Disorganized triage and manual registration slow down patient intake and consultation.

• Misplaced Medical Supplies: Emergency crash carts with unsorted supplies force staff to search for critical tools during life-saving

procedures.

• Process Communication Gaps: Lack of standardized handovers between shifts leads to missing patient history and care delays.

2. KEY TQM FRAMEWORKS APPLIED

• 5S System for Crash Carts: Sort (Seiri), Set in Order (Seiton), Shine (Seiso), Standardize (Seiketsu), and Sustain (Shitsuke) applied to

emergency carts so vital medicines are accessible within 5 seconds.

• Kaizen (Continuous Small Improvements): Incremental daily improvements to streamline patient flow from initial arrival to doctor

consultation.

• Poka-Yoke (Mistake Proofing): Color-coded patient wristbands indicating allergies and medical urgency to prevent clinical errors.

3. THE PDCA CYCLE FOR ER WORKFLOW OPTIMIZATION

1. PLAN (Identify & Map) 2. DO (Implement Changes)

Analyze ER bottleneck data. Target a 40% reduction in patient waiting Introduce digital triage scoring at admission. Implement 5S layout for

time and standardize crash cart tray arrangements across all wards. crash carts and color-coded wristbands for fast-track treatment.

3. CHECK (Monitor Metrics) 4. ACT (Standardize Best Practice)

Audit door-to-doctor time daily. Measure inventory search time during Update standard operating procedures (SOPs) for shift handovers.

emergency calls and gather patient feedback scorecards. Conduct regular 5S audits to sustain operational efficiency.

4. HOSPITAL PROCESS IMPROVEMENT MATRIX

Operational Bottleneck Root Cause Analysis TQM Solution Action Directive

Searching for Medicines: Delayed care Unmarked trays and unorganized 5S Inventory Apply standard visual labels and clear

during resuscitation calls. crash cart drawers. Method acrylic tray dividers.

Registration Delays: Long queues at intake Manual form filling during initial Kaizen & Deploy rapid QR-code / ID scan triage

counters. patient entry. Digitization counters for quick intake.

Medication Errors: Wrong drug dosage or Lack of visual alerts on high-risk Poka-Yoke Mandate color-coded patient bands and

delayed administration. patient files. System allergy warning tags.

Epilogue: Applying TQM principles in healthcare is not merely about administrative efficiency; it directly saves lives. By systematically

executing the PDCA cycle, enforcing 5S, and adopting Kaizen, hospitals can deliver faster, safer, and higher-quality care.

Disclaimer: This document has been created independently as part of the #100DaysOfTQM challenge, based on personal knowledge, quality engineering principles, and

practical research.

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DAY 24

Original source file(s): Day_24_Discharge_TQM_v3_NoDash (1).pdf

Source focus: The Day 24 source material focuses on # 1 0 0 D AY S O F T Q M : D AY 2 4. The source text explicitly discusses the following quality themes and methods: PDCA, FMEA, Kaizen, PFMEA, Root Cause, Audit, RPN.

Section headings identified in the source: FEEDBACK • 1. EXECUTIVE SUMMARY AND PROBLEM CONTEXT • 2. INTEGRATED TQM TOOLKIT FOR DISCHARGE OPERATIONS • 3. EXECUTION ROADMAP: PDCA DISCHARGE PROTOCOL • 1. PLAN (1 Hour Target and Root Cause Mapping) 2. DO (Kanban and Patient Specific Forms) • 3. CHECK (Log Audits and Feedback Analysis) 4. ACT (Comprehensive SOP Manual Creation)

Technical terms detected in the source: PDCA, FMEA, Kaizen, PFMEA, Root Cause, Audit, RPN

Source extract 1: Day_24_Discharge_TQM_v3_NoDash (1).pdf

1 0 0 D AY S O F T Q M : D AY 2 4

DAY 24: STREAMLINING HOSPITAL DISCHARGE AND TAILORED PATIENT

FEEDBACK

Cutting Discharge Time to 1 Hour using PDCA, Gemba Walk, FMEA and Patient Specific QFD

Author: Jagriti Singh | Quality Engineer

Document Objective: Limit hospital discharge turnaround time from 4 to 6 hours down to strictly 1 hour. This operational framework

combines PDCA driven Kaizen, Gemba Walk, Process and System FMEA, Kanban tracking, and Patient Specific Quality Function

Deployment (QFD) to eliminate billing bottlenecks and capture patient centric feedback.

1. EXECUTIVE SUMMARY AND PROBLEM CONTEXT

Extended discharge delays of 4 to 6 hours clog hospital beds and frustrate families. A major operational flaw is treating all patient feedback

as uniform because even patients with the same medical condition experience vastly different operational issues.

• 1 Hour Discharge Target: Streamlining physician summaries, pharmacy clearance, and billing to a strict 60 minute limit.

• Feedback Homogeneity Gap: Standard surveys miss individual patient perspectives, leading to incomplete SOPs that fail to prevent

specific errors.

• Process Blindspots: Lack of real-time status tracking between nursing counters, billing, and pharmacy desks.

2. INTEGRATED TQM TOOLKIT FOR DISCHARGE OPERATIONS

• PDCA Cycle and Kaizen: Core execution engine to enforce the 1 hour discharge protocol across all counters.

• Gemba Walk and FMEA: Observing billing counters directly to identify failure points in processes (PFMEA) and IT software (SFMEA).

• Kanban Visual Status Board: Tracking active discharge files via color coded bins to prevent queue bottlenecks.

• Patient Specific VoP and QFD: Administering tailored physical feedback forms designed around individual patient experiences to build

comprehensive, error proof SOP manuals.

3. EXECUTION ROADMAP: PDCA DISCHARGE PROTOCOL

1. PLAN (1 Hour Target and Root Cause Mapping) 2. DO (Kanban and Patient Specific Forms)

Target a strict 1 hour discharge window. Conduct Gemba Walks and Deploy Kanban visual bins for file tracking. Issue tailored physical 1

execute System and Process FMEA at billing and summary desks. page feedback forms customized to capture individual patient

perspectives.

3. CHECK (Log Audits and Feedback Analysis) 4. ACT (Comprehensive SOP Manual Creation)

Audit billing timestamps against the 1 hour KPI. Analyze unique patient Translate diverse VoP inputs through QFD into a master operational

feedback variations to detect subtle operational flaws. SOP manual, standardizing staff procedures to eliminate recurring

errors.

4. PROCESS IMPROVEMENT AND FAILURE ANALYSIS MATRIX

Discharge Bottleneck Gemba and FMEA Findings TQM Solution Action Directive

Summary and Billing Delays: Idle Process FMEA: Uncoordinated file Kanban Visual Use color coded status bins to enforce a 60

time exceeding 1 hour limit. handoffs across departments. Board minute strict turnaround time.

Billing System Errors: Software System FMEA: High RPN in System FMEA Automate insurance documentation and fix

glitches during final reconciliation. insurance claim processing. and Kaizen software bottlenecks.

Generic Feedback Form: Fails to VoP Gap: Uniform surveys miss Patient Specific Design tailored physical feedback forms and

capture unique patient issues. specific operational friction points. VoP and QFD use insights to build comprehensive SOPs.

Epilogue: Achieving a strict 1 hour discharge cycle transforms hospital operations. By combining Gemba Walks, FMEA, Kanban visual tracking,

and patient specific VoP inputs into a master SOP manual, hospitals eliminate operational errors and deliver truly personalized patient care.

Disclaimer: This document has been created independently as part of the #100DaysOfTQM challenge, based on personal knowledge, quality engineering principles,

and practical research.

DAY 25

Original source file(s): Day 25.pdf

Source focus: The Day 25 source material focuses on #100DAYSOFTQM • DAY 25 MILESTONE. The source text explicitly discusses the following quality themes and methods: AUDIT, ISO Standards, ISO compliance, FMEA, Taguchi, 5S, root cause, SPC, 8D, ISO INTERNATIONAL, ISO 9000, ISO 9001:2015.

Section headings identified in the source: 1. EXECUTIVE OVERVIEW AND 25-DAY MILESTONE • 2. COMPREHENSIVE TQM TOOLS DEPLOYED • 3. ISO INTERNATIONAL STANDARDS DEPLOYED • 4. EXECUTED CASE STUDIES MATRIX (DAY 1 TO 24) • Target Enterprise Industry Sector Core TQM Framework Operational Engineering Impact • 5. MASTER CATALOG: 50 FAMOUS TARGET COMPANIES (NEXT 50 DAYS)

Technical terms detected in the source: AUDIT, ISO Standards, ISO compliance, FMEA, Taguchi, 5S, root cause, SPC, 8D, ISO INTERNATIONAL, ISO 9000, ISO 9001:2015, risk-based, ISO 14001

Source extract 1: Day 25.pdf

#100DAYSOFTQM • DAY 25 MILESTONE

DAY 25: MASTER TQM AUDIT & 50 ENTERPRISE PLAYBOOK

Consolidated Quality Engineering Tools, ISO Standards, and 50 Target Companies

Author: Jagriti Singh | Quality Engineer

Quarter Mark Executive Audit: This master document synthesizes all Total Quality Management tools, ISO compliance

standards, and corporate case studies deployed across Day 1 to Day 24. It details 50 top-tier global companies targeted for

implementation over the next 50 days.

1. EXECUTIVE OVERVIEW AND 25-DAY MILESTONE

Reaching the 25 percent milestone represents a major transformation from theoretical quality management to rigorous operational engineering. Over the first 24 days, complex operational challenges across major global enterprises have been solved using structured quality frameworks, statistical tools, and international compliance standards.

2. COMPREHENSIVE TQM TOOLS DEPLOYED

• Poka Yoke (Error Proofing): Implementing physical/mechanical mechanisms to prevent human error and stop defects at source.

• Failure Mode & Effects Analysis (FMEA): Evaluating potential failure modes, calculating RPNs, and mitigating system vulnerabilities.

• Quality Function Deployment (QFD): Translating Voice of Customer (VoC) into technical design specifications and actionable SOPs.

• Taguchi Methods & Robust Design: Applying parameter design & signal-to-noise ratio optimization to minimize process variation.

• 5S Methodology: Establishing discipline through Sort, Set in Order, Shine, Standardize, and Sustain for layout safety and efficiency.

• Gemba Walk: Direct on-site workplace observations to witness execution, evaluate workflows, and spot operational bottlenecks.

• Kanban Visual Management: Deploying visual inventory boards and pull systems to control WIP and eliminate backlogs.

• Just In Time (JIT) Manufacturing: Synchronizing production directly with demand to eliminate inventory holding costs.

• Ishikawa Fishbone Diagram: Executing root cause analysis across Man, Machine, Material, Method, Measurement, and Environment.

• Control Charts & SPC: Monitoring process stability through Statistical Process Control (X-bar R charts, p-charts) to track variation.

• 8D Problem Solving Methodology: Developing 8 Disciplines corrective action reports to contain defects and prevent recurrence.

• Quality Audit Systems: Designing structured compliance checklists and audit trails across cross-functional operations.

3. ISO INTERNATIONAL STANDARDS DEPLOYED

• ISO 9000: Quality Management Fundamentals and core vocabulary.

• ISO 9001:2015: Quality Management Systems Requirements, risk-based thinking, and continuous improvement metrics.

• ISO 14001: Environmental Management Systems, sustainability, and waste reduction.

• ISO 19011: Guidelines for Auditing Management Systems and auditor competence evaluation.

4. EXECUTED CASE STUDIES MATRIX (DAY 1 TO 24)

Target Enterprise Industry Sector Core TQM Framework Operational Engineering Impact

Tata Motors Automotive Poka Yoke, Taguchi

[Middle content condensed for page fit]

nch Gemba Walk & Queue 5S

44 Goldman Sachs Investment Banking & Tech Algorithmic Quality FMEA

45 JPMorgan Chase Financial Services Compliance ISO 19011 Audit

46 IndiGo Airlines Commercial Aviation Turnaround JIT & Safety Audits

47 Air India Aviation Operations Fleet Maintenance FMEA & 5S

48 Emirates Global Aviation & Ground Passenger VoP & Service QFD

49 Adani Ports & SEZ Infrastructure & Ports Heavy Equipment Maintenance JIT

50 DLF Limited Real Estate & Construction Site Safety ISO 45001 & Audits

6. UNIVERSAL EXPANSION STRATEGY: THE NEXT 50 DAYS ROADMAP

The roadmap for the remaining 50 days is anchored on total industry coverage. Total Quality Management and Quality Engineering principles apply universally. Over the coming weeks, each of these 50 global companies will receive dedicated quality interventions, custom ISO compliance structures, and 8D root cause solutions.

Disclaimer: This document has been independently authored by Jagriti Singh as part of the #100DaysOfTQM initiative, based on personal domain expertise, quality engineering principles, and targeted corporate research.

DAY 26

Original source file(s): Boeing_Day26_Quality_Report_Final.pdf

Source focus: The Day 26 source material focuses on THE BOEING COMPANY — QUALITY ENGINEERING. The source text explicitly discusses the following quality themes and methods: SUPPLIER QUALITY, ISO 9001:2015, ISO 19011, SPC, process capability, Cpk, Six Sigma, audit, traceability, Yield, 8D, Root Cause.

Section headings identified in the source: 1. EXECUTIVE SUMMARY • 2. OPERATIONAL PROBLEM STATEMENT • 3. BENCHMARKING & QUALITY ARCHITECTURE • 4. ENGINEERING IMPACT & TARGET METRICS • Metric Indicator Baseline (Current) Benchmark Target Applied TQM Mechanism • 5. CORRECTIVE ACTION & AUTHORIZATION SIGN-OFF

Technical terms detected in the source: SUPPLIER QUALITY, ISO 9001:2015, ISO 19011, SPC, process capability, Cpk, Six Sigma, audit, traceability, Yield, 8D, Root Cause

Source extract 1: Boeing_Day26_Quality_Report_Final.pdf

THE BOEING COMPANY — QUALITY ENGINEERING

STANDARD OPERATING REPORT | DAY 26: SUPPLIER QUALITY MANAGEMENT

Document ID: QA-BOE-2026-D26 Date: August 22, 2026

Standard: AS9100D / ISO 9001:2015 / ISO 19011 Author: Jagriti Singh | Quality Engineer

Subject: Tier-1 Supplier Non-Conformance & Precision Fastener Risk Mitigation

1. EXECUTIVE SUMMARY

This quality engineering report establishes the supplier non-conformance containment and risk mitigation protocol for commercial aerospace operations at Boeing. By targeting critical precision fasteners and sub-assembly hardware, this process integrates ISO 19011 supplier quality audits, real-time Statistical Process Control (SPC), and automated assembly-line Poka Yoke to ensure zero-defect component delivery, eliminate structural assembly bottlenecks, and enforce strict aerospace compliance.

2. OPERATIONAL PROBLEM STATEMENT

• Target Components: Precision Structural Fasteners & Mechanical Assembly Hardware.

• Core Issue: Dimensional variance, material tolerance drift, and thread specification non-conformance from Tier-1 suppliers.

• Manufacturing Impact: Non-conforming fasteners result in shear test failures, thread stripping, assembly line stoppages, and costly manual rework loops during fuselage and wing section joint integration.

3. BENCHMARKING & QUALITY ARCHITECTURE

• Aerospace Benchmarking Target: Calibrated against AS9100D standard requiring process capability index Cpk ≥ 2.00 (Six Sigma capability, ensuring defect rate threshold below 3.4 Defects Per Million Opportunities / PPM).

• ISO 19011 Supplier Audits: On-site audit protocols verifying vendor tool calibration, raw material traceability, and shop-floor SOP execution prior to dispatch authorization.

• Supplier SPC (X■ & R Charts): Real-time process capability monitoring at supplier plants to identify process drift before parts leave the facility.

• Torque-Limit Poka Yoke: Electronic and mechanical torque-sensing assembly tools deployed on the shop floor to eliminate over-tightening and thread stripping at point-of-use.

4. ENGINEERING IMPACT & TARGET METRICS

Metric Indicator Baseline (Current) Benchmark Target Applied TQM Mechanism

Incoming Rejection Rate 4.2% < 0.00034% ISO 19011 Audits & Supplier SPC (Cpk ≥ 2.00)

First-Pass Yield (FPY) 91.5% 99.99% Torque-Limit Poka Yoke Integration

Assembly Rework Time High Loop Delay 35% Reduction 8D Containment & Standardized Fastening SOPs

5. CORRECTIVE ACTION & AUTHORIZATION SIGN-OFF

Root Cause Protocol: Closed via 8D Problem Solving Methodology Implementation: APPROVED

Tier-1 Aerospace

Authorized By: Jagriti Singh (Quality Engineer) Scope:

Fastener Suppliers

Academic & Professional Disclaimer: This document was prepared by Jagriti Singh as part of the #100DaysOfTQM educational initiative. The

methodologies, quality architecture, and benchmarking models presented herein are derived from standard industrial quality engineering principles

(AS9100D / Six Sigma / ISO 19011) and serve purely as an educational case study.

CASE STUDY SERIES: AEROSPACE QUALITY ENGINEERING | #100DAYSOFTQM Page 1 of 1

DAY 27

Original source file(s): Day 27 Airbus .pdf

Source focus: The Day 27 source material focuses on AIRBUS Standard: AS9115 / ISO 19011. The source text explicitly discusses the following quality themes and methods: ISO 19011, Supplier Quality, Six Sigma, Audit, Cpk, 8D, root cause, process capability.

Section headings identified in the source: Global Supplier Quality and Vendor Performance Management Strategy • 1. EXECUTIVE SUMMARY AND CONTEXT • 2. OPERATIONAL PROBLEM AND ASSEMBLY RISKS • 3. TQM ARCHITECTURE AND QUALITY CONTROL TOOLS • Receiving Bay Poka Yoke (Mistake Proofing Integration): • 4. PROCESS CONTROL AND IMPLEMENTATION MATRIX

Technical terms detected in the source: ISO 19011, Supplier Quality, Six Sigma, Audit, Cpk, 8D, root cause, process capability

Source extract 1: Day 27 Airbus .pdf

Author: Jagriti Singh | Quality Engineer

AIRBUS Standard: AS9115 / ISO 19011

Doc ID: QA AIR 2026 D27

Q UA L IT Y A N D O PE R ATI O N S SYSTE M D I RECTIVE Date: August 2026

100 DAYS OF TQM : CASE STUDY DAY 27/100

Global Supplier Quality and Vendor Performance Management Strategy

Integrated AS9115 Audits, Variation Tracking, and Receiving Poka Yoke Framework

1. EXECUTIVE SUMMARY AND CONTEXT

Airbus operates one of the most complex cross border aerospace supply chain in the world, receiving critical components from hundreds of Tier 1 vendors across France, Germany, Spain, and UK. Ensuring that every incoming sub assembly matches exact design specifications is vital to prevent structural assembly disruptions. This directive outlines a standardized quality framework using

AS9115 audits, dynamic supplier scorecarding, Six Sigma tolerance variation tracking, and material receiving Poka Yoke to ensure

zero non conforming parts reach the final assembly line.

2. OPERATIONAL PROBLEM AND ASSEMBLY RISKS

Minor dimensional variations or specification drift in supplied parts relative to engineering blueprints can cause severe bottlenecks

during structural integration.

• Specification Drift: Inconsistency in supplier manufacturing processes leading to parts that deviate slightly from official Airbus engineering drawings.

• Delayed Corrective Actions: Slow resolution of Supplier Corrective Action Requests (SCAR) creating recurring defect patterns across batches.

• Receiving Errors: Human mistakes at the receiving bay leading to incorrect part orientation or mix ups during lot intake.

3. TQM ARCHITECTURE AND QUALITY CONTROL TOOLS

• AS9115 and ISO 19011 Supplier Audits: Standardized on site auditing protocols to evaluate vendor tool calibration, process capability, and quality management readiness before dispatch.

• 4 Pillar Dynamic Supplier Scorecard: Continuous vendor evaluation tracking Quality (defect rates), Delivery (OTIF), Audit Compliance, and SCAR responsiveness.

• Six Sigma and Cpk Variation Control: Used specifically to measure dimensional variations in incoming parts against Airbus design blueprints to verify if supplier outputs remain strictly within specified tolerance limits.

• Assembly to Vendor Closed Loop Feedback: Real time shopfloor feedback routing component assembly issues back to vendor SCAR logs to trigger immediate 8D root cause investigations.

Receiving Bay Poka Yoke (Mistake Proofing Integration):

Automated barcode scanning connected with digital blueprint specs prevents wrong material logging, while physical keying mechanisms at

receiving jigs stop misaligned or out of spec components from entering storage.

4. PROCESS CONTROL AND IMPLEMENTATION MATRIX

Operational Stage Applied Quality Tool Primary Quality Objective

Vendor Manufacturing AS9115 / ISO 19011 Audits Verify process capability and tool calibration standards.

Material Receiving Receiving Poka Yoke Jigs Prevent intake of incorrect specs or misoriented components.

Incoming QA Six Sigma and Cpk Tracking Monitor dimensional variation against official engineering drawings.

Shop Floor Integration Closed Loop SCAR System Automate defect feedback and enforce vendor 8D corrective actions.

Disclaimer: This case study is developed strictly for educational purposes as part of the 100 Days Of TQM learning challenge. The author is not affiliated with, endorsed by, or

officially connected to Airbus. This analysis addresses potential operational scenarios and quality engineering frameworks from an academic perspective.

DAY 28

Original source file(s): Day 28.Tesla_Deming_Case_Study_Day28_v2.pdf

Source focus: The Day 28 source material focuses on TES L A Framework: Deming SoPK & Kaizen. The source text explicitly discusses the following quality themes and methods: Kaizen, SPC, lean, OEE, yield, 8D.

Section headings identified in the source: TES L A Framework: Deming SoPK & Kaizen • 1. CONTEXT & OPERATIONAL CHALLENGE • 2. DEMING’S 4 PILLARS APPLIED TO TESLA OPERATIONS • 3. DUAL-LEVEL BENCHMARKING FRAMEWORK • 4. PROCESS CONTROL & TOOL MAPPING MATRIX • Focus Domain Applied Quality Tool Implementation & Operational Impact

Technical terms detected in the source: Kaizen, SPC, lean, OEE, yield, 8D

Source extract 1: Day 28.Tesla_Deming_Case_Study_Day28_v2.pdf

Author: Jagriti Singh | Quality Engineer

TES L A Framework: Deming SoPK & Kaizen

Doc ID: QA TSLA 2026 D28

QUALIT Y & OPERATIONS EXCELLENCE FRAMEWORK Date: August 2026

100 DAYS OF TQM : CASE STUDY DAY 28/100

Deming's System of Profound Knowledge (SoPK)

Transforming Gigafactory Culture & Assembly Line Stability via SPC, PDSA, and Benchmarking

1. CONTEXT & OPERATIONAL CHALLENGE

Tesla’s aggressive production targets and rapid iteration cycles often push Gigafactory operations to their limits. Fast design changes and high line speeds create shop-floor anxiety, leading to unreported defect patterns and stability issues across stamping, body-in-white (BIW), and final assembly. Managing quality in such an environment requires moving beyond end-of-line inspections to an integrated approach that harmonizes work culture leadership with hard process controls.

2. DEMING’S 4 PILLARS APPLIED TO TESLA OPERATIONS

• Appreciation for a System: Unifying Gigafactory stamping, battery integration, and software deployments into a single value stream. A minor variation in Gigapress casting directly influences downstream door alignment and automated sealings.

• Knowledge About Variation (SPC & Control Charts): Using Control Charts (X-bar & R Charts, P-Charts) on the assembly line to distinguish between Common Cause (systemic equipment wear) and Special Cause (sensor drift or batch

error) variations, preventing over-adjustment and line instability.

• Theory of Knowledge (PDSA & Kaizen): Replacing unstructured, rapid line tweaks with structured Plan-Do-Study-Act (PDSA) loops and shop-floor Kaizen events to test engineering iterations before scaling across production lines.

• Psychology & Leadership ("Drive Out Fear"): Building a culture where assembly operators can report defects or pull line-stop triggers without fear of penalty, allowing root causes to surface instantly.

3. DUAL-LEVEL BENCHMARKING FRAMEWORK

To elevate both organizational mindset and manufacturing output, a strategic two-tier benchmarking model is integrated:

• Work Culture Benchmarking: Benchmarking psychological safety and frontline defect-reporting protocols against world-class lean automotive systems to eliminate fear and foster continuous improvement.

• Functional Benchmarking: Benchmarking assembly line takt time, overall equipment effectiveness (OEE), and first-time pass yield (FTPY) against top-tier automated precision industries to boost productivity.

4. PROCESS CONTROL & TOOL MAPPING MATRIX

Focus Domain Applied Quality Tool Implementation & Operational Impact

Work Culture & Leadership Culture Benchmarking & Open-Door 8D Drives out fear, encouraging operators to flag defects without penalty.

Assembly Line Variation Monitors weld tolerances and panel gaps to prevent process tampering.

SPC (X-bar & R Charts, P-Charts)

Process Improvement PDSA Cycles & Gemba Kaizen Executes controlled, data-backed trials for all rapid design iterations.

Line Productivity Functional Benchmarking Optimizes takt time and OEE against high-speed automated benchmarks.

Disclaimer: This case study is developed strictly for educational purposes as part of the 100 Days Of TQM learning challenge. The author is not affiliated with, endorsed by, or officially connected to Tesla, Inc. This analysis evaluates industrial quality frameworks from an academic and practical perspective.

DAY 29

Original source file(s): Day 29 Samsung_Quality_Case_Study_v2.pdf

Source focus: The Day 29 source material focuses on SA M S U N G Standards: ISO 9001:2015 | ISO 9004 | ISO 19011. The source text explicitly discusses the following quality themes and methods: ISO 9001:2015, ISO 9004, ISO 19011, Taguchi, Root Cause, Isolating, risk-based, ISO 9004:2018, ISO 9001, ISO 19011:2018, Audit, ISO compliance.

Section headings identified in the source: Standards • 1. STRATEGIC CONTEXT & LEADERSHIP TRANSFORMATION • 2. SHOP-FLOOR EMPOWERMENT & COST OF QUALITY (COQ) • 3. ROBUST ENGINEERING & PRODUCTIVITY ENHANCEMENT • problem-solving methodologies are deployed: • QUALIT Y MANAGEMENT FRAMEWORK & CONCEPT SUMMARY

Technical terms detected in the source: ISO 9001:2015, ISO 9004, ISO 19011, Taguchi, Root Cause, Isolating, risk-based, ISO 9004:2018, ISO 9001, ISO 19011:2018, Audit, ISO compliance, yield, PDCA

Source extract 1: Day 29 Samsung_Quality_Case_Study_v2.pdf

Author: Jagriti Singh | Quality Engineer

SA M S U N G Standards: ISO 9001:2015 | ISO 9004 | ISO 19011

Doc ID: QA-SMSG-2026-D29

QUALIT Y LEADERSHIP & MANAGEMENT TRANSFORMATION FRAMEWORK Date: August 2026

100 DAYS OF TQM : CASE STUDY DAY 29/100

Samsung's Total Quality Transformation & Management Strategy

A Strategic Analysis of Executive Leadership, Worker Empowerment, Taguchi Robust Engineering, and International Quality Standards

1. STRATEGIC CONTEXT & LEADERSHIP TRANSFORMATION

In 1993, Samsung executed one of the most remarkable strategic turnarounds in modern industrial history under Chairman Lee Kun-hee. Shifting the core operational paradigm from volume-driven production to absolute quality supremacy, Chairman Lee declared at the historic Frankfurt Declaration: "Change everything except your wife and children." This wasn't merely a slogan, but

a top-down executive directive recognizing that accepting even a 1% defect rate fundamentally compromises 100% of customer trust.

Strategic Alignment with Deming's 14 Points:

1. Adopt the New Philosophy: Management established that change is an absolute operational law. Accepting the new definition of quality required a complete cultural mindset shift across all management layers.

2. Eliminate Slogans & Exhortations: Transitioning from passive quality goals to active structural reform, enforcing zero tolerance for compromised standards.

2. SHOP-FLOOR EMPOWERMENT & COST OF QUALITY (COQ)

• Worker Empowerment & Andon Line-Stop Authority (Deming's "Drive Out Fear"): Senior executives cannot constantly monitor shop-floor operations. Empowering assembly line operators with complete authority to trigger line stops (Andon system) upon detecting non-conformance eliminates fear, encouraging workers to expose and rectify defects at the source rather than hiding them due to production pressure.

• Risk Management & Failure Cost Trade-off: Allowing non-conforming items to progress down the assembly line incurs escalating costs. Catching defects on the shop floor causes Internal Failure Cost, but letting defective products reach consumers triggers catastrophic External Failure Cost—destroying warranty budgets, brand reputation, customer trust, and long-term retention.

• Zero Defects & Quality Sacrifice (Crosby's Philosophy): In 1995, following defect reports in the SH-770 mobile phone, Chairman Lee ordered the public burning of 150,000 defective devices valued at $50 million in front of 2,000 employees. Aligning with Philip Crosby's Zero Defects ("Do it right the first time"), this extreme decision demonstrated that short-term financial sacrifice is mandatory to protect long-term brand equity and customer faith.

3. ROBUST ENGINEERING & PRODUCTIVITY ENHANCEMENT

• Taguchi's Robust Design & Design of Experiments (DOE): Quality must be designed into the product from inception. By utilizing Taguchi's Design of Experiments (DOE), Samsung optimizes controllable factors to make product performance

immune to uncontrollable "no

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ing engineering resources on vital

responsible for 80% of quality issues. defect categories causing high yield

loss.

ISO 9001:2015 Standard specifying requirements for a formal QMS based on risk Baseline quality certification across all

management and PDCA cycles. global assembly plants and SMT lines.

ISO 9004:2018 Management standard providing guidance for organizational Strategic self-assessment framework for

maturity and sustained long-term success. continuous organizational maturity and

growth.

ISO 19011 Three-Tier Comprehensive audit guideline covering 1st Party (Internal), 2nd Strict multi-layer quality governance

Audit Party (Supplier), and 3rd Party (External) audits. from component suppliers to finished

goods shipping.

Disclaimer: This 2-page case study is developed strictly for educational purposes as part of the 100 Days Of TQM learning challenge. Author: Jagriti Singh, Mechanical

Engineering / Quality Engineer. This analysis evaluates management, TQM, and ISO standards from an academic and industrial engineering perspective.

100 Days Of TQM | Day 29 Case Study Page 2 of 2

DAY 30

Original source file(s): Day 30 Intel_Quality_Case_Study__Final.pdf

Source focus: The Day 30 source material focuses on Jagriti Singh | Quality Engineer. The source text explicitly discusses the following quality themes and methods: ISO 9001, ISO 9004, ISO 45001, ISO 27001, ISO
systems, RCA, Root Cause, ISO SYSTEMS, ISO STANDARDS, ISO 9001:2015, ISO 9004:2018, ISO 45001:2018.

Section headings identified in the source: 1. MANAGEMENT CLARITY & DEMING'S SYSTEMIC RESPONSIBILITY • 2. FRONTLINE EMPOWERMENT & NO-BLAME RCA CULTURE • 3. COST OF QUALITY: THE 1:100 PREVENTION LEVERAGE • The Prevention Cost Economics: • INTEGRATED ISO SYSTEMS & OPERATIONAL TOOL MATRIX • 4. INTEGRATED ISO STANDARDS IN SEMICONDUCTOR FABS

Technical terms detected in the source: ISO 9001, ISO 9004, ISO 45001, ISO 27001, ISO
systems, RCA, Root Cause, ISO SYSTEMS, ISO STANDARDS, ISO 9001:2015, ISO 9004:2018, ISO 45001:2018, ISO STANDARD

Source extract 1: Day 30 Intel_Quality_Case_Study__Final.pdf

Jagriti Singh | Quality Engineer

intel Standards: ISO 9001 | ISO 9004 | ISO 45001 | ISO 27001

Doc Ref: QA INTL 2026 D30

FAB EXCELLENCE, OKRs & DEMING'S SoPK FRAMEWORK Date: 26 August 2026

100 DAYS OF TQM (DAY 30 MILESTONE STUDY, 1 MONTH RECAP)

Intel's OKR Framework & Deming's System of Profound Knowledge

Practical engineering analysis on semiconductor Fab operations, worker empowerment, prevention cost rules, and integrated ISO

systems.

1. MANAGEMENT CLARITY & DEMING'S SYSTEMIC RESPONSIBILITY

In high-tech semiconductor fabs, if management doesn't clarify specs and customer requirements upfront, engineers end up wasting a lot of time on wrong priorities. Intel solved this early using Andy Grove's OKR (Objectives and Key Results) system, setting super clear goals top-down so nobody is guessing what needs to be done on shop floor.

Connecting OKRs with Deming's 85/15 Rule & SoPK:

• Common Cause Ownership (85%): Dr. Deming pointed out that 85% of process issues comes from system design itself, which only management can fix. Clear OKRs remove operational confusion and stabilize common cause variations.

• Special Cause Focus (15%): When daily confusion is gone, engineers actually get bandwidth to track down sudden Special Causes before silicon wafer batches get out of control.

• System of Profound Knowledge (SoPK): Combines understanding systems, managing process variations, using data-driven Key Results, and respecting floor worker psychology.

2. FRONTLINE EMPOWERMENT & NO-BLAME RCA CULTURE

• Empowerment with Accountability: Operators working inside cleanrooms must have direct authority to stop the line if something looks off. If people are scared of getting scolded, they hide defects, and bad parts move forward. Giving worker authority ensures immediate action right where defect happens.

• No-Blame Culture & RCA: When micro-defects happen during chip assembly, blaming operators solves nothing because issues are usually systemic. Intel focuses on structured Root Cause Analysis (RCA), using 5-Why Analysis and Fishbone Diagrams to fix the real technical root cause rather than accusing people.

3. COST OF QUALITY: THE 1:100 PREVENTION LEVERAGE

In chip manufacturing, letting an out-of-control defect pass through assembly can ruin millions of dollars worth silicon wafers in just one shift. Quality engineering economics makes it very clear:

The Prevention Cost Economics:

Spending $1 on Prevention (Clear SOPs, Training & Setup) → Saves $100 in Scrap & Internal/External Failures

Putting small budget into prevention and appraisal prevents massive wafer scrap, customer claims, and brand damage later on.

100 Days Of TQM | Day 30 Milestone Case Study Page 1 of 2

intel Jagriti Singh | Quality Engineer

Doc Ref: QA INTL 2026 D30 | Page 2 of 2

INTEGRATED ISO SYSTEMS & OPERATIONAL TOOL MATRIX

4. INTEGRATED ISO STANDARDS IN SEMICONDUCTOR FABS

• ISO 9001:2015 & ISO 9004:2018 (Baseline QMS + Sustained Success): ISO 9001 gives the basic baseline for process compliance and

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revent

rework, and failure costs later. multi million dollar wafer scrap.

ISO 9001:2015 Basic Quality Management System standard focusing on Standardized cleanroom manufacturing process compliance & SOPs. and process control limits.

ISO 9004:2018 Guidelines for sustained organizational success and Drives regular internal self assessments

continuous maturity. for long term semiconductor leadership.

ISO 45001:2018 Occupational Health & Safety standard keeping worker Safety checks for hazardous cleanroom

safety as top priority. chemicals, gas lines, and robotics.

ISO/IEC 27001:2022 Information Security system for protecting intellectual Secures chip architecture designs, Fab

property and sensitive data. recipes, and proprietary software.

Disclaimer & Notes: Author: Jagriti Singh, Quality Engineer. This case study is developed strictly from an educational perspective for the Day 30 milestone of the

100 Days Of TQM challenge. It aims to analyze industry operational problems and synthesize self-derived quality engineering solutions.

100 Days Of TQM | Day 30 Milestone Case Study Page 2 of 2

DAY 31

Original source file(s): Day 31_Toyota_Case_Study_v2.pdf

Source focus: The Day 31 source material focuses on TOYOTA PRODUCTION SYSTEM & RED X DIAGNOSTICS. The source text explicitly discusses the following quality themes and methods: ROOT CAUSE, isolate, Isolating.

Section headings identified in the source: TOYOTA PRODUCTION SYSTEM & RED X DIAGNOSTICS • 1. PROBLEM IDENTIFICATION: ENGINE ASSEMBLY LINE ALIGNMENT • 2. ROOT CAUSE ELIMINATION: APPLYING SHAININ'S RED X • clear levels: • 3. QUALITY GUARDRAILS: JIDOKA & ANDON INTEGRATION • 4. FAIL-SAFE PREVENTION: PHYSICAL & DIGITAL POKA-YOKE

Technical terms detected in the source: ROOT CAUSE, isolate, Isolating

Source extract 1: Day 31_Toyota_Case_Study_v2.pdf

TOYOTA PRODUCTION SYSTEM & RED X DIAGNOSTICS

Author: Jagriti Singh Date: 27 August 2026 | Ref: QA/TYT/2026/D31

1. PROBLEM IDENTIFICATION: ENGINE ASSEMBLY LINE ALIGNMENT

In engine assembly lines, minor variations in cylinder head bolt tightness often pass basic sensor checks during

initial assembly, but later lead to engine vibration during final testing.

To analyze why standard sensors fail to catch this defect early, the production line workflow requires a thorough

review. In typical line setups, operators experience slight angle resistance while holding heavy tightening tools.

Tool support arm alignment issues, along with slight hand position variations, introduce bolt tightening

inconsistencies.

2. ROOT CAUSE ELIMINATION: APPLYING SHAININ'S RED X

To find the exact cause of variation systematically, Dorian Shainin’s Red X method offers a practical diagnostic

framework. By comparing assembly results across different tools and stations, key process variables isolate into

clear levels:

• Red X (Main Root Cause): Micro-bending in the tool support arm, which reduces actual tightening force by 12

percent at peak torque.

• Pink X (Secondary Cause): Surface roughness differences on bolt seating faces across supplier batches.

• Pale Pink X (Minor Factor): Room temperature variations affecting oil viscosity on assembly parts.

Reinforcing the tool support arm eliminates the main Red X root cause and ensures consistent bolt tightening torque every time.

3. QUALITY GUARDRAILS: JIDOKA & ANDON INTEGRATION

To prevent faulty parts from moving to the next station, Jidoka (built-in quality) logic can be programmed into the station controller. Configuring the tool control unit to track tightening torque in real time ensures that if torque values fall outside set safety limits, the system instantly triggers an Andon alert light and stops the line conveyor, allowing operators to fix the bolt alignment immediately.

4. FAIL-SAFE PREVENTION: PHYSICAL & DIGITAL POKA-YOKE

To prevent human errors completely on the assembly line, a simple two-part Poka-Yoke fail-safe setup is recommended:

• Physical Poka-Yoke: Mounting a self-aligning mechanical socket guide that prevents the wrench from operating unless aligned at an exact 90-degree angle to the engine block.

• Digital Poka-Yoke: Setting up an automated sensor interlock that locks downstream test equipment until the tightening tool sends a verified pass signal.

5. INTEGRATED FRAMEWORK SUMMARY

The table below summarizes the quality engineering principles evaluated in this case study, outlining their core definitions and recommended practical solutions on the shop floor.

Methodology Core Definition Recommended Solution

Genchi Genbutsu Going directly to the shop floor to observe actual physical conditions firsthand Checking physical tool alignment and operator handling dynamics directly instead of relying only on sensor readings

Shainin Red X Isolating the single main variable causing process variation root cause affecting bolt tightness and adding structural reinforcement

Jidoka & Andon Stopping the assembly line immediately when defects occur to contain errors Configuring dynamic torque tracking to automatically halt the conveyor belt and trigger Andon warning lights when errors happen

Poka-Yoke Designing simple physical and digital mechanisms to prevent mistakes and setting digital sensor interlocks for error-proof assembly Installing a 90-degree mechanical socket guide

Educational Disclaimer: This analysis is developed independently for educational purposes as part of the #100DaysOfTQM initiative. All Toyota Production System methodologies described represent standard quality management frameworks applied to a simulated automotive engineering scenario.

DAY 32

Original source file(s): Day 32_Hyundai_Kano_TQM__v3.pdf

Source focus: The Day 32 source material focuses on HYUNDAI MOTOR COMPANY. The source text explicitly discusses the following quality themes and methods: customer satisfaction, Reliability, traceability.

Section headings identified in the source: HYUNDAI MOTOR COMPANY • Kano Model & Customer-Centric Quality Management • 1. PROBLEM IDENTIFICATION: VOICE OF CUSTOMER TO QUALITY REQUIREMENTS • 2. CUSTOMER EXPECTATION ANALYSIS: APPLYING THE KANO MODEL • 3. CUSTOMER DATA MANAGEMENT: KANBAN-BASED VISUAL CONTROL • 4. REQUIREMENT VALIDATION & EXPERT QUALITY REVIEW

Technical terms detected in the source: customer satisfaction, Reliability, traceability

Source extract 1: Day 32_Hyundai_Kano_TQM__v3.pdf

HYUNDAI MOTOR COMPANY

Kano Model & Customer-Centric Quality Management

Author: Jagriti Singh | Quality Engineer Date: 28 August 2026 Ref: QA/HYN/2026/D32

1. PROBLEM IDENTIFICATION: VOICE OF CUSTOMER TO QUALITY REQUIREMENTS

In a customer-driven automotive business, customer expectations are continuously collected through surveys, reviews, complaints,

service interactions, and ownership experiences. However, every customer request cannot automatically be treated as an equally important product or service requirement. Different customers may have different expectations, while two requirements may also conflict when they are considered for practical implementation.

For a Quality Engineer, the challenge is therefore not simply to collect what customers are asking for, but to understand which expectations actually influence customer satisfaction and which requirements should receive higher priority. If every expressed demand is implemented without structured evaluation, unnecessary complexity, additional cost, and implementation difficulties can

increase, which may eventually contribute to quality issues, rework, or corrective work.

The objective is to create a systematic bridge between the Voice of Customer and actual quality or product decisions, while maintaining a practical balance between customer satisfaction, quality, feasibility, and implementation effort.

2. CUSTOMER EXPECTATION ANALYSIS: APPLYING THE KANO MODEL

The Kano Model provides a structured way to understand how different customer requirements influence satisfaction. Instead of treating every customer expectation equally, requirements can be evaluated as Must-Be, One-Dimensional, Attractive, Indifferent,

or Reverse characteristics.

• Must-Be: Basic expectations where the absence of the requirement can create significant dissatisfaction.

• One-Dimensional: Requirements that generally increase customer satisfaction as their performance improves.

• Attractive: Requirements that may not always be explicitly expected but can create additional satisfaction when provided.

• Indifferent: Features that have little meaningful influence on satisfaction regardless of execution.

• Reverse: Characteristics that may actually reduce satisfaction for certain customer groups when implemented.

For Hyundai, the Kano analysis would be developed using customer responses from Kano questionnaires together with customer complaints, reviews, surveys, and service feedback. The purpose is to identify the actual satisfaction impact behind an expressed requirement rather than assuming that every customer request should automatically become a product or service priority.

3. CUSTOMER DATA MANAGEMENT: KANBAN-BASED VISUAL CONTROL

Large volumes of customer information can become difficult to manage when questionnaire responses, complaints, reviews, and

service feedback are collected through different channels. A Kanban-based visual management approach can therefore be used to organize the movement of customer-quality in

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n → QFD

(House of Quality) → Feasibility & Cost-Benefit Analysis → Quality Decision & Implementation

Voice of Customer provides the raw customer expectations through questionnaires, complaints, reviews, and service feedback. The Kano Model evaluates how these expectations influence satisfaction, while a Kanban-based visual workflow helps maintain systematic movement and traceability of the collected information.

After Kano analysis, experienced cross-functional review and customer re-validation can be used where necessary. The prioritized

requirements can then enter QFD/House of Quality, where customer expectations are translated into quality and product requirements. Finally, cost-benefit and feasibility analysis supports the decision regarding which requirements should become actual quality or product priorities.

The objective is not to reject customer requirements or to implement every request blindly. The objective is to listen carefully, understand objectively, prioritize systematically, and implement responsibly.

100 Days Total Quality Engineering Challenge | Day 32 Page 2 of 2

DAY 33

Original source file(s): Day 33_100.pdf

Source focus: The Day 33 source material focuses on Q UA L I T Y E N G I N E E R I N G C A S E S T U DY D O C U M E N T. The source text explicitly discusses the following quality themes and methods: SERVICE QUALITY, PDCA, customer satisfaction, Inspection.

Section headings identified in the source: Q UA L I T Y E N G I N E E R I N G C A S E S T U DY D O C U M E N T • MAHINDRA & MAHINDRA • SERVICE QUALITY & CUSTOMER WAITING-TIME REDUCTION • Executive Context & Scope: • This case study presents an independent Quality Engineering analysis of customer waiting-time • 1. PROBLEM IDENTIFICATION: WAITING TIME ACROSS THE SERVICE JOURNEY

Technical terms detected in the source: SERVICE QUALITY, PDCA, customer satisfaction, Inspection

Source extract 1: Day 33_100.pdf

Q UA L I T Y E N G I N E E R I N G C A S E S T U DY D O C U M E N T

MAHINDRA & MAHINDRA

SERVICE QUALITY & CUSTOMER WAITING-TIME REDUCTION

Executive Context & Scope:

This case study presents an independent Quality Engineering analysis of customer waiting-time management and process optimization within an automotive service environment. By combining Total Quality Management (TQM) principles, detailed customer journey mapping, root-cause diagnostics (5 Why & Shainin Red X), and structured PDCA execution, this report outlines a practical framework for reducing operational delays while enhancing overall customer satisfaction.

Author: Jagriti Singh | Quality Engineer

Series & Day: Day 33 | #100DaysOfTQM

Document Ref: QA/M&M/2026/D33

Topic Area: Service Operations, Queue Management & Quality Control

#100DaysOfTQM | Day 33 | Ref: QA/M&M/2026/D33 Page 1 of 3

MAHINDRA & MAHINDRA

SERVICE QUALITY & CUSTOMER WAITING-TIME REDUCTION — PAGE 2

1. PROBLEM IDENTIFICATION: WAITING TIME ACROSS THE SERVICE JOURNEY

Customer waiting time in an automotive service centre is rarely created by a single problem. A delay may begin while getting an

appointment, during workshop check-in, while waiting for the vehicle to be attended, because of uneven workload, lack of required

parts or resources, or because the expected completion time is not communicated clearly. A delay at one stage can also affect the

customers and activities that follow.

For this reason, waiting time should not be treated only as a scheduling problem. It should be studied as part of the complete service

process. The objective is to understand where delays are being created, how they move through the service journey, and how they

finally affect the customer's overall service experience.

For this case, the focus is not only on making the service faster. The aim is to reduce customer waiting time while improving the overall

service experience, without disturbing the parts of the process that are already working effectively.

2. UNDERSTANDING THE COMPLETE SERVICE PROCESS

Improvement should begin by understanding the process before attempting to change it. The complete customer journey can be

mapped from the first service appointment request to the final feedback and follow-up. This makes it easier to see where customers

are waiting, where work is getting delayed, and where coordination between different service activities may be creating a gap.

The proposed service flow begins with Service Appointment Request, followed by Appointment Confirmation, Customer Arrival at

Workshop, Vehicle Reception and Check-in, Initial Vehicle Inspection, Service Requirement Confirmation, Job Card Creation, Parts

and Material Availability Check, Technician Assignment, Service or Repair Work, Quality Check, Final Vehicle Check, Service

Completion, Billing and Payment, Customer Notification, Vehicle Handover, Customer Feedback, Complaint Resolution if required, and

finally Feedback Analysis and Continuous Improvement.

Process mapping

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customer feedback is combined with actual waiting-time data to validate whether the

service improvement worked from both the process and customer perspectives.

Standardization / SOP: Standardization converts a proven and stable improvement into a consistent working method. In this case, the

improved process is monitored first and, only after sustainable results are confirmed, converted into a formal SOP.

“From waiting time to customer experience, quality improvement begins with understanding the process.”

EDUCATIONAL DISCLAIMER: This case study has been independently developed for educational and learning purposes as part of the #100DaysOfTQM initiative.

It applies TQM and Quality Engineering concepts to a Mahindra & Mahindra service-quality context using independent analysis and publicly available context. It

does not represent an official Mahindra & Mahindra project, internal company process or company-approved solution. The author is not representing or claiming

affiliation with Mahindra & Mahindra through this case study.

#100DaysOfTQM | Day 33 | Ref: QA/M&M/2026/D33 Page 3 of 3

DAY 34

Original source file(s): Day 34 Maruti Suzuki .pdf

Source focus: The Day 34 source material focuses on MARUTI SUZUKI. The source text explicitly discusses the following quality themes and methods: service quality, audit, isolated, PDCA, SIX SIGMA, ISO
13053, ISO 18404, Lean, ISO 13053.

Section headings identified in the source: MARUTI SUZUKI • IMPROVEMENT APPROACH & NET WORK STANDARDIZATION • The purpose is to move from repeated local correction towards understanding whether a common cause • TQM TOOLS & METHODS APPLIED

Technical terms detected in the source: service quality, audit, isolated, PDCA, SIX SIGMA, ISO
13053, ISO 18404, Lean, ISO 13053

Source extract 1: Day 34 Maruti Suzuki .pdf

MARUTI SUZUKI

QUALITY ENGINEERING & TOTAL QUALITY MANAGEMENT CASE STUDY #100DaysOfTQM

CONSISTENT QUALITY ACROSS 6,000+ SERVICE TOUCHPOINTS

DAY: 34 DATE: 30 August 2026 REF: QA/MS/2026/D34 AUTHOR: Jagriti Singh | Quality Engineer

CASE CONTEXT & OBJECTIVE: This is an independently developed educational Quality Engineering case study based on publicly available

information and TQM principles. It explores how a large service network can maintain a defined level of service quality while allowing flexibility for

local conditions. It does not represent an official Maruti Suzuki project or internal company study, nor does it claim that every service touchpoint

currently has a quality issue.

01 THE QUALITY CHALLENGE ACROSS A LARGE SERVICE NETWORK

Maintaining consistent service quality becomes a different challenge when a company operates through a very large number of service

touchpoints. A customer visiting a service centre in a major city may have access to different infrastructure, resources and support compared

with a customer visiting a smaller city or town. These differences can naturally affect how a service is delivered.

Location is therefore one factor that can influence consistency, but it is not the only one. Customer expectations and the way customers

judge their service experience can also differ. At the same time, differences in service methods, resource availability, training and local

practices can create variation between touchpoints.

The challenge is not to make every service centre completely identical. The more practical objective is to ensure that every customer

receives a defined level of quality, while allowing reasonable flexibility according to local conditions. This case therefore looks at a simple

TQM question: how can a large service network maintain a common quality expectation while still responding to the realities of different

locations?

02 ESTABLISHING A COMMON QUALITY STANDARD

The first step is to define a common minimum quality standard that every applicable service touchpoint is expected to meet. This standard

provides a common direction across the network and creates a clear reference for measuring service performance.

However, standardization should not mean that every location must operate in exactly the same way. A smaller city may have different

infrastructure or resource limitations compared with a major city. Therefore, the standard should define what level of quality must be

maintained, while allowing suitable flexibility in how that level is achieved.

This approach creates a balance between consistency and local reality. The objective is not to remove every difference between locations,

but to prevent those differences from creating an unacceptable difference in the customer's basic service experience.

03 MEASURING CONSISTENCY THROUGH MULTIPLE SOURCES

Once a common quality standard is defined, the next requirement is to measure whether it is actually being maintained. A single

measurement cannot always provide

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audits and KPIs to verify that process changes translate into

real service improvement.

ISO 13053 & ISO 18404: International standards providing guidance on Six Sigma methodology implementation and Lean/Six Sigma

competencies across the network.

Standardization & SOP: Converts proven, stable improvements into formal Standard Operating Procedures to anchor high quality into daily

operations.

“Quality becomes consistent when standards are shared, performance is measured, and improvement is continuously

carried across the network.”

Educational Disclaimer: This case study has been independently developed for educational and learning purposes as part of the #100DaysOfTQM initiative. It applies TQM and

Quality Engineering concepts to a Maruti Suzuki service-quality context using independent analysis and publicly available information. It does not represent an official Maruti

Suzuki project, internal company process or company-approved solution. The author is not representing or claiming affiliation with Maruti Suzuki through this case study.

Ref: QA/MS/2026/D34 | #100DaysOfTQM Page 2 of 2

DAY 35

Original source file(s): Day 35.pdf

Source focus: The Day 35 source material focuses on RELIANCE INDUSTRIES. The source text explicitly discusses the following quality themes and methods: SUPPLIER QUALITY, SPC, AUDIT, INSPECTION, ROOT CAUSE, process capability, Cpk, ISO 2859-1, LEAN, ISO INTEGRATION, 5S, ISO 9001.

Section headings identified in the source: RELIANCE INDUSTRIES • SUPPLY CHAIN & SUPPLIER QUALITY ASSURANCE • EXECUTIVE CASE STUDY: VENDOR QUALITY & DELIVERY (OTIF) OPTIMIZATION • DIRECT ON-SITE GEMBA WALK FINDINGS (VENDOR FACTORY FLOOR AUDIT): • WORKPLACE LAYOUT IN-LINE INSPECTION EQUIPMENT MAINTENANCE • TOTAL WAREHOUSE REJECTIONS GARMENT DIMENSIONAL MISMATCHES BARCODE SCANNING FAILURES

Technical terms detected in the source: SUPPLIER QUALITY, SPC, AUDIT, INSPECTION, ROOT CAUSE, process capability, Cpk, ISO 2859-1, LEAN, ISO INTEGRATION, 5S, ISO 9001, ISO 45001, ISO 13053

Source extract 1: Day 35.pdf

RELIANCE INDUSTRIES

SUPPLY CHAIN & SUPPLIER QUALITY ASSURANCE

Document Ref: RIL/SQA/2026/D35 Author: Jagriti Singh (Quality Engineer) Date: 31 August 2026

Program: 100-Day TQM Challenge (Day 35)

EXECUTIVE CASE STUDY: VENDOR QUALITY & DELIVERY (OTIF) OPTIMIZATION

Target Entity: Reliance Retail Limited | Methodology: Gemba Walk, Shainin RedX, Advanced SPC & Regulatory Integration

01 OPERATIONAL CONTEXT & EXECUTIVE OVERVIEW

Reliance Retail manages a massive procurement ecosystem supplied by thousands of garment and packaging vendors across India.

Operating at this scale creates a constant challenge: variations on vendor shop floors directly translate into defective incoming shipments

at Distribution Centers (DCs) and delivery delays. To address stock-outs and excess return logistics, this case study establishes a

practical Quality Engineering framework designed to bring inward DC defect rates down to below 1%, enforce root-cause problem

solving at vendor facilities, and build a scalable supplier governance model.

02 PROBLEM STATEMENT & GEMBA WALK OBSERVATIONS

CORE CHALLENGE: Frequent inward shipment rejections caused by inaccurate garment dimensions, incorrect product labels, and

damaged transit packaging across retail fulfillment lines.

DIRECT ON-SITE GEMBA WALK FINDINGS (VENDOR FACTORY FLOOR AUDIT):

WORKPLACE LAYOUT IN-LINE INSPECTION EQUIPMENT MAINTENANCE

Disorganized packing areas causing Lack of mid-production quality checkpoints, Overused barcode printers producing

accidental mixing of clothing sizes and letting minor stitching and cutting variations faded, unscan-able labels; worn-out

wrong barcode labels on boxes. pass through unnoticed into final bulk packing. cutting machinery drifting off specification

limits.

03 ROOT CAUSE IDENTIFICATION & QUALITY DIAGNOSTICS

A. Fishbone Analysis & Pareto Prioritization (80/20 Rule):

Initial data collection revealed that 83% of total warehouse rejections were driven by just two major issues: garment dimensional

mismatches (52%) and barcode scanning failures (31%).

83% 52% 31%

TOTAL WAREHOUSE REJECTIONS GARMENT DIMENSIONAL MISMATCHES BARCODE SCANNING FAILURES

B. Shainin RedX Methodology (Pinpointing the Dominant Cause):

RedX (Primary Root Cause):

Thermal drift and calibration loss in vendor laser cutting machines, causing fabric panels to shrink off-spec during extended high-speed

production runs.

Pink X (Secondary Root Cause):

Degraded printheads on vendor barcode label printers, resulting in low-contrast labels that fail automated warehouse scanners.

C. Statistical Process Control (SPC Execution):

Rolled out continuous X-bar and R Control Charts at vendor cutting and stitching stations to monitor daily dimension variations, locking

process capability targets to Cpk > 1.33.

Process Capability Target: Continuous SPC monitoring enforces strict statistical control across critical vendor cutting operations, ensuring

stable manufacturing output aligned with Cpk > 1.33 quality thresholds.

04 DUAL-GATE QUALITY

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fies vendor quality inspectors and auditors to ensure technical

competence across production lines.

MSME ZED Scheme Government of India Incentive Connects small suppliers with Government Zero Defect Zero Effect

subsidies (60–80% cost support) to upgrade quality infrastructure.

06 APPLIED TQM TOOLS & POLICY MAPPING SUMMARY

Problem-Solving Chain:

Gemba Walk → Cause-and-Effect Analysis → Pareto Charting → Shainin RedX → SPC Control Charts.

Defect Prevention Chain:

5S Workplace Cleanliness → Conveyor Poka-Yoke Automation → Dual-Gate (PSI + DC Inward) Inspections.

Governance & Sustainability Synergy:

International Quality & Safety Standards (ISO 9001, 45001, 13053, 18404) tied directly to MSME ZED Scheme (60–80% cost support) to

build a zero-defect supply chain.

DISCLAIMER & ACADEMIC DECLARATION

This document has been authored by Jagriti Singh (Quality Engineer) as part of the 100-Day TQM Challenge (Day 35) submission dated 31 August

2026. Developed exclusively for educational and analytical purposes. The author is an independent researcher not affiliated with Reliance Industries

Limited.

DAY 36

Original source file(s): Day 36 _100.pdf

Source focus: The Day 36 source material focuses on TATA STEEL LIMITED. The source text explicitly discusses the following quality themes and methods: Six Sigma, DMAIC, yield, audit, isolate, ISO SYSTEMS, root cause, isolated, ISO frameworks, inspection, ISO Optimize.

Section headings identified in the source: TATA STEEL LIMITED • Engineer) • 1. Executive Summary and Research Objective • 2. Problem Statement • 3. Methodology and Step-by-Step Implementation • philosophies with advanced analytical tools:

Technical terms detected in the source: Six Sigma, DMAIC, yield, audit, isolate, ISO SYSTEMS, root cause, isolated, ISO frameworks, inspection, ISO Optimize

Source extract 1: Day 36 _100.pdf

TATA STEEL LIMITED

Quality Assurance and Management Division | Research Case Study

Document Ref ID: TTL/TQM/100/2026/D36 Author: Jagriti Singh (Quality

Engineer)

Tools Used: Deming's Profound Knowledge, Prevention Cost Rule, Gemba Walk, Pareto Analysis, Date: September 1, 2026

Fishbone, Shainin Red X, Six Sigma DMAIC

1. Executive Summary and Research Objective

Modern integrated heavy manufacturing industries operate in hyper-competitive environments where process variations directly impact structural integrity, financial yield, and corporate reputation. This research document investigates a conceptual framework for addressing chronic surface scaling and micro-cracks in Hot Rolled coils.

Rather than presenting an on-site operational audit of Tata Steel, this study serves as an academic and professional simulation modeling how Tata Steel's pioneering Total Quality Management (TQM) philosophy can be systematically deployed to diagnose, isolate, and eliminate complex metallurgical bottlenecks. The primary objective is to transition from reactive appraisal toward a rigorous, data-backed proactive quality ecosystem.

2. Problem Statement

Within heavy hot strip milling processes, surface defects such as oxide scaling, uneven thermal distribution, and minor micro-cracks can introduce severe structural compromises. Historically, when rejection rates fluctuate upwards, plants often experience inflated internal re-rolling expenses and downstream customer rejections.

The core challenge addressed in this research is how an organization can structurally trace, analyze, and permanently eliminate such multi-variable process anomalies without disrupting continuous mass-production workflows.

3. Methodology and Step-by-Step Implementation

To resolve the simulated manufacturing defect, a structured, multi-tiered framework was formulated by integrating classic quality

philosophies with advanced analytical tools:

• STEP A: CULTURAL FOUNDATION (DEMING'S SYSTEM OF PROFOUND KNOWLEDGE)

Rather than resorting to a traditional blame culture directed at frontline operators, management establishes a systemic perspective. By appreciating the interactions across departments, analyzing natural versus assignable process variations, and cultivating psychological safety, teams can transparently report raw data without fear of penalty.

• STEP B: FINANCIAL JUSTIFICATION (THE PREVENTION COST RULE)

Adopting an economic lens, the framework leverages foundational quality economics. Investing capital upfront into preventive

measures such as automated sensor calibration, strict thermal profiling, and precise descaling header maintenance costs

a fraction of downstream correction, where prevention saves significantly on internal failure and external customer failure.

• STEP C: GROUND-LEVEL DIAGNOSIS (GEMBA WALK AND PARETO ANALYSIS)

Quality engineers simulate stepping onto the shop floor (Gemba) to observe physical realities firsthand. Subsequently, a Pareto

analysis applies the foundational di

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on

Gemba Walk Align boardroom strategies with actual shop-floor Ground-Level Process Observation

realities.

Pareto Chart and Fishbone Isolate vital few defects and structure 4M root Diagnostic Root-Cause Analysis

causes.

Shainin Red X Statistically pin down the single dominant Advanced Statistical

manufacturing variable. Troubleshooting

Six Sigma DMAIC and ISO Optimize process parameters and lock changes into Process Excellence and

standard workflows. Standardization

Disclaimer and Contextual Note: This document and the associated analysis have been prepared strictly as a professional research exercise

and represent Day 36 of an independent 100 Days TQM Challenge. Jagriti Singh is an independent Quality Engineer conducting this study

for professional development and portfolio enhancement. This document is neither commissioned by, affiliated with, nor officially

endorsed by Tata Steel Limited. All operational scenarios modeled herein are educational simulations inspired by industry-standard quality

management methodologies.

TTL/TQM/100/2026/D36 | Independent Research Study Page 2 of 2

DAY 37

Original source file(s): Day 37.pdf

Source focus: The Day 37 source material focuses on JSW STEEL. The source text explicitly discusses the following quality themes and methods: Reliability, INSPECTION, audit.

Section headings identified in the source: JSW STEEL • Supply Chain Reliability and Vendor Quality Management Research Solution • 1. Executive Overview and Research Context • 2. Operational Problem Definition • jeopardize continuous mill schedules: • 3. Series Reliability in Supply Chain Flow

Technical terms detected in the source: Reliability, INSPECTION, audit

Source extract 1: Day 37.pdf

JSW STEEL

Supply Chain Reliability and Vendor Quality Management Research Solution

Document Reference ID: JSW SCM RESEARCH 2026 Independent Researcher & Author: Jagriti Singh

Industry Focus: JSW Steel Plants at Vijayanagar and Dolvi Date: September 2 2026

1. Executive Overview and Research Context

Through my independent research into large scale industrial manufacturing operations and my academic knowledge of Quality

Engineering and Total Quality Management, I have formulated a robust supply chain reliability framework tailored for JSW Steel

operations at Vijayanagar and Dolvi. Continuous steel production requires absolute synchronization across all material intakes. My

proprietary analytical solution addresses chronic supply chain bottlenecks by integrating advanced reliability metrics and structured

maintenance philosophies to eliminate external vulnerabilities.

2. Operational Problem Definition

Through secondary research and operational analysis of JSW Steel supply workflows, I identified two primary failure modes that

jeopardize continuous mill schedules:

• Late Deliveries of Raw Materials: Critical manufacturing inputs such as iron ore coking coal and packaging supplies frequently suffer

from severe transit and dispatch delays from external vendors, directly risking plant line starvation.

• Inconsistent Material Quality: Repeated chemical composition shifts and physical dimension variations in incoming material batches

severely compromise downstream processing standards and finished product integrity.

3. Series Reliability in Supply Chain Flow

To eliminate these vulnerabilities, I have structured the entire material intake pipeline into a rigorous Series Reliability model. In this

framework, every operational stage acts as a dependent link where failure at any single point disrupts the entire workflow:

• STEP A: VENDOR SOURCING AND DISPATCH

The initial control checkpoint where raw materials are prepared and dispatched by vendors. Enforcing strict dispatch compliance here

establishes a high baseline reliability score.

• STEP B: BULK TRANSPORTATION AND TRANSIT

Logistics movement via rail and road corridors. Tracking transit velocity closely guarantees precise material arrival timelines at plant

gates.

• STEP C: PLANT GATE INSPECTION AND QUALITY SCREENING

Mandatory quality screening upon arrival. Any detected chemical or physical variation results in immediate batch rejection, preventing

defective inputs from entering shop floors.

• STEP D: WAREHOUSE STORAGE AND SHOP FLOOR FEED

Final secure routing into active production lines, ensuring uninterrupted material availability for heavy rolling mills.

4. Reliability Metrics: MTBF and MTTF Implementation

To quantify and control operational downtime across our supply network, I have deployed precise reliability metrics separating reparable

and non reparable failure domains:

• MTBF Mean Time Between Failures for Reparable Elements: I apply this metric strictly to operational systems that can be restore

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d storage junctures to

minimize total operational cost.

6. Applied TQM Quality Tools

In developing this independent research solution for JSW Steel, I have embedded the following core Total Quality Management

methodologies:

• Series Reliability Modeling: To mathematically evaluate end to end supply chain vulnerability.

• MTBF and MTTF Analysis: To segregate reparable operational downtime from non reparable batch rejections.

• Vendor Quality Audits & SLA Enforcement: To eliminate raw material variation at source.

• Hybrid Maintenance Framework (PM, PdM, RCM): To ensure asset availability and zero line starvation.

Authoritative Research Statement: This case study represents my independent professional research and engineering solution developed by Jagriti Singh for

JSW Steel supply chain operations at Vijayanagar and Dolvi. All strategies, reliability frameworks, and metric implementations detailed herein are authored

exclusively based on advanced quality engineering principles and academic research.

JSW SCM Case Study | Independent Research and Quality Engineering Solution Page 2 of 2

DAY 38

Original source file(s): DAY 38 Google_tqm_case_study.pdf

Source focus: The Day 38 source material focuses on Google Ref: GGL/TQM/100/2026/D38. The source text explicitly discusses the following quality themes and methods: reliability, customer satisfaction, isolation, PDCA, audit.

Section headings identified in the source: FROM CUSTOMER EXPECTATIONS TO CUSTOMER DELIGHT • MUST-BE PERFORMANCE ATTRACTIVE • QUALITY METHODS APPLIED • FINAL QUALITY INSIGHT

Technical terms detected in the source: reliability, customer satisfaction, isolation, PDCA, audit

Source extract 1: DAY 38 Google_tqm_case_study.pdf

Google Ref: GGL/TQM/100/2026/D38

Date: 03 September 2026

QUALIT Y ENGINEERING RESEARCH CASE STUDY Author: Jagriti Singh | Quality Engineer

FROM CUSTOMER EXPECTATIONS TO CUSTOMER DELIGHT

A Kano Model–Based Customer-Centric Quality Improvement Framework

01 | CUSTOMER-CENTRIC QUALITY

Customer expectations in digital products and services do not remain static. Users evaluate quality through more than

basic functionality; reliability, ease of use, responsiveness and overall experience can influence how they

perceive value. In such an environment, customer feedback should not be viewed only as a mechanism for resolving

complaints.

From a Quality Engineering perspective, customer feedback can become an input for understanding where quality expectations are being met, where performance can improve and where additional value may be created. This case study explores a structured approach in which customer signals are translated into quality priorities and carried into a continuous improvement cycle.

02 | UNDERSTANDING CUSTOMER EXPECTATIONS

Customer requirements can have very different effects on satisfaction. Some requirements are considered basic by users and their absence can create dissatisfaction. Other requirements influence satisfaction according to their level of performance, while certain features can create additional value when they go beyond normal expectations.

The Kano Model provides a useful way to distinguish between these different types of requirements. Combining Kano thinking with customer feedback, usage signals and emerging expectations allows quality priorities to be viewed from a customer perspective rather than treating every feedback item in the same way.

An important consideration is that customer expectations evolve. A feature that creates additional satisfaction today may become an expected part of the experience over time. This makes periodic reassessment important within a customer-centric quality system.

03 | CUSTOMER VOICE AND QUALITY SIGNALS

A customer-centric quality approach needs reliable inputs. Direct Voice of Customer can come from reviews, ratings, surveys, complaints and feature requests. These signals describe what customers are experiencing and what they expect to change.

Customer behaviour can provide another perspective. Usage patterns and recurring points of friction may reveal needs that are not always expressed directly. Emerging requests and changing preferences can also indicate where expectations are moving.

The proposed framework therefore considers customer feedback together with relevant usage signals and emerging expectations. The purpose is not to assume that one source represents the complete customer experience, but to bring different signals together before making quality decisions.

04 | KANO MODEL AS A QUALITY PERSPECTIVE

The Kano Model classifies customer requirements according to their relationship with satisfaction. Must-Be requirements represent basic expectations. Their presence may not create

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ysis — Prioritising recurring quality concerns

Voice of Customer (VoC) — Customer needs and feedback PDCA — Structured continuous improvement

Customer Segmentation — Understanding user group differences Before–After Validation — Evaluating improvement outcomes

Feedback Analysis — Identifying recurring customer signals Continuous Improvement — Sustaining the learning cycle

FINAL QUALITY INSIGHT

“Customer focus becomes quality improvement when customer signals are understood, prioritised, validated through outcomes and carried into the next improvement cycle.”

Independent professional research exercise developed independently as part of Jagriti Singh’s #100DaysOfTQM learning series. This case study represents an

independently proposed Quality Engineering approach and is not an official Google project, internal Google document, audit, endorsement or affiliation with

Google. All concepts and perspectives are formulated purely from public quality engineering methodologies and independent professional knowledge.

GGL/TQM/100/2026/D38 | Day 38 | #100DaysOfTQM Jagriti Singh | Quality Engineer

DAY 39

Original source file(s): Day 39 Microsoft .pdf

Source focus: The Day 39 source material focuses on Ref: MSFT/TQM/100/2026/D39. The source text explicitly discusses the following quality themes and methods: isolating, Root Cause, isolate, reliability, TAGUCHI, PDCA, FMEA, audit.

Section headings identified in the source: MICROSOFT • Windows Update Quality Improvement • EXECUTIVE SUMMARY ANALYTICAL PROGRESSION • contributors to isolate the vital few failure modes from the • systematically reduce update completion duration while strictly • diagnostics to identify the single dominant parameter

Technical terms detected in the source: isolating, Root Cause, isolate, reliability, TAGUCHI, PDCA, FMEA, audit

Source extract 1: Day 39 Microsoft .pdf

Ref: MSFT/TQM/100/2026/D39

MICROSOFT

Jagriti Singh | Quality Engineer

QUALIT Y ENGINEERING CASE STUDY 04 September 2026

Windows Update Quality Improvement

EXECUTIVE SUMMARY ANALYTICAL PROGRESSION

This independent quality engineering research case study A rigorous three-tier diagnostic sequence provides the

investigates the complex operational dynamics surrounding analytical backbone for isolating systemic bottlenecks:

Windows Update completion efficiency. Long and

• Root Cause Analysis: Maps operational symptoms to

unpredictable update completion times create a fragmented fundamental systemic triggers across diverse hardware

user experience, discourage prompt adoption of critical configurations.

patches, and frequently coexist with transient update failures

• Pareto Analysis: Quantifies major time-consuming

and error states. The primary quality objective is to

contributors to isolate the vital few failure modes from the

systematically reduce update completion duration while strictly

trivial many.

preserving system reliability, transactional safety, and overall

• Shainin Red X: Deploys advanced variance-tracking

user confidence.

diagnostics to identify the single dominant parameter

Positioned strictly as an independent analytical exercise driving update duration instability.

applying Total Quality Management principles, this study

examines how structured statistical and engineering TARGET SETTING

frameworks can diagnose, control, and optimize software Improvement targets must be established through empirical

delivery pipelines without relying on proprietary internal baseline measurement rather than arbitrary managerial

telemetry. mandates. Baseline performance metrics captured across

representative test fleets inform realistic, statistically sound

QUALITY PROBLEM

targets governed by stringent quality and reliability thresholds.

The core concern extends beyond the absolute time required

to apply software packages. The critical issue is excessive TAGUCHI ROBUST DESIGN AND DOE

duration coupled with high statistical variance across Taguchi's robust design principles and Design of Experiments

heterogeneous hardware landscapes. Unpredictable update are adapted to evaluate how environmental noise factors and

windows disrupt operational continuity and correlate directly control variables impact update performance. By testing

with user reluctance and forced termination behaviors, which multiple factor combinations simultaneously, engineers

in turn trigger boot failures and support overhead. identify parameter settings that make the update process

immune to hardware variability.

UPDATE PROCESS

The end-to-end software delivery pipeline follows a sequential PDCA IMPROVEMENT CYCLE

lifecycle: The Plan-Do-Check-Act cycle governs iterative refinement:

System Preparation → Download → Installation → Restart • Plan: Identify high-latency stages, establish baselines, and

→ Configuration → Completion define reliability criteria.

• Do: Execute controlled pilot experime

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-ASSISTED PREDICTIVE PREVENTION

As a conceptual future enhancement, machine learning

models analyze historical telemetry and local system states to

predict failure patterns before execution:

Detect Risk → Automatic Diagnosis → Preventive Action →

Re-check → Continue Update

The guiding design maxim is clear: Stop the failure, not

necessarily the update.

Disclaimer: This document is an independent professional research exercise by Jagriti Singh based on publicly available concepts, established

Quality Engineering and TQM methodologies, and independent analytical reasoning. It is not an official Microsoft document, internal Microsoft project,

Microsoft audit, or representation of Microsoft's internal processes, systems, performance data, or quality practices. The Windows Update scenario

described here is a proposed quality-improvement framework and should not be interpreted as a claim regarding Microsoft's current internal

operations. No proprietary, confidential, or non-public Microsoft information has been used.

MSFT/TQM/100/2026/D39 | Independent Quality Engineering Study Page 2 of 2

DAY 40

Original source file(s): Day 40.pdf

Source focus: The Day 40 source material focuses on UI QUALITY ENGINEERING · FROM USER EXPERIENCE TO CONTINUOUS. The source text explicitly discusses the following quality themes and methods: Root Cause, audit.

Section headings identified in the source: A PPLE • QUALITY IMPROVEMENT • This case study explores how Quality Engineering principles can be SUPPORT • improving the experience through user feedback and behavioural CRITICAL ACTION • Permanent storage changes will affect • Confirm & Proceed

Technical terms detected in the source: Root Cause, audit

Source extract 1: Day 40.pdf

A PPLE

UI QUALITY ENGINEERING · FROM USER EXPERIENCE TO CONTINUOUS

QUALITY IMPROVEMENT

DAY 40/100 · 05 SEPTEMBER 2026 | Jagriti Singh | Quality Engineer APL/TQM/100/2026/D40

01 — CASE OBJECTIVE 03 — USER DECISION

This case study explores how Quality Engineering principles can be SUPPORT

embedded into UI/UX design to create interfaces that are not only Instead of relying only on a simple “Are

visually clear, but also reliable, user-centred, measurable and you sure?” confirmation message, the

continuously improvable. interface provides relevant information

The proposed framework treats the user interface as a quality system. or contextual questions before a critical

Instead of considering UI quality only in terms of visual appearance, action.

the approach focuses on helping users make informed decisions, INFORMATION → UNDERSTANDING

preventing avoidable errors, enabling recovery when failures occur, → USER DECISION → ACTION

validating design changes through evidence, and continuously

improving the experience through user feedback and behavioural CRITICAL ACTION

signals. CONFIRMATION

Permanent storage changes will affect

all synced devices immediately.

02 — QUALITY PROBLEM

Confirm & Proceed

How can UI/UX be designed and managed as a measurable quality

system where user decisions, error prevention, recovery, usability and

The user remains in control rather than being

continuous improvement are treated as interconnected quality

pushed toward a predetermined choice.

requirements?

USER DECISION SUPPORT ERROR PREVENTION

ERROR RECOVERY USABILITY VALIDATION

CONTINUOUS QUALITY

04 — ERROR PREVENTION & RECOVERY

If an unexpected error occurs during an important action,

RECOVERY STATE LOGIC

the proposed interface first attempts automatic recovery

wherever technically possible. The user receives clear Retry Cancel Recover Session

feedback about the current state.

AUTOMATIC RECOVERY → CLEAR FEEDBACK → USER-

CONTROLLED RECOVERY

05 — UI QUALITY AS AN ENGINEERING PROCESS

USER REQ UI/UX FRICTION ROOT

→ DESIGN

→ PROTOTYPE → USABILITY

→ ID

→ CAUSE

→ IMPROVEMENT

→ VALIDATION

APPLE UI QUALITY ENGINEERING · CASE STUDYPAGE 1 OF 2

A PPLE

UI QUALITY ENGINEERING · FROM DESIGN TO QUALITY

06 — VALIDATING UI QUALITY 09 — CONTINUOUS QUALITY LOOP

A design should not be considered successful simply Validated improvements are incorporated into design

because it looks cleaner or more modern. The proposed standards and future development practices through a

framework validates improvement through measurable continuous feedback and monitoring loop.

outcomes. STANDARDISE → MONITOR → SIGNALS → DEVIATION →

IMPROVE → RE-TEST

-34% 99.2%

TASK TIME SUCCESS RATE

10 — USER BEHAVIOUR SIGNALS

Real user behaviour provides additional signals about

interface quality. Continuous monitoring helps identify

07 — A/ B TESTING

repeated interaction difficulties, recurring errors, and

Comparing alternative interface designs using defined high-friction journeys before they become major

quality criteria across completion time, error frequency, usability issues.

user behaviour and usability outcomes.

Design A: Baseline Design B: Proposed

11 — QUALITY ENGINEERING × UI/ UX

QE: Measurement · Root Cause · Validation · Risk

Thinking

08 — USER SEGMENTATION & KANO

UI/UX: Interaction · Usability · Accessibility · Experience

User feedback is segmented to identify common

patterns and underlying requirements. Kano perspective Result: User Need → Design → Measure → Analyse →

Improve

classifies must-be expectations from performance

attributes.

Must-Be Performance Attractive

KEY QUALITY INSIGHT

“UI quality is not only about how an interface looks. It is about how reliably it helps users make decisions, complete tasks, prevent

errors, recover from failures and continuously improve through evidence.”

Disclaimer: Independent professional research exercise by Jagriti Singh. Not an official Apple project, Apple internal document, product

specification, audit, endorsement, partnership or affiliation with Apple Inc.

DAY 40/100 · #100DaysOfTQM · JAGRITI SINGHPAGE 2 OF 2

DAY 41

Original source file(s): Day 41.pdf

Source focus: The Day 41 source material focuses on M E TA #100DaysOfTQM • Day 41. The source text explicitly discusses the following quality themes and methods: FMEA, RPN, TAGUCHI, isolated, KAIZEN, PDCA, audit.

Section headings identified in the source: AI-DRIVEN PREDICTIVE & ROBUST QUALITY ENGINEERING • A Quality Engineering Framework for Personalised Content Systems • supports preventive improvement before problems become • ANALY TICAL SIGNAL PROGRESSION ARCH ITECTURE • Emerging • Mass User AI / LLM Predictive FMEA / RPN Preventive

Technical terms detected in the source: FMEA, RPN, TAGUCHI, isolated, KAIZEN, PDCA, audit

Source extract 1: Day 41.pdf

M E TA #100DaysOfTQM • Day 41

AI-DRIVEN PREDICTIVE & ROBUST QUALITY ENGINEERING

A Quality Engineering Framework for Personalised Content Systems

Ref: META/TQM/100/2026/D41 Author: Jagriti Singh Role: Quality Engineer Date: 06 September 2026

01 — CASE OBJECTIVE 02 — QUALITY PROBLEM

Modern content platforms operate at an enormous scale, A personalised content system may gradually become less
where user interests, behaviour and expectations relevant when recommendations fail to adapt to changing
continuously change. A quality approach for such an interests or repeatedly surface similar content. Three key
environment cannot depend only on fixed user segments or quality concerns are addressed within this framework:
periodic manual review. • Relevance: Content may no longer match the user's
This case study proposes an AI-driven Quality current interests.
Engineering framework that continuously interprets large- • Repetition: Similar content may appear repeatedly,
scale user signals, identifies emerging quality risks and reducing diversity.
supports preventive improvement before problems become
• Adaptability: The system may respond slowly when
widespread.
behaviour changes.
The objective is not simply to recommend what should be At Meta's scale, manually creating and updating large user
improved. The proposed system also considers how an segments becomes inefficient. The proposed approach uses
improvement should be implemented, what could go wrong AI-assisted analysis of behavioural and feedback signals
during implementation, and how those risks can be reduced to identify patterns dynamically.
beforehand.
03 — AI-DRIVEN SIGNAL ANALYSIS 04 — FROM DETECTION TO PREDICTIVE
Instead of relying on fixed structures, the system QUALITY
continuously analyses multiple dimensions of user The proposed framework extends beyond reactive problem
interaction: detection by leveraging Predictive Quality methodologies.
• Explicit Signals: Likes, follows, feedback and direct user AI continuously analyses quality signals to identify:
interactions. • Risk Identification: Potential relevance degradation and
• Behavioural Signals: Content viewed, skipped, re- repetition patterns.
watched and engagement duration. • Change Tracking: Emerging shifts in user behaviour and
• Emerging Signals: Changes in interests and newly preferences.
developing preference patterns. • Impact Evaluation: Possible future failures and proposed
An AI/LLM-based analysis layer combines these signals improvement risks.
to identify meaningful changes at scale without replacing This connects naturally with Philip Crosby's Zero Defects
human engineering judgement. philosophy: emphasizing prevention rather than correction
after failure occurs.
ANALY TICAL SIGNAL PROGRESSION ARCH ITECTURE
Emerging
Mass User AI / LLM Predictive FMEA / RPN Preventive
Signals
→ Analysis
→ Behaviour & → Quality

→ Risk Evaluation → Action

Need Detection

05 — FMEA-BASED RISK PREDICTION

Before implementing any proposed improvement,

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MMARY

The proposed framework unifies core methodologies: AI/LLM Analysis for signal interpretation, Predictive Quality for early degradation identification, FMEA for risk evaluation, Zero Defects prevention-oriented thinking, Taguchi Robust Design & DOE for systematic experimentation, PDCA structure, progressive validation, and Kaizen .

KEY QUALITY INSIGHT

“Quality at scale is not achieved by predicting what users want alone. It is achieved by detecting change early, designing robust solutions, preventing implementation failures, validating progressively and continuously improving through evidence.”

Disclaimer: This case study is an independent professional research exercise by Jagriti Singh. It is a conceptual application of Quality Engineering

methodologies to a large-scale personalised content environment and is not an official Meta project, internal Meta analysis, audit, specification, endorsement,

partnership or affiliation. The framework represents independent professional reasoning and proposed methodology.

Ref: META/TQM/100/2026/D41 | Jagriti Singh (#100DaysOfTQM) Page 2 of 2

DAY 42

Original source file(s): Day 42.pdf

Source focus: The Day 42 source material focuses on DIGITAL PRODUCT QUALITY ENGINEERING. The source text explicitly discusses the following quality themes and methods: isolated, PDCA, ROOT CAUSE, TAGUCHI, FMEA, RPN, KAIZEN, audit.

Section headings identified in the source: DIGITAL PRODUCT QUALITY ENGINEERING • A Quality Engineering Framework for Reliable and Continuously Improving Creative Workflows • ANALYTICAL PROGRESSION FRAMEWORK • THREE STAGE VALIDATION STRUCTURE • WIDER ROLLOUT • CONTINUOUS IMPROVEMENT VISUAL

Technical terms detected in the source: isolated, PDCA, ROOT CAUSE, TAGUCHI, FMEA, RPN, KAIZEN, audit

Source extract 1: Day 42.pdf

ADOBE.

DIGITAL PRODUCT QUALITY ENGINEERING

A Quality Engineering Framework for Reliable and Continuously Improving Creative Workflows

Ref: ADB/TQM/100/2026/D42 Author: Jagriti Singh Role: Quality Engineer Date: 07 September 2026 Day: 42/100

01 CASE OBJECTIVE

Creative software environments involve complex workflows where users may perform multiple actions before completing an intended task. From a Quality Engineering perspective, quality should not be evaluated only through software functionality or visual appearance. It should also consider whether users can complete creative workflows efficiently, reliably and consistently while avoiding unnecessary errors and rework.

This case study proposes a Quality Engineering approach that combines user behaviour, workflow data and direct feedback to identify quality concerns within creative workflows. The improvement process then moves through structured analysis, experimentation, risk evaluation, controlled validation and continuous improvement.

02 QUALITY PROBLEM

A creative workflow can lose quality when users repeatedly encounter delays, errors, unnecessary steps or rework during task completion. Such problems may not always originate from a single visible feature. They can emerge from interactions between workflow stages, user behaviour, system conditions and the complexity of the task itself.

The proposed approach therefore focuses on understanding where quality problems occur, why they occur and how strongly they affect workflow performance. Instead of improving a feature based only on isolated feedback, the complete workflow is examined using data, user signals and Quality Engineering methods.

03 USER BEHAVIOUR, WORKFLOW DATA AND FEEDBACK

The first stage combines three sources of quality information: user behaviour, workflow data and direct user feedback. Behavioural signals can reveal where users experience friction, repeat actions or spend additional time. Workflow data can help identify stages associated with delay, errors or rework, while direct feedback provides context behind those observations.

These signals are analysed together rather than treating one source as the complete representation of quality. Once a significant quality concern is identified, the improvement process enters a structured PDCA cycle to support systematic planning, experimentation, evaluation and refinement.

04 ROOT CAUSE ANALYSIS

After identifying a major quality concern, Pareto Analysis can be used to determine which workflow problems contribute most significantly to the overall issue. This prevents improvement efforts from being spread across numerous minor problems when a smaller number of dominant contributors may have a greater impact.

The major contributor can then be investigated using 5 Why and Fishbone Analysis to explore possible causes across relevant categories. Shainin Red X can subsequently support identification of the dominant cause responsible for the observed variation or performance problem. The obje

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ITY ENGINEERING AND DIGITAL PRODUCT QUALITY

This framework connects user centred product thinking with Quality Engineering methodology. User behaviour and feedback provide quality signals, analytical tools identify major contributors, Taguchi Robust Design supports experimentation, FMEA evaluates risks, PDCA structures improvement and controlled validation verifies performance across different conditions.

"Quality in a digital creative workflow is not achieved by improving one feature alone. It is achieved by understanding user behaviour, identifying dominant causes, designing robust solutions, evaluating implementation risks, validating across variation and continuously improving through evidence."

Disclaimer: This case study is an independent professional research exercise by Jagriti Singh. It presents a conceptual application of Quality

Engineering methodologies to a digital creative workflow environment and is not an official Adobe project, internal Adobe analysis, audit,

specification, endorsement, partnership or affiliation.

Ref: ADB/TQM/100/2026/D42 | Jagriti Singh Page 2 of 2

DAY 43

Original source file(s): Day 43.pdf

Source focus: The Day 43 source material focuses on AI SYSTEM RELIABILITY AND ROBUST QUALITY ENGINEERING. The source text explicitly discusses the following quality themes and methods: RELIABILITY, ROOT CAUSE, isolate, FMEA, TAGUCHI, isolated, KAIZEN, PDCA, audit.

Section headings identified in the source: NVIDIA • AI SYSTEM RELIABILITY AND ROBUST QUALITY ENGINEERING • Artificial intelligence systems operate under constantly changing ANALYSIS • Engineering analysis to move beyond isolated operational • incidents and uncover broader systemic patterns indicating • thresholds triggering it and whether the pattern points toward a

Technical terms detected in the source: RELIABILITY, ROOT CAUSE, isolate, FMEA, TAGUCHI, isolated, KAIZEN, PDCA, audit

Source extract 1: Day 43.pdf

NVIDIA

AI SYSTEM RELIABILITY AND ROBUST QUALITY ENGINEERING

A Quality Engineering Framework for Predictive, Preventive and Continuously Improving AI Systems

Ref: NVIDIA/TQM/100/2026/D43 Author: Jagriti Singh Role: Quality Engineer Date: 08 September 2026 Day: 43/100

01 CASE OBJECTIVE 04 ROOT CAUSE AND DOMINANT FACTOR

Artificial intelligence systems operate under constantly changing ANALYSIS

workloads, operating conditions and usage patterns. Under such Once a significant quality pattern is identified, Pareto Analysis

dynamic variation, a system may perform exceptionally in one determines which failure contributors exert the greatest influence

environment while exhibiting inconsistent behaviour in another. on the overall problem. The major contributor is subsequently

From a rigorous Quality Engineering perspective, reliability must investigated using Five Why Analysis and Fishbone Analysis to

therefore be evaluated far beyond a single successful result. The explore potential root causes across operational dimensions.

primary objective of this case study is to develop a conceptual Shainin Red X methodologies can then isolate the exact

framework that uses system behaviour, performance data, dominant factor responsible for the observed variation. This

operating conditions and failure signals to identify quality systematic progression prevents quality improvement efforts

patterns and progressively improve system robustness. The from treating symptoms as root causes, guiding the investigation

framework focuses on prevention, experimentation, risk from broad signals toward precise, evidence based

evaluation, controlled validation and continuous learning. understanding.

02 SYSTEM QUALITY PROBLEM 05 SYSTEM FMEA

An artificial intelligence system can experience performance Alongside root cause investigation, System FMEA provides a

instability when operating conditions or workload characteristics structured methodology to evaluate potential failure modes

shift. Quality concerns often manifest through performance across the broader artificial intelligence architecture. Potential

variation, unexpected failures, reliability degradation or failure modes are examined across Severity, Occurrence and

alterations in system behaviour. Detection parameters, generating a Risk Priority Number

Rather than prematurely redesigning system architecture, the perspective that prioritizes areas requiring enhanced preventive

proposed approach first examines available evidence to controls.

understand when inconsistency occurs and which conditions System FMEA is not restricted to historical failures; it actively

correlate with it. System behaviour, performance metrics, anticipates prospective failure modes before they manifest as

environmental parameters and failure signals are synthesized to operational disruptions.

isolate meaningful quality patterns. The improvement process

SYSTEM SIGNALS → FAILURE PATTERNS → PARETO → 5 WHY

then enters a structured Plan Do Check Act cycle, esta

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essively and continuously learning from

evidence."

UNEXPECTED FAILURE → ROOT CAUSE ANALYSIS → FMEA

REASSESSMENT → SOLUTION RESTRUCTURING → RE TEST → Disclaimer: This case study is an independent professional research exercise by Jagriti

CONTROLLED VALIDATION Singh. It presents a conceptual application of Quality Engineering methodologies to an AI

system environment and is not an official NVIDIA project, internal NVIDIA analysis, audit,

specification, endorsement, partnership or affiliation.

The overarching objective is to prevent identical anomalies from

becoming recurring operational liabilities.

10 PREDICTIVE QUALITY AND CONTINUOUS

LEARNING

Following successful validation, system quality must be

monitored continuously because workloads and operating

conditions evolve. Predictive Quality utilizes live system signals

to detect early indications of potential degradation or future

failure.

This shifts operations from reactive correction to proactive

prevention, enabling engineers to apply controls before minor

issues escalate.

NVIDIA / TQM / 100 / 2026 / D43 | Jagriti Singh Page 2 of 2

DAY 44

Original source file(s): Day 44.pdf

Source focus: The Day 44 source material focuses on I N DE PE N DE N T Q UA L IT Y E N G I N E E R I N G RES E A RC H • REF : O PE NA I / TQ M / 1 0 0 / 2 0 2 6 / D 4 4. The source text explicitly discusses the following quality themes and methods: audit, isolating, ROOT CAUSE, isolate, Isolates, inspection, FMEA, RPN, PDCA, KAIZEN, Taguchi.

Section headings identified in the source: patching superficial symptoms: • EXPECTED OUTCOME DEFINITION • FITNESS FOR USE EVALUATION • QUALITY CRITERIA LOCK • risks must receive immediate engineering intervention regardless • STANDARDISATION

Technical terms detected in the source: audit, isolating, ROOT CAUSE, isolate, Isolates, inspection, FMEA, RPN, PDCA, KAIZEN, Taguchi

Source extract 1: Day 44.pdf

I N DE PE N DE N T Q UA L IT Y E N G I N E E R I N G RES E A RC H • REF : O PE NA I / TQ M / 1 0 0 / 2 0 2 6 / D 4 4

OpenAI | AI RESPONSE QUALITY ENGINEERING

A Quality Engineering Framework for Reliable, Context-Aware and Continuously Improving AI Responses

Author: Jagriti Singh Role: Quality Engineer Date: 09 Sep 2026 Topic: AI TQM Framework

IMPORTANT NOTICE: Independent professional research by Jagriti Singh. NOT an official OpenAI project, internal analysis, audit, specification, product document,

endorsement, partnership or affiliation.

01 — CASE OBJECTIVE 04 — RESPONSE QUALITY EVALUATION

This independent research explores how classical Quality Effective quality management mandates proactive evaluation

Engineering and Total Quality Management (TQM) principles can rather than reactive complaint handling. This framework

be systematically embedded into AI response architectures. The proposes a three-tier evaluation architecture:

central engineering question investigated is: “How can an AI 1. Automated Quality Checks: Algorithmic validation against

system consistently generate reliable, relevant and context- syntax, schema and factual retrieval markers.

appropriate responses across different users, prompts and 2. AI-Based Self-Evaluation: Secondary model instances

situations?” auditing logical consistency, tone, and constraint adherence.

The core objective is to transition AI systems from heuristic text 3. Human Validation: Expert review for high-risk, ambiguous, or

generation toward a rigorous quality management paradigm. safety-critical edge cases.

This requires understanding user requirements, evaluating Identified failure modes include incorrect data retrieval, irrelevant

response quality, isolating root causes, assessing risk, reasoning, incomplete coverage, misleading confidence markers,

engineering robust improvements, and maintaining continuous context mismatch, and failure to fulfil the core user requirement.

learning loops.

Foundational to this framework is Joseph Juran’s concept of

Fitness for Use. Quality is not restricted to literal token 05 — ROOT CAUSE INVESTIGATION

correctness; an AI response must satisfy the user's operational When response failures occur, systematic diagnostic tools are

context and intent. deployed to isolate underlying systemic defects rather than

patching superficial symptoms:

• Pareto Analysis: Identifies vital few failure categories

02 — DEFINING AI RESPONSE QUALITY responsible for 80% of quality loss.

AI response quality is evaluated against the multi-dimensional • 5 Why Analysis: Iteratively drills down from symptom to

needs of the user within a specific operational context. Within prompt/architecture flaws.

this framework, quality is measured across distinct dimensions: • Fishbone Analysis: Maps potential failure sources across

• Accuracy: Factual and logical correctness. data, weights and prompt templates.

• Relevance: Direct alignment with the core query. • Shainin Red X: Isolates the dominant root vari

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ct production users.

ensure response stability across noisy, unconstrained user

inputs.

11 — CONTINUOUS LEARNING &

STANDARDISATION

08 — VIRTUAL SIMULATION & PLANNING

CONTINUOUS MONITORING & FEEDBACK

Before deploying improvements at scale, an AI-based virtual

simulation conceptually evaluates prospective architectural ↓

changes: EMERGING FAILURE & DRIFT DETECTION

PROPOSED IMPROVEMENT ↓

↓ PDCA / KAIZEN & STANDARDISED LEARNING

VIRTUAL SIMULATION (FAILURE & VARIATION)

12 — QUALITY ENGINEERING × AI

RESOURCE & LATENCY IMPACT ASSESSMENT

Intersection: QE Measurement, Root Cause, Risk & Prevention

↓ × AI User Signals, Context Variation & Continuous Learning.

CONTROLLED IMPLEMENTATION “AI quality is achieved when the system understands user intent,

Ensures engineers evaluate not just “What should improve?” but “What evaluates quality, prevents failures, validates improvements, and

could fail during rollout?” continuously adapts.”

DISCLAIMER & REF: Independent research by Jagriti Singh. OpenAI/TQM/

100/2026/D44.

OpenAI | AI Response Quality Engineering — Ref: OpenAI/TQM/100/2026/D44 Page 2 of 2

DAY 45

Original source file(s): Day 45.pdf

Source focus: The Day 45 source material focuses on JPMORGAN CHASE & CO.. The source text explicitly discusses the following quality themes and methods: reliability, isolate, PDCA, inspection, FMEA, Six Sigma.

Section headings identified in the source: Enhancing Cybersecurity Quality and Operational Resilience • 1. EXECUTIVE CONTEXT AND CORE PHILOSOPHY • 2. PROBLEM STATEMENT AND SELECTION RATIONALE • FOCUS AREA OPERATIONAL CONTEXT AND QUALITY SIGNIFICANCE • zero defect approach reduces vulnerability probabilities and mitigates operational • experiences across complex technology infrastructures and high volume transaction

Technical terms detected in the source: reliability, isolate, PDCA, inspection, FMEA, Six Sigma

Source extract 1: Day 45.pdf

JPMORGAN CHASE & CO.

QUALITY ENGINEERING CASE STUDY | DAY 45 | REFERENCE ID: QE-100D-045

Enhancing Cybersecurity Quality and Operational Resilience

Author: Jagriti Singh | Role: Quality Engineer | Date: 10 September 2026

Disclaimer: This case study has been developed as an independent learning exercise for the 100 Days Quality Engineering Challenge. It

applies Quality Engineering and Total Quality Management principles to publicly discussed challenges relevant to JPMorgan Chase. This work

is not affiliated with, sponsored by, endorsed by, or officially connected with JPMorgan Chase and should not be interpreted as an official

statement or recommendation from the company.

1. EXECUTIVE CONTEXT AND CORE PHILOSOPHY

Quality Engineering is traditionally associated with mechanical engineering and manufacturing environments. However, its underlying philosophy encompassing prevention, risk reduction, customer focus, process improvement, error reduction, reliability, consistency, and root-cause elimination is universally applicable. While industry terminology adapts across domains, the foundational framework remains exceptionally powerful. Just as Kanban governs digital workflows and predictive maintenance translates to digital systems, Quality Engineering principles can optimize cybersecurity and operational resilience in large-scale financial institutions like JPMorgan Chase.

2. PROBLEM STATEMENT AND SELECTION RATIONALE

Large financial institutions operate complex technology ecosystems and handle massive volumes of sensitive financial information. Ensuring absolute security and continuous service availability is a monumental challenge where conventional reactive security is insufficient. This case study focuses on two deeply interconnected operational pillars: cybersecurity and operational resilience.

FOCUS AREA OPERATIONAL CONTEXT AND QUALITY SIGNIFICANCE

Cybersecurity Large financial institutions manage vast volumes of sensitive data. A prevention oriented

zero defect approach reduces vulnerability probabilities and mitigates operational

disruptions.

Operational Resilience Ensures continuous service availability, process reliability, and consistent customer

experiences across complex technology infrastructures and high volume transaction

networks.

3. QUALITY ENGINEERING PROBLEM-SOLVING FRAMEWORK

A structured and sequential methodology is deployed to systematically identify, analyze, and resolve operational and security vulnerabilities without relying on ad-hoc interventions.

PHASE METHODOLOGY AND PRACTICAL APPLICATION

Step 1: Identification Distinguish critical high impact security risks from surrounding operational symptoms.

Step 2: 5 Why Analysis Systematically probe underlying triggers to uncover systemic root causes rather than surface symptoms.

Step 3: Ishikawa Diagram Categorize contributing cybersecurity vulnerabilities across people, process, technology, and policy dimensions.

Step 4: Pareto Analysis

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egression

and operational drift. Through recurring improvement loops and active feedback channels, JPMorgan Chase can sustain long-term resilience, fostering an enduring organizational culture centered on continuous quality enhancement.

7. EXPECTED IMPACT AND SUCCESS MEASURES

IMPACT DOMAIN PROPOSED SUCCESS INDICATORS AND OUTCOMES

Cybersecurity Reduction in preventable security incidents, enhanced threat detection service speed, and strengthened preventive barriers.

Operational Resilience Improved service continuity, reduced operational errors, and optimized recovery capabilities across transaction networks.

Business and Customer Trust Elevated client confidence, streamlined compliance outcomes, and reinforced systemic financial stability.

8. CLOSING INSIGHT

"Quality Engineering is not restricted to manufacturing or mechanical engineering. While terminology adapts across industries, the foundational philosophy of prevention, risk reduction, customer focus, process reliability, and continuous improvement remains universally transformative for modern financial institutions."

Page 2 of 2

DAY 46

Original source file(s): Day 46.pdf

Source focus: The Day 46 source material focuses on PERFORMANCE EXCELLENCE THROUGH QUALITY. The source text explicitly discusses the following quality themes and methods: PDCA, Kaizen, isolated, audit.

Section headings identified in the source: PERFORMANCE EXCELLENCE THROUGH QUALITY • MANAGEMENT • An Independent Application of the Malcolm Baldrige Criteria for Performance Excellence • Top management involvement is essential because Total • authority to identify and address quality issues within • testing provide evidence of performance before wider

Technical terms detected in the source: PDCA, Kaizen, isolated, audit

Source extract 1: Day 46.pdf

VISA

PERFORMANCE EXCELLENCE THROUGH QUALITY

MANAGEMENT

An Independent Application of the Malcolm Baldrige Criteria for Performance Excellence

Reference: VISA/TQM/100/2026/D46 Author: Jagriti Singh | Quality Engineer Date: 11 September 2026

Core Purpose: This independent research case study explores how the Malcolm Baldrige Criteria for Performance Excellence can be applied to a conceptual Visa organizational context to connect Leadership, Strategy, Customers, Measurement, Analysis and Knowledge Management, Workforce, Operations, and Results into one continuous quality system.

01 | LEADERSHIP 02 | STRATEGY

Leadership is fundamental to applying Total Quality Vision and mission cannot remain statements written by
Management within an organization because leadership management. They must be translated into continuous
determines the direction in which the organization moves. action through day-to-day operations. Strategy should
Vision, mission, and organizational objectives cannot be therefore be treated as a continuously assessed process.
meaningfully established without strong leadership and a An organization should identify the gap between its current
clear understanding of what the organization wants to position and the position it wants to achieve, identify risks,
achieve. and establish preventive actions.

For this independent Visa case study, leadership is viewed Business risk and acceptable cost exposure should be
not as authority, control, or the ability to simply instruct considered. Quality improvement cannot be separated from
employees, but as the ability to create direction, understand business feasibility. The objective is to achieve better quality
people, encourage participation, and build an environment while maintaining a sustainable balance between quality,
in which quality becomes a shared responsibility. cost, and business growth.

A strong leader should understand employee opinions, Vision and mission require coordination across the
encourage participation, and create implementation organization. If conflicts arise between departments, the
practices that support employee development. Employee objective should be to resolve the conflict rather than allow
growth and organizational growth are interconnected. internal competition to weaken organizational performance.
Leadership should therefore focus on empowerment, Strategy must also be communicated so employees
learning, and Continuous Improvement rather than fear, understand its purpose, expected impact, and how their
pressure, or excessive control. work contributes to achieving the organizational vision.

Top management involvement is essential because Total
Quality Management cannot remain limited to the Quality 04 | MEASUREMENT, ANALYSIS AND
Department. Employees should receive appropriate KNOWLEDGE MANAGEMENT
authority to identify and address quality issues within
defined responsibilities, creating genuine ownership of An improvement cannot be considered successful simply
quality.

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monstrates how Malcolm Baldrige can be
experience and feedback should contribute to their independently applied as a systems-level Quality Management
development, creating consistency without eliminating future perspective.
improvement.

Disclaimer: Independent professional research exercise by Jagriti Singh.
This case study independently applies the Malcolm Baldrige Criteria to a
07 | RESULTS AND PERFORMANCE EVALUATION conceptual Visa organizational context. It is not an official Visa project,
internal analysis, audit, specification, endorsement, partnership, or
Final Results should be evaluated through measurable
affiliation. No confidential or proprietary Visa information has been used.

performance indicators. The organization should examine

impact magnitude, percentage of improvement, whether

customer outcomes changed, and whether meaningful

development occurred.

Benchmarking can be used to understand how

performance compares with reference points. Results

should generate learning and feed future strategy.

VISA / TQM / 100 / 2026 / D46 — Independent Research Case Study Page 2 of 2

DAY 47

Original source file(s): Day 47.pdf

Source focus: The Day 47 source material focuses on AMAZON | BOTTLENECK TO ROBUST. The source text explicitly discusses the following quality themes and methods: TAGUCHI, FMEA, PDCA, Kaizen, isolated, audit.

Section headings identified in the source: QUALITY • ROBUST DESIGN AND DOE • The objective of this case study is to explore how Quality CONDITION ANALYSIS • The Quality Engineering challenge is therefore to understand the • stable when relevant operating variation and noise factors are • conditions influencing that constraint and develop a robust

Technical terms detected in the source: TAGUCHI, FMEA, PDCA, Kaizen, isolated, audit

Source extract 1: Day 47.pdf

AMAZON | BOTTLENECK TO ROBUST

QUALITY

AN ADVANCED QUALITY ENGINEERING FRAMEWORK USING THEORY OF CONSTRAINTS, TAGUCHI

ROBUST DESIGN AND DOE

Ref: AMZ/TQM/100/2026/D47 Author: Jagriti Singh | Quality Engineer Date: 12 September 2026 (Day 47)

SECTION 01: CASE OBJECTIVE SECTION 04: HISTORICAL DATA & OPERATING

The objective of this case study is to explore how Quality CONDITION ANALYSIS

Engineering principles can be used to improve performance in a Once the critical constraint is identified, historical performance

complex large scale process by first identifying the system data and relevant operating conditions are analysed. The

constraint and then designing an improvement that remains objective is to understand when the constraint becomes

robust under variation. significant, what factors influence its performance and whether

The framework combines Theory of Constraints with Taguchi the observed limitation remains consistent across different

Robust Design and Design of Experiments. FMEA is then used to conditions.

evaluate implementation risks, followed by controlled validation This analysis creates the foundation for experimental investigation

and continuous improvement through PDCA and Kaizen. rather than relying on assumptions or isolated observations.

The central idea is that improving quality should not focus only on Relevant factors should be identified before experimentation so

making one activity faster or more efficient. The improvement that the subsequent DOE is based on meaningful process

should strengthen the overall system while remaining stable when knowledge.

operating conditions change.

SECTION 05: FROM CONSTRAINT TO

SECTION 02: QUALITY PROBLEM EXPERIMENTATION

In a large scale digital and operational environment, performance After the critical constraint and relevant influencing factors are

can be affected by multiple interconnected stages. A delay or understood, Taguchi Robust Design and Design of Experiments

limitation in one stage can influence the performance of the entire can be used to systematically investigate the relationship between

system. process factors and performance.

Improving every stage simultaneously may consume resources Instead of changing factors randomly, controlled experimentation

without addressing the factor that is actually limiting overall allows multiple conditions to be evaluated in a structured manner.

performance. The objective is not simply to find the best result under one ideal

condition. The objective is to identify a combination that remains

The Quality Engineering challenge is therefore to understand the

stable when relevant operating variation and noise factors are

complete process, identify the most critical constraint, analyse the

present.

conditions influencing that constraint and develop a robust

improvement that can perform consistently across relevant CONSTRAINT → FACTORS → DOE → ROBUST DESIGN →

variation. VALIDATION

SECTION 03: PROCESS ASSESSMENT &

CORE QUALITY CONCEPT

CONSTRAINT IDENTIFICAT

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mits

FAILURE MODE → SEVERITY → OCCURRENCE →

DETECTION → PREVENTIVE CONTROL performance, investigates influencing factors, designs robust

solutions, evaluates risk, validates under variation and

SECTION 09: CONTROLLED PILOT & VALIDATION continuously improves through evidence.

The validated solution should not immediately be introduced

across the complete system. A controlled pilot is first conducted

under defined conditions. Performance is then measured across

different relevant operating conditions and compared with the

established baseline.

CONTROLLED PILOT → VARIATION VALIDATION →

RESTRUCTURE & RETEST

Disclaimer: This document is an independent professional research exercise by Jagriti Singh. It is a conceptual Quality Engineering framework developed for

academic and professional learning purposes using publicly understood Quality Management and Quality Engineering methodologies. It is not an official Amazon

project, internal Amazon analysis, internal Amazon specification, audit, operational document, endorsement, partnership or affiliation.

AMZ / TQM / 100 / 2026 / D47 Page 2 of 2

DAY 48

Original source file(s): Day 48 .pdf

Source focus: The Day 48 source material focuses on NETFLIX Ref: NFLX/TQM/100/2026/D48. The source text explicitly discusses the following quality themes and methods: Root Cause, Taguchi, PDCA, Kaizen, reliability, audit.

Section headings identified in the source: CORE QUALITY CONCEPT • MUST BE PERFORMANCE ATTRACTIVE • KEY QUALITY INSIGHT • METHODOLOGY APPLIED:

Technical terms detected in the source: Root Cause, Taguchi, PDCA, Kaizen, reliability, audit

Source extract 1: Day 48 .pdf

NETFLIX Ref: NFLX/TQM/100/2026/D48

Jagriti Singh | Quality Engineer

PERSONALIZED DISCOVERY QUALITY ENGINEERING 13 September 2026 | Day 48 of 100 Days of TQM

CORE QUALITY CONCEPT

User Need and Expectation × Voice of Customer × QFD × CTQ × Kano Model × Root Cause Analysis × Shainin Red X × Taguchi Robust

Design × DOE × Controlled Validation × PDCA × Kaizen

SECTION 01 CASE OBJECTIVE

The objective of this case study is to explore how Quality Engineering principles can be applied to improve personalised content discovery in

a large scale digital streaming environment. The focus is not simply on increasing the amount of content consumed. The objective is to

understand whether users can discover relevant content efficiently while the system continuously adapts to changing interests, viewing

behaviour and expectations. The proposed framework treats recommendation quality as an engineering quality problem that can be

understood through user requirements, measurable quality characteristics, root cause analysis, robust experimentation and continuous

improvement.

SECTION 02 QUALITY PROBLEM

In a large content environment, users may encounter recommendations that do not match their current interests, expectations or viewing

context. User interests can change over time, while different users may interpret relevance differently. A Quality Engineering challenge is

therefore to understand what users actually need, convert those needs into measurable quality requirements and determine why

recommendation quality may not consistently meet those requirements. The objective is to reduce irrelevant discovery experiences while

improving the reliability and usefulness of personalised content discovery.

SECTION 03 USER NEED AND EXPECTATION

The first step is to understand the actual user requirement rather than immediately changing the recommendation system. The framework

begins by identifying what quality means from the user perspective. Relevant questions include whether the user can find suitable content

efficiently, whether recommendations reflect current interests and whether the discovery experience requires unnecessary effort. This

creates the foundation for defining measurable quality requirements before technical improvement begins.

USER NEED → EXPECTATION → QUALITY REQUIREMENT → MEASURABLE OUTCOME

SECTION 04 VOICE OF CUSTOMER AND BEHAVIOURAL SIGNALS

User expectations should not be understood through one source alone. The proposed approach combines direct user feedback with

behavioural quality signals such as viewing behaviour, search patterns, skipping behaviour, content completion and other relevant interaction

signals. Combining these signals provides a broader understanding of what users find relevant, where friction occurs and how expectations

may change over time. The purpose is not simply to collect more data. The purpose is to convert user signals into meaningful quality

information.

SECTION 05 QFD AND CRITICAL TO QUALITY

Once user requirements are understo

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lity is not achieved by producing a relevant suggestion once. It is achieved when user requirements are understood,

translated into measurable quality characteristics, classified according to user expectations, analysed through evidence, improved through robust

experimentation, validated under variation and continuously refined through feedback and learning.

METHODOLOGY APPLIED:

Voice of Customer • Quality Function Deployment • Critical to Quality • Kano Model • Pareto Analysis • 5 Why Analysis • Fishbone Analysis • Shainin Red X • Taguchi

Robust Design • Design of Experiments • Controlled A B Testing • PDCA • Kaizen • Continuous Monitoring

DISCLAIMER: This document is an independent professional research exercise by Jagriti Singh. It is a conceptual Quality Engineering framework developed for academic and

professional learning purposes using publicly understood Quality Management and Quality Engineering methodologies. It is not an official Netflix project, internal Netflix analysis,

internal Netflix specification, audit, operational document, endorsement, partnership or affiliation.

DAY 49

Original source file(s): Day 49.pdf

Source focus: The Day 49 source material focuses on MCDONALD'S DAY 049 / 100. The source text explicitly discusses the following quality themes and methods: ROOT CAUSE, isolate, Audit, FMEA, RPN, PDCA, KAIZEN, isolates.

Section headings identified in the source: C O R P O R AT E Q U A L I T Y E N G I N E E R I N G C A S E S T U D Y • ENHANCING RESTAURANT-LEVEL PROCESS CONSISTENCY & CUSTOMER EXPERIENCE • 2. Pareto Analysis

(80/20 Principle) • continuously improved unless the organization understands • Applied post-prioritization to isolate the single dominant root • from standard operating procedures (SOPs) to drive

Technical terms detected in the source: ROOT CAUSE, isolate, Audit, FMEA, RPN, PDCA, KAIZEN, isolates

Source extract 1: Day 49.pdf

MCDONALD'S DAY 049 / 100

C O R P O R AT E Q U A L I T Y E N G I N E E R I N G C A S E S T U D Y

ENHANCING RESTAURANT-LEVEL PROCESS CONSISTENCY & CUSTOMER EXPERIENCE

AUTHOR: Jagriti Singh ROLE: Quality Engineer DATE: 14 September 2026 REF ID: QE-100D-049

01 | THE QUALITY CHALLENGE 03 | FROM PROBLEMS TO REAL ROOT CAUSE

McDonald’s operates at an enormous global scale, where When audits and VOC reveal multiple operational issues, the

customer expectation demands not simply good food, but a goal is structured prioritization rather than solving everything

consistent, dependable restaurant experience. simultaneously.

Key drivers include speed, accuracy, friendliness, and quality. 1. Fishbone (Ishikawa) Analysis

However, restaurant-level experience still varies due to: Categorizes potential causes behind service variation across 6

• Service and waiting time fluctuations key operational dimensions:

• Order accuracy inconsistencies Man Crew training, fatigue, communication

• Peak-hour operational bottlenecks

Machine POS responsiveness, kitchen equipment layout

• Process gaps between standard & ground execution

Method Order assembly workflow, station handoffs

PROBLEM STATEMENT Material Packaging, ingredient staging, inventory flow

How can McDonald’s strengthen restaurant-level

Measurement Order accuracy tracking, hold-time monitors

process consistency, reduce avoidable service

variation, and deliver a more reliable customer Environment Peak-hour kitchen heat, aisle noise, space layout

experience across locations?

2. Pareto Analysis (80/20 Principle)

Applied as a prioritization concept to separate the "vital few"

defects from the "trivial many", focusing engineering effort

02 | QUALITY ENGINEERING APPROACH

where impact is highest.

Prevention & Understanding First: A process cannot be 3. Shainin Red X Methodology

continuously improved unless the organization understands

Applied post-prioritization to isolate the single dominant root

what is actually going wrong at ground level.

cause (the Red X) responsible for critical variation, enabling

Step 1: Voice of Customer (VOC) + System & Process Audit targeted intervention.

• VOC Collection: Gather structured feedback to capture

customer dissatisfaction, expectations, and recurring pain

points. Translate actionable requirements into process VISUAL QE IDENTIFICATION FLOW

improvements.

• System & Process Audits: Evaluate process adherence,

VOC & AUDITS → FISHBONE → PARETO (80/20) →

service execution, order accuracy, food quality controls, and

hygiene practices. SHAININ RED X

• Audit Objective: Identify where actual execution deviates

from standard operating procedures (SOPs) to drive

prevention.

McDonald’s Quality Engineering Case Study | 100 Days QE Challenge Page 1 of 2

MCDONALD'S | QUALITY ENGINEERING EXECUTION REF ID: QE-100D-049

04 | SOLUTION DESIGN & INNOVATION 10 | COMPLETE QE METHODOLOGY FLOW

Finding the root cause is only half the problem. The process

must be redesigned without shifting constraints elsewhere

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ality

visible in actual restaurant execution. Customer feedback shows

experience; audits reveal deviation; Fishbone & Pareto structure

problems; Shainin isolates critical causes; TOC targets bottlenecks;

07 | CONTROL & CONTINUOUS IMPROVEMENT TRIZ resolves trade-offs; FMEA anticipates risk; PDCA & Kaizen

sustain performance. That is execution-driven Quality Engineering.

PDCA + Kaizen + Continuous Monitoring

• Plan: Define problem, root cause, and ≥98% accuracy

target. 12 | Disclaimer

Independent Quality Engineering case study created for professional development.

• Do: Execute controlled pilot in selected locations. Based on public domain concepts and TQM principles. Not affiliated with, sponsored, or

• Check: Analyze POS data, digital tracking, and audit endorsed by McDonald’s. Proposed analyses reflect the author’s independent work.

metrics.

• Act: Standardize successful SOPs; correct deviations.

Kaizen ensures ongoing incremental gains, while digital POS

monitoring alerts management to drift.

Jagriti Singh | Quality Engineer | Day 49 (14 Sept 2026) Ref ID: QE-100D-049 | Page 2 of 2

DAY 50

Original source file(s): Day 50.pdf

Source focus: The Day 50 source material focuses on MASTERCARD. Ref: MST/TQM/100/2026/D50. The source text explicitly discusses the following quality themes and methods: reliability, PDCA, Kaizen, audit.

Section headings identified in the source: ADAPTIVE PAYMENT QUALITY ENGINEERING • STRONGER SECURITY • Engineering principles can be applied to a large scale LOWER CUSTOMER FRICTION • The central challenge is to strengthen protection against ADAPTIVE QUALITY SOLUTION • cyber threats and fraudulent activity while reducing the • PROBLEM relevant risk signals while maintaining reliability and

Technical terms detected in the source: reliability, PDCA, Kaizen, audit

Source extract 1: Day 50.pdf

MASTERCARD. Ref: MST/TQM/100/2026/D50

Jagriti Singh | Quality Engineer

ADAPTIVE PAYMENT QUALITY ENGINEERING

15 September 2026 | Day 50 of 100 Days of TQM

SECTION 01 / CASE OBJECTIVE

STRONGER SECURITY

The objective of this case study is to explore how Quality + (CONTRADICTION) +

Engineering principles can be applied to a large scale LOWER CUSTOMER FRICTION

global payment ecosystem where security, transaction ↓

reliability, processing performance and customer TRIZ CONTRADICTION RESOLUTION

experience must work together. ↓

The central challenge is to strengthen protection against ADAPTIVE QUALITY SOLUTION

cyber threats and fraudulent activity while reducing the

possibility of unnecessary impact on genuine transactions SECTION 04 / IDEAL FINAL RESULT

and maintaining a seamless customer experience. The ideal final result is a payment quality system that

The proposed framework uses TRIZ as the central problem provides strong protection against suspicious activity while

solving perspective to address the contradiction between allowing genuine transactions to move through the payment

stronger security and lower customer friction. experience with minimal unnecessary friction. The system

should not treat every transaction identically.

SECTION 02 / INTEGRATED QUALITY

Instead, security intensity should adapt according to

PROBLEM relevant risk signals while maintaining reliability and

A global payment ecosystem must handle multiple quality customer experience as quality requirements. The objective

requirements at the same time. Transactions should be is not to reduce security. The objective is to make security

secure against evolving cyber threats and fraudulent more adaptive and context aware.

behaviour. At the same time, genuine transactions should

not be unnecessarily blocked, transaction processing LOW RISK → SEAMLESS PROCESSING

should remain reliable and consistent, and customers SUSPICIOUS → ADDITIONAL VERIFICATION

should experience a smooth payment journey. HIGH RISK → STRONGER SECURITY CONTROLS

Strengthening one requirement can create an unintended

negative effect on another requirement. The Quality SECTION 05 / ADAPTIVE RISK BASED

Engineering challenge is therefore not to optimise security, QUALITY

speed or customer experience independently. It is to The proposed solution combines the ideas of adaptive

develop an integrated quality approach that balances these security, additional verification and real time response. Low

requirements within one adaptive system. risk transactions can follow a more seamless processing

path.

SECURITY + FRAUD PROTECTION + TRANSACTION

When suspicious signals appear, an additional verification

RELIABILITY

layer can be introduced rather than immediately blocking

+ PROCESSING PERFORMANCE + CUSTOMER EXPERIENCE

↓ the transaction. When stronger risk indicators are present,

INTEGRATED PAYMENT QUALITY the system can increase the intensity of security controls to

protect the ecosystem while reducing unnecessary

SECTION 03 / TRIZ &

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icious processing performance and customer experience are

transactions receive appropriate additional verification and managed as interconnected quality requirements.

whether higher risk situations trigger stronger controls

before wider implementation.

CONTROLLED TESTING → DIFFERENT RISK CONDITIONS

MEASURE → ANALYSE → REFINE → RETEST

METHODOLOGY: TRIZ Contradiction Analysis, Ideal Final Result, Adaptive Risk Based Quality, Real Time Signal Analysis, Quality Measurement,

Controlled Validation, PDCA, Kaizen, Continuous Improvement.

MST/TQM/100/2026/D50 | Jagriti Singh Page 2 of 3

DISCLAIMER: This document is an independent professional research exercise by Jagriti Singh. It is a conceptual Quality Engineering framework

developed for academic and professional learning purposes using publicly understood Quality Management, TRIZ and Quality Engineering

methodologies. It is not an official Mastercard project, internal Mastercard analysis, internal Mastercard specification, audit, operational document,

endorsement, partnership or affiliation.

MST/TQM/100/2026/D50 | Jagriti Singh Page 3 of 3

DAY 51

Original source file(s): Day 51.pdf

Source focus: The Day 51 source material focuses on AMERICAN EXPRESS. The source text explicitly discusses the following quality themes and methods: Reliability, FMEA, yield, isolated, isolate, RCA, PDCA, Kaizen, Root Cause, audit.

Section headings identified in the source: AMERICAN EXPRESS • Adaptive Payment Quality Engineering • Customer Experience • 01. INTEGRATED QUALITY PROBLEM 04. CRITICAL QUALITY AREAS • 02. PROCESS MAPPING & QUALITY REQUIREMENTS QUALITY ARCHITECTURE FLOW • The initial phase of the proposed methodology maps the entire payment Process Mapping

Technical terms detected in the source: Reliability, FMEA, yield, isolated, isolate, RCA, PDCA, Kaizen, Root Cause, audit

Source extract 1: Day 51.pdf

AMERICAN EXPRESS

Adaptive Payment Quality Engineering

An Integrated Quality Engineering Framework for Transaction Reliability, Security, Processing Performance and

Customer Experience

Date: 16 September 2026 Timeline: Day 51 / 100 Author: Jagriti Singh | Quality Engineer Ref: AMEX/TQM/100/2026/D51

Core Case Study: How can a payment ecosystem simultaneously improve transaction reliability, strengthen security, maintain

reliable processing performance and preserve a smooth customer experience through an integrated adaptive quality system?

01. INTEGRATED QUALITY PROBLEM 04. CRITICAL QUALITY AREAS

A large-scale payment ecosystem operates across multiple interconnected Synthesizing process mapping, multi-source data

quality dimensions. Transaction failures directly undermine customer trust analysis, Pareto rankings, and FMEA prioritization

and institutional credibility, while stronger security controls can introduce enables the precise identification of critical

additional verification steps or operational friction. Furthermore, processing intervention areas. Rather than attempting to optimize

delays degrade user experience, and evolving fraud or cyber risks threaten every parameter simultaneously, the framework

baseline reliability. Therefore, the quality challenge must be addressed as targets specific nodes where quality enhancements

an interconnected system involving reliability, security, processing yield the highest systemic impact across reliability,

performance, and customer experience rather than treating each security, processing performance, and customer

dimension as an isolated problem. experience.

02. PROCESS MAPPING & QUALITY REQUIREMENTS QUALITY ARCHITECTURE FLOW

The initial phase of the proposed methodology maps the entire payment Process Mapping

lifecycle to understand how distinct stages and operational factors ↓

influence one another. The primary objective is to uncover Critical Quality Requirements

interdependencies and establish Critical Quality Requirements. Key ↓

parameters include transaction reliability, security effectiveness, Data Analysis

processing consistency, processing time, customer experience, and ↓

institutional trust. These rigorous requirements establish the baseline Pareto Analysis & FMEA

foundation for empirical measurement and continuous improvement. ↓

Critical Quality Areas

03. DATA-DRIVEN QUALITY ANALYSIS

Following the definition of Critical Quality Requirements, comprehensive 05. TRQ BASED ADAPTIVE APPROACH

operational data is analyzed prior to solution deployment. This analysis Presented as an independent conceptual Quality
combines transaction ledgers, fraud and security signals, processing Engineering framework, this solution utilizes adaptive
latency metrics, failure and decline patterns, and direct customer feedback. controls driven by transactional risk and
Pareto Analysis is applied to isolate the most significant contributors to environmental conditions. The fundamental principle
quality breakdowns, wh
[Middle content condensed for page fit]
cross diverse risk scenarios. If 5. Pareto + 10. RCA &
operational anomalies emerge, a structured feedback loop is triggered FMEA Refinement
immediately: Result → Analysis → Root Cause Analysis → Solution
Refinement → Retesting. Full standardization occurs only after exhaustive
validation. “Quality in a payment ecosystem is not achieved
by maximizing a single dimension. It is achieved
by continuously balancing security, reliability,
processing performance and customer experience
through data driven and adaptive quality
improvement.”
This framework treats payment quality as a system-
level engineering problem where risk, reliability,
performance, and customer impact are evaluated
together.

Methodology Note: Independent professional research and conceptual Quality Engineering exercise. This framework is an analytical application of Quality

Engineering and TQM principles and does not represent an official American Express project, internal analysis, audit, specification, partnership, endorsement, or

affiliation.

AMEX / TQM / 100 / 2026 / D51 | Jagriti Singh, Quality Engineer Page 2 of 2

DAY 52

Original source file(s): Day 52.pdf

Source focus: The Day 52 source material focuses on G O L D MA N SAC H S. The source text explicitly discusses the following quality themes and methods: reliability, isolated, RCA, Root Cause, PDCA, Kaizen, audit.

Section headings identified in the source: G O L D MA N SAC H S • DATA QUALITY & OPERATIONAL RESILIENCE • A Quality Engineering Framework for Reliable Decision Making • CORE QUALITY PROBLEM • 01. INTEGRATED QUALITY PROBLEM • 02. DATA QUALITY REQUIREMENTS

Technical terms detected in the source: reliability, isolated, RCA, Root Cause, PDCA, Kaizen, audit

Source extract 1: Day 52.pdf

G O L D MA N SAC H S

DATA QUALITY & OPERATIONAL RESILIENCE

A Quality Engineering Framework for Reliable Decision Making

DAY 52 / 100 | 17 September 2026 | Jagriti Singh | Quality Engineer | Ref: GS/TQM/100/2026/D52

CORE QUALITY PROBLEM

How can a financial organization improve decision quality and operational resilience when decisions depend on data that must remain

accurate, complete, consistent and available even during operational disruptions?

Interconnected Dimensions: Decision Quality • Operational Resilience • Data Quality Engineering

01. INTEGRATED QUALITY PROBLEM

Financial decision making depends heavily on the quality, integrity, and availability of operational information. If enterprise data is inaccurate,

incomplete, or inconsistent across systems, the reliability of critical business decisions can be severely compromised. At the same time, any

unexpected disruption to a primary data source or operational processing system can make timely and reliable decision making

exceptionally difficult. Therefore, the core Quality Engineering challenge is to build an architectural and procedural system where data quality

and operational resilience reinforce reliable decision making together, treating them as an interconnected ecosystem rather than isolated

functional problems.

02. DATA QUALITY REQUIREMENTS

To establish a solid analytical foundation, this framework defines three core data quality dimensions that govern all incoming and processed

information streams across enterprise pipelines:

ACCURACY COMPLETENESS CONSISTENCY

Data must correctly and precisely represent the Required information attributes must not be Information must remain fully aligned and

true operational and market reality without missing or truncated when needed for reconcilable across all relevant internal and

systemic distortion. downstream decision processes. external systems.

These three dimensions establish the rigorous baseline required for reliable downstream financial decisions and automated risk workflows.

03. DATA VALIDATION AND CROSS SYSTEM CONSISTENCY

The proposed Quality Engineering approach mandates that critical data must pass structured validation gates before being utilized in high-

impact decision processes. Cross-system consistency checks actively identify situations where disparate platforms contain conflicting

information attributes. The ultimate purpose is not simply to flag an incorrect data value, but to verify whether information remains entirely

trustworthy across the complete end-to-end operational lifecycle.

DATA INPUT → VALIDATION → ACCURACY CHECK → COMPLETENESS CHECK → CROSS-SYSTEM CONSISTENCY → DECISION-READY

DATA

04. OPERATIONAL RESILIENCE

Reliable decision making also depends critically on data and processing systems remaining continuously available when required. If a

primary data source or operational system experiences a temporary outage, the framework triggers a validated alternative data source or

fallback pathway. The decision process cont

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11. FINAL QUALITY ENGINEERING FRAMEWORK

INTEGRATED PROBLEM → DATA REQS → ACCURACY/COMPLETENESS/CONSISTENCY → VALIDATION → CROSS-SYSTEM CHECKS →

RESILIENCE → ALTERNATIVE PATHWAYS → CONTROLLED CONTINUITY → RECONCILIATION → DECISION QUALITY → CONTINUOUS

MONITORING → RESILIENCE TESTING → RCA → CORRECTIVE ACTION → RETESTING → STANDARDISE → PDCA + KAIZEN

“Reliable decisions depend not only on the decision process, but also on the quality, consistency and availability of the data

supporting it.”

Quality Engineering in a complex financial environment must connect data quality, operational resilience, and decision reliability into a unified

architectural fabric rather than optimizing them in functional silos.

Methodology Note: Independent professional research and conceptual Quality Engineering exercise. This framework is an analytical application of Quality Engineering and TQM

principles and does not represent an official Goldman Sachs project, internal analysis, audit, specification, partnership, endorsement, or affiliation.

Page

GS/TQM/100/2026/D52 | Jagriti Singh | Quality Engineer 2 of

2

DAY 53

Original source file(s): Day 53.pdf

Source focus: The Day 53 source material focuses on Deloitte.. The source text explicitly discusses the following quality themes and methods: reliability, isolated, ROOT CAUSE, isolates, isolate, PDCA, Kaizen, RCA, audit.

Section headings identified in the source: AI IMPLEMENTATION QUALITY ENGINEERING • accuracy while concurrently fitting reliably into the overarching • PROCESS DATA • primary objective is to pinpoint exactly where AI intervention • PARETO ANALYSIS (Vital Few) • AI INTEGRATION POINTS & DEPENDENCIES

Technical terms detected in the source: reliability, isolated, ROOT CAUSE, isolates, isolate, PDCA, Kaizen, RCA, audit

Source extract 1: Day 53.pdf

Deloitte.

AI IMPLEMENTATION QUALITY ENGINEERING

A Quality Engineering Framework for Reliable AI Integration, Human Usability and Continuous Improvement

DAY 53 / 100 | 18 September 2026 | Jagriti Singh | Quality Engineer REF: DEL/TQM/100/2026/D53

Core Case Study Challenge: How can an AI-based business solution be integrated into an existing workflow while maintaining AI

accuracy, process reliability, and human usability? This framework combines three interconnected dimensions: AI Accuracy, Business

Process Integration, and Human Usability, treating AI integration not as three isolated issues, but as a unified Quality Engineering

problem.

SECTION 01: INTEGRATED QUALITY PROBLEM SECTION 03: PROCESS DATA ANALYSIS

Implementing an artificial intelligence solution into an existing Following workflow mapping, empirical process data is thoroughly

business environment is fundamentally not just a technology analyzed to uncover where quality defects actually originate.

challenge. Even when an underlying machine learning model Rather than relying on speculative assumptions, the framework

produces highly accurate outputs, embedding that model into an evaluates historical process performance data, error frequency

established workflow often introduces unforeseen process errors, patterns, workflow cycle delays, end-user feedback, direct AI

operational bottlenecks, user confusion, or adoption friction. output observations, and task completion behaviors. This data-

Therefore, the core Quality Engineering challenge is to ensure driven foundation ensures subsequent engineering efforts target

that the AI solution maintains its mathematical and predictive genuine operational pain points.

accuracy while concurrently fitting reliably into the overarching

business process and remaining entirely intuitive, transparent,

SECTION 04: ROOT CAUSE ANALYSIS

and usable for human operators. To address underlying failures, the framework deploys a robust

Total Quality Management root cause methodology. Pareto

SECTION 02: PROCESS MAPPING & INTEGRATION Analysis isolates the vital few contributors responsible for the

The initial operational step requires mapping the complete end- majority of quality defects. 5 Why and Fishbone Analysis

to-end workflow before making any modifications or introducing systematically explore underlying systemic causes across

the AI solution. This comprehensive map identifies current machinery, methods, materials, and man-power. Finally, Shainin

workflow stages, human decision gates, AI interaction nodes, data Red X is applied as a rigorous statistical technique to isolate the

movement pathways, dependencies, potential failure points, dominant variable driving variation.

temporal delays, and specific user interaction requirements. The

PROCESS DATA

primary objective is to pinpoint exactly where AI intervention

influences existing process quality.

PARETO ANALYSIS (Vital Few)

EXISTING BUSINESS WORKFLOW ↓

↓ 5 WHY & FISHBONE (Underlying Causes)

PROCESS MAPPING ↓

↓

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AMEWORK

INTEGRATED PROBLEM → PROCESS MAPPING → AI INTEGRATION POINTS → PROCESS DATA → PARETO → 5 WHY / FISHBONE → SHAININ RED X →

CRITICAL AREAS → CONTROLLED PILOT → MEASUREMENT → REFINEMENT → CONTROLLED ROLLOUT → PDCA / KAIZEN → CONTINUOUS

MONITORING

“AI implementation quality is not defined by model accuracy alone. It is achieved when accurate AI outputs, reliable business

processes and human usability work together as one quality system.”

Quality Engineering provides a rigorous, structured methodology to validate AI solutions not merely through isolated technical

benchmarks, but through holistic process performance, user behavioral adaptation, controlled experimentation, and relentless

continuous improvement.

Methodology Note: Independent professional research and conceptual Quality Engineering exercise. This framework is an analytical application of Quality

Engineering and TQM principles and does not represent an official Deloitte project, internal analysis, audit, specification, client engagement, partnership,

endorsement, or affiliation.

DEL / TQM / 100 / 2026 / D53 Page 2 of 2

DAY 54

Original source file(s): Day 54.pdf

Source focus: The Day 54 source material focuses on Q UA LC O M M. The source text explicitly discusses the following quality themes and methods: Yield, root cause, isolated, RCA, isolating, isolates, PDCA, Kaizen, Cp, Cpk, reliability, audit.

Section headings identified in the source: Q UA LC O M M • SEMICONDUCTOR QUALITY ENGINEERING • CORE QUALITY ENGINEERING PROBLEM • 01. INTEGRATED SEMICONDUCTOR QUALITY 04. CORRECTIVE ACTION & INNOVATIVE • PROBLEM ALTERNATIVE PATHWAY • 02. PROCESS VARIATION AND QUALITY 05. PROCESS RESILIENCE

Technical terms detected in the source: Yield, root cause, isolated, RCA, isolating, isolates, PDCA, Kaizen, Cp, Cpk, reliability, audit

Source extract 1: Day 54.pdf

Q UA LC O M M

SEMICONDUCTOR QUALITY ENGINEERING

Process Variation, Yield Improvement and Resilient Manufacturing

Day: 54 / 100 Date: 19 September 2026 Author: Jagriti Singh | Quality Engineer Ref: QCOM/TQM/100/2026/D54

CORE QUALITY ENGINEERING PROBLEM

How can semiconductor manufacturing quality be improved by controlling process variation, reducing defects, improving yield and

designing resilient alternative pathways when process problems occur?

Framework Pillars: Process Variation | Defect Reduction | Yield Improvement | Resilient Manufacturing

01. INTEGRATED SEMICONDUCTOR QUALITY 04. CORRECTIVE ACTION & INNOVATIVE

PROBLEM ALTERNATIVE PATHWAY

Semiconductor manufacturing requires highly controlled micro- Once the dominant root cause is isolated, engineering

environments and precise chemical-mechanical processes solutions must transcend simple parameter tuning. This
where minute fluctuations in critical process conditions can framework integrates three mandatory tiers: rigorous corrective
heavily influence defect generation, electrical performance action on the primary variable, controlled experimental
parameters and overall wafer fabrication yield. validation, and the parallel creation of an innovative alternative
The fundamental Quality Engineering challenge is therefore process pathway.
not merely inspecting and sorting out defective dies after This alternative pathway serves as a validated contingency
fabrication concludes, but proactively understanding process route designed to maintain throughput and defect-free
variation, identifying dominant root causes, and instituting fabrication if the primary process node experiences recurring
validated technical improvements that render the instability. Rather than relying on unverified workarounds, the
semiconductor manufacturing line fundamentally more robust alternative route undergoes strict qualification to build
against environmental and mechanical drift. structural resilience.

02. PROCESS VARIATION AND QUALITY 05. PROCESS RESILIENCE

Process variation must be rigorously analyzed prior to Resilient semiconductor manufacturing mandates that
executing large-scale corrective interventions. In advanced production lines maintain a deterministic, pre-validated
semiconductor fabrication, variation arises across complex response mechanism when critical manufacturing parameters
equipment behavior, chamber conditions, chemical purity, undergo unexpected shifts or thermal-chemical degradation.
thermal profiles and mechanical handling parameters.

PROCESS ISSUE → RCA →

The core objective is isolating which specific variables actively

contribute to catastrophic defect formation or systematic yield CORRECTIVE + ALTERNATIVE → EXPERIMENT →

loss versus background operational noise. VALIDATION → STANDARDISE

PROCESS DATA → VARIATION ANALYSIS → This structured protocol ensures recurring process disruptions

CRITICAL VARIABLES → DEFECT PATTERNS → are systematically mitigated, dampening variance propagation

across subse

[Middle content condensed for page fit]

ing instabilities into structured opportunities for permanent

refinement across all auxiliary tooling. defect reduction, yield enhancement, and operational resilience.

Methodology Note: Independent professional research and conceptual Quality

09. RETESTING AND STANDARDISATION

Engineering exercise. This framework is an analytical application of Quality

When validation reveals unexpected yield excursions or Engineering and TQM principles and does not represent an official Qualcomm project,

internal analysis, audit, specification, partnership, endorsement or affiliation.

secondary defects, full-scale deployment is halted. The system

executes a structured retesting loop to refine solutions prior to

organizational lock-in.

RESULT → ANALYSIS → RCA → REFINEMENT →

RETEST → STANDARDISE

10. CONTINUOUS MONITORING

Semiconductor quality engineering is iterative. Continuous

monitoring identifies emerging defect signatures, thermal drift,

and tool wear. When anomalies arise, the framework loops

back to root cause analysis.

QCOM / TQM / 100 / 2026 / D54 | Jagriti Singh | Quality Engineer Page 2 of 2

DAY 55

Original source file(s): Day 55 cisco_network_qe_case_study_day55.pdf

Source focus: The Day 55 source material focuses on NETWORK QUALITY ENGINEERING. The source text explicitly discusses the following quality themes and methods: Reliability, isolated, RCA, isolate, ISOLATION, ROOT CAUSE, Isolating, Taguchi, PDCA, Kaizen, audit.

Section headings identified in the source: NETWORK QUALITY ENGINEERING • network quality extends far beyond binary connection • NETWORK PERFORMANCE DATA • The primary Quality Engineering challenge is to FAILURE PATTERNS & PARETO ANALYSIS • understand how these interdependent dimensions interact • and to isolate the specific operational factors that trigger

Technical terms detected in the source: Reliability, isolated, RCA, isolate, ISOLATION, ROOT CAUSE, Isolating, Taguchi, PDCA, Kaizen, audit

Source extract 1: Day 55 cisco_network_qe_case_study_day55.pdf

CISCO

NETWORK QUALITY ENGINEERING

An Integrated Framework for Network Reliability, Performance, Security and Resilience

DAY 55 / 100 | 20 SEPTEMBER 2026 | JAGRITI SINGH | QUALITY ENGINEER | REF: CISCO/TQM/100/2026/D55

Core QE Problem: How can a networking environment maintain reliable performance, low latency, low packet loss and

strong security across changing traffic loads and operating conditions? This framework treats Network Reliability,

Performance, Latency, Packet Loss, Cybersecurity, and Network Resilience as interconnected quality dimensions rather

than isolated problems.

SECTION 01: INTEGRATED PROBLEM SECTION 03: DATA-DRIVEN RCA FLOW

Modern networking environments operate under To identify operational bottlenecks, raw telemetry is

continuously fluctuating traffic loads, rigorous operating channeled through an advanced analytical diagnostic

conditions, and strict security requirements. Consequently, sequence.

network quality extends far beyond binary connection

NETWORK PERFORMANCE DATA

availability. It encompasses reliability, performance

consistency, packet delivery efficiency, latency bounds, ↓

robust security controls, and system resilience under OPERATING CONDITIONS

dynamic stress. ↓

The primary Quality Engineering challenge is to FAILURE PATTERNS & PARETO ANALYSIS

understand how these interdependent dimensions interact

and to isolate the specific operational factors that trigger

5 WHY + FISHBONE ANALYSIS

performance degradation or reliability breakdown.

SECTION 02: PERFORMANCE CONDITIONS

SHAININ RED X ISOLATION

Network performance inherently varies with traffic surges,

bandwidth saturation, and environmental shifts. A rigorous

evaluation framework must analyze behavior across DOMINANT ROOT CAUSE

diverse operational states rather than depending on

skewed aggregate averages.

Evaluated variables include peak traffic loads, queue depth

utilization, environmental operating states, performance

variation metrics, packet drop patterns, and asynchronous

security events. This analytical focus uncovers root failure

precursors without relying on proprietary internal telemetry.

SECTION 04: ROBUST PERFORMANCE SECTION 05: VARIATION EVALUATION

Isolating the root cause is foundational, but the engineered Network fixes must be stress-tested across varied

solution must maintain efficacy amidst operational shifts. operational profiles to guarantee baseline stability before

We integrate Taguchi Robust Design and Design of enterprise deployment.

Experiments (DOE) to evaluate multi-factor interactions

NORMAL LOAD → HIGH TRAFFIC → CHANGING CONDITIONS →

systematically. PERFORMANCE MEASUREMENT → ROBUSTNESS EVALUATION

Taguchi principles evaluate how proposed network Improvements are validated against strict quality indicators

adjustments behave in the presence of noise factors and including jitter bounds, end-to-end latency, and packet loss

environmental variance, optimizing system resilience rather ratios.

than tuning purely for ideal lab conditions.

CISCO /

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bility, security, and resilience are treated

as a unified matrix.

SECTION 08: CONTINUOUS MONITORING

Quality assurance extends past validation. Continuous

network telemetry monitoring detects latent degradation,

rising latency signatures, packet loss anomalies, and

security-related behavior shifts early.

““Network quality is not defined by performance under one ideal condition. It is demonstrated when

reliability, performance and security remain controlled as operating conditions change.””

Quality Engineering delivers a systematic discipline to isolate dominant root causes, engineer robust architectures, and maintain

continuous operational oversight rather than depending on reactive troubleshooting.

Methodology Note: Independent professional research and conceptual Quality Engineering exercise. This framework is an analytical application of

Quality Engineering and TQM principles and does not represent an official Cisco project, internal analysis, audit, specification, partnership,

endorsement or affiliation.

CISCO / TQM / 100 / 2026 / D55 | Jagriti Singh | Quality Engineer Page 2 of 2

DAY 56

Original source file(s): Day 56 accenture_qe_case_study.pdf

Source focus: The Day 56 source material focuses on A C C E N T U R E • 1 0 0 DAY S T Q M & Q UA L I T Y E N G I N E E R I N G C H A L L E N G E. The source text explicitly discusses the following quality themes and methods: isolated, PDCA, SPC, isolate, Isolates, root cause.

Section headings identified in the source: 1. THE QUALITY CHALLENGE • 2. PROPOSED QUALITY ENGINEERING APPROACH • deployed across the project lifecycle: • Fishbone / Ishikawa Analysis Core Diagnostic Distinction • operational categories: a strict separation of analytical tools: • Fishbone structures possible causes but requires empirical

Technical terms detected in the source: isolated, PDCA, SPC, isolate, Isolates, root cause

Source extract 1: Day 56 accenture_qe_case_study.pdf

A C C E N T U R E • 1 0 0 DAY S T Q M & Q UA L I T Y E N G I N E E R I N G C H A L L E N G E

“Early Defect Detection & Prevention in Software and AI Projects”

Author: Jagriti Singh Role: Quality Engineer Date: 21 September 2026 Ref. ID: QE-56

1. THE QUALITY CHALLENGE

Modern software and artificial intelligence (AI) implementations frequently encounter recurring defects, escaped defects,

shifting requirements, and failures that surface late in the software development lifecycle (SDLC). Within an enterprise

technology-services environment operating at Accenture’s scale, uncovering defects exclusively during final User

Acceptance Testing (UAT) or post-deployment significantly inflates remediation costs and jeopardizes overall delivery

standards.

“How can defects be identified earlier, their critical root causes isolated, and recurring defects prevented before they reach

production?”

Note: This document represents an illustrative Quality Engineering case scenario designed for an enterprise technology-

services scale, rather than an internal Accenture defect disclosure.

2. PROPOSED QUALITY ENGINEERING APPROACH

To shift from reactive defect detection to proactive defect prevention, a structured end-to-end TQM methodology is

deployed across the project lifecycle:

Problem Identification Cause Analysis Pareto Prioritisation Shainin Red X PDCA Improvement Pilot & SPC Control

Problem Identification Pareto Analysis (80/20 Rule)

Systematically capture and categorize software and AI Analyze historical defect logs to isolate the vital few

defects (e.g., hallucination anomalies, prompt defect categories contributing to the majority of quality regressions, integration failures, API boundary escapes. Pareto prioritization ensures engineering mismatches). Quantify their downstream impact on effort is directed where it yields the maximum risk quality metrics, rework hours, and client experience. reduction.

Fishbone / Ishikawa Analysis Core Diagnostic Distinction

Organize potential defect drivers into structured To avoid diagnostic ambiguity, the framework enforces operational categories: a strict separation of analytical tools:

• Requirements: Ambiguous user stories, unhandled • Fishbone: Generates and categorizes possible edge cases. causes.

• Code: Logic gaps, technical debt, concurrency • Pareto: Prioritizes high-impact defect categories. bugs. • Red X: Isolates the exact dominant root cause

• Testing: Inadequate test coverage, stale test data. through experimentation.

• Data / Environment: Skewed training data, config drift.

• Process / People: Handoff friction, training gaps.

Fishbone structures possible causes but requires empirical validation to isolate root causes.

Accenture | QE-56 Case Study (Jagriti Singh) Page 1 of 2

3. SHAININ RED X — CRITICAL CAUSE IDENTIFICATION

Once Pareto analysis identifies the major defect category, the investigation proceeds to Shainin methodology. While Pareto highlights major contributors, complex AI and software environments o

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sion.

6. COMPLETE QUALITY LOOP

Detect → Analyse → Prioritise → Isolate → Improve → Pilot → Validate → Monitor → Standardise → Continuously Improve

“The objective is not simply to find defects faster, but to move the quality process towards defect prevention and sustained process improvement.”

7. EXPECTED QUALITY IMPACT

Operational Outcomes

• Earlier identification of critical quality and AI risks

• Significant reduction in recurring software defects

• Lower production defect escape rates

Strategic Outcomes

• Reduced rework and optimized engineering bandwidth

• Enhanced process stability and data-driven decisions

• Stronger organizational continuous-improvement culture

Disclaimer: This is not an original Accenture project, internal Accenture case study, or representation of Accenture's proprietary quality processes or

data. This case study has been independently created by Jagriti Singh as part of a 100 Days TQM & Quality Engineering Challenge to demonstrate

the practical application of Quality Engineering and TQM methodologies.

Accenture | QE-56 Case Study (Jagriti Singh) Page 2 of 2

DAY 57

Original source file(s): Day 57_Oracle_QE_Case_Study_.pdf

Source focus: The Day 57 source material focuses on O R AC L E C A S E S T U DY Ref. ID: QE-57. The source text explicitly discusses the following quality themes and methods: isolating, PDCA, SPC, root cause, isolate, Isolates.

Section headings identified in the source: Early Defect Detection & Prevention in Software & Cloud Applications • 1. ILLUSTRATIVE QUALITY ENGINEERING CHALLENGE • Diagnostic Core Questions Contextual Scope • from reactive firefighting to predictive defect prevention across • 2. PROBLEM-SOLVING FRAMEWORK • 3. FISHBONE / ISHIKAWA — FINDING POSSIBLE CAUSES

Technical terms detected in the source: isolating, PDCA, SPC, root cause, isolate, Isolates

Source extract 1: Day 57_Oracle_QE_Case_Study_.pdf

O R AC L E C A S E S T U DY Ref. ID: QE-57

Early Defect Detection & Prevention in Software & Cloud Applications

Jagriti Singh | Quality Engineer | Date: 22 September 2026

1. ILLUSTRATIVE QUALITY ENGINEERING CHALLENGE

Software and cloud applications can experience recurring defects, production incidents, escaped defects, and failures

discovered late in the development lifecycle. The Quality Engineer's central problem is not merely detecting a bug, but

systematically isolating root causes and preventing recurrences.

Diagnostic Core Questions Contextual Scope

• Where does the defect originate in the pipeline? This study models an enterprise cloud microservices

• Why did existing unit/integration controls fail to intercept it? deployment pipeline. The objective is shifting quality practices

from reactive firefighting to predictive defect prevention across

• Which systemic causes contribute most significantly?

continuous integration cycles.

• Is the corrective action genuinely sustaining process stability?

Note: This framework illustrates methodology application. It does not claim that Oracle experiences these specific internal problems or

statistics.

2. PROBLEM-SOLVING FRAMEWORK

A structured, multi-stage TQM workflow links identification to standardisation:

Problem Identification → Fishbone Analysis → Pareto Prioritisation → Shainin Red X → PDCA Improvement → Pilot

→ Validation → SPC / Monitoring → Control & Standardisation

3. FISHBONE / ISHIKAWA — FINDING POSSIBLE CAUSES

A Quality Engineer structures potential causes of recurring cloud application defects across 8 major operational branches:

1. Requirements: Ambiguous API payload specs or missing multi-tenant concurrency criteria.

2. Code: Concurrency race conditions, unhandled asynchronous callback exceptions.

3. Testing: Inadequate mocking of distributed transactional database states.

4. Data: Schema migration discrepancies between staging and production clusters.

5. Infrastructure: Network latency spikes, misconfigured Kubernetes ingress controllers.

6. Deployment: Rolling update sequencing failures, missing environment parity.

7. Process: Incomplete code review checklists for asynchronous modules.

8. Human Factors: Context switching across overlapping sprint deliverables.

Fishbone generates and organises potential causes; it does not prove actual root cause.

4. PARETO ANALYSIS — PRIORITISING THE PROBLEM

After collecting incident data, Pareto Analysis determines which defect categories contribute most to the quality bottleneck.

Defect Category Incidents Cum. %

Async Concurrency Failures 42 42%

Schema Migration Errors 28 70%

Config / Ingress Mismatch 15 85%

API Spec Ambiguities 10 95%

Other Minor Causes 5 100%

Illustrative Data — Not Oracle Internal Data.

Fishbone identifies possible causes; Pareto prioritises where to focus engineering effort. Note: Not all quality problems strictly follow the

80/20 rule.

QE-57 | 100 Days TQM & Quality Engineering Challenge Page 1 of 2

O R AC L E C A S E S T U

[Middle content condensed for page fit]

ical Process Control (SPC) charts on transaction latency to ensure stability.

CONTROL: Integrated automated quality gates in CI/CD pipeline blocking builds lacking backpressure tests.

8. COMPLETE QE LOOP & EXPECTED IMPACT

DETECT → ANALYSE → PRIORITISE → ISOLATE → IMPROVE → PILOT → VALIDATE → MONITOR → CONTROL → CONTINUOUS

IMPROVEMENT

Expected Quality Impact

• Earlier detection in CI pipeline | • Reduced escape rates

• Lower rework & engineering toil | • Enhanced cloud stability

• Data-driven quality gates | • Stronger TQM maturity

“The objective is to shift quality from reactive defect detection towards data-driven defect prevention and sustained process improvement.”

Disclaimer: This is not an original Oracle project, internal Oracle case study, or representation of Oracle's proprietary processes, systems or data. This

case study has been independently created by Jagriti Singh as part of a 100 Days TQM & Quality Engineering Challenge to demonstrate the practical

application of Quality Engineering and TQM methodologies.

QE-57 | 100 Days TQM & Quality Engineering Challenge Page 2 of 2

DAY 58

Original source file(s): Day 58 walmart_quality_case_study.pdf

Source focus: The Day 58 source material focuses on End-to-End Retail Quality Improvement | Case Study #58. The source text explicitly discusses the following quality themes and methods: customer satisfaction, isolate, reliability, isolated, Root Cause, Isolation, FMEA, RPN, SPC, PDCA.

Section headings identified in the source: WALMART • 1. QUALITY CHALLENGE • Core Problem Statement • 2. END-TO-END QUALITY FLOW • 3. FISHBONE ANALYSIS (CAUSES) 4. PARETO ANALYSIS • Fishbone diagrams structure potential root causes (PRIORITISATION)

Technical terms detected in the source: customer satisfaction, isolate, reliability, isolated, Root Cause, Isolation, FMEA, RPN, SPC, PDCA

Source extract 1: Day 58 walmart_quality_case_study.pdf

WALMART

End-to-End Retail Quality Improvement | Case Study #58

Author: Jagriti Singh (Quality Engineer) Date: 23 September 2026 Ref. ID: QE-58

1. QUALITY CHALLENGE

Illustrative Quality Engineering Case Study — Not Internal Walmart Data

Retail operational quality is dictated by interconnected failure domains: supplier variation, supply chain

transit delays, inventory stock-outs, and checkout friction. A breakdown at any single node cascades

downstream, directly degrading customer satisfaction and retention.

Core Problem Statement

"How can an end-to-end Quality Engineering framework identify major quality risks, isolate critical

causes and improve reliability across the retail value chain?"

2. END-TO-END QUALITY FLOW

Quality must be evaluated across the complete value chain rather than in isolated silos.

Supplier → Supply Chain → Inventory → Store / Checkout → Customer Experience

3. FISHBONE ANALYSIS (CAUSES) 4. PARETO ANALYSIS

Fishbone diagrams structure potential root causes (PRIORITISATION)

across operational categories. Note: Fishbone Pareto analysis identifies which failure categories

organises possibilities; it does not prove root contribute most significantly to operational losses.

causes. Fishbone generates potential causes; Pareto

prioritises major contributors.

Key Ishikawa Categories

• Supplier: Batch inconsistency, lead-time Failure Category Contribution Cumulative

variance.

Inventory Data Inaccuracy 42% 42%

• Logistics: Transit delays, cold-chain breaches.

• Inventory/Data: ERP sync errors, inaccurate Supplier Lead-Time

28% 70%

stock counts. Variance

• Store Operations: Delayed shelf replenishment,

In-Store Replenishment

scan errors. 18% 88%

Delays

Checkout / POS Errors 12% 100%

*Illustrative Data — Not Walmart Internal Data. Relationship is

non-fixed.

QE-58 | 100 Days TQM & Quality Engineering Challenge — Jagriti Singh Page 1 of 2

WALMART — CASE STUDY (PAGE 2)

Risk Assessment, Root Cause Isolation & Control Framework

5. FMEA — PROACTIVE RISK ASSESSMENT

FMEA evaluates both System FMEA (interconnected value chain risks) and Process FMEA (specific

operational vulnerabilities).

Process / Step Failure Mode Potential Effect S O D RPN

Inventory Sync ERP vs Shelf Mismatch Stock-out / Lost Sale 7 6 5 210

Supplier Intake Non-compliant Batch Quality Reject / Delay 8 4 6 192

Checkout / POS Barcode Scan Failure Customer Queue Delay 5 5 4 100

RPN = Severity × Occurrence × Detection. *Illustrative RPN values — Not Walmart Internal Data.

6. SHAININ RED X 8. PILOT → VALIDATION → SPC →

Used after Pareto/FMEA to isolate the dominant CONTROL

cause of critical failures by comparing contrasting Pilot & Validation: Test countermeasure in a

operational conditions or batches, rather than mere controlled environment. Track KPIs: stock-out

brainstorming. frequency, supply-chain delay, supplier defect rate,

and customer complaints (No fake data claimed).

7. PDCA IMPROVEMENT

SPC & Control: Use Control Charts to monitor

• Plan: Define critical problem & countermeasure.

• Do: Implement improvement on controlled scale. stability. Establish standard operating procedures

• Check: Compare pre/post quality metrics. and trigger corrective actions upon process drift.

• Act: Standardise successful fixes.

9. COMPLETE QUALITY ENGINEERING FRAMEWORK

IDENTIFY → ANALYSE → PRIORITISE → ASSESS RISK → ISOLATE → IMPROVE → PILOT → VALIDATE →

MONITOR → CONTROL

Fishbone → Pareto → FMEA/RPN → Shainin Red X → PDCA → Pilot → Validation → SPC → Control

10. POTENTIAL OUTCOMES KEY LEARNING

• Better visibility of quality risks & failure modes

“Retail quality is not controlled at a single point. It is

• Improved supply-chain reliability & inventory

built across the entire value chain—from supplier and

availability

supply chain to inventory, store operations and

• Reduced product defects & consistent checkout

customer experience.”

experience

• Data-driven continuous improvement

Disclaimer: This is not an original Walmart project, internal Walmart case study, or representation of Walmart’s proprietary processes, systems

or data. Independently created by Jagriti Singh as part of a 100 Days TQM & Quality Engineering Challenge to demonstrate practical Quality

Engineering methodologies.

QE-58 | 100 Days TQM & Quality Engineering Challenge — Jagriti Singh Page 2 of 2

DAY 59

Original source file(s): Day 59 honeywell_quality_case_study.pdf

Source focus: The Day 59 source material focuses on HONEYWELL. The source text explicitly discusses the following quality themes and methods: Reliability, supplier quality, isolated, FMEA, PDCA, SPC, RPN, Isolate, root cause, isolating.

Section headings identified in the source: HONEYWELL • Predictive Quality & Process Reliability in Industrial Operations • 1. THE QUALITY CHALLENGE • Core Problem Statement • 2. QUALITY ENGINEERING TOOLCHAIN FLOW • 1. 2. 5 WHY 3. PARETO 4. FMEA 5. RED X 6. POKA 7. PDCA 8. PILOT 9. SPC 10.

Technical terms detected in the source: Reliability, supplier quality, isolated, FMEA, PDCA, SPC, RPN, Isolate, root cause, isolating

Source extract 1: Day 59 honeywell_quality_case_study.pdf

DAY 5 9 | 1 0 0 DAYS TQ M & Q UA L IT Y E N G I N E E R I N G C HA L L E N G E

HONEYWELL

Predictive Quality & Process Reliability in Industrial Operations

Author: Jagriti Singh (Quality Engineer) Date: 24 September 2026 Ref. ID: QE-59

1. THE QUALITY CHALLENGE

Illustrative Case Study — Not Honeywell Internal Data

Examining how industrial organizations manage manufacturing process variation, product and supplier quality variation, automation

and system related failures, and process reliability across complex production lifecycles.

Core Problem Statement

“How can critical quality risks be identified early, dominant causes isolated, and process reliability improved before failures recur

in industrial environments?”

2. QUALITY ENGINEERING TOOLCHAIN FLOW

1. 2. 5 WHY 3. PARETO 4. FMEA 5. RED X 6. POKA 7. PDCA 8. PILOT 9. SPC 10.

FISHBONE Drill Down Prioritise RPN Isolate Cause YOKE Improvement Test Scale Stability CONTROL

Possible Risk Error Proof Sustain

Causes Assessment

3. HOW THE TOOLS CONNECT

Tool / Method Primary Engineering Function Operational Value

Structures brainstorming and exposes underlying failure

Fishbone and 5 Why Identify possible causes and drill down to root cause

triggers.

Pareto and FMEA Prioritise major contributors and assess risks (Severity Directs engineering resources toward high-impact system

RPN x Occurrence x Detection) and process failure modes.

Shainin Red X and Pinpoints exact variation drivers and builds physical and

Isolate dominant cause and error-proof the process

Poka Yoke systemic countermeasures.

PDCA, Pilot, SPC and Implement, test, monitor stability and sustain Ensures engineered solutions permanently control

Control standardization process drift and sustain reliability.

4. POTENTIAL OUTCOMES KEY TAKEAWAY

• Earlier identification of critical quality and process risks.

“Predictive quality is not only about detecting failures

• Reduced recurring defects and unexpected operational

early. It is about identifying variation, isolating dominant

failures.

causes, and building controls that prevent recurrence

• Enhanced process reliability across industrial automation.

permanently.”

• Improved supplier and raw material quality control.

• Minimized process variation and stronger preventive culture.

Disclaimer: This is an independently created illustrative case study, not an original Honeywell project or representation of Honeywell’s proprietary processes, systems

or data. Created as part of a 100 Days TQM & Quality Engineering Challenge by Jagriti Singh (Day 59/100).

QE-59 | Day 59/100 — Jagriti Singh Page 1 of 1

DAY 60

Original source file(s): Day 60 siemens_tqm_qe_case_study.pdf

Source focus: The Day 60 source material focuses on Industrial Quality & Process Reliability. The source text explicitly discusses the following quality themes and methods: Reliability, isolated, Root Cause, FMEA, RPN, ISOLATION, isolate, Kaizen, PDCA, SPC, supplier quality, inspection.

Section headings identified in the source: SIEMENS • Industrial Quality & Process Reliability • 1. EXECUTIVE CASE OVERVIEW • 2. PROBLEM STATEMENT • 3. END-TO-END QUALITY CHAIN 4. ROOT-CAUSE ANALYSIS • Supplier / Input Quality Fishbone (Ishikawa) Analysis

Technical terms detected in the source: Reliability, isolated, Root Cause, FMEA, RPN, ISOLATION, isolate, Kaizen, PDCA, SPC, supplier quality, inspection

Source extract 1: Day 60 siemens_tqm_qe_case_study.pdf

SIEMENS

100 DAYS TQM & QUALIT Y ENGINEERING CHALLENGE (DAY 60/100)

Industrial Quality & Process Reliability

Case Study: End-to-End Quality Engineering Analysis Ref ID: QE-60 | Date: 25 September 2026 Author: Jagriti Singh | Quality Engineer

1. EXECUTIVE CASE OVERVIEW

This document presents an illustrative industrial Quality Engineering scenario investigating interconnected manufacturing process variations, production defects, supplier input quality inconsistencies, and automation/system reliability. In modern industrial ecosystems, anomalies originating upstream in supplier materials, core manufacturing processes, or operational automation can propagate downstream, ultimately undermining final product consistency, performance, and field reliability.

Note: This is an independently developed illustrative technical case study and does not represent an actual Siemens internal project.

2. PROBLEM STATEMENT

Industrial Production Challenge: An advanced industrial manufacturing system is experiencing recurring quality variation and component defects. The operational issue is non-isolated; multi-variable interactions between supplier input variations, equipment process instability, automation control behaviour, and latent product quality risks contribute to compounded downstream failures.

Core Objective: Systematically identify dominant root causes, reduce recurring defects and variance, strengthen prevention mechanisms, validate corrective actions, and establish sustainable statistical process control.

3. END-TO-END QUALITY CHAIN 4. ROOT-CAUSE ANALYSIS

Supplier / Input Quality Fishbone (Ishikawa) Analysis

Structured investigation framework categorizing potential failure modes across six dimensions:

Risk: Material property drift & specification non-conformance

Man | Machine | Method | Material | Measurement |

Manufacturing Process Environment

Risk: Thermal/mechanical variation & equipment wear 5 Why Methodology

Iterative interrogative technique drilling past symptoms to system-

Automation / System

level causes:

↓ Observed Defect → Immediate Cause → Deeper Cause →

Risk: Sensor calibration drift & PLC logic timing variance System Cause → Root Cause

Product Quality & Reliability

5. CAUSE PRIORITISATION (PARETO 6. RISK ASSESSMENT (FMEA)

ANALYSIS) System FMEA and Process FMEA evaluate potential

Multiple potential contributors identified during investigation vulnerabilities at both architecture and operational levels:

are evaluated using Pareto analysis. By ranking defect

RPN = Severity (S) × Occurrence (O) × Detection (D)

frequencies and financial impact, quality engineers separate

vital few contributors from trivial many. Technical Note: RPN acts as a prioritisation aid and is evaluated

alongside engineering criticality and customer impact.

Purpose: Focus limited engineering resources on high-

impact variables, prevent unfocused corrective actions, and

ensure maximum reduction in overall process variance.

"The objective is not simply to identify many possible

[Middle content condensed for page fit]

investigation Poka-Yoke Prevent errors

Pareto Prioritise contributors Kaizen Improve continuously

FMEA Assess system/process risk PDCA / Pilot Implement & validate

Validation Verify effectiveness SPC & Control Monitor & sustain improvement

13. POTENTIAL QUALITY OUTCOMES 14. KEY QUALITY ENGINEERING INSIGHT

• Reduced recurring defect rates & process variance

"Quality should be built into the process through

• Enhanced process stability & product consistency

prevention, risk assessment, root-cause elimination,

• Stronger supplier quality assurance & controls

validation and statistical control — not dependent only on

• Advanced automation reliability & defect prevention

final inspection."

Disclaimer: This is an independently developed illustrative case study created by Jagriti Singh as part of the 100 Days TQM & Quality

Engineering Challenge (Day 60/100). It is not an official Siemens project, internal Siemens case study, or representation of Siemens proprietary

processes, systems, data, or results.

Siemens Case Study | 100 Days TQM & Quality Engineering Challenge — Ref: QE-60 Page 2 / 2

DAY 61

Original source file(s): Day 61 Rolls_Royce_TQM_Case_Study_QEv2.pdf

Source focus: The Day 61 source material focuses on ROLLS -ROYCE — ADVANCED AEROSPACE MANUFACTURING QUALITY. The source text explicitly discusses the following quality themes and methods: Reliability, Inspection, Isolate, PROCESS CAPABILITY, Cp, Cpk, FMEA, RPN, Root Cause, isolating, Kaizen, PDCA.

Section headings identified in the source: ENGINEERING • Component Reliability • 1. CASE CONTEXT & PROBLEM OVERVIEW • 2. MEASUREMENT SYSTEM ANALYSIS (MSA) 3. PROCESS CAPABILITY ANALYSIS • Distinction: Process Stability ensures predictable operation • 4. ROOT-CAUSE ANALYSIS (FISHBONE & 5 WHY) 5. RISK & FAILURE-MODE ANALYSIS (FMEA)

Technical terms detected in the source: Reliability, Inspection, Isolate, PROCESS CAPABILITY, Cp, Cpk, FMEA, RPN, Root Cause, isolating, Kaizen, PDCA, SPC, Control Plan

Source extract 1: Day 61 Rolls_Royce_TQM_Case_Study_QEv2.pdf

ROLLS -ROYCE — ADVANCED AEROSPACE MANUFACTURING QUALITY

ENGINEERING

1 0 0 DAYS T Q M & Q UA L I T Y E N G I N E E R I N G C H A L L E N G E — DAY 6 1 / 1 0 0

Precision Aerospace Manufacturing: Controlling Variation, Measurement Risk &

Component Reliability

Company: Rolls-Royce (Illustrative Scenario) Prepared by: Jagriti Singh | Quality Engineer

Theme: Advanced Aerospace Manufacturing QE Date: 26 September 2026 | Ref ID: QE-61

1. CASE CONTEXT & PROBLEM OVERVIEW

This technical case study examines advanced quality engineering methodologies applied to critical precision aerospace engine components. Performance is influenced by material/supplier variation, precision machining instability, dimensional variation, measurement uncertainty, and component non-conformance.

Supplier / Material → Manufacturing Process → Measurement & Inspection → Component Quality → Engine Reliability

Quality Problem Definition: A precision aerospace machining process experiences variation in critical characteristics. The investigation determines whether variation originates from material, process, measurement, or variable interactions. Objective: Detect →

Characterise → Prioritise → Isolate → Correct → Validate → Control.

2. MEASUREMENT SYSTEM ANALYSIS (MSA) 3. PROCESS CAPABILITY ANALYSIS

Positioned before process capability evaluations to determine if Evaluates whether a stable process consistently meets specified variation stems from the process or measurement error. requirements after confirming measurement reliability.

• Repeatability & Reproducibility (GR&R): Equipment and • Stability Check: Verifying absence of special causes via operator variation checks. control charts.

• Bias, Stability & Resolution: Calibration offset, temporal • Capability Assessment: Quantifying spread relative to limits drift, and discrimination. (Cp, Cpk conceptually).

Distinction: Process Stability ensures predictable operation

Observed Variation → MSA Check → Separate Gauge Error

over time, whereas Process Capability evaluates whether

Note: “Capability conclusions are only meaningful when the predictable spread fits within engineering tolerances.

measurement system is sufficiently capable.”

4. ROOT-CAUSE ANALYSIS (FISHBONE & 5 WHY) 5. RISK & FAILURE-MODE ANALYSIS (FMEA)

Structured investigation to uncover underlying mechanical and Systematic risk evaluation differentiating design and

operational mechanisms driving non-conformances. manufacturing risks to prioritize engineering actions.

• Fishbone (Ishikawa): Structures contributors across 6M: • Design FMEA: Evaluates potential failure modes at system/

Man, Machine, Method, Material, Measurement, Environment. design level.

• 5 Why Analysis: Deepens investigation from symptom to root • Process FMEA: Evaluates how manufacturing steps create

physical/systemic cause. or transfer risk.

RPN: Severity × Occurrence × Detection = RPN

Symptom → 6M Causes → Evidence → Root Cause

Note: RPN is a prioritisation aid; safety criticality and customer

impact mu

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ES & KEY INSIGHT

• PLAN: Define problem, evidence, and targets. • Improved measurement confidence & reduced variation.

• DO: Execute controlled pilot implementation. • Enhanced process capability & defect prevention.

• CHECK: Analyse post-implementation evidence. • Robust risk prioritization & sustainable controls.

• ACT: Standardize validated improvements.

“In aerospace manufacturing, controlling quality requires more

SPC & Control Plan: Monitor critical characteristics via control than detecting defects — measuring correctly, understanding

charts, supported by reaction plans and defined frequencies. variation, isolating dominant causes, validating improvements, and

maintaining control over critical characteristics.”

Disclaimer: Illustrative case study independently developed by Jagriti Singh as part of the 100 Days TQM & Quality Engineering Challenge. This is not an official Rolls-Royce project,

internal Rolls-Royce case study, or representation of Rolls-Royce proprietary processes, systems, data or results.

Rolls-Royce | Advanced Aerospace Manufacturing QE | Ref: QE-61 Page 2 / 02

DAY 62

Original source file(s): Day 62 emirates_quality_engineering_case_study_2.pdf

Source focus: The Day 62 source material focuses on QE-62 | PAGE 1 OF 2. The source text explicitly discusses the following quality themes and methods: Reliability, Root Cause, FMEA, RPN, PDCA, SPC.

Section headings identified in the source: EMIRATES • End-to-End Aviation Operational Quality & Reliability Case Study • 1. CASE STUDY TITLE & CONTEXT • Title: End-to-End Aviation Operational Quality & Reliability • 2. PROBLEM STATEMENT 3. SIPOC ANALYSIS • Hypothetical Scenario: Airline hub experiences variation SUPPLIERS INPUTS PROCESS OUTPUTS CUSTOMER

Technical terms detected in the source: Reliability, Root Cause, FMEA, RPN, PDCA, SPC

Source extract 1: Day 62 emirates_quality_engineering_case_study_2.pdf

QE-62 | PAGE 1 OF 2

EMIRATES

End-to-End Aviation Operational Quality & Reliability Case Study

THEME: TQM & QUALITY ENGINEERING | AUTHOR: JAGRITI SINGH, QUALITY ENGINEER | DATE: 27 SEPTEMBER 2026

IMPORTANT NOTICE: Independently created illustrative Quality Engineering case study, NOT an actual Emirates internal project or proprietary dataset.

1. CASE STUDY TITLE & CONTEXT

Title: End-to-End Aviation Operational Quality & Reliability

Context: Modern aviation networks rely on seamless operational synchronisation. This case study examines interconnected

operational challenges across Flight Operations → Ground Operations → Aircraft Turnaround → Baggage Handling → Passenger

Service. Variation or failures at one stage propagate downstream into operational and customer-quality issues.

2. PROBLEM STATEMENT 3. SIPOC ANALYSIS

Hypothetical Scenario: Airline hub experiences variation SUPPLIERS INPUTS PROCESS OUTPUTS CUSTOMER

across turnaround times, ground handling, baggage Ground Aircraft, Arrival → On-time Passengers

movement, and coordination. Objective: establish root causes, teams, pax, Turnaround → departure, &

implement corrective improvements, and sustain quality maintenance, baggage, Boarding safe stakeholders

IT info turnaround

without fabricated numbers.

4. PROCESS MAPPING & VARIATION POINTS

Aircraft Arrival → Ground Ops → Baggage Handling → Catering / Cabin → Boarding → Departure

Note: Highlighted points represent potential quality variation nodes (timing drift, handoff friction, resource contention) – not actual Emirates failure points.

5. FISHBONE ANALYSIS (ISHIKAWA) 6. 5 WHY ROOT-CAUSE ANALYSIS

Exploratory Areas: 1. Symptom: Flight departure delayed during boarding.

• People: Coordination & communication gaps. 2. Immediate: Transfer baggage reconciliation delayed.

• Process: Sequencing & timing variation. 3. Process: Late bag arrival from inbound flights.

• Equipment: GSE & hydraulic availability. 4. Coordination: Inbound delay not synced with sorting.

• Technology: Real-time tracking latency. 5. Underlying: Lack of automated alert triggers.

• Materials: Supply staging delays. 6. Root Cause: Absence of integrated real-time feedback loop.

• Environment: Weather & gate congestion.

7. PARETO ANALYSIS 8. SYSTEM / PROCESS FMEA

Multiple Contributors → Pareto Analysis → Vital Few Critical RPN = Severity × Occurrence × Detection

Contributors (80/20 Rule). Focuses quality resources on Assesses broad network service risks and specific turnaround

significant systemic contributors instead of random variations. failure modes. (RPN is a risk-prioritisation aid, not sole decision

basis).

Ref. ID: QE-62 | TQM & Quality Engineering Challenge | Jagriti Singh Page 1 of 2

QE-62 | PAGE 2 OF 2

EMIRATES

Risk Prioritisation, Improvement, Control & Quality Framework

THEME: TQM & QUALITY ENGINEERING | AUTHOR: JAGRITI SINGH, QUALITY ENGINEER | DATE: 27 SEPTEMBER 2026

9. COMPLETE ANALYTICAL CHAIN & PRIORITISATION

SIPOC → Process Map → Fishbone → 5 Why → Pareto → FMEA/RPN → Root Cause

10. PDCA IMPROVEMENT 11. PILOT & VALIDATION

• Plan: Define operational problem & objective. Tests change under controlled operational conditions.

• Do: Pilot improvement on controlled basis. Assesses process consistency, reliability, customer impact,

• Check: Compare indicators before/after. and sustainability without fabricating KPIs.

• Act: Standardise successful changes.

12. KPI / SPC MONITORING 13. CONTROL & STANDARDISATION

Monitors turnaround consistency, process adherence, baggage Sustained via SOPs, process controls, defined responsibilities,

quality, and early warnings. Illustrative monitoring dimensions reaction plans, periodic reviews, and continuous improvement.

only.

14. COMPLETE QUALITY ENGINEERING METHODOLOGY FRAMEWORK

METHODOLOGY STAGE PURPOSE & OPERATIONAL APPLICATION

SIPOC & Process Map Define boundaries and visualise end-to-end operational workflow steps.

Fishbone & 5 Why Explore multi-factor causes and drill down to underlying process mechanisms.

Pareto & FMEA Prioritise vital contributors and assess operational failure risks (RPN).

PDCA, Pilot & Validation Implement, test under controlled conditions, and verify improvement effectiveness.

KPI/SPC, Control & Monitor performance, establish robust controls, and drive continuous improvement.

Standardisation

15. POTENTIAL OUTCOMES 16. KEY QUALITY ENGINEERING INSIGHT

• Reduced operational variation & consistent turnarounds. "In aviation operations, quality is an end-to-end system.

• Better coordination & reduced recurring failures. Reliable customer experience depends on controlling

• Stronger preventive quality culture & reliability. variation across interconnected processes."

DISCLAIMER: Illustrative case study independently developed by Jagriti Singh for the 100 Days TQM & Quality Engineering Challenge. Not an official Emirates

project.

Ref. ID: QE-62 | TQM & Quality Engineering Challenge | Jagriti Singh Page 2 of 2

DAY 63

Original source file(s): Day 63 BorgWarner_TQM_Case_Study (1).pdf

Source focus: The Day 63 source material focuses on BORGWARNER TECHNICAL CASE STUDY. The source text explicitly discusses the following quality themes and methods: Reliability, root cause, FMEA, RPN, Isolation, Audit, RCA, isolate, CAPA, isolating, inspection.

Section headings identified in the source: BORGWARNER TECHNICAL CASE STUDY • Jagriti Singh • 1. EXECUTIVE SUMMARY AND CASE CONTEXT • 2. FMEA METHODOLOGY AND RISK PRIORITISATION • 3. ILLUSTRATIVE FAILURE MODE ANALYSIS • 4. LIMITATIONS OF FMEA IN LIFECYCLE ANALYSIS

Technical terms detected in the source: Reliability, root cause, FMEA, RPN, Isolation, Audit, RCA, isolate, CAPA, isolating, inspection

Source extract 1: Day 63 BorgWarner_TQM_Case_Study (1).pdf

DAY 6 3 / 1 0 0 | TQ M & Q UA L I T Y E N G I N E E R I N G C H A L L E N G E

BORGWARNER TECHNICAL CASE STUDY

Jagriti Singh

Automotive Component Reliability, Risk Prioritisation and Failure Prevention Quality Engineer

28 Sept 2026 | Ref: QE-63

1. EXECUTIVE SUMMARY AND CASE CONTEXT

Modern automotive powertrains demand rigorous engineering reliability. In this illustrative engineering evaluation, an automotive sub

assembly exhibits functional instabilities across its operational lifecycle, manifesting as premature infant mortality failures, random mid life

operational breakdowns, and wear out degradation.

The core objective of this Quality Engineering study is to establish a robust analytical framework to identify failure modes, quantify risk

priorities, model time to failure distributions, and execute systematic root cause mitigation.

2. FMEA METHODOLOGY AND RISK PRIORITISATION

Failure Mode and Effects Analysis (FMEA) serves as the foundational engineering tool to proactively evaluate potential vulnerabilities

before operational deployment. The risk matrix evaluates three core dimensions:

• Severity (S): Assessment of the clinical and operational consequence of a failure mode on system performance.

• Occurrence (O): Statistical frequency or probability that a specific failure cause will manifest.

• Detection (D): Capability of current quality controls to intercept the failure mode prior to customer impact.

Risk Priority Number (RPN) = Severity (S) × Occurrence (O) × Detection (D)

RPN dictates engineering resource allocation for risk mitigation. (Note: Numerical RPN thresholds are maintained per internal standard

operating procedures and omitted from this illustrative framework).

3. ILLUSTRATIVE FAILURE MODE ANALYSIS

Failure Mode: Structural or functional degradation under dynamic Potential Investigation Vectors (Non proprietary):

vehicular loading conditions. • Component dimensional and metallurgical variance

Potential Effect: Sub optimal torque transmission and accelerated • Cyclic thermal and mechanical overload conditions

assembly mechanical wear. • Lubrication degradation and boundary friction

4. LIMITATIONS OF FMEA IN LIFECYCLE ANALYSIS

While FMEA successfully establishes static risk hierarchies, it operates independently of temporal dynamics. It does not answer a critical

reliability engineering question: “When failures manifest in the field, what chronological pattern do they follow over operational hours?”

Bridging this operational gap requires transitioning from static risk scoring to stochastic reliability modelling via Weibull Analysis.

FMEA for Static Risk Identification | Weibull Analysis for Dynamic Lifecycle Behaviour

5. WEIBULL ANALYSIS AND LIFE DATA MODELLING

Weibull Analysis utilizes life data distributions governed by the shape parameter (β) to model failure behavior across operating hours:

$\beta < 1$ — Infant Mortality $\beta \approx 1$ — Random Failures $\beta > 1$ — Wear Out Phase

Early life failures driven by manufacturing or Constant failure rate

[Middle content condensed for page fit]

sis Statistical modelling of time-to-failure data and lifecycle regime classification.

Ishikawa (Fishbone) Diagram Structured brainstorming of potential investigation vectors and root causes.

5-Why Analysis Iterative interrogation technique to trace symptoms to root assignable causes.

CAPA Framework Systematic execution of corrective and preventive engineering actions.

14. CONCLUDING ENGINEERING TAKEAWAY

“FMEA establishes a structured baseline for risk prioritisation and failure mode identification, whereas Weibull Analysis decodes temporal failure

behaviour across the operational lifecycle. Integrating both methodologies provides a comprehensive framework for proactive quality engineering

and reliability enhancement.”

Technical case study independently authored and developed by Jagriti Singh as part of the 100 Days TQM & Quality Engineering Challenge. Not an official BorgWarner project or representation of

proprietary BorgWarner enterprise systems, manufacturing data, or internal operational procedures.

Ref. ID: QE-63 | Jagriti Singh | TQM & Quality Engineering Challenge Page 2 of 2

DAY 64

Original source file(s): Day 64 skyroot_quality_engineering__v4.pdf

Source focus: The Day 64 source material focuses on S K Y R O O T A E R O S PAC E 100 DAYS QUALIT Y ENGINEERING CHALLENGE • DAY 64. The source text explicitly discusses the following quality themes and methods: Reliability, traceability, FMEA, Root Cause, Kaizen, PDCA, isolated, SPC, Process Capability, inspection.

Section headings identified in the source: Quality engineers must rigorously control and monitor six core dimensions: • assemblies and electronic components for lot traceability and • lifecycle: • Reliability • FMEA (Failure Mode & Effects Analysis) Pareto Analysis • FTA (Fault Tree Analysis) FRACAS

Technical terms detected in the source: Reliability, traceability, FMEA, Root Cause, Kaizen, PDCA, isolated, SPC, Process Capability, inspection

Source extract 1: Day 64 skyroot_quality_engineering__v4.pdf

S K Y R O O T A E R O S PAC E 100 DAYS QUALIT Y ENGINEERING CHALLENGE • DAY 64

Launch Vehicle Quality, Failure Analysis & Reliability Engineering

An illustrative Quality Engineering case study for aerospace manufacturing and launch-system reliability

Date: 29 September 2026

Notice: “This is an illustrative Quality Engineering case study based on standard aerospace quality practices. It does not represent Skyroot Aerospace's proprietary internal processes or data.”

1 Quality Challenge: Launch-Vehicle Subsystem Scenario

In aerospace manufacturing and propulsion systems, launch vehicles demand absolute precision. Consider an illustrative quality scenario involving a high-pressure propellant valve assembly or avionics separation mechanism within a launch vehicle subsystem.

Quality engineers must rigorously control and monitor six core dimensions:

• Material & Component Variation: Ensuring raw alloy tensile • Test Failures: Analyzing anomalies during cryogenic proof strength, grain structure, and fastener integrity meet strict pressure testing, hot-fire bench tests, and vibration simulation. specifications. • System-Level Reliability Risks: Mitigating cascading failure

• Manufacturing & Process Variation: Controlling CNC modes across propulsion, avionics, and structural separation machining tolerances, welding heat-input, and cleanroom interfaces. assembly parameters. • Launch-Readiness Risks: Ensuring flight hardware meets

• Supplier & Component Quality: Vetting outsourced sub- cumulative pedigree requirements prior to countdown. assemblies and electronic components for lot traceability and

zero-defect compliance.

2 Quality Engineering Approach: Core Toolchain

To systematically address aerospace risks, engineers utilize a structured framework of quality tools deployed across the product lifecycle:

FMEA → Pareto → FTA → FRACAS → Root Cause → Process Var. → Corrective Action → Kaizen/PDCA → Testing → Reliability

FMEA (Failure Mode & Effects Analysis) Pareto Analysis

Identify what can fail, why it can fail, effect of failure, and prioritise Identify the small number of failure categories or contributors risks using Severity, Occurrence and Detection. responsible for a large share of observed problems.

FTA (Fault Tree Analysis) FRACAS

Start with critical top-level failure and work backwards to identify Record actual failure, investigate, assign corrective action, verify combinations of causes using logical relationships (AND/OR). effectiveness and formally close the failure record.

3 How the Tools Work Together

The synergy of quality engineering tools creates a closed-loop reliability feedback system from proactive risk assessment to corrective action:

Potential Risk → FMEA (Proactive Risk Ranking)

Observed Failure Pattern → Pareto Analysis (Focus Areas)

Critical Failure → Fault Tree Analysis (FTA - Logical Root Paths)

Actual Failure Event → FRACAS (Tracking & Investigation)

Root Cause → Corrective Action (Permanent Resolution)

Day 64 | 100 Days Quality

[Middle content condensed for page fit]

tion is mandatory:

• Functional Testing: Verifying baseline operational performance, electrical continuity, and actuation timing.

• Environmental & Condition-Based Testing: Subjecting hardware to thermal vacuum, vibration spectra, and acoustic loading matching

flight profiles.

• Verification Against Requirements: Ensuring hardware performance satisfies all design margin criteria.

• Repeat Testing & Documentation: Conducting multi-cycle endurance testing post-corrective action and compiling complete

qualification data packages.

7 Final Quality Engineering Framework

RISK IDENTIFICATION → FMEA → PARETO → FTA → FRACAS

ROOT CAUSE & PROCESS VARIATION ANALYSIS

CORRECTIVE ACTION → KAIZEN → PDCA → TESTING & VALIDATION →

RELIABILITY MONITORING

KEY TAKEAWAY

“Quality Engineering in aerospace is not only about detecting defects. It is about identifying risks before failure, understanding

why failures occur, controlling process variation, verifying corrective actions, and continuously improving reliability.”

Day 64 | 100 Days Quality Engineering Challenge — Skyroot Aerospace Case Study Page 2 of 2

DAY 65

Original source file(s): day_65_nasa_systems_engineering.pdf

Source focus: Day 65 source document titled NASA MISSION ASSURANCE DAY 6 5 • TEC H N I CA L CAS E STU DY.

NASA MISSION ASSURANCE DAY 6 5 • TEC H N I CA L CAS E STU DY

Systems Engineering, Risk Management & Mission Assurance

How Quality Engineering helps control risk, manage changes, verify systems and improve mission reliability.

Date: 30 September 2026 Author: Jagriti Singh Domain: Aerospace Quality & Systems

Educational Note: This is an educational, illustrative case study based on publicly available systems-engineering and quality principles. It does not

describe NASA's confidential or proprietary internal processes.

1. THE QUALITY CHALLENGE

A space mission is not made of one component. It is a combination of many connected systems such as:

Structure + Power + Communication + Sensors + Software + Thermal Systems + Control Systems

A problem in one subsystem can affect other parts of the mission. Therefore, Quality Engineering must look at the complete system, not only

at individual components.

Component → Subsystem → Spacecraft → Mission

Systems Engineering: Means looking at the complete system and understanding how its different parts work together to achieve the

mission objective.

2. RISK IDENTIFICATION

Before a mission moves forward, engineers need to ask: “What can go wrong?”

POSSIBLE RISKS INCLUDE RISK ANALYSIS & APPROACH

• Component failure or sensor error Risk Analysis: Identifying possible problems early,

• Software error / bus interruption understanding impact, and deciding priority.

• Power interruption / Communication loss Identify → Analyse → Control → Verify

• Manufacturing variation

• Incorrect design or specification

• Uncontrolled engineering changes

3. FAULT TREE ANALYSIS — FTA

What is FTA? A method used to start with a major failure and work backwards to find its possible causes. Instead of asking “What

component failed?” we ask “What different conditions could have caused this major failure?”

TOP EVENT: Spacecraft loses required function

• Power System Problem (OR) | • Communication System Problem (OR) | • Control System Problem

AND vs OR Logic: OR: Any one condition causes failure. | AND: Two or more conditions occur together.

Why FTA is useful: Helps engineers understand how major failures occur, identify failure paths, and focus on critical causes.

4. CONFIGURATION MANAGEMENT

Making sure that the correct and approved version of a design, drawing, software, specification or process is being used.

Design v1 → Change → Version 2 → Approval → Testing → Release

Why important? Prevents confusion between design, manufacturing, and testing teams through traceability.

DAY 65 — Jagriti Singh | NASA Mission Assurance Page 1 of 2

5. VERIFICATION VS VALIDATION

These two words sound similar, but they answer different questions in aerospace quality engineering.

VERIFICATION VALIDATION

Question: “Did we build the product according to specified Question: “Does the product actually work for its intended

requirements?” purpose?”

Checks requirements via inspection, analysis, calculation, Looks at system suitability for its intended operating

test

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. Problems identified must be: Recorded → Investigated → Corrected →

Retested.

7. ANOMALY ANALYSIS

What is an anomaly? Something unexpected happens during testing or operation (behavior differs from prediction).

Quality Approach: Anomaly Detected → Record → Collect Evidence → Root Cause → Corrective Action → Retest → Effectiveness

Check.

Important principle: Fixing the symptom is not enough; understand why it happened to prevent recurrence.

8. FINAL QUALITY ENGINEERING FRAMEWORK

NASA Mission Quality Approach:

System Requirements → Risk ID → Systems Eng → FTA → Config Control → Verification → Validation → Anomaly Analysis → Corrective Action →

Retesting → Effectiveness → Mission Assurance

KEY TAKEAWAY

Quality Engineering in space is about understanding the complete system, identifying risks early, controlling changes, testing carefully,

investigating anomalies, and ensuring mission readiness.

“Build it correctly, control every important change, test it with evidence, learn from failures, and continuously improve reliability.”

DAY 65 — Jagriti Singh | NASA Mission Assurance Page 2 of 2

DAY 66

Original source file(s): Day 66_ISRO_Quality_Management_Case_Study_2Pages.pdf

Source focus: The Day 66 source material focuses on Challenges in Quality Management. The source text explicitly discusses the following quality themes and methods: Reliability, isolation, supplier quality, inspection, FMEA, RCA, CAPA, audit, isolated, Root Cause, isolate, SPC.

Section headings identified in the source: Challenges in Quality Management • Understanding the Quality Challenge • Quality Engineering Approach • 1. Requirements → 2. Risk Identification • 3. Failure Analysis → 4. Reliability Assessment • 5. Verification & Qualification → 6. Quality Assurance

Technical terms detected in the source: Reliability, isolation, supplier quality, inspection, FMEA, RCA, CAPA, audit, isolated, Root Cause, isolate, SPC

Source extract 1: Day 66_ISRO_Quality_Management_Case_Study_2Pages.pdf

ISRO

Challenges in Quality Management

1 ST OCTO B E R • DAY 6 6 • 1 0 0 DAYS QUA L IT Y MA NAG E M E NT CHA L L E NG E

Disclaimer: This is an independent educational case study created as part of my “100 Days Quality Management Challenge.” ISRO is a separate

organisation, and this case study is not an official ISRO project, report, collaboration, or representation of ISRO. The discussion is based on publicly

available information and general Quality Management / Quality Engineering principles. No confidential, proprietary, or internal ISRO information is being

used.

Main Topic: Challenges in Quality Management — ISRO

“How Quality Engineering and Reliability Principles Help Manage Quality Risks in Space Missions”

Space missions represent one of the most unforgiving engineering environments. Unlike terrestrial products, space systems operate under

extreme physical stresses and feature zero margin for human intervention once launched. This case study examines rigorous quality

management frameworks and reliability methodologies.

Understanding the Quality Challenge

Quality management in space engineering is uniquely demanding due to compounding complexity and operational isolation. A single minor

component failure can cascade into total mission loss.

• Extremely high reliability requirements: Systems must • Component-to-system risk escalation: Single-point failures

function flawlessly for years in orbit without maintenance. require strict redundancy and fault tolerance.

• Complex hardware/software interactions: Integrated avionics • Configuration & change control: Every engineering change

and guidance algorithms demand synchronization. across supply chains must be traceable.

• Launch & space stresses: Extreme acoustic vibrations, high G- • Strict verification & qualification: Extensive ground testing

forces, thermal vacuum cycles, and radiation. must simulate lifetime environments.

• Zero repair capability: Post-launch physical intervention is • External supplier quality control: Multi-tier subcontracting

impossible; faults must be anticipated upfront. requires rigorous incoming inspection and audits.

Quality Engineering Approach

To systematically manage these risks, quality engineering applies an end-to-end lifecycle framework:

1. Requirements → 2. Risk Identification

What? Establish baseline specs and identify failure modes via FMEA/FTA. Why? Maps critical vulnerabilities prior to manufacturing.

3. Failure Analysis → 4. Reliability Assessment

What? Evaluate degradation, stress limits, and probabilistic reliability margins. Why? Quantifies survivability under launch/orbital stresses.

5. Verification & Qualification → 6. Quality Assurance

What? Conduct environmental testing (vibration, thermal vacuum) and audits. Why? Validates hardware against worst-case environments.

7. Anomaly Analysis → 8. Corrective Action → 9. Retesting

What? Investigate deviations via RCA, apply CAPA, and retest. Why? Confirms anomalies are fully resolved prior to flight.

10. M

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APA

How This Relates to ISRO's Quality Framework

Based on publicly available and verifiable information, ISRO's quality assurance framework emphasizes rigorous independent oversight. A

prominent mechanism highlighted in public documentation is the Integrated Product Assurance Board (IPAB), an apex-level review

mechanism overseeing end-to-end quality, reliability, and safety across mission elements, including engineering analyses, non-

conformances, environmental tests, and lessons learned.

Key Quality Management Takeaways

1. Quality must be built into the system from the beginning, not inspected only at the end.

2. Reliability and failure analysis help identify risks before they become mission-level problems.

3. Testing becomes valuable when its results are converted into evidence-based quality decisions.

4. Corrective action is incomplete until its effectiveness is verified.

5. Lessons learned help prevent recurrence across future missions.

100 Days Quality Management Challenge | Day 66 | 1st October Independent educational case study. Not an official ISRO project or representation.

DAY 67

Original source file(s): Day 67 SpaceX_Quality_Management_Case_Study__v2.pdf

Source focus: The Day 67 source material focuses on Failure, Learning & Continuous Quality Improvement 2nd October | Day 67. The source text explicitly discusses the following quality themes and methods: reliability, FMEA, Root Cause.

Section headings identified in the source: SPACEX • THE QUALITY CHALLENGE QUALITY ENGINEERING APPROACH • Recurrence • Core QE Tools & Contributions: • FMEA Systematically evaluates • FTA Deductive fault tree analysis to

Technical terms detected in the source: reliability, FMEA, Root Cause

Source extract 1: Day 67 SpaceX_Quality_Management_Case_Study__v2.pdf

SPACEX

Failure, Learning & Continuous Quality Improvement 2nd October | Day 67

Disclaimer: This is an independent educational case study created as part of my “100 Days Quality Management Challenge.” SpaceX is a separate

organisation. This is not an official SpaceX project, report, collaboration, or representation. The analysis uses publicly available information and general

Quality Engineering principles. No confidential or proprietary SpaceX information is used.

THE QUALITY CHALLENGE QUALITY ENGINEERING APPROACH

In reusable launch systems, a failure or unexpected deviation during INVESTIGATION & RESOLUTION

testing is not simply a pass/fail event. It can provide important engineering WORKFLOW

evidence about design, hardware, software, process and reliability. Detect • Collect Evidence • Analyse • Identify Root

The Quality Engineering objective is to identify the failure mechanism, Cause • Correct • Re-test • Verify • Prevent

Recurrence

correct the underlying cause, verify the corrective action and prevent

recurrence.

Core QE Tools & Contributions:

FMEA Systematically evaluates

ILLUSTRATIVE QUALITY SCENARIO — NOT AN ACTUAL potential failure modes and

SPACEX INCIDENT severity.

FTA Deductive fault tree analysis to

During repeated testing of a reusable launch-system component, trace failure pathways.

performance gradually changes across test cycles and an unexpected

Root Cause Uncovers fundamental

failure or deviation occurs.

Analysis underlying systemic triggers.

Core Investigative Questions: Reliability Assesses lifecycle degradation

Analysis and failure probabilities.

What happened? • Why did it happen? • What is the risk? •

What needs to change? • Did the corrective action work? Verification & Confirms requirements and

Validation corrective efficacy.

Corrective Implements permanent

PROPOSED QUALITY SOLUTION Action engineering fixes.

Configuration Tracks hardware and software

1. Detect & Document: Capture the failure, test conditions and objective evidence.

Control baseline modifications.

2. Verify the Data: Confirm the measurement and test setup are reliable.

Lessons Institutionalizes knowledge

3. Analyse the Failure: Use FMEA, FTA, reliability data and root-cause analysis Learned across programs.

as appropriate.

4. Correct the Cause: Improve the design, process, software/hardware interaction

or test method.

WHY THIS APPROACH MATTERS FOR

5. Re-test & Verify: Repeat relevant verification and qualification activities.

REUSABLE LAUNCH SYSTEMS

6. Prevent Recurrence: Update risk assessments, configuration records and

lessons learned. Publicly available SpaceX information describes an

iterative development approach involving testing,

learning from test outcomes and making engineering

improvements.

Iterative Feedback Loop:

Test • Failure/Deviation • Data • Root

Cause • Design/Process Improvement • Re-

test • Reliability Improvement

Note: This does not claim an internal SpaceX QA procedure, nor

does it use proprietary statistics.

EXPECTED QUALITY IMPACT

■ Earlier identification of failure risks

■ Better understanding of failure mechanisms

■ Verified corrective actions

■ Reduced recurrence risk and improved reliability

“A failure during testing is not just a setback — when analysed correctly, it becomes engineering data for building a more

reliable system.”

100 Days Quality Management Challenge | Day 67 | 2nd October

Independent educational case study — Not an official SpaceX project, publication, collaboration, or representation.

DAY 68

Original source file(s): day_68_case_study (1).pdf

Source focus: The Day 68 source material focuses on Supplier Quality Management & Prevention-Based Quality 3RD OCTOBER • DAY 68. The source text explicitly discusses the following quality themes and methods: Supplier Quality, RCA, inspection, process capability, Root Cause, SPC, Audit, FMEA, reliability.

Section headings identified in the source: THE QUALITY MANAGEMENT THEORY PROPOSED QUALITY SOLUTION • 4. Improve: Implement corrective/preventive actions • THE INDUSTRIAL QUALITY PROBLEM • 5. Verify: Use data to confirm capability and process stability • Semiconductor manufacturing equipment depends on • extreme precision components and a complex global

Technical terms detected in the source: Supplier Quality, RCA, inspection, process capability, Root Cause, SPC, Audit, FMEA, reliability

Source extract 1: day_68_case_study (1).pdf

ASML

Supplier Quality Management & Prevention-Based Quality 3RD OCTOBER • DAY 68

Disclaimer: This is an independent educational case study created as part of my “100 Days Quality Management Challenge.” ASML is a separate

organisation. This is not an official ASML project, report, collaboration, or representation. The case study uses publicly available information and general

Quality Engineering principles. No confidential or proprietary ASML information is used.

THE QUALITY MANAGEMENT THEORY PROPOSED QUALITY SOLUTION

Supplier Quality Management + Prevention-Based Supplier Process → Detect Variation → Capability Assessment →

Quality RCA → Corrective Action → Verify → Continuous Monitoring

Quality should not depend only on detecting defective 1. Detect: Monitor supplier process data instead of relying only on

components after they arrive. Supplier processes incoming inspection.

should be monitored and improved proactively so that 2. Analyse: Investigate variation, stability, and process capability.

variation and potential nonconformities are controlled 3. Identify Root Cause: Determine if issues stem from

before they affect the final system. parameters, equipment, material, method, or measurement.

4. Improve: Implement corrective/preventive actions

collaboratively with the supplier.

THE INDUSTRIAL QUALITY PROBLEM

5. Verify: Use data to confirm capability and process stability

Illustrative Quality Scenario — Not an Actual ASML Incident improvements.

A critical component supplied for a high-precision 6. Prevent Recurrence: Continue monitoring and update supplier

semiconductor manufacturing system meets the risk assessments.

current specification during incoming inspection.

However, supplier process data shows: WHY THIS MATTERS IN HIGH-PRECISION

• Increasing process variation EQUIPMENT

• Fluctuating process behaviour

Semiconductor manufacturing equipment depends on

• Borderline process capability

extreme precision components and a complex global

• Recurring nonconformities over time

supplier ecosystem. Unchecked variation introduces severe

The immediate problem can be contained, but simply risk:

replacing defective components does not eliminate the Supplier Variation → Component Risk → System Performance

underlying risk. Risk

Core Question: How can we detect and control Proactive supplier quality management identifies quality

supplier-process risk before it becomes a risks early rather than depending solely on final inspection.

recurring product-quality problem? (Note: Based solely on publicly available information; does

not reflect internal ASML procedures.)

QUALITY TOOLS USED

EXPECTED QUALITY IMPACT

SPC → Process Capability → Trend Analysis → RCA →

Supplier Audit → Corrective Action → Verification → FMEA • &d5; Supplier-related defects

SPC: Detect process variation & instability. • &d5; Process variation

Capability Analysis: Determine if process meets requirements • &d5; Recurring nonconformities

consistently.

• &u5; Component reliability & consistency

RCA: Identify why variation/nonconformity occurs.

Corrective Action: Address underlying cause. Better supplier performance visibility + earlier risk detection +

Effectiveness Verification: Confirm improvement sustainability. stronger prevention.

“A supplier part can meet today's specification and still carry tomorrow's quality risk if process variation is not understood and controlled.”

100 Days Quality Management Challenge | Day 68 | 3rd Independent educational case study — Not an official ASML project, publication or

October representation.

DAY 69

Original source file(s): Day_69_Editorial_Theme (1) (1).pdf

Source focus: Day 69 source document titled L’ O R É A L.

L’ O R É A L

P R O D U C T C O N S I S T E N C Y & Q U A L I T Y AT S C A L E

4 T H O C TO B E R | DAY 6 9

DISCLAIMER: This is an independent educational case study created as part of my “100 Days Quality Management Challenge.” L’Oréal is a

separate organisation. This is not an official L’Oréal project, report, collaboration, or representation. The analysis applies general Quality

Engineering principles using publicly available information.

Q UA L IT Y C HA L L E N G E

In large-scale cosmetic production, maintaining consistent viscosity, texture, appearance and performance across

batches is an important quality challenge.

A batch may remain within specification while showing a gradual increase in variation or process drift. If this trend is

not detected early, it can eventually affect product consistency and increase quality risk.

Quality Engineering Question:

How can process variation be detected and controlled before it becomes a consumer-facing quality issue?

Q UA L IT Y E N G I N E E R I N G TO O LS

MSA — Verify measurement reliability SPC — Monitor variation & trends Cp/Cpk — Evaluate capability &

centering

Trend Analysis — Detect gradual drift RCA — Identify underlying causes FMEA — Assess & prioritize risks

PRO P O S E D Q UA L I T Y A PPROAC H

MEASURE → MONITOR → DETECT → ANALYSE → CORRECT → VERIF Y → PREVENT

• 1. Measurement Reliability: Validate the measurement system before making process decisions.

• 2. Process Monitoring: Use SPC and trend analysis to identify unusual variation or gradual drift.

• 3. Capability Assessment: Use Cp/Cpk to evaluate process spread and centering against specifications.

• 4. Root Cause & Corrective Action: Investigate significant variation and address the underlying process cause.

• 5. Verification & Prevention: Verify corrective-action effectiveness and continue monitoring to prevent recurrence.

E X P E C T E D Q U A L I T Y I M PA C T

• Improved batch-to-batch consistency • Fewer recurring quality issues

• Earlier detection of process drift • Better process predictability

• Reduced process variation • Stronger prevention-based quality

« “Quality at scale is not achieved by inspecting more. It is achieved by controlling variation before it becomes

a defect.” »

I N D U S T R Y: B E A U T Y & C O S M E T I C S • Q U A L I T Y F O C U S : P R O C E S S Q U A L I T Y • P R O D U C T

C O N S I S T E N CY • VA R I AT I O N C O N T R O L

DAY 70

Original source file(s): Day 70 disney_quality_case_study (1).pdf

Source focus: The Day 70 source material focuses on T H E WA LT D I S N E Y C O M PA N Y. The source text explicitly discusses the following quality themes and methods: SPC, customer satisfaction, service quality, RCA, FMEA, PDCA.

Section headings identified in the source: T H E WA LT D I S N E Y C O M PA N Y • E N TE RTA I N M E N T Q UA L IT Y & E X PE R I E N C E CO N S I STE N CY • QUALITY CHALLENGE QUALITY ENGINEERING TOOLS • PROPOSED QUALITY APPROACH • S TA N DA R D I S E • Capture: Collect customer feedback and identify Improve: Implement targeted corrective

Technical terms detected in the source: SPC, customer satisfaction, service quality, RCA, FMEA, PDCA

Source extract 1: Day 70 disney_quality_case_study (1).pdf

T H E WA LT D I S N E Y C O M PA N Y

E N TE RTA I N M E N T Q UA L IT Y & E X PE R I E N C E CO N S I STE N CY

5th October | Quality Engineer — Jagriti Singh | Day 70

QUALITY CHALLENGE QUALITY ENGINEERING TOOLS

Entertainment businesses operate through multiple VOC — Customer Expectations

interconnected touchpoints — content, digital platforms, guest

experiences, service delivery, and customer interactions. Process Mapping — Touchpoints

When variation occurs across these touchpoints, it can affect

SPC & Trend Analysis — Variation

experience consistency, customer satisfaction, and overall

service quality. Pareto Analysis — Prioritisation

Quality Engineering Question: RCA — Root Causes

How can experience variation be identified, analysed, and

controlled to deliver a more consistent customer FMEA — Risk Assessment

experience?

PDCA — Continuous Improvement

PROPOSED QUALITY APPROACH

C A P T U R E → M E A S U R E → A N A LY S E → P R I O R I T I S E → I M P R OV E → V E R I F Y →

S TA N DA R D I S E

Capture: Collect customer feedback and identify Improve: Implement targeted corrective

1 5

critical experience factors. improvements.

Measure: Define measurable indicators for key Verify: Evaluate whether improvements are

2 6

quality characteristics. effective and sustained.

Analyse: Monitor trends and identify unusual Standardise:

3

experience variation. 7 Incorporate successful improvements into daily

Prioritise: Use Pareto and RCA to focus on major standard process.

4

contributors.

EXPECTED QUALITY IMPACT

✓ More consistent customer experience ✓ Better process predictability

✓ Earlier detection of process variation ✓ Improved service quality

✓ Reduced recurring issues ✓ Stronger continuous improvement

Q UA L I T Y TA K E AWAY

“Quality in entertainment is not only about delivering a great experience once — it is about consistently

delivering the intended experience across every important touchpoint.”

DISCLAIMER: This is an independent educational case study created as part of my “100 Days Quality Management Challenge.” The Walt Disney Company is

a separate organisation. This is not an official Disney project, report, collaboration, or representation. The analysis applies general Quality Engineering

principles using publicly available information.

Industry: Entertainment Providers | Quality Focus: Experience Quality • Service Consistency • Process Jagriti Singh |

Improvement • Customer Feedback Day 70

DAY 71

Original source file(s): Day71 (1).pdf

Source focus: The Day 71 source material focuses on 6th October | Quality Engineer | Jagriti Singh | Day 71. The source text explicitly discusses the following quality themes and methods: SERVICE QUALITY, SPC, RCA, FMEA, PDCA, reliability, Root Cause.

Section headings identified in the source: MARR IOT T INTER NATIONAL • SERVICE QUALITY & CUSTOMER EXPERIENCE • QUALITY CHALLENGE • QUALITY ENGINEERING TOOLS • VOC Process SPC & Trend Pareto Analysis RCA FMEA PDCA • expectations & Identify critical variation service issues root causes service failures process

Technical terms detected in the source: SERVICE QUALITY, SPC, RCA, FMEA, PDCA, reliability, Root Cause

Source extract 1: Day71 (1).pdf

6th October | Quality Engineer | Jagriti Singh | Day 71

MARR IOT T INTER NATIONAL

SERVICE QUALITY & CUSTOMER EXPERIENCE

Disclaimer: This is an independent educational case study created as part of my “100 Days Quality Management Challenge.” Marriott International is a separate

organisation. This is not an official Marriott International project, report, collaboration, or representation. The analysis applies general Quality Engineering

principles using publicly available information.

QUALITY CHALLENGE

Hospitality involves multiple interconnected service processes: reservations, check-in, room preparation, housekeeping, guest

requests, and service recovery. Even when a service process meets its defined standard, variation in service time, process

execution, or customer interactions can create inconsistent guest experiences.

Quality Engineering Question: How can service variation be identified, analysed, and controlled to deliver a more

consistent guest experience?

QUALITY ENGINEERING TOOLS

VOC Process SPC & Trend Pareto Analysis RCA FMEA PDCA

Understand guest Mapping Monitor service time Prioritise recurring Identify underlying Anticipate potential Drive continuous

expectations & Identify critical variation service issues root causes service failures process

feedback service touchpoints improvement

PROPOSED QUALITY APPROACH

1. CAPTURE 2. MEASURE 3. ANALYSE 4. PRIORITISE 5. IMPROVE 6. VERIFY 7. STANDARDISE

Guest feedback & Service indicators & Trends & variation Pareto Analysis & RCA Targeted corrective Check variation Sustain & integrate into

service characteristics issue frequency across shifts for major factors actions for root causes reduction & service processes

effectiveness

EXPECTED QUALITY IMPACT

QUALITY TAKEAWAY

✓ More consistent guest experience “Service quality is not only about meeting a standard. It is about

✓ Earlier detection of service variation

consistently delivering the intended experience across every

critical customer touchpoint.”

✓ Reduced recurring complaints

✓ Better process predictability

✓ Improved service reliability

✓ Stronger continuous improvement culture

APPLIED QUALITY & TQM TOOLS SUMMARY

• Voice of Customer (VOC): Captures guest feedback and • Root Cause Analysis (RCA / 5 Whys): Uncovers underlying

expectations. operational issues.

• Process Mapping: Visualises critical touchpoints across service • Failure Mode and Effects Analysis (FMEA): Mitigates potential

workflows. process failures.

• Statistical Process Control (SPC) & Trend Analysis: Tracks service • PDCA Cycle (Plan-Do-Check-Act): Ensures continuous, structured

time variation. process improvement.

• Pareto Analysis (80/20 Rule): Identifies vital few causes behind • Standardisation (SOPs): Locks in operational gains and sustains

complaints. quality standards.

Industry: Hospitality | Quality Focus: Service Quality, Customer Experience, 100 Days Quality Management Challenge | Day 71

Process Variation, Continuous Improvement

DAY 72

Original source file(s): Day 72 de_beers_quality_case_study_v2 (1).pdf

Source focus: The Day 72 source material focuses on Industry: Retail Luxury Goods & Jewelry 100 Days Quality Management Challenge | Day 72. The source text explicitly discusses the following quality themes and methods: Traceability, RCA, FMEA, SPC, CAPA, Isolates.

Section headings identified in the source: QUALITY CHALLENGE EXPECTED QUALITY IMPACT • Variation between processing stages or recurring quality issues can • QUALITY ENGINEERING QUESTION • QUALITY TAKEAWAY • preventing quality risks throughout the • identification & quality data variation RCA & FMEA

Technical terms detected in the source: Traceability, RCA, FMEA, SPC, CAPA, Isolates

Source extract 1: Day 72 de_beers_quality_case_study_v2 (1).pdf

Industry: Retail Luxury Goods & Jewelry 100 Days Quality Management Challenge | Day 72

Quality Focus: Process Quality • Traceability • Variation Control • Risk Management Quality Engineer — Jagriti Singh

DE BEERS GROUP 7th October | Day 72

D IA M O N D Q UA L IT Y, TR AC E AB I L IT Y & PROC ES S CO N TRO L

Disclaimer: This is an independent educational case study created as part of my “100 Days Quality Management Challenge.” De Beers Group is a

separate organisation. This is not an official De Beers Group project, report, collaboration, or representation. The analysis applies general Quality

Engineering principles using publicly available information.

QUALITY CHALLENGE EXPECTED QUALITY IMPACT

Diamond processing involves multiple stages where consistency, ✓ Better process consistency

identification, grading and process control are important quality

✓ Earlier detection of quality variation

considerations.

✓ Stronger traceability

Variation between processing stages or recurring quality issues can

✓ Reduced recurring quality issues

affect downstream quality, process efficiency and traceability.

✓ Better process predictability

QUALITY ENGINEERING QUESTION

✓ Improved quality-risk management

How can process variation, quality risks and traceability-

related issues be identified and controlled throughout the ✓ Stronger prevention-based quality

diamond value chain?

QUALITY TAKEAWAY

PROPOSED QUALITY APPROACH “Quality is not only about inspecting the

final product. It is about controlling

TRACE → MEASURE → MONITOR → ANALYSE variation, maintaining traceability and

preventing quality risks throughout the

Maintain Collect critical Detect abnormal Apply Pareto,

identification & quality data variation RCA & FMEA

process.”

process history

CORRECT → VERIFY → CONTROL

Address underlying Confirm corrective Standardise & monitor

causes effectiveness continuously

CORE QUALITY TOOLS

FMEA Process Mapping

Identify and prioritise potential failure Understand critical processing stages

modes

SPC & Trend Analysis Traceability Controls

Monitor process stability and variation Maintain product/process history

RCA Pareto Analysis

Identify underlying causes Prioritise significant quality problems

CAPA

Implement and verify corrective/preventive actions

SUMMARY OF APPLIED QUALITY TOOLKIT

1. RISK & FLOW 2. MONITORING 3. PROBLEM SOLVING 4. ACTION & CONTROL

FMEA & Process Mapping: SPC & Traceability: Utilises control RCA & Pareto Analysis: Isolates CAPA System: Drives standardized

Identifies high-risk processing charts to detect process drift early the vital 20% of defect causes (e.g. corrective actions to eliminate

stages and establishes standard while locking digital assets to cutting variance) using 5-Why and repeat defects and sustain

input-output criteria for grading physical batch serials. Fishbone frameworks. continuous process control.

consistency.

DAY 73

Original source file(s): Day 73 Schneider_Electric_Quality_Case_Study_Day73 (1).pdf

Source focus: The Day 73 source material focuses on SCHNEIDER ELECTRIC. The source text explicitly discusses the following quality themes and methods: RELIABILITY, SPC, Cp, Cpk, supplier quality, FMEA, RCA, CAPA, inspection, process capability, Root Cause.

Section headings identified in the source: SCHNEIDER ELECTRIC • INDUSTRIAL QUALITY & RELIABILITY ENGINEERING • QUALITY TARGET QUALITY ENGINEERING TOOLS • QUALITY PROBLEM FMEA FTA / Failure Analysis • Identify and prioritize potential process/ Investigate recurring or intermittent • Reliability Testing & Verification

Technical terms detected in the source: RELIABILITY, SPC, Cp, Cpk, supplier quality, FMEA, RCA, CAPA, inspection, process capability, Root Cause

Source extract 1: Day 73 Schneider_Electric_Quality_Case_Study_Day73 (1).pdf

SCHNEIDER ELECTRIC

INDUSTRIAL QUALITY & RELIABILITY ENGINEERING

8th October | Quality Engineer — Jagriti Singh | Day 73 (100 Days Quality Management Challenge)

DISCLAIMER: This is an independent educational case study created as part of my “100 Days Quality Management Challenge.” Schneider Electric is a separate organisation. This is not an

official Schneider Electric project, report, collaboration, or representation. The analysis applies general Quality Engineering principles using publicly available information and does not use

confidential company information.

QUALITY TARGET QUALITY ENGINEERING TOOLS

Critical electrical and industrial products, with focus on manufacturing SPC & Trend Analysis Capability — Cp/Cpk

process consistency, product reliability, supplier quality and prevention- Monitor process stability, mean shifts Evaluate process spread and centering

based quality. and variation. vs spec limits.

QUALITY PROBLEM FMEA FTA / Failure Analysis

Identify and prioritize potential process/ Investigate recurring or intermittent

product risks. failure mechanisms.

• Process data may remain within specification while showing

gradual mean shift or increasing variation. RCA CAPA / Corrective Action

• Critical supplier components may meet final inspection Identify underlying causes rather than Implement and control robust corrective

treating symptoms. actions.

requirements but show intermittent field failures.

• Borderline process capability can increase future defect and

Reliability Testing & Verification

reliability risk. Confirm that improvements and design changes perform reliably in operational

• Corrective actions must be verified for sustained effectiveness, environments.

not only short-term improvement.

PROPOSED QUALITY APPROACH

• Design changes require assessment of both manufacturing

capability and reliability risk.

Monitor → Detect → Analyse → Risk → Correct → Verify → Sustain

PROCESS & RELIABILITY DIAGNOSTICS 1. Monitor: Use SPC & trend data to detect process shifts/variation.

2. Detect: Identify abnormal trends, failures or capability concerns.

3. Analyse: Evaluate process capability, failure patterns and risk.

UCL

4. Identify Risk: Apply FMEA/FTA/RCA to determine root causes.

CL 5. Correct: Implement targeted corrective & preventive actions.

6. Verify: Re-test process capability, reliability and effectiveness.

LCL 7. Sustain: Continue monitoring to ensure long-term stability.

EXPECTED IMPACT

• Improved process stability • Reduced process variation

• Earlier detection of risks • Improved supplier reliability

• Reduced recurring failures • Better CAPA effectiveness

• Prevention-based quality • Predictable manufacturing

“Quality is not only about meeting specifications. It is about controlling variation, managing risk and verifying that improvements remain

effective over time.”

My approach to the case:

Data → Variation → Risk → Root Cause → Corrective Action → Verification → Prevention

This quality solution is developed based on my own knowledge and understanding of Quality Engineering principles. I have used the tools, concepts and approaches that I have learned

and practiced to analyse the problem and propose this solution.

Industry: Energy Management • Industrial Automation • Electrification

Quality Focus: Process Quality • Reliability • Supplier Quality • Risk Management • Continuous Improvement

Quality Engineer — Jagriti Singh | Day 73

DAY 74

Original source file(s): Day 74 Quality_Engineering_Across_Industries_ (1).pdf

Source focus: The Day 74 source material focuses on QUALITY ENGINEERING ACROSS INDUSTRIES. The source text explicitly discusses the following quality themes and methods: Taguchi, SPC, process capability, reliability, Cp, Cpk, FMEA, yield, CAPA, PFMEA, 8D, DFMEA.

Section headings identified in the source: QUALITY ENGINEERING ACROSS INDUSTRIES • DISCLAIMER • EXECUTIVE SUMMARY • Core Objectives & Insights: • INDUSTRY-WISE QUALITY ENGINEERING APPLICATIONS • Industry & Known Examples Typical Quality Challenge Relevant QE Tools Intended Quality Outcome

Technical terms detected in the source: Taguchi, SPC, process capability, reliability, Cp, Cpk, FMEA, yield, CAPA, PFMEA, 8D, DFMEA, Traceability, Inspection

Source extract 1: Day 74 Quality_Engineering_Across_Industries_ (1).pdf

100 DAYS QUALITY MANAGEMENT CHALLENGE

QUALITY ENGINEERING ACROSS INDUSTRIES

Key Learnings, Tools, Theories & Practical Applications

9 October 2026 | Day 74 Report Prepared by: Quality Engineer | Jagriti Singh

"Over the course of this challenge, Quality Management and Quality Engineering principles have been explored across multiple industries. This report consolidates the

key engineering concepts, tools, methodologies and practical approaches used to understand quality challenges, analyse risks, control variation and propose

sustainable improvements."

DISCLAIMER

This report is an independent educational summary prepared as part of the "100 Days Quality Management Challenge." It consolidates general Quality Engineering principles, industry-

related learning and independently developed educational analyses. All organisations mentioned are separate entities. This report is not an official publication, project, report, collaboration

or representation of any organisation. The proposed applications illustrate general Quality Engineering approaches and should not be interpreted as verified internal practices or actual

failures of the organisations discussed.

EXECUTIVE SUMMARY

The 100 Days Quality Management Challenge was initiated to rigorously investigate the cross-industry application of Quality Management and Quality Engineering

principles. Over 74 consecutive days, this educational initiative has bridged theoretical frameworks with realistic engineering scenarios.

Core Objectives & Insights:

• Industry Exploration: Investigated how quality challenges differ fundamentally across electronics, automotive, aerospace, consumer goods, digital systems, and

service operations.

• Tool Selection Framework: Developed systematic criteria for selecting appropriate analytical tools based on process physics, data availability, and operational risk.

• Technical Synthesis: Connected management theories (Deming, Juran, Taguchi) with quantitative engineering methods (SPC, MSA, Weibull Analysis, DOE).

• Risk & Variation Mastery: Focused on characterising process capability, identifying root causes, mitigating system risk, and ensuring process reliability over time.

• Independent Problem-Solving: Formulated a structured, repeatable methodology to diagnose complex quality problems and propose sustainable engineering

remedies.

"The objective of Quality Engineering is not merely to detect defects, but to understand variation, manage risk, improve processes and prevent

recurrence."

INDUSTRY-WISE QUALITY ENGINEERING APPLICATIONS

Note: Known examples explored during the challenge. Educational synthesis, not verified internal procedures.

Industry & Known Examples Typical Quality Challenge Relevant QE Tools Intended Quality Outcome

A. Electronics & Semiconductor Sub-micron process variation, SPC, Cp/Cpk, MSA, FMEA, Failure Eliminate yield loss, guarantee manufacturing

Intel, ASML, Siemens, Schneider Electric component drift, multi-tier supplier Analysis, CAPA consistency, ensur

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INDIAN PSU & GOVERNMENT QUALITY ENGINEERING

The final quarter of the challenge transitions from global commercial industries to evaluating mission-critical public sector undertakings (PSUs), government authorities, and

national infrastructure projects in India.

Key Sectors & Domains to be Explored: Heavy Engineering (BHEL), Aerospace & Defence (HAL, BEL, DRDO), Railways (Indian Railways, RDSO), Energy & Power (NTPC,

PowerGrid), Oil & Gas (ONGC, IOCL), Steel & Metallurgy (SAIL), Shipbuilding (Mazagon Dock), Infrastructure (NHAI), and Accreditation/Standards (BIS, NABL).

Methodology: Every upcoming case study will apply the same rigorous independent framework: Organisation Context → Quality Problem → Risk → Tool Selection → Analysis →

Proposed Solution → Expected Impact.

"The focus now shifts from exploring quality across industries to applying Quality Engineering thinking to India's critical public-sector and government systems."

DAY 74 COMPLETE | NEXT: DAY 75 | 100 | INDIAN PSU & GOVERNMENT QUALITY ENGINEERING

100 Days Quality Management Challenge | Day 74 | Jagriti Singh Page 4 of 4

DAY 75

Original source file(s): Day 75 BHEL_Integrated_Heavy_Engineering_Quality_Case_Study_Final (1).pdf

Source focus: The Day 75 source material focuses on BHARAT HEAVY ELECTRICALS LIMITED. The source text explicitly discusses the following quality themes and methods: Risk-Based, reliability, NDT, traceability, inspection, SPC, isolate, Audit, root cause, Process Capability, Cp, Cpk.

Section headings identified in the source: BHARAT HEAVY ELECTRICALS LIMITED • Integrated Heavy Engineering Quality • Heavy Engineering and the Importance of Manufacturing Quality • Hypothetical Manufacturing Scenario • Quality Feature Potential Risk Possible Causes Verification Method Quality Control • From Observed Variation to Verified Cause

Technical terms detected in the source: Risk-Based, reliability, NDT, traceability, inspection, SPC, isolate, Audit, root cause, Process Capability, Cp, Cpk, FMEA, Control Plan

Source extract 1: Day 75 BHEL_Integrated_Heavy_Engineering_Quality_Case_Study_Final (1).pdf

BHARAT HEAVY ELECTRICALS LIMITED

BHEL

Integrated Heavy Engineering Quality

Manufacturing Quality, Process Control & Risk-Based Quality Assurance

Date: 10 October 2026 | Day 75 Author: Jagriti Singh Role: Quality Engineer

SECTION 01: INDUSTRY AND QUALITY CONTEXT

Heavy Engineering and the Importance of Manufacturing Quality

Bharat Heavy Electricals Limited (BHEL) is a flagship engineering and manufacturing enterprise in India that produces heavy equipment for core sectors like power generation, transmission, railways, heavy industry, oil & gas, and defense. These critical industrial systems operate under high pressure, high temperature, and heavy mechanical stress for decades. Therefore, maintaining strict manufacturing quality, structural safety, and high operating reliability is essential.

In heavy manufacturing, small variations lead to severe consequences. A single dimensional error or micro-crack in a supercritical steam turbine rotor, generator stator casing, or high-pressure boiler drum can cause sudden equipment failure, expensive plant shutdowns, or severe safety hazards. Quality engineers use precise dimensional checks, metallurgical testing, non-destructive examination (NDT), and complete part traceability to ensure parts perform safely from day one.

Focusing on defect prevention rather than simple post-production inspection reduces scrap and rework costs while improving operational lifetime. Using data-driven process control helps quality engineers find and eliminate deviations long before components reach shop-floor assembly.

SECTION 02: ENGINEERING PROBLEM STATEMENT

Hypothetical Manufacturing Scenario

IMPORTANT NOTICE: This is a hypothetical educational study to demonstrate quality engineering methods. It does not describe any real internal report, defect, or official event at BHEL.

During routine multi-axis CNC turning and boring of a precision-machined component, operators observed dimensional variation across consecutive production batches. Critical outer diameters and concentricity features were drifting close to or

exceeding their specified tolerance limits. This creates a high risk of assembly fitting issues, expensive shop-floor rework, or total component rejection.

To fix this problem effectively, the engineering team must run a structured root-cause investigation rather than making unverified manual adjustments. They need to analyze machine heat expansion, cutting tool wear, workholding clamping forces, raw material hardness variations, and gauge calibration accuracy.

SECTION 03: QUALITY RISKS AND CRITICAL PARAMETERS

Controlling critical parameters keeps heavy machinery safe and reliable over long operating lifecycles. The table below outlines key quality features, failure risks, contributing factors, verification methods, and quality controls.

BHEL Integrated Heavy Engineering Quality Case Study | Day 75 Page 1 of 6

Quality Feature Potential Risk Possible Causes Verification Method Quality Control

Dimensional Parts do not fi

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Bharat Heavy Electricals Limited (BHEL). Official Website & Corporate Publications. https://www.bhel.com

• ISO 9001:2015. Quality Management Systems: Requirements. International Organization for Standardization.

• Montgomery, D. C. (2019). Introduction to Statistical Quality Control (8th ed.). John Wiley & Sons.

• AIAG & VDA. (2019). FMEA Handbook. Automotive Industry Action Group.

• AIAG. (2010). Measurement Systems Analysis (MSA) Reference Manual (4th ed.).

Disclaimer: This is an independent educational case study prepared to demonstrate the application of Quality Engineering principles in a heavy-engineering context.

It is not an official BHEL publication and is not endorsed by BHEL. The manufacturing problem and proposed analysis are hypothetical unless explicitly identified as

verified public information. All proposed solutions are educational recommendations and should be evaluated against applicable specifications, standards, risk

requirements, and authorised engineering procedures before real-world use.

BHEL Integrated Heavy Engineering Quality Case Study | Day 75 Page 6 of 6

DAY 76

Original source file(s): HAL_Day_76_Quality_Engineering_Report (1).pdf

Source focus: The Day 76 source material focuses on H I N D U S TA N A E R O N A U T I C S L I M I T E D (H A L) • I N T E G R AT E D A E R O S PA C E. The source text explicitly discusses the following quality themes and methods: Inspection, Reliability, Risk-Based, Traceability, FMEA, root cause, CAPA, SPC, audit, Process Capability, Cp, Cpk.

Section headings identified in the source: Q UA L I T Y • Integrated Aerospace Quality Engineering • Aerospace quality management encompasses several critical pillars: • approach through: • 3. Risk Assessment: Anticipating failure modes and evaluating severity, occurrence, and detection. • 4. Reliable Measurement: Ensuring measurement systems are precise and calibrated for their specific

Technical terms detected in the source: Inspection, Reliability, Risk-Based, Traceability, FMEA, root cause, CAPA, SPC, audit, Process Capability, Cp, Cpk, RPN, RCA

Source extract 1: HAL_Day_76_Quality_Engineering_Report (1).pdf

H I N D U S TA N A E R O N A U T I C S L I M I T E D (H A L) • I N T E G R AT E D A E R O S PA C E

Q UA L I T Y

Integrated Aerospace Quality Engineering

Manufacturing Quality, Inspection, Reliability & Risk-Based Quality Management

Date: 11 October 2026 (Day 76) Author: Jagriti Singh Role: Quality Engineer Challenge: 100 Days QMQE

SECTION 01: Industry and Quality Context

Why Quality Engineering Matters in Aerospace Manufacturing: In aerospace manufacturing, product safety, mission success, and regulatory compliance depend entirely on uncompromising quality standards. Aerospace systems operate under extreme thermal, mechanical, and environmental loads, leaving zero margin for unmitigated failure. Quality Engineering provides the systematic framework required to ensure every component satisfies its rigorous design, material, manufacturing, inspection, and functional requirements.

Aerospace quality management encompasses several critical pillars:

• Manufacturing Quality: Ensuring machining, joining, assembly, and special processes consistently conform to established engineering parameters.

• Dimensional Accuracy: Maintaining tight geometric tolerances to ensure precise mechanical fit and functional interoperability.

• Material Properties: Verifying chemical composition, microstructural integrity, and mechanical strength against applicable specifications.

• Inspection and Testing: Applying rigorous dimensional, non-destructive, and functional testing to verify product conformity.

• Process Control: Monitoring manufacturing processes to maintain stability and prevent variation before nonconformities occur.

• Component Reliability: Designing and manufacturing parts to perform intended functions reliably throughout their operational lifecycle.

• Product Traceability: Maintaining unbroken documentation links from raw material heats through production steps to final assembly.

• Defect Prevention: Proactively identifying vulnerabilities rather than relying solely on reactive sorting.

• Documented Verification: Generating objective evidence to demonstrate compliance with all engineering requirements.

While final inspection is essential for verifying finished parts, inspection alone cannot establish every aspect of product conformity. Latent internal flaws, material degradation during processing, or subtle cumulative variations may escape end-line visual or dimensional checks. Quality Engineering supports a comprehensive lifecycle approach through:

1. Prevention of Nonconformities: Addressing potential process drift before defective parts are produced.

2. Process Monitoring: Utilizing statistical tools to maintain control over critical manufacturing operations.

3. Risk Assessment: Anticipating failure modes and evaluating severity, occurrence, and detection.

4. Reliable Measurement: Ensuring measurement systems are precise and calibrated for their specific application.

5. Evidenc

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6. Automotive Industry Action Group (AIAG) — Measurement Systems Analysis (MSA) Reference Manual (4th Edition).

7. Society of Automotive Engineers (SAE) — AS9100D: Quality Management Systems — Requirements for Aviation, Space, and Defense Organizations.

Note: These references represent recognized industry and international standards. Bibliographic details are verified; specific standards do not necessarily apply directly to every internal operational procedure of HAL.

LEGAL & EDUCATIONAL DISCLAIMER

"This is an independent educational case study prepared to demonstrate the application of Quality Engineering principles in a hypothetical aerospace manufacturing context. It is not an official HAL publication and is not endorsed by HAL. The manufacturing scenario, analysis, and proposed solutions are educational and do not describe an actual HAL incident or internal procedures. All real-world decisions must follow applicable engineering specifications, standards, risk requirements, and authorised procedures."

DAY 77

Original source file(s): Day_77_DRDO_Integrated_Defence_Quality_Engineering (1).pdf

Source focus: The Day 77 source material focuses on DEFENCE RESEARCH AND DEVELOPMENT ORGANISATION. The source text explicitly discusses the following quality themes and methods: Reliability, root cause, yield, Traceability, FMEA, Cp, Cpk, audit, Risk-based, ISO 9001, ISO 31000.

Section headings identified in the source: DEFENCE RESEARCH AND DEVELOPMENT ORGANISATION • INTEGRATED DEFENCE QUALITY ENGINEERING • Author: Jagriti Singh • Role: Quality Engineer • INDEPENDENT EDUCATIONAL CASE STUDY • HOW TO USE THIS REPORT

Technical terms detected in the source: Reliability, root cause, yield, Traceability, FMEA, Cp, Cpk, audit, Risk-based, ISO 9001, ISO 31000

Source extract 1: Day_77_DRDO_Integrated_Defence_Quality_Engineering (1).pdf

100 DAYS QUALITY MANAGEMENT CHALLENGE

DRDO

DEFENCE RESEARCH AND DEVELOPMENT ORGANISATION

INTEGRATED DEFENCE QUALITY ENGINEERING

Research Quality • Experimental Testing • Reliability Assessment

Failure Investigation • System Validation

DAY 77 | 12 OCTOBER 2026

Author: Jagriti Singh

Role: Quality Engineer

INDEPENDENT EDUCATIONAL CASE STUDY

A practical learning document connecting quality engineering principles with a hypothetical system-testing problem.

Not an official DRDO publication.

TECHNICAL LEARNING • EVIDENCE-BASED REASONING • ENGINEERING JUDGEMENT

HOW TO USE THIS REPORT

This report separates public organisational context, general engineering principles and hypothetical case-study details.

• Organisational context: Refer to DRDO's official public website for general information.

• Engineering principles: Technical explanations are general educational material, not a description of DRDO's internal methods.

• Hypothetical details: The scenario is invented for learning and does not represent a real DRDO incident or test.

• Decision discipline: A suspected cause is not a confirmed root cause; completing an action does not prove effectiveness.

Section Purpose

1. Defence R&D; quality context Quality, verification, validation and reliability.

2. Hypothetical problem Define a test-performance variation without assuming its cause.

3. Critical quality characteristics Requirements, evidence and acceptance logic.

4. Tools and analysis Method selection, interpretation and limitations.

5. Root-cause investigation Compare plausible explanations with evidence.

6. Risk assessment Identify consequences and mitigation.

7. Engineering solution Match action to supported cause.

8. Validation Review action effectiveness and requirements evidence.

9. Interview preparation 25 original research-oriented questions with answers.

10. References and disclaimer Traceable further reading and clear scope.

Core principle: sound engineering decisions require defined requirements, trustworthy measurements, documented evidence and appropriate technical judgement.

1. DEFENCE R&D; QUALITY CONTEXT

Defence research and development applies scientific and engineering methods to develop, evaluate and improve technologies. Quality engineering supports clear requirements, meaningful experiments, trustworthy measurements, risk awareness and evidence-based conclusions. This report discusses general principles; it does not describe DRDO's internal procedures.

Research-stage quality and production-stage quality

At the research stage, teams may be evaluating a concept, learning how factors affect performance or determining whether a design meets a defined need. Test methods, assumptions and conditions require careful control. Production-stage quality also emphasises repeatable manufacture, supplier controls and consistent conformity across units. These activities overlap, but the engineering questions are not identical.

Term Meaning Illustrative example

Quality Assurance (QA) Planned activit

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ntially relevant learning references, not a claim that every listed standard is mandatory or applicable to every DRDO programme. Verify edition and project-specific applicability before use.

Disclaimer

This document is an independent educational case study prepared by Jagriti Singh as part of the 100 Days Quality Management Challenge. It is not an official DRDO publication and is not endorsed by DRDO. The engineering scenario and any illustrative data are hypothetical and do not describe an actual DRDO project, incident, internal procedure or test result. Technical concepts are discussed for educational purposes. Actual engineering decisions must follow applicable requirements, approved procedures, safety obligations and authorised technical documentation.

FINAL TAKEAWAY: Quality Engineering in research is not simply about obtaining a passing result. It is about defining the question, designing a suitable test, trusting the measurement, investigating variation, managing risk and supporting conclusions with defensible evidence.

DAY 77 | DRDO | INDEPENDENT EDUCATIONAL CASE STUDY Page 9

DAY 78

Original source file(s): Day_78_BEL_Integrated_Electronics_Quality_Engineering (1).pdf

Source focus: The Day 78 source material focuses on BHARAT ELECTRONICS LIMITED. The source text explicitly discusses the following quality themes and methods: Reliability, Risk-Based, Inspection, Traceability, FMEA, Cp, Cpk, isolation, root cause, ISO 9001, ISO 31000.

Section headings identified in the source: BHARAT ELECTRONICS LIMITED • INTEGRATED ELECTRONICS QUALITY ENGINEERING • Author: Jagriti Singh • Role: Quality Engineer • INDEPENDENT EDUCATIONAL CASE STUDY • HOW TO USE THIS REPORT

Technical terms detected in the source: Reliability, Risk-Based, Inspection, Traceability, FMEA, Cp, Cpk, isolation, root cause, ISO 9001, ISO 31000

Source extract 1: Day_78_BEL_Integrated_Electronics_Quality_Engineering (1).pdf

100 DAYS QUALITY MANAGEMENT CHALLENGE

BEL

BHARAT ELECTRONICS LIMITED

INTEGRATED ELECTRONICS QUALITY ENGINEERING

Electronic Systems Testing • Reliability • Failure Analysis

Risk-Based Quality Assurance • Validation

DAY 78 | 13 OCTOBER 2026

Author: Jagriti Singh

Role: Quality Engineer

INDEPENDENT EDUCATIONAL CASE STUDY

A learning document using a hypothetical electronic-system testing scenario. Not an official BEL publication.

TECHNICAL LEARNING • EVIDENCE-BASED ANALYSIS • ENGINEERING JUDGEMENT

HOW TO USE THIS REPORT

This report separates general engineering principles from hypothetical scenario details.

• Organisational context: consult BEL's official public website for general company information.

• Technical concepts: general engineering learning material, not BEL internal procedures.

• Hypothetical details: the scenario and numerical example are invented; they do not describe an actual BEL incident.

• Decision discipline: a suspected cause is not a verified cause; action completion is not proof of effectiveness.

Section Purpose

1. Electronics quality context Quality, reliability, test integrity and validation.

2. Hypothetical problem Define performance variation without assuming cause.

3. Critical characteristics Requirements, risks and evidence.

4. Tools and technical analysis Methods, examples and limitations.

5. Failure investigation Evaluate plausible causes.

6. Risk assessment Prioritise risks and mitigation.

7. Proposed solution Match action to evidence.

8. Validation Review results and effectiveness.

9. Interview preparation 20 original questions with answers.

10. References and disclaimer Further reading and scope.

Core principle: electronic-system quality decisions require defined requirements, controlled testing, suitable measurements and

documented evidence.

1. ELECTRONICS QUALITY ENGINEERING CONTEXT

Bharat Electronics Limited (BEL) is an Indian public-sector enterprise associated with defence electronics and related technology

areas. Consult BEL's official website for company-specific facts. This report does not describe BEL's internal procedures or any

actual project.

Electronic-system quality engineering considers whether components, assemblies and integrated systems meet requirements

under specified conditions. Depending on the product, this may involve functional and environmental testing, reliability assessment,

configuration control and failure analysis.

Key concepts

Concept Meaning Practical significance

Quality Assurance (QA) Planned activities intended to provide confidence that requirements Plans, responsibilities, reviews and records.

will be met.

Quality Control (QC) Activities used to assess whether an output meets requirements. Inspection and test results evaluated against criteria.

Functional testing Checks selected functions under defined conditions. Confirms only the behaviours actually tested.

Reliability Probability of required function over a stated time and conditions. Needs defined failure criteria and suitable

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sult the applicable verified

edition where relevant.

These are potentially relevant learning references, not a claim that every listed standard applies to every BEL product. Verify edition

and project-specific applicability.

Disclaimer

This document is an independent educational case study prepared by Jagriti Singh as part of the 100 Days Quality Management

Challenge. It is not an official BEL publication and is not endorsed by BEL. The scenario and illustrative numbers are hypothetical

and do not describe an actual BEL incident, product failure, project or internal procedure. Technical concepts are discussed for

educational purposes. Actual engineering decisions must follow applicable requirements, approved procedures, safety obligations

and authorised technical documentation.

FINAL TAKEAWAY: Electronics Quality Engineering requires more than a pass/fail reading. Reliable conclusions depend on valid

test conditions, suitable measurement methods, controlled configurations, systematic fault investigation and documented

evidence.

DAY 78 | BEL | INDEPENDENT EDUCATIONAL CASE STUDY Page 8

DAY 79

Original source file(s): Day_79_Indian_Railways_Quality_Engineering (1) (1).pdf

Source focus: The Day 79 source material focuses on INDIAN RAILWAYS. The source text explicitly discusses the following quality themes and methods: Reliability, Inspection, traceability, root cause, FMEA, audit, isolated.

Section headings identified in the source: INDIAN RAILWAYS • Practical Quality Engineering • An Integrated Engineering Case Study • 2. Introduction to Quality Engineering in Indian Railways • reliability techniques: • 3. Hypothetical Engineering Case Study

Technical terms detected in the source: Reliability, Inspection, traceability, root cause, FMEA, audit, isolated

Source extract 1: Day_79_Indian_Railways_Quality_Engineering (1) (1).pdf

INDIAN RAILWAYS

Reliability, Preventive Maintenance and

Practical Quality Engineering

An Integrated Engineering Case Study

Challenge: 100 Days Quality Management Challenge

Day: 79

Date: 14 October 2026

Prepared by: Jagriti Singh, Quality Engineer

Day 79 | 100 Days Quality Management Challenge Page 1

2. Introduction to Quality Engineering in Indian Railways

Quality Engineering plays a foundational role in ensuring the safe, reliable, and efficient operation of railway

systems. In heavy engineering sectors like railways, operations involve thousands of interconnected

mechanical and electrical components operating under demanding physical conditions. Quality engineering

principles provide a structured approach to design, manufacturing, assembly, and maintenance, ensuring that

assets perform reliably throughout their operational lifecycle.

In railway maintenance engineering, key domains benefit significantly from structured quality control and

reliability techniques:

• Mechanical Components: Ensuring structural integrity, dimensional precision, and material durability for high-stress parts under cyclic loading.

• Wheels and Brakes: Managing wear tolerances, thermal stresses, and frictional characteristics to guarantee safe stopping distances and structural wheel integrity.

• Coach Maintenance: Implementing scheduled inspections, structural checks, and vibration monitoring to maintain passenger comfort and structural safety.

• Inspection and Measurement: Using calibrated gauges and precision instruments to verify that components conform strictly to approved drawings and specifications.

• Preventive Maintenance: Servicing and replacing components based on operating hours or wear trends before unexpected functional failures occur.

• Equipment Reliability: Analysing failure patterns to improve mean time between failures and reduce overall operational downtime.

• Safety and Documentation: Maintaining rigorous compliance records, traceability logs, and standard operating procedures to uphold absolute safety standards.

By integrating these practices, maintenance workshops can systematically reduce repeated component failures, improve fleet reliability, and ensure uncompromising passenger safety.

3. Hypothetical Engineering Case Study

Note: The following scenario is a hypothetical engineering case study created purely for educational purposes to demonstrate the practical application of quality tools. It does not describe an actual Indian Railways incident or internal railway workshop procedure.

Scenario: A regional railway maintenance workshop notices that a specific mechanical component within a coach suspension sub-assembly is wearing out significantly earlier than its designed service life. When the component fails, the workshop maintenance team replaces it with a new spare part. However, the exact same premature wear and failure symptom reappears on multiple coaches after a similar operating interval. The maintenance team wants to investigate the root ca

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ndian Railways Official Website. Ministry of Railways, Government of India. Available at: https://indianrailways.gov.in

• Research Designs and Standards Organisation (RDSO). Indian Railways Maintenance and Standardization Guidelines. Available at: https://rdso.indianrailways.gov.in

• Montgomery, D. C. (2019). Introduction to Statistical Quality Control (8th ed.). John Wiley & Sons.

• Juran, J. M., & Gryna, F. M. (2004). Juran's Quality Handbook (5th ed.). McGraw-Hill.

Disclaimer:

"This is an independent educational case study prepared by Jagriti Singh. It is not an official Indian Railways publication and is not endorsed by Indian Railways. The engineering situation is hypothetical and does not describe an actual incident or internal railway procedure. The interview questions are independently prepared practice

questions and are not claimed to be actual recruitment interview questions. Real railway work must follow applicable

approved drawings, specifications, maintenance manuals, safety requirements, and authorised procedures."

Day 79 | 100 Days Quality Management Challenge Page 10

DAY 80

Original source file(s): Day_80_NTPC_Quality_Engineering_Report (1).pdf

Source focus: The Day 80 source material focuses on NTPC LIMITED. The source text explicitly discusses the following quality themes and methods: Reliability, root cause, inspection, isolation, RCA, FMEA, traceability, audit.

Section headings identified in the source: NTPC LIMITED • INTEGRATED POWER PLANT • QUALITY ENGINEERING • Investigation and Risk Based Quality Management • A comprehensive engineering case study and structured quality • methodology report examining rotating machinery reliability and

Technical terms detected in the source: Reliability, root cause, inspection, isolation, RCA, FMEA, traceability, audit

Source extract 1: Day_80_NTPC_Quality_Engineering_Report (1).pdf

NTPC LIMITED

INTEGRATED POWER PLANT

QUALITY ENGINEERING

Equipment Reliability, Preventive Maintenance, Failure

Investigation and Risk Based Quality Management

A comprehensive engineering case study and structured quality

methodology report examining rotating machinery reliability and

systematic root cause problem solving in thermal power plants.

Document Details

Challenge: 100 Days Quality Management Challenge (Day 80)

Date: 15 October 2026

Author: Jagriti Singh

Professional Identity: Quality Engineer

NTPC Limited | Day 80 - Quality Management Challenge Page 1 of 14

SECTION 2: INTRODUCTION TO NTPC AND QUALITY

ENGINEERING

NTPC Limited is India's largest power utility, playing a vital role in meeting the nation's energy demands

through thermal, hydro, solar, and nuclear power generation. Operating large-scale thermal power plants

requires continuous operation of complex mechanical equipment under demanding conditions of high

temperature, high pressure, and continuous loading. In such heavy industrial environments, equipment

reliability is directly linked to operational safety, plant efficiency, and overall energy generation costs.

Quality Engineering in power generation is not merely a paperwork exercise or a final inspection step. It is a

systematic engineering discipline focused on preventing defects, maintaining machine health, and ensuring

that every mechanical asset performs reliably throughout its lifecycle. Power plants rely heavily on a diverse

range of mechanical components that must operate in seamless coordination.

Key Mechanical Systems in Power Generation

• Pumps: Centrifugal and positive displacement pumps handle critical fluid transfers, including boiler feed water, condenser cooling, and chemical dosing.

• Motors: Electric motors provide the primary mechanical drive for pumps, fans, conveyors, and pulverizers.

• Bearings: Rolling element and journal bearings support rotating shafts, reducing friction and maintaining precise shaft positioning.

• Valves: Control and isolation valves regulate fluid flow, pressure, and temperature across steam and water cycles.

• Heat Exchangers: Condensers, feedwater heaters, and closed-loop coolers transfer thermal energy efficiently between fluid streams.

• Mechanical Components: Couplings, seals, shafts, and gearboxes transmit power and maintain pressure boundaries.

• Equipment Inspection: Systematic physical and non-destructive checks detect early signs of wear, corrosion, and structural degradation.

• Preventive Maintenance: Planned servicing activities prevent unexpected breakdowns through scheduled overhauls and component replacements.

• Maintenance Records: Historical logs provide vital data on equipment performance, failure frequency, and repair history.

• Safety and Reliability: Rigorous engineering standards protect plant personnel, prevent catastrophic failures, and ensure continuous grid stability.

NTPC Limited | Day 80 - Quality Management Challenge Page 2 of 14

Engineering Perspective: Q

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ry of Power. Publications and Sector Reports. Available from: https://

powermin.gov.in

• Mobley, R. Keith. An Introduction to Predictive Maintenance. 2nd ed. Butterworth-Heinemann, 2002.

• Svensson, Göran. Total Quality Management and Operational Reliability in Industrial Systems. Industrial Press, 2018.

• Bloch, Heinz P., and Geitner, Fred K. Machinery Failure Analysis and Troubleshooting. 4th ed. Gulf Professional Publishing, 2012.

SECTION 13: DISCLAIMER

Disclaimer:

This is an independent educational case study prepared by Jagriti Singh. It is not an official NTPC publication and is not endorsed by NTPC Limited. The engineering scenario is hypothetical and does not describe an actual NTPC incident or internal procedure. The interview questions are independently prepared educational practice questions and are not claimed to be actual NTPC recruitment interview questions. Real engineering work must follow applicable equipment specifications, approved

maintenance manuals, safety requirements, and authorised procedures.

NTPC Limited | Day 80 - Quality Management Challenge Page 14 of 14

DAY 81

Original source file(s): Day_81_ONGC_Integrated_Oil_and_Gas_Quality_Engineering_NoDash (1).pdf

Source focus: The Day 81 source material focuses on INTEGRATED OIL AND GAS QUALITY. The source text explicitly discusses the following quality themes and methods: Reliability, Inspection, Risk-Based, Traceability, NDT, yield, Root Cause, RPN, audit, ISO 9001:2015, ISO 31000:2018.

Section headings identified in the source: INTEGRATED OIL AND GAS QUALITY • ENGINEERING • Quality Management • Organisation: Oil and Natural Gas Corporation Author: Jagriti Singh • Limited (ONGC) • 1. Organisation and Industry Context

Technical terms detected in the source: Reliability, Inspection, Risk-Based, Traceability, NDT, yield, Root Cause, RPN, audit, ISO 9001:2015, ISO 31000:2018

Source extract 1: Day_81_ONGC_Integrated_Oil_and_Gas_Quality_Engineering_NoDash (1).pdf

INTEGRATED OIL AND GAS QUALITY

ENGINEERING

Pipeline Integrity, Corrosion Monitoring, Equipment Reliability and Risk Based

Quality Management

Organisation: Oil and Natural Gas Corporation Author: Jagriti Singh

Limited (ONGC)

Challenge: 100 Days Quality Management Identity: Quality Engineer

Challenge (Day 81)

Date: 16 October 2026 Document Type: Independent Educational Case

Study

1. Organisation and Industry Context

Oil and Natural Gas Corporation Limited (ONGC) is a premier Indian public sector energy enterprise operating under the Ministry of Petroleum and Natural Gas, Government of India. As India's largest crude oil and natural gas corporation, ONGC contributes a significant share of domestic oil and gas production. Operating across complex upstream and midstream environments, including offshore platforms, deep-water drilling rigs, onshore gathering stations, and extensive cross-country pipeline networks, demands uncompromising standards in quality engineering, safety, and mechanical reliability.

In the oil and gas sector, Quality Engineering (QE) extends far beyond standard manufacturing quality control. It encompasses the systematic assurance of structural integrity, material suitability, corrosion resistance, and equipment reliability throughout the complete asset lifecycle. A failure in an upstream production facility or a transmission pipeline can lead to catastrophic environmental consequences, severe safety hazards, substantial financial losses, and critical energy supply disruptions. Consequently, robust quality frameworks are integral to every operational phase.

Key Focus Areas and Upstream and Midstream Quality Assurance:

• Pipeline Integrity: Ensuring continuous containment and safe transport of hydrocarbons under high

pressure and corrosive conditions.

• Corrosion Monitoring: Implementing proactive detection mechanisms for internal and external degradation phenomena.

• Equipment Inspection: Systematic non-destructive testing and dimensional verification of pressure vessels, valves, and rotating machinery.

• Mechanical Reliability: Maximizing operational uptime and preventing unplanned outages through rigorous preventive maintenance regimes.

• Risk-Based Management: Prioritizing engineering resources and inspection frequencies based on quantitative and qualitative risk evaluations.

ONGC and Day 81 and Integrated Oil and Gas Quality Engineering Page 1 of 11

• Traceability and Documentation: Maintaining rigorous material test reports, welding logs, and inspection histories to substantiate engineering decisions.

Note on Distinction: While industry-wide standards, such as ISO, ASTM, and NACE, establish baseline engineering methodologies, specific operational procedures, threshold limits, and asset management protocols implemented by ONGC are governed by internal corporate standards, statutory regulations set by the Oil Industry Safety Directorate (OISD), and Petroleum and Natural Gas Regulatory Board (PNGRB) guidelines in India.

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tems.

Association for Materials Protection and Performance.

• Fontana, M. G., Corrosion Engineering, 3rd Edition, McGraw-Hill Education.

• Juran, J. M., and Joseph A. De Feo, Juran's Quality Handbook, 6th Edition, McGraw-Hill.

Disclaimer:

This document is an independent educational case study prepared by Jagriti Singh as part of the 100 Days Quality Management Challenge. It is not an official Oil and Natural Gas Corporation publication and is not endorsed by the corporation.

The pipeline corrosion scenario is hypothetical and does not describe an actual incident or internal procedure.

The technical interview questions are independently prepared for learning and interview preparation. They are not represented as actual recruitment interview questions.

All engineering examples are educational. Real pipeline inspection, integrity assessment, operation, repair, and maintenance decisions must follow applicable regulations, engineering codes, approved specifications, safety requirements, and authorised procedures.

ONGC and Day 81 and Integrated Oil and Gas Quality Engineering Page 11 of 11

DAY 82

Original source file(s): Day_82_GAIL_Integrated_Pipeline_Quality_Engineering (1).pdf

Source focus: The Day 82 source material focuses on GAIL (INDIA) LIMITED. The source text explicitly discusses the following quality themes and methods: reliability, SPC, Inspection, NDT, FMEA, ISO quality, root cause, CP, isolation, Traceability, RCA, RPN.

Section headings identified in the source: GAIL (INDIA) LIMITED • INTEGRATED PIPELINE QUALITY • ENGINEERING • Assessment and Preventive Maintenance • Author: Jagriti Singh • Scope: Natural Gas Transmission & Pipeline Integrity Management

Technical terms detected in the source: reliability, SPC, Inspection, NDT, FMEA, ISO quality, root cause, CP, isolation, Traceability, RCA, RPN, Isolated, Risk-Based

Source extract 1: Day_82_GAIL_Integrated_Pipeline_Quality_Engineering (1).pdf

DAY 82 / 100 CHALLENGE

GAIL (INDIA) LIMITED

INTEGRATED PIPELINE QUALITY

ENGINEERING

Pipeline Integrity, Pressure Variation, Potential Leakage Investigation, Risk

Assessment and Preventive Maintenance

Document ID: 100D-QM-DAY82-2026

Author: Jagriti Singh

Date: 17 October 2026

Context: 100 Days Quality Management Challenge

Scope: Natural Gas Transmission & Pipeline Integrity Management

1. EXECUTIVE SUMMARY & ORGANISATIONAL CONTEXT

1.1 About GAIL (India) Limited

GAIL (India) Limited is India’s flagship natural gas company, operating under the Ministry of Petroleum and Natural Gas, Government of India. As a Maharatna Public Sector Enterprise, GAIL plays a pivotal role in the country’s energy landscape. Official corporate disclosures (accessible via https://gailonline.com/) highlight that GAIL operates an extensive cross-country natural gas pipeline network spanning over 16,000 kilometers. This infrastructure handles the transmission of natural gas connecting gas sources, liquefied natural gas (LNG) import terminals, power generation facilities, fertilizer plants, city gas distribution (CGD) networks, and industrial consumers across India.

1.2 Criticality of Pipeline Integrity and Pressure Monitoring

In natural gas transmission, maintaining pipeline structural integrity and operating within designed pressure envelopes is essential. Natural gas is transported under high pressure (typically between 50 bar to 90 bar in cross-country trunklines). A loss of containment or structural breach introduces catastrophic process safety risks, including high-pressure gas release, fire, explosion, severe environmental degradation, operational downtime, and significant commercial losses.

Real-time pressure monitoring serves as the primary health indicator of a gas pipeline system. Transient pressure drops, unexplained flow rate imbalances, or abnormal pressure fluctuations may indicate severe underlying operational issues—ranging from physical gas leakage, valve malfunctions, measurement instrumentation drift, line pack variations, to compressor station operational shifts. Continuous monitoring combined with proactive preventive maintenance ensures grid reliability and operational safety.

1.3 Role of Quality Engineering in Pipeline Operations

Quality Engineering provides a structured, data-driven framework to maintain process control, asset reliability, and safety compliance across natural gas transmission operations. In pipeline integrity management, Quality Engineering plays an indispensable role through the following systematic functions:

• Identifying Abnormal Conditions: Utilizing statistical process control (SPC) and automated SCADA alarms to detect subtle deviations from normal baseline operating limits before failure escalates.

• Collecting Reliable Inspection Data: Establishing standard operating procedures for field data collection, recalibrating instruments, and executing non-destructive testing (NDT) to ensure data integrity.

• Investigating Root Causes:

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case study prepared by Jagriti Singh as part of the 100 Days Quality Management Challenge. It is not an official GAIL publication and is not endorsed by GAIL (India) Limited.

The engineering scenario described herein is strictly hypothetical and does not represent an actual GAIL incident, operational breach, or confidential internal procedure.

Day 82: 100 Days Quality Management Challenge | GAIL (India) Limited Page 10 of 11

The proposed solutions and recommendations are educational exercises designed to demonstrate Quality Engineering principles and are not claimed to have been implemented by GAIL.

The technical interview questions included in this report are independently created for educational study and are not official GAIL recruitment examination questions.

Actual industrial engineering decisions, pipeline maintenance actions, and safety procedures must always follow applicable statutory regulations, official engineering standards, approved internal procedures, and authorized safety assessments.

Day 82: 100 Days Quality Management Challenge | GAIL (India) Limited Page 11 of 11

DAY 83

Original source file(s): Day_83_SAIL_Integrated_Steel_Process_Quality_Engineering (1).pdf

Source focus: The Day 83 source material focuses on STEEL AUTHORITY OF INDIA LIMITED (SAIL). The source text explicitly discusses the following quality themes and methods: Root Cause, reliability, yield, Inspection, Traceability, SPC, isolate, Isotope, audit, Process Capability, Cp, Cpk.

Section headings identified in the source: STEEL AUTHORITY OF INDIA LIMITED (SAIL) • INTEGRATED STEEL PROCESS QUALITY ENGINEERING • Crucial quality characteristics dependent on rigorous process engineering include: • Methodological Note on Organisational Context • Hypothetical Scenario: Gauge Variation in Hot Rolled Steel Strip Production • Key Engineering Questions & Investigative Foundations

Technical terms detected in the source: Root Cause, reliability, yield, Inspection, Traceability, SPC, isolate, Isotope, audit, Process Capability, Cp, Cpk, RCA, FMEA

Source extract 1: Day_83_SAIL_Integrated_Steel_Process_Quality_Engineering (1).pdf

STEEL AUTHORITY OF INDIA LIMITED (SAIL)

100 DAYS QUALIT Y MANAGEMENT CHALLENGE • DAY 83

INTEGRATED STEEL PROCESS QUALITY ENGINEERING

Process Variation, Statistical Process Control, Root Cause Analysis and Continuous Improvement

Date: 18 October 2026 Author: Jagriti Singh

SECTION 1: INTRODUCTION TO SAIL AND STEEL PROCESS QUALITY

Steel Authority of India Limited (SAIL) is one of the largest steel-making companies in India and one of the Maharatna Central Public Sector Enterprises (CPSEs). Operating major integrated steel plants at Bhilai, Rourkela, Durgapur, Bokaro, and Burnpur (Isco), alongside special steel plants at Salem, Durgapur, and Bhadravathi, SAIL manufactures a comprehensive portfolio of flat, long, and tubular steel products catering to strategic infrastructure, railway, defence, automotive, and construction sectors.

In integrated steel production, Quality Engineering serves as the structural backbone ensuring structural reliability, mechanical integrity, and dimensional consistency across millions of tonnes of steel output. Unlike simple assembly manufacturing, steelmaking involves complex thermo-mechanical processing—transitioning from raw iron ore, coke, and flux through liquid ironmaking (Blast Furnace), steel refining (Basic Oxygen Furnace/Electric Arc Furnace), Continuous Casting, and Hot/Cold Rolling Mills. At each thermodynamic and mechanical processing stage, variations in operating parameters directly manifest in the final metallurgical and dimensional attributes of the finished steel product.

Crucial quality characteristics dependent on rigorous process engineering include:

• Process Consistency: Thermal uniformity in reheating furnaces, stable roll gap settings, and controlled cooling speeds across the run-out table.

• Chemical Composition: Precise alloying additions and tight control over residual elements (such as Carbon, Manganese, Phosphorus, Sulphur, Silicon) during secondary steelmaking.

• Dimensional Accuracy: Gauge uniformity (thickness), width, flatness, crown, and edge profile across the entire length and width of plate or sheet products.

• Mechanical Properties: Tensile strength, yield strength, percentage elongation, impact toughness, and bendability governed by grain size and microstructural phases.

• Surface Quality: Freedom from scale pit marks, scratches, seams, slivers, laminations, and rust.

• Process Monitoring: Real-time tracking of roll force, inter-stand tension, casting speed, and temperature profiles via automatic gauge control (AGC) systems.

• Inspection and Testing: Non-destructive testing (ultrasonic, eddy current) and destructive testing (tensile, Charpy impact, bend tests) conducted systematically.

• Equipment Condition: Roll wear, hydraulic responsiveness, bearing alignment, and thermal expansion of mill stands.

• Material Traceability: Heat-wise tracking, slab-to-coil identification numbers, and digital logging across manufacturing execution systems.

• Compliance with Product Specifications: Al

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rk, 2019.

6. AIAG (Automotive Industry Action Group): Statistical Process Control (SPC) Reference Manual, 2nd Edition.

AIAG, Southfield, MI.

100 Days Quality Management Challenge • Day 83 Page 14 of 15

7. AIAG & VDA: Failure Mode and Effects Analysis (FMEA) Handbook, 1st Edition, 2019.

SECTION 12: DISCLAIMER

This is an independent educational case study prepared by Jagriti Singh as part of the 100 Days Quality Management Challenge. The engineering scenario and any illustrative numerical data are hypothetical and do not describe an actual SAIL incident, actual production results, or an internal SAIL procedure. The proposed actions are educational recommendations and are not represented as actions implemented by SAIL. The interview questions are independently developed for learning and preparation and are not claimed to be actual SAIL recruitment interview questions. Real manufacturing decisions must follow the applicable product specifications, approved procedures, safety requirements, and authorised engineering assessments.

100 Days Quality Management Challenge • Day 83 Page 15 of 15

DAY 84

Original source file(s): NALCO_Day84_Quality_Engineering_Report (1).pdf

Source focus: The Day 84 source material focuses on N AT I O N A L A LU M I N I U M C O M PA NY L I M I TE D (N A LC O). The source text explicitly discusses the following quality themes and methods: Root Cause, yield, Inspection, Reliability, Traceability, CAPA, Isolated, SPC, process capability, FMEA, RPN, audit.

Section headings identified in the source: N AT I O N A L A LU M I N I U M C O M PA NY L I M I TE D (N A LC O) • TECHNICAL REPORT • INTEGRATED ALUMINIUM PROCESS • QUALITY ENGINEERING • Continuous Improvement • AUTHOR

Technical terms detected in the source: Root Cause, yield, Inspection, Reliability, Traceability, CAPA, Isolated, SPC, process capability, FMEA, RPN, audit, isolate, ISO 9001:2015

Source extract 1: NALCO_Day84_Quality_Engineering_Report (1).pdf

N AT I O N A L A LU M I N I U M C O M PA NY L I M I TE D (N A LC O)

100 Days Quality Management Challenge • Day 84

TECHNICAL REPORT

INTEGRATED ALUMINIUM PROCESS

QUALITY ENGINEERING

Surface Defect Investigation, Process Control, Root Cause Analysis and

Continuous Improvement

AUTHOR

Jagriti Singh

DATE

19 October 2026

FOCUS AREA

Primary Aluminium Processing & Surface Engineering Quality Control

Independent Educational Quality Management Report. Compiled for technical evaluation and process improvement study.

1. NALCO & Aluminium Quality Engineering

1.1 About National Aluminium Company Limited (NALCO)

National Aluminium Company Limited (NALCO) is a Navratna Central Public Sector Enterprise (CPSE) under the Ministry of Mines, Government of India. Established in 1981, NALCO is an integrated group with diversified operations in mining, metal, and power. The company operates high-capacity bauxite mines, an alumina refinery at Damanjodi, Odisha, and an aluminium smelter alongside a captive power plant at Angul, Odisha. NALCO produces high-quality primary aluminium products, including ingots, wire rods, billets, cast strips, and rolled products, catering to domestic and international markets (Official website: https://nalcoindia.com/).

1.2 Importance of Quality Engineering in Aluminium Production

Aluminium processing demands strict quality engineering across all production stages—from bauxite extraction and Bayer refining to Hall-Héroult smelting, casting, rolling, and extrusion. Key quality parameters include:

• Surface Quality: Prevention of scratches, blisters, oxides, inclusions, mechanical marks, and handling indentations.

• Dimensional Accuracy: Tight tolerances on width, thickness, gauge variation, flatness, and profile geometry.

• Material Properties: Chemical composition limits (e.g., Fe, Si ratio), grain structure, tensile strength, yield strength, and percentage elongation.

• Process Consistency: Thermal management during casting/rolling, bath ratio control in smelting, and controlled cooling.

• Inspection & Measurement Reliability: Calibration of optical surface inspection, ultrasonic testing, and dimensional gauges.

• Traceability & Corrective Action: End-to-end heat-lot traceability and closed-loop corrective action system (CAPA).

1.3 Quality Beyond Final Inspection

Quality cannot be inspected into a product; it must be built into the manufacturing process. Relying solely on final inspection is inefficient and costly, leading to scrap, rework, delayed deliveries, and uncontained defect risks. Consistent process control, calibrated measurement systems, systematic root cause analysis, and evidence-based decision-making ensure long-term product integrity and operational reliability.

Note on Verified Information: All company organizational details presented above are sourced from public information on NALCO's official website. No access to internal NALCO procedures, private inspection logs, or confidential plant operational data is claimed or i

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tion for Standardization, Geneva,

Switzerland.

4. Montgomery, D. C. (2019): Introduction to Statistical Quality Control, 8th Edition, John Wiley & Sons, New York.

5. ASM Handbook, Volume 2B: Properties and Selection: Aluminium Alloys. ASM International, Materials Park, Ohio.

12. Disclaimer

This is an independent educational case study prepared by Jagriti Singh as part of the 100 Days Quality Management Challenge. The engineering scenario and any illustrative data are hypothetical and do not describe an actual NALCO incident, actual production results, or an internal NALCO procedure. The proposed actions are educational recommendations and are not

represented as actions implemented by NALCO. The interview questions are independently developed for educational learning and

preparation and are not claimed to be actual NALCO recruitment interview questions. Real manufacturing decisions must follow applicable product specifications, approved procedures, safety requirements, and authorised engineering assessments.

NALCO Day 84 • Integrated Aluminium Process Quality Engineering Page 12 of 12

DAY 85

Original source file(s): Day_85_CSL_Integrated_Shipbuilding_Quality_Engineering (1).pdf

Source focus: The Day 85 source material focuses on Integrated Shipbuilding Quality Engineering. The source text explicitly discusses the following quality themes and methods: Inspection, Risk-Based, reliability, NDT, yield, Traceability, Root Cause, RCA, FMEA, RPN, audit, NCR.

Section headings identified in the source: Integrated Shipbuilding Quality Engineering • Management • ORGANISATION REPORT DATE AUTHOR • Hypothetical Educational Scenario • Common Weld Imperfections • 1. Visual Testing (VT)

Technical terms detected in the source: Inspection, Risk-Based, reliability, NDT, yield, Traceability, Root Cause, RCA, FMEA, RPN, audit, NCR, ISO 5817, ISO 9001:2015

Source extract 1: Day_85_CSL_Integrated_Shipbuilding_Quality_Engineering (1).pdf

100 DAYS QUALITY MANAGEMENT CHALLENGE — DAY 85

Integrated Shipbuilding Quality Engineering

Welding Quality, Dimensional Inspection, Structural Integrity and Risk-Based Quality

Management

ORGANISATION REPORT DATE AUTHOR

Cochin Shipyard Limited (CSL) 20 October 2026 Jagriti Singh

Section 1: Introduction

Cochin Shipyard Limited (CSL) is one of India's premier shipbuilding and ship repair facilities, located in Kochi, Kerala. Incorporated in 1972, CSL has established itself as a pioneer in construction of complex maritime vessels, ranging from commercial bulk carriers and platform supply vessels to defence ships, including India's first indigenous aircraft carrier (INS Vikrant). Operating under the administrative control of the Ministry of Ports, Shipping and Waterways, Government of India, CSL handles large-scale marine engineering projects that require meticulous quality standards and structural reliability.

In shipbuilding, quality engineering is paramount. Marine structures are continuously subjected to severe operational environments, including cyclic hydrostatic loading, wave-induced dynamic stress, impact loads, and aggressive corrosive seawater environments. A single undetected structural flaw, material deficiency, or weld defect can propagate under cyclic fatigue, potentially resulting in catastrophic structural failure, water ingress, loss of vessel integrity, environmental hazards, or loss of human life. Therefore, quality engineering in shipbuilding serves not merely as an inspection gate, but as an integrated preventive discipline ensuring structural compliance, operational safety, and long-term durability across the ship's entire design lifespan.

Section 2: Shipbuilding Quality Engineering

Quality Engineering in welded marine structures encompasses systemic controls across design, procurement, fabrication, non-destructive testing, and final validation. Key domains include:

• Welding Quality: Ensuring sound metallurgical bonding, appropriate heat input control, correct choice of welding consumables, strict compliance with approved Welding Procedure Specifications (WPS), and qualified welder performance to avoid structural embrittlement or defect formation.

• Dimensional Accuracy: Managing thermal distortion during heavy plate cutting and multi-pass welding. Preserving precise hull geometry, block alignment, deck flatness, and margin allowances is essential to avoid fit-up stress concentrations during hull block assembly.

• Inspection and Testing: Deploying systematic non-destructive testing (NDT) regimens—ranging from surface visual inspection to volumetric ultrasonic and radiographic evaluations—to verify sound joint integrity against class society rules (e.g., DNV, IRS, ABS, Lloyd's Register).

• Structural Integrity: Verifying that fabricated joints meet minimum yield strength, toughness, fatigue life, and load distribution criteria across critical strength decks, shell plating, bulkheads, and engine foundations.

• Traceability of Insp

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elding — Fusion-welded joints in steel, nickel, titanium and their alloys — Quality levels for imperfections.

4. ISO 17635: Non-destructive testing of welds — General rules for metallic materials.

5. American Welding Society (AWS) D1.1/D1.1M: Structural Welding Code — Steel.

6. Indian Register of Shipping (IRS) / DNV: Rules for Classification of Steel Ships and Offshore Structures.

DISCLAIMER:

This is an independent educational case study prepared for learning and professional development. The welding scenario is hypothetical and is not presented as an actual incident, internal procedure, or official technical report of Cochin Shipyard Limited.

This document is not affiliated with or endorsed by CSL. Actual inspection, acceptance, repair, and release decisions must follow the applicable drawings, specifications, approved procedures, contractual requirements, and authorised engineering decisions.

Day 85: Integrated Shipbuilding Quality Engineering | CSL Case Study Page 10 of 11

Prepared by

Jagriti Singh

Day 85: Integrated Shipbuilding Quality Engineering | CSL Case Study Page 11 of 11

DAY 86

Original source file(s): Day_86_GRSE_Integrated_Marine_Quality_Engineering (1).pdf

Source focus: The Day 86 source material focuses on INTEGRATED MARINE QUALITY ENGINEERING. The source text explicitly discusses the following quality themes and methods: Supplier Quality, Inspection, Traceability, Root Cause, Risk-Based, yield, RCA, audit, reliability, Process Capability, NDT, FMEA.

Section headings identified in the source: INTEGRATED MARINE QUALITY ENGINEERING • Management • Garden Reach Shipbuilders & Engineers Limited (GRSE) is a premier Indian defense shipyard under the • Receiving inspection of fabricated marine components involves a structured verification protocol: • Illustrative Example (Hypothetical): • 3. Why did the fixture shift? A locating pin was worn beyond its allowable wear limit.

Technical terms detected in the source: Supplier Quality, Inspection, Traceability, Root Cause, Risk-Based, yield, RCA, audit, reliability, Process Capability, NDT, FMEA, isolate, NCR

Source extract 1: Day_86_GRSE_Integrated_Marine_Quality_Engineering (1).pdf

INTEGRATED MARINE QUALITY ENGINEERING

Supplier Quality Assurance, Material Inspection, Traceability, Root Cause Analysis and Risk-Based Quality Management

CHALLENGE: Day 86 / 100 ORGANISATION: GRSE Ltd. DATE: 21 October 2026 AUTHOR: Jagriti Singh

SECTION 1: INTRODUCTION

Garden Reach Shipbuilders & Engineers Limited (GRSE) is a premier Indian defense shipyard under the administrative control of the Ministry of Defence. Established as a commercial vessel repair yard and later acquired by the Government of India, GRSE has grown into a key contributor to India's maritime defense capability, manufacturing warships, survey vessels, offshore patrol vessels, and specialized engineering products.

Shipbuilding is a highly complex engineering domain where quality management plays a pivotal role. Modern naval and commercial vessels rely on thousands of fabricated mechanical components, structural assemblies, and specialized equipment sourced from external suppliers. In marine applications, structural integrity, dimensional accuracy, material soundness, and traceability are non-negotiable. A defect or dimensional nonconformity in a sub-assembly can lead to fitment failures during hull integration, operational delays, increased cost, or catastrophic failure under extreme marine operating conditions. Thus, rigorous quality engineering ensures compliance with design specifications, safety standards, and defense classification society rules.

SECTION 2: MARINE QUALITY ENGINEERING

Quality Engineering in shipbuilding encompasses the systematic application of engineering principles, inspection methods, and statistical tools across the lifecycle of ship design, procurement, fabrication, and integration.

• Material Quality: Ensuring that physical, chemical, and mechanical properties (e.g., yield strength, toughness, corrosion resistance) conform to applicable marine material standards.

• Dimensional Inspection: Verifying geometric and dimensional compliance against design drawings using calibrated precision measuring tools to ensure perfect alignment during assembly.

• Supplier Quality Assurance (SQA): Establishing proactive oversight of external vendors to ensure their manufacturing processes, quality controls, and deliverables meet technical standards.

• Material Traceability: Maintaining an unbroken chain of documentation linking every physical component back to its raw material heat number, mill test certificate, and inspection batch.

• Inspection Documentation: Recording, verifying, and preserving structured quality records to establish legal and technical compliance before component acceptance.

• Prevention of Recurring Defects: Applying structured root-cause analysis (RCA) and corrective actions to eliminate systemic quality flaws.

• Compliance with Drawings and Specifications: Ensuring every part strictly conforms to approved engineering drawings, revision status, and contractual codes.

SECTION 3: ENGINEERING PROBLEM STATEMENT

Hypothetical Scenario Disclaimer:

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• ISO 9001:2015 - Quality Management Systems — Requirements

• ISO 10005:2018 - Quality Management — Guidelines for Quality Plans

• ISO 19011:2018 - Guidelines for Auditing Management Systems

• Juran, J. M., & De Feo, J. A., Juran's Quality Handbook: The Complete Guide to Performance Excellence, McGraw-Hill Education.

• Montgomery, D. C., Introduction to Statistical Quality Control, John Wiley & Sons.

DISCLAIMER: This is an independent educational case study prepared for learning and professional development. The supplier quality and dimensional inspection scenario is hypothetical and is not presented as an actual incident, internal procedure, or official technical report of Garden Reach Shipbuilders & Engineers Limited. This document is not affiliated with or endorsed by GRSE. Actual inspection, acceptance, rejection, rework, and release decisions must follow the applicable drawings, specifications,

contractual requirements, approved procedures, and authorised engineering decisions.

Prepared by

Jagriti Singh

Day 86: Integrated Marine Quality Engineering | Jagriti Singh Page 8 of 8

DAY 87

Original source file(s): Day_87_MDL_Integrated_Shipbuilding_Quality_Engineering (1).pdf

Source focus: The Day 87 source material focuses on Organisation: Mazagon Dock Shipbuilders Limited (MDL). The source text explicitly discusses the following quality themes and methods: NDT, Inspection, traceability, yield, ISO 5817, RELIABILITY, Root Cause, RCA, FMEA, RPN, Audit, NCR.

Section headings identified in the source: Organisation: Mazagon Dock Shipbuilders Limited (MDL) • INTEGRATED SHIPBUILDING QUALITY ENGINEERING • Prevention • 1. INTRODUCTION • integrity and dimensional precision directly impacts: • 2. STRUCTURAL FABRICATION QUALITY

Technical terms detected in the source: NDT, Inspection, traceability, yield, ISO 5817, RELIABILITY, Root Cause, RCA, FMEA, RPN, Audit, NCR, isolates, SPC

Source extract 1: Day_87_MDL_Integrated_Shipbuilding_Quality_Engineering (1).pdf

Organisation: Mazagon Dock Shipbuilders Limited (MDL)

100 DAYS QUALITY MANAGEMENT CHALLENGE • DAY 87

Date: 22 October 2026 | Author: Jagriti Singh

INTEGRATED SHIPBUILDING QUALITY ENGINEERING

Structural Fabrication, Welding Distortion, Dimensional Accuracy, NDT Interpretation and Defect

Prevention

1. INTRODUCTION

Mazagon Dock Shipbuilders Limited (MDL), known as "Shipbuilders to the Nation," is one of India's premier defense sector shipyards under the Ministry of Defence. Operating from Mumbai, MDL specializes in constructing warships, submarines, offshore platforms, and commercial vessels. The shipyard has a distinguished legacy of manufacturing complex naval platforms, requiring extreme precision and adherence to strict defense specifications.

In naval shipbuilding, structural fabrication quality and dimensional accuracy are paramount. Warships and submarines operate under extreme dynamic loads, cyclic hydrostatic pressures, and harsh marine environments. Ensuring structural integrity and dimensional precision directly impacts:

• Structural Alignment & Stability: Precise hull geometry ensures hydrodynamic efficiency, propulsion shafting alignment, and proper weight distribution.

• Fatigue Life & Marine Safety: High weld quality prevents localized stress concentrations that lead to catastrophic fatigue failure under wave action or underwater shock.

• Modular Assembly Fit-Up: Modern shipbuilding uses block assembly methods. Out-of-tolerance blocks cause severe assembly delays, re-work costs, and joint mismatch during final integration.

2. STRUCTURAL FABRICATION QUALITY

Achieving superior fabrication quality requires rigorous technical control across several core mechanical engineering pillars:

• Dimensional Accuracy: Verification of overall block dimensions, panel flatness, and frame spacing against approved construction drawings to ensure seamless integration.

• Weld Quality: Sound weld deposits free from harmful discontinuities, adhering to specified strength, ductility, and toughness requirements.

• Fabrication Sequence: Planned assembly and welding progression designed to balance heat input, minimize residual stress build-up, and control overall distortion.

• Joint Fit-Up & Structural Alignment: Control of root gap, bevel angle, root face, and edge alignment (hi-lo) prior to welding, preventing lack of penetration or excessive melt-through.

• Inspection & Testing: Non-destructive evaluation (NDE) and stage-wise dimensional verifications at defined quality hold-points.

• Defect Prevention & Documentation: Proactive quality controls, qualified procedures (WPS/PQR), welder qualification, and complete material traceability records.

3. ENGINEERING PROBLEM STATEMENT (HYPOTHETICAL CASE STUDY)

Educational Scenario Notice

This case study is a hypothetical educational exercise designed to illustrate quality engineering principles in shipbuilding. It does not represent an actual incident, internal inspection record, or operational failure at Mazagon Dock

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l, nickel, titanium and their alloys — Quality levels for

imperfections.

• American Welding Society (AWS) D1.1/D1.1M, Structural Welding Code — Steel.

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• Juran, J. M., & De Feo, J. A. (2010). Juran's Quality Handbook: The Complete Guide to Performance Excellence (6th ed.). McGraw-Hill.

Educational Disclaimer: This is an independent educational case study prepared for learning and professional development as part of

the "100 Days Quality Management Challenge" (Day 87). The fabrication scenario described herein is hypothetical and is not presented

as an actual incident, internal procedure, or official technical report of Mazagon Dock Shipbuilders Limited (MDL). This document is not

affiliated with or endorsed by MDL. Actual inspection, acceptance, repair, and structural release decisions must follow applicable

engineering drawings, specifications, contractual requirements, approved procedures, and authorised engineering decisions.

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Jagriti Singh

Day 87: Quality Management Challenge | MDL Case Study Page 8 of 8

DAY 88

Original source file(s): Day_88_IOCL_Refinery_Quality_Engineering (1).pdf

Source focus: The Day 88 source material focuses on Integrated Refinery Quality Engineering. The source text explicitly discusses the following quality themes and methods: Root Cause, isolations, audit, Isolate, RCA, FMEA, Reliability, RPN, SPC, ISO 9001:2015.

Section headings identified in the source: Integrated Refinery Quality Engineering • ORGANISATION AUTHOR • Indian Oil Corporation Limited (IOCL) Jagriti Singh • 1. Introduction • 2. Practical Case Study (Educational Scenario) • 3. Important Quality Tools

Technical terms detected in the source: Root Cause, isolations, audit, Isolate, RCA, FMEA, Reliability, RPN, SPC, ISO 9001:2015

Source extract 1: Day_88_IOCL_Refinery_Quality_Engineering (1).pdf

100 DAYS QUALITY MANAGEMENT CHALLENGE — DAY 88

Integrated Refinery Quality Engineering

Petroleum Product Quality Testing, Process Control and Root Cause Analysis

ORGANISATION AUTHOR

Indian Oil Corporation Limited (IOCL) Jagriti Singh

1. Introduction

In modern petroleum refining, Quality Engineering is critical to ensuring operational safety, environmental compliance,

and consistent product performance. Petroleum products such as High-Speed Diesel (HSD), Motor Spirit (Petrol), and

Aviation Turbine Fuel (ATF) must adhere strictly to predefined physical and chemical specifications.

Meeting stringent quality standards guarantees engine safety, prevents equipment damage, and minimises harmful

emissions. Rigorous quality control, continuous process monitoring, and standardized laboratory testing are vital

safeguards against off-spec production and costly re-refining or operational downtime.

2. Practical Case Study (Educational Scenario)

Scenario: During routine batch testing of High-Speed Diesel (HSD) from a refinery rundown tank, sample test

results exhibit unexpected variation in Flash Point measurements, with one sample failing to meet the minimum

specified requirement of 35°C.

• What the Quality Problem Is: Off-specification Flash Point variation in an HSD rundown batch, indicating potential

light-end contamination or localized temperature control instability during distillation.

• Why It Matters: A reduced flash point increases fuel volatility and flammability risks during storage and transport, compromising safety standards and violating regulatory specs.

• What the Engineer Should Check: Inspect recent fractionating column operating parameters (top/side stream temperatures, stripping steam rates), check rundown line valve isolations, and review laboratory testing equipment calibration logs.

• How the Possible Cause Can Be Investigated: Conduct cross-verification testing using a calibrated secondary flash point tester, perform gas chromatography to detect light-end hydro-carbon ingress, and audit the distillation unit stream logs for sudden pressure or reflux fluctuations.

• What Corrective Action May Be Appropriate: Isolate the affected intermediate tankage, adjust the side-stripping column steam flow rate to strip off excess light components, and re-route the off-spec batch for controlled reprocessing or re-blending.

• How to Verify Whether the Action Worked: Draw statistically representative composite samples across top, middle, and bottom tank levels post-recirculating/re-blending, confirming flash point compliance across three consecutive standardized tests.

100 Days Quality Management Challenge | Day 88 Page 1 of 4

3. Important Quality Tools

1. Check Sheet

Purpose: A structured, real-time data collection form used to record frequency and patterns of quality defects or measurement variations during testing.

Case Study Example: Used by lab technicians to log daily flash point test variations, sampling locations, operator shifts, and testing a

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ash point.

Q14: What constitutes effective verification of corrective action?

A: Re-testing multiple representative samples post-correction over time to confirm consistent compliance and absence of recurrence.

Q15: Why are standard operating procedures (SOPs) essential for refinery quality labs?

A: SOPs standardize testing sequences and sample handling across all technician shifts, minimizing operator-dependent measurement error.

6. References & Disclaimer

References

• Official Indian Oil Corporation Limited (IOCL) Corporate Website (www.iocl.com) - Overview of Refining Operations & Quality

Control.

• ISO 9001:2015 Quality Management Systems — Requirements.

• Montgomery, D. C., Statistical Quality Control: A Modern Introduction, 7th Edition, John Wiley & Sons.

Disclaimer: This is an independent educational case study created for learning purposes. The scenario is hypothetical and does

not represent an actual IOCL incident, internal procedure, or official IOCL publication. This work is not affiliated with or endorsed

by IOCL.

100 Days Quality Management Challenge | Day 88 Page 4 of 4

DAY 89

Original source file(s): Day_89_BPCL_Petroleum_Quality_Engineering-v2 (1).pdf

Source focus: The Day 89 source material focuses on Integrated Petroleum Quality Engineering. The source text explicitly discusses the following quality themes and methods: Root Cause, inspection, RCA, FMEA, RPN, customer satisfaction, Reliability, Isolating, isolation, CAPA, SPC, ISO 9001:2015.

Section headings identified in the source: Integrated Petroleum Quality Engineering • 1. INTRODUCTION • 2. PRACTICAL EDUCATIONAL CASE STUDY • Hypothetical Case Scenario: Flash Point Deviation in Diesel Fuel • 3. IMPORTANT QUALITY TOOLS • are applied:

Technical terms detected in the source: Root Cause, inspection, RCA, FMEA, RPN, customer satisfaction, Reliability, Isolating, isolation, CAPA, SPC, ISO 9001:2015

Source extract 1: Day_89_BPCL_Petroleum_Quality_Engineering-v2 (1).pdf

100 DAYS QUALITY MANAGEMENT CHALLENGE DAY 89

Integrated Petroleum Quality Engineering

Product Quality Testing, Process Control and Root Cause Analysis

Organisation Focus: Bharat Petroleum Corporation Limited (BPCL) | Author: Jagriti Singh

1. INTRODUCTION

Petroleum products ranging from High-Speed Diesel (HSD) and Motor Spirit (MS) to Aviation Turbine Fuel (ATF) and industrial lubricants must stringently adhere to precise chemical and physical specifications defined by standards organizations (such as the Bureau of Indian Standards - BIS). These specifications govern properties like flash point, viscosity, sulfur content, density, and distillation parameters.

Quality assurance in petroleum engineering ensures operational safety, compliance with environmental emission norms, engine efficiency, and customer trust. Continuous product quality testing acts as a critical line of defense, validating that processing units operate within target control limits and identifying contamination or operational drift before product offloading or distribution.

2. PRACTICAL EDUCATIONAL CASE STUDY

Hypothetical Case Scenario: Flash Point Deviation in Diesel Fuel

During routine laboratory testing of a newly refined batch of High-Speed Diesel (HSD) prior to storage tank transfer, the tested sample exhibited a Flash Point of 31°C, falling below the minimum standard safety limit of 35°C.

1. What the Problem Is: The HSD batch failed the minimum flash point quality specification, indicating an unexpected concentration of volatile light-end hydrocarbons in the fuel mixture.

2. Why It Matters: A low flash point significantly elevates the fire hazard risk during fuel storage, transportation, and handling. Furthermore, it indicates off-specification product performance that could lead to engine vapor lock or early auto-ignition.

3. What the Engineer Should Check First: The quality engineer must immediately re-sample and re-test the product using a calibrated Pensky-Martens Closed Cup tester to rule out testing, sampling, or lab-instrument error. Concurrently, quarantine the storage batch to prevent blending with clean fuel.

4. How to Investigate Possible Causes: Conduct a structured review across the supply chain: check the distillation column operating conditions (stripping steam rates, top/draw-off temperatures), inspect pipeline manifolds and valve alignments for cross-contamination from lighter fractions (e.g., petrol/naphtha), and inspect sampling containers for residual solvent residues.

5. What Corrective Action May Be Appropriate: Adjust the distillation unit operating parameters (increasing stripping column reboiler temperature or stripping steam flow to drive off light components). For the quarantined off-spec batch, re-process it through the stripper unit or perform controlled blending under lab supervision with a heavy component batch.

6. How to Verify Whether the Action Worked: Draw multi-level representative samples (top, middle, bottom) from the treated/reprocess

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action was successful following an off-spec event?

Verification is achieved by re-sampling the corrected product, analyzing key quality parameters against standard limits, and monitoring process trends over time to confirm process stability.

Q15: Why is instrument calibration critical for quality laboratory accreditation?

Calibration links test measurements to certified standard reference materials, ensuring test accuracy, legal metrology compliance, and international test result acceptance.

6. REFERENCES & DISCLAIMER

• Bharat Petroleum Corporation Limited (BPCL) Official Website: www.bharatpetroleum.in

• ISO 9001:2015 Quality Management Systems Requirements

• Juran, J.M., & De Feo, J.A. (2010). Juran's Quality Handbook: The Complete Guide to Performance Excellence (6th ed.). McGraw-Hill.

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DAY 90

Original source file(s): Day_90_HPCL_Equipment_Reliability_Quality (1).pdf

Source focus: The Day 90 source material focuses on Hindustan Petroleum Corporation Limited (HPCL). The source text explicitly discusses the following quality themes and methods: Reliability, Root Cause, inspection, ISO 1940, RCA, FMEA, RPN, ISO 9001:2015.

Section headings identified in the source: Hindustan Petroleum Corporation Limited (HPCL) • Maintenance • Author: Jagriti Singh • 1. INTRODUCTION • 2. PRACTICAL CASE STUDY (EDUCATIONAL SCENARIO) • Scenario Description

Technical terms detected in the source: Reliability, Root Cause, inspection, ISO 1940, RCA, FMEA, RPN, ISO 9001:2015

Source extract 1: Day_90_HPCL_Equipment_Reliability_Quality (1).pdf

100 DAYS QUALIT Y MANAGEMENT CHALLENGE • DAY 90

Hindustan Petroleum Corporation Limited (HPCL)

Equipment Reliability and Maintenance Quality: Pump Vibration, Root Cause Analysis & Preventive Maintenance

Author: Jagriti Singh

1. INTRODUCTION

In continuous process industries such as petroleum refining and processing, high rotating equipment availability and reliability are essential for safe, steady, and profitable operations. Centrifugal pumps handle volatile liquids, hydrocarbons, and utility streams under demanding thermal and pressure conditions.

Unplanned failures or recurring equipment deterioration can result in process upsets, safety hazards, environmental leaks, and substantial production losses. Implementing robust preventive maintenance (PM) practices alongside disciplined root cause failure investigations ensures that mechanical equipment maintains operational integrity throughout its lifecycle.

2. PRACTICAL CASE STUDY (EDUCATIONAL SCENARIO)

Scenario Description

A heavy-duty centrifugal pump deployed in a refinery processing unit develops repeated high-vibration alerts and performance degradation, even after routine maintenance overhauls and bearing replacements.

• What the Problem Is: The pump exhibits excessive vibration amplitude during steady-state operation, accompanied by reduced delivery head and minor mechanical seal weeping. The recurring nature of the problem indicates that routine maintenance is addressing symptoms rather than the underlying cause.

• Why Repeated Vibration Needs Immediate Attention: Persistent vibration accelerates mechanical fatigue across pump components, damages seal faces, degrades bearing races, loosens foundation anchor bolts, and raises the risk of catastrophic shaft fracture or fluid containment loss.

• What the Engineer Should Check First: Prior to invasive dismounting, the engineer should conduct non-intrusive physical and visual checks: foundation anchor bolt tightness, baseplate rigidity, piping strain, oil levels and contamination, overall operating temperature, pump hydraulic operating point (flow rate vs. BEP), and spectrum vibration measurement across drive-end and non-drive-end bearings.

• How Maintenance Records & Operating Conditions Help: Reviewing historical log sheets, vibration spectral trends, past work orders, lubrication records, and process parameters (suction pressure, fluid viscosity, operating temperature) helps determine whether the vibration correlates with specific throughput changes, thermal expansion, or recent maintenance interventions.

• How to Investigate the Possible Root Cause (Evidence-Based Approach): Avoid assuming a single default cause (e.g., automatically replacing bearings). Perform systematic diagnostic testing: FFT vibration spectrum analysis to distinguish 1X (unbalance), 2X (misalignment), or high-frequency (bearing defect/cavitation) peaks; laser alignment check under cold and hot thermal growth conditions; internal clearance inspection; impeller dynamic bal

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overhauls that risk introducing maintenance-induced defects.

Q15: What steps should be taken if a pump continues to vibrate immediately after shaft alignment?

A: Verify baseplate rigidity/soft foot, re-check thermal growth offsets, perform vibration spectral analysis to check for unbalance or

resonance, and inspect for external piping forces on casing.

6. REFERENCES & DISCLAIMER

References:

• Hindustan Petroleum Corporation Limited (HPCL) Official Website: https://www.hindustanpetroleum.com

• ISO 9001:2015 - Quality Management Systems — Requirements. International Organization for Standardization.

• Bloch, H. P., & Geitner, F. K. (2012). Practical Machinery Management for Process Plants: Machinery Component Maintenance and

Repair (4th ed.). Gulf Professional Publishing.

Disclaimer: This is an independent educational case study based on a hypothetical scenario. It does not represent an actual HPCL incident,

internal procedure, or official HPCL publication. This work is not affiliated with or endorsed by HPCL.

DAY 91

Original source file(s): Day_91_AAI_Airport_Equipment_Quality_Management (1).pdf

Source focus: The Day 91 source material focuses on Airport Equipment Reliability & Quality Management. The source text explicitly discusses the following quality themes and methods: Reliability, Inspection, Root Cause, RCA, FMEA, RPN, ISO 9001:2015.

Section headings identified in the source: Airport Equipment Reliability & Quality Management • SCENARIO: RECURRENT MECHANICAL MALFUNCTION IN GROUND SUPPORT EQUIPMENT • 1. Operational Problem Identification • 2. Operational & Safety Importance • 3. Initial Engineering Verification & Inspection • 4. Operating Conditions & Maintenance Records Analysis

Technical terms detected in the source: Reliability, Inspection, Root Cause, RCA, FMEA, RPN, ISO 9001:2015

Source extract 1: Day_91_AAI_Airport_Equipment_Quality_Management (1).pdf

100 DAYS QUALIT Y MANAGEMENT CHALLENGE • DAY 91

Airport Equipment Reliability & Quality Management

Preventive Maintenance, Failure Investigation and Corrective Action

Organisation: Airports Authority of India (AAI) | Author: Jagriti Singh

■ 1. INTRODUCTION

In modern airport operations, equipment reliability and preventive maintenance are critical to ensuring operational safety,

passenger safety, and punctuality. Airport Ground Support Equipment (GSE)—such as baggage tractors, pushback tugs,

passenger boarding stairs, and aircraft fueling units—operates in demanding, time-critical environments. A mechanical

breakdown can disrupt flight schedules, block taxiways, or compromise safety around active aircraft.

Implementing robust quality engineering practices—such as scheduled preventive maintenance, structured failure

investigations, and verified corrective actions—minimizes unexpected downtime, extends equipment operational life, and

maintains strict adherence to aviation safety protocols.

■ 2. PRACTICAL CASE STUDY (EDUCATIONAL SCENARIO)

SCENARIO: RECURRENT MECHANICAL MALFUNCTION IN GROUND SUPPORT EQUIPMENT

1. Operational Problem Identification

A fleet of airport baggage towing tractors experiences repeated hydraulic system overheating and pressure loss during peak turnaround operations. Remarkably, these failures persist even shortly after routine scheduled maintenance, causing unexpected operational delays.

2. Operational & Safety Importance

Baggage tractors transport critical luggage cargo across busy ramp zones. Hydraulic loss compromises steering and braking responsiveness, posing severe safety risks to ground crew, aircraft, and airfield infrastructure.

3. Initial Engineering Verification & Inspection

Engineers must first inspect basic mechanical parameters: fluid levels, filter cleanliness, seal integrity, and signs of external leakage. Immediate checks should confirm whether hydraulic fluid specifications match manufacturer guidelines and whether fluid contamination is present.

4. Operating Conditions & Maintenance Records Analysis

Reviewing historical logbooks helps correlate failures with specific operating conditions (e.g., extreme ambient summer temperatures, continuous engine idling during flight delays, or specific operator shifts). Maintenance records reveal whether past servicing used non-genuine seals or incorrect hydraulic oil viscosity.

5. Root Cause Investigation

A structured investigation reveals that routine maintenance involved changing oil but failed to flush thermal bypass valves. Accumulation of micro-particles caused bypass valves to stick, restricting oil flow to the heat exchanger during heavy duty cycles. Hence, a single surface symptom (fluid overheating) resulted from a secondary failure mechanism (valves stuck due to fluid contamination).

6. Corrective & Preventive Action Implementation

Corrective actions include flushing the hydraulic cooling circuit, replacing contaminated bypass valves, upgrading to high-efficienc

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cal failure investigation on heavy machinery?

Ans: Strict adherence to Lockout/Tagout (LOTO) protocols, pressure relief of pneumatic/hydraulic lines, and securing moving mechanical elements to ensure technician safety.

Q15: What is the role of continuous monitoring in Quality Assurance for airport ground equipment?

Ans: Continuous monitoring tracks real-time operating metrics (vibration, temp, pressure) to detect micro-trends, allowing predictive interventions before catastrophic field failure occurs.

6. References

• Airports Authority of India (AAI) Official Portal: www.aai.aero

• ISO 9001:2015 Quality Management Systems — Requirements.

• Mobley, R. Keith. An Introduction to Predictive Maintenance. Elsevier / Butterworth-Heinemann.

Disclaimer: This is an independent educational case study based on a hypothetical scenario. It does not represent an actual AAI incident, internal procedure, or official AAI publication. This work is not affiliated with or endorsed by AAI.

100 Days Quality Management Challenge | Day 91 Page 4 of 4

DAY 92

Original source file(s): Day_92_India_Post_Service_Quality_Management (1) (1).pdf

Source focus: The Day 92 source material focuses on Service Quality Management and Process Improvement. The source text explicitly discusses the following quality themes and methods: Service Quality, Root Cause, isolated, RCA, FMEA, customer satisfaction, reliability, ISO 9001, ISO 9001:2015.

Section headings identified in the source: Service Quality Management and Process Improvement • Organisation: India Post Author: Jagriti Singh • 1. INTRODUCTION • 2. PRACTICAL CASE STUDY (HYPOTHETICAL) • Records to Review: • Corrective & Improvement Actions:

Technical terms detected in the source: Service Quality, Root Cause, isolated, RCA, FMEA, customer satisfaction, reliability, ISO 9001, ISO 9001:2015

Source extract 1: Day_92_India_Post_Service_Quality_Management (1) (1).pdf

100 DAYS QUALIT Y MANAGEMENT CHALLENGE • DAY 92

Service Quality Management and Process Improvement

Delivery Delays, Root Cause Analysis and Corrective Action

Organisation: India Post Author: Jagriti Singh

1. INTRODUCTION

In postal and logistics services, service quality and reliable delivery are critical for maintaining customer trust and operational efficiency. Timely consignment processing ensures public satisfaction and consistency across mail networks. Process improvement methodologies enable organizations to identify operational bottlenecks, eliminate waste, reduce processing errors, and establish standardized workflows to meet service level agreements (SLAs).

2. PRACTICAL CASE STUDY (HYPOTHETICAL)

Scenario Overview: A delivery hub observes repeated delays in consignment dispatches, causing customer complaints and missed delivery targets.

Problem Statement: Consignments passing through a regional post office hub experience an average delivery lag of 24–48 hours beyond the standard transit window during peak weekday hours.

Importance of Addressing Delivery Delays: Delivery delays harm service credibility, increase customer inquiry volumes, overload storage space at hubs, and cause downstream scheduling disruptions across transport routes.

Records to Review:

• Consignment logbooks and electronic tracking system timestamps.

• Sorting center shift logs, attendance registers, and volume arrival reports.

• Vehicle dispatch schedules, transit manifests, and hand-over receipts.

Investigation Approach (Sorting, Workload, & Process Delays):

• Sorting Errors: Check misrouting logs to see if packages are placed into wrong delivery bags due to unclear address labels or manual sorting mistakes.

• Workload Spikes: Compare incoming mail volume data against shift staffing levels to check for capacity imbalances

during peak hours.

• Process Bottlenecks: Map consignment flow from vehicle unloading to final bag sealing to pinpoint waiting times at scanning or sorting stages.

Identifying Underlying Causes: Evidence-based data collection is used rather than assumptions. By analyzing check sheets and process logs, investigators distinguish between isolated delays (e.g., vehicle breakdown) and systemic failures (e.g., inadequate sorting capacity or unclear sorting guidelines).

Corrective & Improvement Actions:

• Implement pre-sorting checks and barcode scanning updates to minimize manual sorting errors.

• Adjust shift rosters to align staffing levels with incoming peak consignment hours.

• Standardize bag sealing and vehicle loading workflows to minimize idle handling time.

Verifying Improvement Effectiveness: Track delivery turnaround times and misrouting counts for 30–60 days post-implementation. Compare metrics against baseline data using control charts or daily tracking logs to ensure stable, long-term performance.

Day 92 | Quality Management Challenge Page 1 of 3

3. IMPORTANT QUALITY TOOLS

1. Check Sheet: A structured, manual data-collection form designed t

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tainment action versus a permanent corrective action?

A: Containment is a quick temporary fix (e.g., authorising overtime to clear a backlog), while corrective action fixes the systemic root cause long-term.

Q15: How does ISO 9001 support continuous improvement in postal operations?

A: It provides a structured Quality Management System framework emphasizing customer focus, process approach, data-driven decisions, and continual improvement.

6. REFERENCES AND DISCLAIMER

References:

• Official India Post Website: www.indiapost.gov.in

• ISO 9001:2015 Quality Management Systems — Requirements. International Organization for Standardization.

• Juran, J. M., & De Feo, J. A. (2010). Juran's Quality Handbook: The Complete Guide to Performance Excellence (6th ed.). McGraw-Hill Education.

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Day 92 | Quality Management Challenge Page 3 of 3

DAY 93

Original source file(s): day_93_quality_engineering (1).pdf

Source focus: The Day 93 source material focuses on Integrated Quality Engineering for Safety-Critical. The source text explicitly discusses the following quality themes and methods: FMEA, Root Cause, inspection, Traceability, NDT, RCA, Isolated, CAPA, NCR, audit, reliability, risk-based.

Section headings identified in the source: Integrated Quality Engineering for Safety-Critical • Components • 1. Introduction — Why Quality Matters in Safety-Critical Engineering • Quality engineering in safety-critical systems relies on three core tenets: • 2. Hypothetical Case Study Scenario • 3. FMEA — Identifying Potential Failures

Technical terms detected in the source: FMEA, Root Cause, inspection, Traceability, NDT, RCA, Isolated, CAPA, NCR, audit, reliability, risk-based, ISO 9001:2015, ISO 19443:2018

Source extract 1: day_93_quality_engineering (1).pdf

100 DAYS QUALITY ENGINEERING CHALLENGE — DAY 93

Integrated Quality Engineering for Safety-Critical

Components

Bringing Together FMEA, Non-Destructive Testing, Root Cause Analysis, and Quality Assurance

Educational Disclaimer & Scenario Notice: This document is created for educational and technical training purposes as part of the 100 Days Quality Engineering Challenge. The case study presents a hypothetical engineering scenario involving a research reactor auxiliary system. It is strictly illustrative and does not represent an actual incident, inspection record, or operational failure at BARC (Bhabha Atomic Research Centre) or any specific facility. Official BARC procedures, regulatory mandates, and technical guidelines remain the sole authority for real-world applications.

1. Introduction — Why Quality Matters in Safety-Critical Engineering

In general engineering, a minor defect might result in a temporary product return or minor inconvenience. In safety-critical engineering—such as nuclear power, aerospace, defense, or high-pressure energy systems—the margin for error is effectively zero. A safety-critical component is any part whose failure could cause equipment destruction, severe environmental contamination, system shutdown, or risk to human life.

Quality engineering in safety-critical systems relies on three core tenets:

• Prevention over Detection: Preventing a defect during design and manufacturing is far safer and less costly than detecting it after fabrication—or worse, discovering it during operation.

• Defense in Depth: Final inspection alone is never sufficient. A component can pass external visual checks while harboring subsurface stress cracks or volumetric voids. Quality must be built into every step using an integrated framework.

• Complete Traceability: Every component must have an unbroken, verified paper trail linking raw material heat numbers, process parameters, non-destructive evaluation (NDE) records, and final sign-offs.

2. Hypothetical Case Study Scenario

Scenario Overview: A primary coolant bypass manifold component, manufactured from austenitic stainless steel (SS 316L), was fabricated for a hypothetical nuclear thermal research loop. The component operates under high hydrostatic pressure (15 MPa) and elevated temperature (310°C).

The Quality Concern: During routine post-weld Liquid Penetrant Testing (PT) of the manifold flange attachment weld, a small, linear indication measuring 2.8 mm in length was observed along the heat-affected zone (HAZ).

Inspection Dilemma: The primary fabrication inspector noted that 2.8 mm falls near the threshold of allowable surface indications under standard non-critical guidelines. However, making an unverified acceptance decision without investigating the depth, volumetric nature, or underlying cause introduces unacceptable operational risk. Accepting an unknown linear flaw in a high-pressure loop could lead to stress corrosion cracking (SCC)

propagation during thermal cycling.

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; Verification

proves the fix held over time.

• Traceability Rule: Unverified heat numbers or missing heat-treatment logs render a component unsafe regardless of physical dimensions.

15. References and Further Reading

1. ISO 19443:2018: Quality management systems — Specific requirements for the application of ISO 9001:2015 by organizations in the supply chain of the nuclear energy sector. International Organization for Standardization.

2. ASME Boiler and Pressure Vessel Code (BPVC) Section V: Nondestructive Examination. American Society of Mechanical Engineers.

3. ASNT SNT-TC-1A: Personnel Qualification and Certification in Nondestructive Testing. American Society for Nondestructive Testing.

4. AIAG & VDA FMEA Handbook: Failure Mode and Effects Analysis — Design FMEA & Process FMEA. Automotive Industry Action Group & Verband der Automobilindustrie.

5. IAEA Safety Standards Series No. GS-G-3.1: Application of the Management System for Facilities and Activities. International Atomic Energy Agency, Vienna.

Day 93: Integrated Quality Engineering | Safety-Critical Components Page 18 of 18

DAY 94

Original source file(s): Day_94_Quality_Engineering_NABL_Case_Study-v3 (1).pdf

Source focus: Day 94 source document titled 1 0 0 DAYS Q UA L I T Y E N G I N E E R I N G C H A L L E N G E — DAY 9 4.

1 0 0 DAYS Q UA L I T Y E N G I N E E R I N G C H A L L E N G E — DAY 9 4

Building Reliable Laboratory Results: An

Integrated Approach to Quality

Management and Measurement Assurance

NABL-Inspired Case Study on ISO/IEC 17025, Calibration, Measurement Uncertainty,

and Laboratory Quality Improvement

Document Objective: Technical Educational Case Study & Advanced Quality Engineering Guide

Target Audience: Quality Engineers, Metrologists, Laboratory Managers, and Technical Interview Candidates

Core Standards: ISO/IEC 17025:2017, NABL Specific Criteria, ISO 9001

Author: Quality Engineering Excellence Initiative

Notice: This document is an educational case study inspired by laboratory accreditation principles. It does not represent actual NABL events, official audits, or specific commercial laboratory data.

1. Document Objective & Context

This technical educational case study forms Day 94 of the 100 Days Quality Engineering Challenge. The central objective is to explain how a testing and calibration laboratory can transform variable, inconsistent testing data into highly reliable, metrologically sound results through an integrated Laboratory Quality Management System (LQMS).

Using the National Accreditation Board for Testing and Calibration Laboratories (NABL) as the organizational context, this guide illustrates the practical interplay between ISO/IEC 17025 requirements, calibration management, measurement uncertainty, Measurement System Analysis (MSA), personnel competence, and proficiency testing.

Educational Disclaimer: This case study is a hypothetical educational model created for professional training and interview preparation. It is inspired by established accreditation frameworks (ISO/IEC 17025 and NABL guidance). It does not present real-world

security incidents, confidential client data, or official NABL audit findings.

4. Introduction to NABL & Laboratory Accreditation

What is NABL?

The National Accreditation Board for Testing and Calibration Laboratories (NABL) is an autonomous body operating under the guidance of the Quality Council of India (QCI). NABL provides formal third-party recognition of the technical competence of testing and calibration laboratories, medical laboratories, reference material producers (RMPs), and proficiency testing providers (PTPs) in accordance with international standards.

Understanding Laboratory Accreditation vs. ISO 9001 Certification

A common misconception among early-career engineers is confusing ISO 9001 certification with laboratory accreditation under ISO/IEC 17025. The distinction is critical:

Parameter ISO 9001 Certification NABL Accreditation (ISO/IEC 17025)

Core Focus Quality Management System (QMS) pro■ Technical competence, valid testing meth■

cesses and customer satisfaction. odologies, and measurement accuracy.

Evaluation Basis Conformity to broad management system Demonstrated technical capability evalu■

requirements across any organization. ated by qualified technical assessors in

specific te

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ry.

Intermediate Checks In-house periodic performance checks between external calibra■

tions using check-standards and control charts.

Decision Rules & Guardbanding Documented rules for declaring pass/fail near specification limits considering expanded measurement uncertainty.

Key Formula Reference

• Combined Standard Uncertainty: $u_c = \sqrt{(u_1^2 + u_2^2 + ... + u_n^2)}$

• Expanded Uncertainty: $U = k \cdot u_c$ (typically k = 2 for 95% coverage probability)

• PT Z-Score: $Z = (x - X) / \sigma$

22. References & Standards Framework

• ISO/IEC 17025:2017: General requirements for the competence of testing and calibration laboratories.

• NABL-100: General Information Brochure on NABL Accreditation Services.

• JCGM 100:2008 (GUM): Evaluation of measurement data — Guide to the expression of uncertainty in measurement.

• ILAC-G8:09/2019: Guidelines on Decision Rules and Statements of Conformity.

• ISO 5725: Accuracy (trueness and precision) of measurement methods and results.

• NABL-163: Policy for Participation in Proficiency Testing Activities.

100 Days Quality Engineering Challenge — Day 94 Page 15 of 15

DAY 95

Original source file(s): Day_95_DVC_Case_Study (1).pdf

Source focus: The Day 95 source material focuses on Improving Power Plant Reliability Through Integrated Quality. The source text explicitly discusses the following quality themes and methods: Reliability, FMEA, RCA, Root Cause, Risk-Based, inspection, RPN, ISO Turbine, ISO 10816, Audit, ISO VG, PDCA.

Section headings identified in the source: Improving Power Plant Reliability Through Integrated Quality • Engineering and Maintenance Management • Thermal Asset Quality • 1. DOCUMENT OBJECTIVE • 4. INTRODUCTION TO DVC AND POWER GENERATION • 5. HYPOTHETICAL CASE STUDY SCENARIO

Technical terms detected in the source: Reliability, FMEA, RCA, Root Cause, Risk-Based, inspection, RPN, ISO Turbine, ISO 10816, Audit, ISO VG, PDCA, NDT, Isolation

Source extract 1: Day_95_DVC_Case_Study (1).pdf

100 DAYS QUALITY ENGINEERING CHALLENGE — DAY 95

Improving Power Plant Reliability Through Integrated Quality

Engineering and Maintenance Management

Integrating FMEA, RCA, Reliability-Centred Maintenance, Condition Monitoring and Continuous Improvement

Organisation Context: Damodar Valley Corporation (DVC) | Author: Jagriti Singh (Quality Engineer) | Domain: Power Generation Reliability &

Thermal Asset Quality

1. DOCUMENT OBJECTIVE

This document forms Day 95 of the 100 Days Quality Engineering Challenge. Its central objective is to provide an in-depth, technically rigorous educational case study demonstrating how core Quality Engineering principles—traditionally associated with manufacturing and assembly—can be integrated into heavy thermal power plant operations to boost equipment reliability, mitigate recurring failures, optimize maintenance costs, and enforce workplace safety.

Using Damodar Valley Corporation (DVC) as an organizational context, this case study illustrates the systematic cross-functional deployment of Failure Mode and Effects Analysis (FMEA), Root Cause Analysis (RCA), Reliability-Centred Maintenance (RCM), Condition Monitoring, Vibration Analysis, and Risk-Based Maintenance.

Educational Disclaimer: This document is an educational case study inspired by heavy thermal power generation engineering. The scenarios, plant operational data, vibration spectra readings, and failure histories presented herein are entirely hypothetical. They do not represent actual historical incidents, confidential performance statistics, or official operational procedures of Damodar Valley Corporation (DVC).

PART A — CASE STUDY

4. INTRODUCTION TO DVC AND POWER GENERATION

Damodar Valley Corporation (DVC), established in 1948 as India's first multipurpose river valley project under the Damodar Valley Corporation Act (Act No. XIV of 1948), plays a pivotal role in power generation, flood control, irrigation, and water supply across eastern India (primarily West Bengal and Jharkhand). DVC operates a vast network of thermal power stations (e.g., Mejia, Koduwa, Chandrapura, Durgapur, Raghunathpur) and hydroelectric stations, contributing significantly to the regional grid stability and industrial infrastructure.

In base-load thermal power generation, electricity is generated continuously through the thermodynamic Rankine cycle. High-pressure steam produced in boilers drives steam turbines, which power synchronous generators. Because thermal plants run continuously under severe thermal, mechanical, and environmental stress, equipment reliability is paramount:

• Thermal & Mechanical Stresses: Boilers operate under extreme pressures (>170 bar) and temperatures (>540°C), while turbines rotate at 3000 RPM. Minor defects rapidly propagate into catastrophic structural failures.

• Cost of Unplanned Outages: Forced outages lead to severe financial penalties, revenue losses from unserved energy, high peak-demand power replacement costs, and thermal stress cycling duri

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w.dvc.gov.in

• ISO 10816-3: Mechanical vibration — Evaluation of machine vibration by measurements on non-rotating parts — Part 3: Industrial machines. International Organization for Standardization.

• ISO 55001: Asset management — Management systems — Requirements. International Organization for Standardization.

• SAE JA1011: Evaluation Criteria for Reliability-Centered Maintenance (RCM) Processes. SAE International.

• Moubray, John (1997): Reliability-Centered Maintenance (RCM II). Industrial Press Inc.

• Bloch, Heinz P. & Geitner, Fred K. (2012): Machinery Failure Analysis and Troubleshooting (Practical Machinery Management for Process Plants). Butterworth-Heinemann.

• Mobley, R. Keith (2002): An Introduction to Predictive Maintenance (2nd Edition). Elsevier Science.

• Bureau of Energy Efficiency (BEE), India: Guide Books for National Certification Examination for Energy Managers and Energy Auditors — Thermal Power Utilities.

DAY 96

Original source file(s): Day 96 hydropower_case_study_ (1).pdf

Source focus: The Day 96 source material focuses on Enhancing Hydropower Infrastructure Reliability. The source text explicitly discusses the following quality themes and methods: Reliability, FMEA, root cause, ISO code, Risk-Based, RPN, inspection, ISO
14, RCA, NDT, ISO 1940, Traceability.

Section headings identified in the source: Enhancing Hydropower Infrastructure Reliability • Through Integrated Quality Engineering • A Technical Case Study & Comprehensive Reliability Framework for Hydropower Facilities • Hydropower (Public Context) Monitoring • 1. Document Objective • 2. Introduction to NHPC Limited & Hydropower Infrastructure

Technical terms detected in the source: Reliability, FMEA, root cause, ISO code, Risk-Based, RPN, inspection, ISO 14, RCA, NDT, ISO 1940, Traceability, ISO 14414, ISO 10816-5

Source extract 1: Day 96 hydropower_case_study_ (1).pdf

100 DAYS QUALITY ENGINEERING CHALLENGE — DAY 96

Enhancing Hydropower Infrastructure Reliability

Through Integrated Quality Engineering

A Technical Case Study & Comprehensive Reliability Framework for Hydropower Facilities

Target Sector: Infrastructure & Case Benchmark: NHPC Limited Focus: FMEA, Vibration, Dam Hydropower (Public Context) Monitoring

1. Document Objective

This case study serves as an exhaustive educational reference module developed for Day 96 of the 100 Days Quality Engineering Challenge. Its main goal is to demonstrate how modern Quality Engineering (QE) principles, data-driven reliability methodologies, structural health monitoring, and systematic maintenance frameworks apply directly to large-scale hydropower infrastructure.

The document breaks down complex engineering concepts into structured, accessible explanations without sacrificing technical rigor. It is designed specifically for Mechanical Engineering students, Quality Engineers, GATE/ESE aspirants, and power sector professionals seeking practical knowledge in reliability-centered maintenance, asset integrity, instrumentation interpretation, and root cause analysis.

2. Introduction to NHPC Limited & Hydropower Infrastructure

2.1 Company Profile & Role in India's Energy Sector

NHPC Limited (formerly National Hydroelectric Power Corporation) is India's premier public sector enterprise in the field of hydropower development. Incorporated in 1975, NHPC is a Schedule 'A' Miniratna Enterprise under the Ministry of Power, Government of India. It possesses end-to-end capabilities ranging from concept to commissioning of large-scale hydro projects, as well as diversification into solar and wind power. NHPC plays a critical role in balancing India's national power grid by providing essential peaking power, grid stability, and renewable green energy.

2.2 Working Principle of a Hydropower Plant

Hydropower plants convert the potential energy of stored water into mechanical rotational energy using water turbines, which is subsequently converted into electrical energy by synchronous generators.

Potential Energy ($PE = m \cdot g \cdot H$) → Kinetic Energy ($KE = \frac{1}{2} m \cdot v^2$) → Mechanical Rotational Energy → Electrical Energy ($P = \eta \cdot \rho \cdot g \cdot Q \cdot H$)

Where:

• P = Generated Electrical Power (Watts)

100 Days QE Challenge | Day 96: NHPC Reliability Engineering Case Study Page 1 of 11

• η = Overall Efficiency of the plant (Turbine × Generator × Transformer)

• ρ = Density of water (~1000 kg/m^3)

• g = Acceleration due to gravity (9.81 m/s^2)

• Q = Volumetric Flow Rate of water (m^3/s)

• H = Net Head / Effective water drop height (m)

2.3 Major Mechanical & Civil Components

Component Primary Function Key Quality & Reliability Considerations

Dam & Reservoir Retains water, creates artificial head, and Seepage control, pore pressure management,

regulates flow. structural stability against sliding and overturning.

Penstock High-pressure steel/concrete conduit carrying Transient pressure surges (water hammer), weld

w

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10. References & Technical Standards

1. NHPC Limited: Publicly available Corporate Sustainability Reports & Hydro Operations Overview

(www.nhpcindia.com).

2. Central Electricity Authority (CEA): Technical Standards for Construction of Electric Plants and Electric Lines

Regulations, Ministry of Power, Govt. of India.

3. Central Dam Safety Organization (CDSO) / CWC: Guidelines for Instrumentation of Large Dams, Central Water

Commission, New Delhi.

4. IS 12800 (Part 3): Guidelines for Selection of Hydraulic Turbines for Hydroelectric Power Houses, Bureau of

Indian Standards (BIS).

5. ISO 10816-5: Mechanical vibration — Evaluation of machine vibration by measurements on non-rotating parts —

Part 5: Machine sets in hydraulic power generation and pump-storage plants.

6. ISO 13374: Condition monitoring and diagnostics of machines — Data processing, communication and

presentation.

7. Mobley, R. K. (2002): An Introduction to Predictive Maintenance, 2nd Edition, Elsevier Butterworth-Heinemann.

100 Days QE Challenge | Day 96: NHPC Reliability Engineering Case Study Page 11 of 11

DAY 97

Original source file(s): Day 97 nhai_quality_case_study_v6 (1).pdf

Source focus: The Day 97 source material focuses on 1 0 0 DAYS Q UA L I T Y E N G I N E E R I N G C H A L L E N G E — DAY 9 7 CA S E S T U DY. The source text explicitly discusses the following quality themes and methods: FMEA, NCR, Root Cause, RCA, Control Plan, Inspection, Audit, Lean, Traceability, CAPA, RPN, ISO 9001.

Section headings identified in the source: Enhancing Highway Infrastructure Quality Through • Integrated Construction Quality Engineering • 1. INTRODUCTION TO THE NATIONAL HIGHWAYS AUTHORITY OF INDIA • 1.1 Establishment and Statutory Mandate • 1.2 Core Responsibilities and Operational Scope • 1.3 Importance of Quality Assurance in Highway Megaprojects

Technical terms detected in the source: FMEA, NCR, Root Cause, RCA, Control Plan, Inspection, Audit, Lean, Traceability, CAPA, RPN, ISO 9001, PDCA, ISO 9001:2015

Source extract 1: Day 97 nhai_quality_case_study_v6 (1).pdf

1 0 0 DAYS Q UA L I T Y E N G I N E E R I N G C H A L L E N G E — DAY 9 7 CA S E S T U DY

Enhancing Highway Infrastructure Quality Through

Integrated Construction Quality Engineering

Organisation: NHAI Domain: Construction Quality Engineering Document Ref: QE-100D-D97-NHAI

Notice & Disclaimer: This technical case study is authored as part of an educational quality engineering challenge. While Section 1 and Section 8 reference public domain structural standards and operational roles of the National Highways Authority of India (NHAI), the case study in Section 3 & 7 is a hypothetical educational scenario designed solely to demonstrate Quality Control (QC), Quality Assurance (QA), Failure Mode and Effects Analysis (FMEA), Non-Conformance Reporting (NCR), and Root Cause Analysis (RCA) methodologies. It does NOT represent an actual incident, failure, or official report endorsed by NHAI.

1. INTRODUCTION TO THE NATIONAL HIGHWAYS AUTHORITY OF INDIA

(NHAI)

1.1 Establishment and Statutory Mandate

The National Highways Authority of India (NHAI) was constituted by an Act of Parliament, the National Highways Authority of India Act, 1988. It was formally operationalized in February 1995 as an autonomous agency under the administrative control of the Ministry of Road Transport and Highways (MoRTH), Government of India. NHAI is responsible for the development, maintenance, and management of the National Highways network across India, comprising over 140,000 km of primary arteries critical to national logistics and passenger mobility.

1.2 Core Responsibilities and Operational Scope

• Infrastructure Execution: Planning, designing, constructing, and expanding multi-lane expressways, access-controlled corridors, bypasses, and national highway networks.

• Project Delivery Models: Executing projects via Engineering, Procurement, and Construction (EPC), Hybrid Annuity Model (HAM), and Build-Operate-Transfer (BOT) frameworks.

• Asset Management & Toll Operations: Managing operational corridors, electronic toll collection (FASTag infrastructure), and long-term pavement maintenance.

• Quality Management & Technical Audits: Formulating quality control guidelines, enforcing technical specifications aligned with the Indian Roads Congress (IRC) and MoRTH, and conducting third-party quality audits.

1.3 Importance of Quality Assurance in Highway Megaprojects

Highway pavements are multi-layered structural systems subjected to aggressive axle loading, dynamic tire pressures, cyclic temperature variations, and environmental stress (moisture ingress). A robust Quality Assurance (QA) program ensures that construction materials, volumetric proportions, field compaction, and layer thicknesses satisfy design standards. High quality assurance directly minimizes premature failure modes—such as fatigue cracking, rutting, and subgrade pumping—thereby extending structural design life (typically 15–20 years for flexible pavements and 30+ years for rigid pavements) and optimizing Life Cycle

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Aggregate < 30% Wear & abrasion resistance

Aggregate Impact Value Aggregate < 24% Impact & shock load resistance

Flakiness & Elongation Aggregate < 35% (Combined) Particle shape suitability

Bitumen Penetration Bitumen 40 – 50 dmm (VG-40) Hardness / consistency at 25°C

Bitumen Softening Point Bitumen ≥ 50°C (VG-40) High-temperature stability limit

Pavement Failure Quick Reference

Distress Mode Visual Sign Primary Structural Root Cause

Rutting Longitudinal depression in wheel path High air voids, soft binder, subgrade shear strain

Fatigue Cracking Alligator pattern cracking Repeated tensile stress at bottom of binder layer

Bleeding Shiny film of free bitumen on surface Excess binder content (V_a < 2\%), soft bitumen

Day 97: Quality Engineering Challenge | NHAI Case Study Page 15 of 16

Distress Mode Visual Sign Primary Structural Root Cause

Ravelling Dislodging of coarse aggregate particles Lean binder, cold compaction, high air voids

*** End of Technical Case Study Document — Day 97 Quality Engineering Challenge ***

Day 97: Quality Engineering Challenge | NHAI Case Study Page 16 of 16

DAY 98

Original source file(s): Day 98 CPWD_Quality_Engineering_Case_Study-v2 (1).pdf

Source focus: The Day 98 source material focuses on Strengthening Construction Quality and Structural Reliability. The source text explicitly discusses the following quality themes and methods: Reliability, audit, Inspection, NDT, Traceability, Yield, FMEA, root cause, RPN, isolation, RCA, isolates.

Section headings identified in the source: Strengthening Construction Quality and Structural Reliability • Through Integrated Quality Engineering • A Technical Case Study & Comprehensive Quality Management Framework for Public Infrastructure • Assurance • 1. Introduction to CPWD & Organizational Quality Governance • 1.1 Historical Background & Role in Public Infrastructure

Technical terms detected in the source: Reliability, audit, Inspection, NDT, Traceability, Yield, FMEA, root cause, RPN, isolation, RCA, isolates, NCR, CAPA

Source extract 1: Day 98 CPWD_Quality_Engineering_Case_Study-v2 (1).pdf

100 DAYS QUALITY ENGINEERING CHALLENGE — DAY 98

Strengthening Construction Quality and Structural Reliability

Through Integrated Quality Engineering

A Technical Case Study & Comprehensive Quality Management Framework for Public Infrastructure

Target Organisation: Central Public Works Department (CPWD), Government of India | Theme: Construction Quality Management & Structural

Assurance

EDUCATIONAL DISCLAIMER & NOTICE: This case study document is prepared as an academic and professional learning resource

for Day 98 of the 100 Days Quality Engineering Challenge. Section 3 and Section 7 feature a hypothetical government building

construction scenario created strictly for analytical and educational purposes. It does not represent an actual failure, incident, audit

report, or official publication of the Central Public Works Department (CPWD). All references to CPWD specifications, BIS standards,

and NBC guidelines reflect publicly verifiable engineering standards.

1. Introduction to CPWD & Organizational Quality Governance

1.1 Historical Background & Role in Public Infrastructure

The Central Public Works Department (CPWD), operating under the Ministry of Housing and Urban Affairs (MoHUA), Government of India, is the nation's premier public sector construction management authority. Established in July 1854 when Lord Dalhousie organized a central agency for executing public works, CPWD has evolved over 170 years into a comprehensive engineering organization capable of delivering complex architectural, structural, electrical, and HVAC infrastructure projects.

CPWD serves as the principal technical advisor and execution agency for the Government of India. Its operational jurisdiction spans across multi-story administrative complexes, educational institutions, medical research institutes, border infrastructure, residential townships, and national monuments. CPWD plays a pivotal role in national development by implementing sustainable construction practices, standardizing national building specifications, and ensuring structural longevity across diverse seismic and environmental zones.

1.2 Major Responsibilities & Project Life Cycle Management

CPWD's mandate encompasses the complete life cycle of public assets, structured around three core operational pillars:

• Building & Infrastructure Construction: Execution of capital projects from pre-feasibility studies, architectural conceptualization, and structural design to site construction and hand-over.

• Asset Maintenance & Upgradation: Day-to-day preventive, routine, and specialized structural maintenance of government assets to ensure operational continuity and serviceability.

• Consultancy & Project Management Services: Technical consultancy to autonomous bodies, public sector undertakings (PSUs), and foreign governments under bilateral infrastructure assistance initiatives.

1.3 Quality Assurance (QA) and Quality Control (QC) Framework

Quality governance in CPWD is institutionalized through a multi-tier Qual

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ot in-situ

Compression concrete.

Rebound Hammer IS 13311 Part 2 Rapid non-destructive surface hardness scan. Measures surface layer (< 30mm) only.

UPV Test IS 13311 Part 1 Assess internal homogeneity & defect voids. Sensitive to steel density & moisture.

Day 98 QE Challenge | CPWD Construction Quality Management Page 13 of 14

Governing

Test Method Primary Engineering Purpose Key Limitation

Standard

Core Testing IS 516 Part 4 Direct compressive test of in-situ structural Destructive; requires structural re-analysis.

concrete.

10.3 Common Construction Defects & Root Causes

• Delayed Concrete Cracking: Excessive w/c ratio, inadequate 14-day wet curing, thermal gradients, misplaced top rebar.

• Structural Honeycombing: Lack of needle vibration, high rebar congestion, unworkable concrete slump.

• Basement/Toilet Leakage: Missing 90° corner coves, unprimed substrates, skipped 48-hr ponding test, punctured

membrane.

• Rebar Corrosion & Spalling: Insufficient concrete clear cover (< 20mm), carbonation, chloride ingress.

Day 98 QE Challenge | CPWD Construction Quality Management Page 14 of 14

DAY 99

Original source file(s): Day 99 UPSC_Quality_Engineering_Case_Study_v3.pdf

Source focus: The Day 99 source material focuses on 1 0 0 D AY S Q U A L I T Y E N G I N E E R I N G C H A L L E N G E D AY 9 9. The source text explicitly discusses the following quality themes and methods: FMEA, RPN, reliability, Audit, inspection, Traceability, isolation, Root Cause, RCA, isolate, isolated, Risk-Based.

Section headings identified in the source: Applying Quality Engineering Principles to • Large-Scale Competitive Examination Systems • A Technical Case Study & Operational Framework for Process Excellence in Public Administration • ORGANISATION Union Public Service Commission (UPSC) & Public Examination Sector • THEME Examination Process Quality & Administrative Process Quality Management • 1. Introduction to UPSC

Technical terms detected in the source: FMEA, RPN, reliability, Audit, inspection, Traceability, isolation, Root Cause, RCA, isolate, isolated, Risk-Based, ISO 9001, PDCA

Source extract 1: Day 99 UPSC_Quality_Engineering_Case_Study_v3.pdf

Day 99: Quality Engineering Challenge

1 0 0 D AY S Q U A L I T Y E N G I N E E R I N G C H A L L E N G E D AY 9 9

Applying Quality Engineering Principles to

Large-Scale Competitive Examination Systems

A Technical Case Study & Operational Framework for Process Excellence in Public Administration

ORGANISATION Union Public Service Commission (UPSC) & Public Examination Sector

THEME Examination Process Quality & Administrative Process Quality Management

DOCUMENT TYPE Technical Case Study, Engineering Guide & Assessment Resource

IMPORTANT ACCURACY & SCOPE NOTE: This case study is developed as an educational and technical resource for Quality Engineers, process consultants, engineering students, and public administration aspirants. References to the Union Public Service Commission (UPSC) reflect its constitutional mandates and publicly established operational roles. All operational failure scenarios, Failure Mode and Effects Analysis (FMEA) tables, Risk Priority Numbers (RPN), and incident analyses are strictly hypothetical and created solely to demonstrate the practical application of Quality Engineering methodologies. No security breaches, procedural failures, or administrative non-conformities are asserted against the UPSC.

1. Introduction to UPSC

1.1 Establishment and Constitutional Position

The Union Public Service Commission (UPSC) is India's premier central recruiting agency. Established under Article 315 of the Constitution of India, the Commission functions as an independent constitutional authority mandated to conduct examinations for appointments to the services of the Union. The constitutional status ensures independence from executive influence, providing a foundation of impartiality, structural integrity, and public trust.

1.2 Constitutional Provisions Related to UPSC

The primary constitutional framework governing the Commission spans Articles 315 to 323 under Part XIV of the

Constitution:

• Article 315: Mandates the constitution of a Public Service Commission for the Union and for each State.

• Article 316: Governs the appointment, tenure, and conditions of service of the Chairman and Members of the Commission.

• Article 320: Specifies the duties and functions of the Commission, including conducting examinations for appointments to Civil Services, All India Services, Defense Services, and Technical Services.

• Article 321: Authorizes the extension of Commission functions by Parliament.

• Article 323: Requires the Commission to submit an annual report of its performance and advisory recommendations directly to the President of India.

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Day 99: Quality Engineering Challenge

1.3 Major Responsibilities and Examination Functions

The Commission handles a complex, multi-tiered portfolio of competitive examinations and administrative selections. Major functions include:

• Conducting direct recruitment through national competitive examinations (e.g., Civil Services Examination, Engineering Services Examination, Combined Medical Service

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99: Quality Engineering Challenge

bodies can build resilient operational systems. Ultimately, quality management in public administration serves a vital goal: ensuring that every candidate is evaluated under standardized, secure, transparent, and flawless conditions.

16. References

• Constitution of India: Articles 315 to 323, Part XIV (Services Under the Union and the States).

• Union Public Service Commission (UPSC): Official Public Web Portal (Government of India) - u.p.s.c.gov.in (Operational mandates, examination notifications, and public advisories).

• Bureau of Indian Standards (BIS): IS/ISO 9001:2015 Quality Management Systems Requirements.

• International Organization for Standardization: ISO 31000:2018 Risk Management Guidelines.

• Juran, J. M., & De Feo, J. A.: Juran's Quality Handbook: The Complete Guide to Performance Excellence, 7th Edition, McGraw-Hill Education.

• Montgomery, D. C.: Introduction to Statistical Quality Control, 8th Edition, John Wiley & Sons.

• American Society for Quality (ASQ): The Certified Quality Engineer Handbook, ASQ Quality Press.

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PART III — CROSS-INDUSTRY PERSPECTIVES

7. What Changes Across Industries

The day-wise collection spans component manufacturing, automotive, steel, retail, software and digital platforms, aerospace, defence, energy, transport, infrastructure, laboratory quality, and service delivery. The engineering principles overlap, but the critical-to-quality characteristics differ. A manufacturing process may prioritise dimensional conformity, process capability, supplier controls, and defect prevention. A digital service may prioritise availability, response time, security, data correctness, and user experience. An infrastructure project must consider materials, workmanship, inspection, durability, and lifecycle reliability.

8. Prevention, Detection and Response

A mature quality system combines prevention, detection, and response. Prevention includes robust design, supplier qualification, process planning, FMEA, standard work, and training. Detection includes inspection, testing, audits, SPC, and monitoring. Response includes containment, root-cause analysis, corrective action, verification, and lessons learned. Over-reliance on final inspection can allow process weaknesses to persist.

9. Documentation, Traceability and Organisational Learning

Quality records make decisions reviewable. Reports, inspection records, audit trails, calibration evidence, nonconformance records, corrective-action files, and traceability systems help establish what was planned, what happened, what was measured, and what action followed. Documentation should support control and learning rather than become a paperwork exercise.

10. Digitalisation and the Future of Quality

The source set includes digital platforms, AI-related quality, semiconductor manufacturing, network performance, and data quality. These areas reinforce the need to validate data, monitor systems, test edge cases, manage change, protect information, and maintain resilience. Digital tools can improve visibility and speed, but their outputs still require sound requirements, trustworthy data, validation, and accountability.

PART IV — BEYOND THE 100 DAYS: WHAT QUALITY ENGINEERING TAUGHT ME

One hundred days of structured exploration can change the way an engineer looks at a process. A process is no longer just a sequence of operations; it becomes a system of inputs, methods, people, measurements, risks, decisions, and outcomes. Quality is visible in the design of the work, the control of variation, the reliability of equipment, the quality of supplied materials, the clarity of requirements, and the discipline with which problems are investigated.

The journey across industries highlights that there is no single quality tool that solves every problem. An 8D report is useful when a defined problem requires structured containment and corrective action. FMEA helps anticipate risk before failure occurs. SPC supports the interpretation of process behaviour over time. OEE helps organise losses in equipment effectiveness. Lean methods examine flow and waste. Supplier quality systems extend process discipline beyond organisational boundaries. Service quality methods focus attention on the experience and reliability of delivery.

A second lesson is the importance of evidence. A conclusion should be traceable to observations, measurements, documented requirements, or clearly stated assumptions. Hypothetical scenarios can be useful for learning, but they must remain labelled as hypothetical. Proposed improvements should not be reported as achieved results unless there is verified evidence.

A third lesson is that quality is cross-functional. Engineering, operations, procurement, maintenance, testing, management, and customer-facing teams influence the final outcome. Quality culture develops when people can report problems, investigate causes, act on evidence, and prevent recurrence.

Finally, the challenge reinforces the value of continuous learning. Studying a new industry each day does not make an engineer an expert in every sector; it builds curiosity, a wider vocabulary of methods, and the habit of asking better questions. The most

useful outcome is not a list of company names, but a more disciplined way to observe processes, assess risk, evaluate evidence, and pursue improvement.

A Personal Engineering Takeaway

Quality is not simply the last checkpoint before a product or service reaches its user. It is the result of decisions made throughout the system. It is designed, built, measured, reviewed, and improved. Engineering excellence grows when every stage is treated as an opportunity to prevent failure and strengthen reliability.

CONCLUSION AND FUTURE SCOPE

The source collection presents Quality Engineering as a broad discipline that connects technical control with organisational learning. Across the documented days, recurring themes include prevention, risk assessment, process capability, supplier quality, inspection, reliability, service consistency, data quality, and continuous improvement.

Future learning can deepen the work through verified standards research, comparison with publicly available technical references, practical data analysis, validated industrial examples, and clearer separation between published organisational practices and educational application frameworks. Further editions may also include diagrams, process maps, control-chart examples, FMEA worksheets, audit checklists, and carefully sourced references.

The central conclusion is that quality is a continuous engineering responsibility. It is strengthened through clear requirements, capable processes, reliable measurement, risk-based decisions, effective problem solving, and a willingness to learn.

Glossary and Abbreviations

SOURCE INVENTORY AND COVERAGE AUDIT

The supplied ZIP contains 99 PDF files. After filename and extracted-text reconciliation, original source PDFs are mapped to all 99 day numbers. Multiple PDFs associated with a day are shown separately. This is a coverage inventory, not a claim that all documents have been independently fact-checked against the named organisations.

Day 01: Day 01 _8D_Report_Automotive_Engine_Shaft (1) (1).pdf

Day 02: Day 02_8D_Official_Quality_Report_v2 (1).pdf

Day 03: Day 03_Full Analysis 8D Report(1) (1).pdf

Day 04: Day04_spc_capability_report_final (1).pdf

Day 05: Day 05_ ISRO Process Audit Report.pdf

Day 06: Day 06 _COQ_Analysis_Report_Jagriti_Singh (1).pdf | Day_69_Editorial_Theme (1) (1).pdf | day_65_nasa_systems_engineering.pdf

Day 07: Day_07_Taguchi_QLF_Quality_Report_Jagriti (1).pdf

Day 08: Day 08_Supplier_Quality_Rating_Report_Day08_Updated (1).pdf

Day 09: Day09_OEE_Report_Nestle_Targeted (1).pdf | Day_94_Quality_Engineering_NABL_Case_Study-v3 (1).pdf

Day 10: Day 10_Quality_Engineering_Master_PlaybookFinal_v3.pdf.pdf

Day 11: Day11_Universal_Supplier_Quality_SOP_Clean_White.pdf

Day 12: Day12_FAI_Decision_and_Audit_Dashboard.pdf

Day 13: Day13_Executive_TQM_Gurus_and_Strategic_Frameworks_v2.pdf

Day 14: Day14_Lean_TQM_Framework_Jagriti_Singh.pdf

Day 15: Day15_Lean_TQM_Master_Reference_Jagriti_Singh.pdf

Day 16: Day16_TQM_Zomato_Swiggy_Case_Study_v2.pdf

Day 17: Day 17_Nykaa_PFMEA_Case_Study.pdf

Day 18: Day18_Paytm_FinTech_Quality_Framework.pdf

Day 19: Day19_TQM_Framework_Jagriti_Singh.pdf

Day 20: Day_20_Case_Study_SQC1_TQM_v3.pdf

Day 21: Chapter_21_Clean_Book.pdf

Day 22: Day_22_Full_One_Page.pdf

Day 23: Day_23_Healthcare_TQM.pdf

Day 24: Day_24_Discharge_TQM_v3_NoDash (1).pdf

Day 25: Day 25.pdf

Day 26: Boeing_Day26_Quality_Report_Final.pdf

Day 27: Day 27 Airbus .pdf

Day 28: Day 28.Tesla_Deming_Case_Study_Day28_v2.pdf

Day 29: Day 29 Samsung_Quality_Case_Study_v2.pdf

Day 30: Day 30 Intel_Quality_Case_Study__Final.pdf

Day 31: Day 31_Toyota_Case_Study_v2.pdf

Day 32: Day 32_Hyundai_Kano_TQM__v3.pdf

Day 33: Day 33_100.pdf

Day 34: Day 34 Maruti Suzuki .pdf

Day 35: Day 35.pdf

Day 36: Day 36 _100.pdf

Day 37: Day 37.pdf

Day 38: DAY 38 Google_tqm_case_study.pdf

Day 39: Day 39 Microsoft .pdf

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Day 42: Day 42.pdf

Day 43: Day 43.pdf

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Day 49: Day 49.pdf

Day 50: Day 50.pdf

Day 51: Day 51.pdf

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Day 55: Day 55 cisco_network_qe_case_study_day55.pdf

Day 56: Day 56 accenture_qe_case_study.pdf

Day 57: Day 57_Oracle_QE_Case_Study_.pdf

Day 58: Day 58 walmart_quality_case_study.pdf

Day 59: Day 59 honeywell_quality_case_study.pdf

Day 60: Day 60 siemens_tqm_qe_case_study.pdf

Day 61: Day 61 Rolls_Royce_TQM_Case_Study_QEv2.pdf

Day 62: Day 62 emirates_quality_engineering_case_study_2.pdf

Day 63: Day 63 BorgWarner_TQM_Case_Study (1).pdf

Day 64: Day 64 skyroot_quality_engineering__v4.pdf

Day 65: day_65_nasa_systems_engineering.pdf

Day 66: Day 66_ISRO_Quality_Management_Case_Study_2Pages.pdf

Day 67: Day 67 SpaceX_Quality_Management_Case_Study__v2.pdf

Day 68: day_68_case_study (1).pdf

Day 69: Day_69_Editorial_Theme (1) (1).pdf

Day 70: Day 70 disney_quality_case_study (1).pdf

Day 71: Day71 (1).pdf

Day 72: Day 72 de_beers_quality_case_study_v2 (1).pdf

Day 73: Day 73 Schneider_Electric_Quality_Case_Study_Day73 (1).pdf

Day 74: Day 74 Quality_Engineering_Across_Industries_ (1).pdf

Day 75: Day 75 BHEL_Integrated_Heavy_Engineering_Quality_Case_Study_Final (1).pdf

Day 76: HAL_Day_76_Quality_Engineering_Report (1).pdf

Day 77: Day_77_DRDO_Integrated_Defence_Quality_Engineering (1).pdf

Day 78: Day_78_BEL_Integrated_Electronics_Quality_Engineering (1).pdf

Day 79: Day_79_Indian_Railways_Quality_Engineering (1) (1).pdf

Day 80: Day_80_NTPC_Quality_Engineering_Report (1).pdf

Day 81: Day_81_ONGC_Integrated_Oil_and_Gas_Quality_Engineering_NoDash (1).pdf

Day 82: Day_82_GAIL_Integrated_Pipeline_Quality_Engineering (1).pdf

Day 83: Day_83_SAIL_Integrated_Steel_Process_Quality_Engineering (1).pdf

Day 84: NALCO_Day84_Quality_Engineering_Report (1).pdf

Day 85: Day_85_CSL_Integrated_Shipbuilding_Quality_Engineering (1).pdf

Day 86: Day_86_GRSE_Integrated_Marine_Quality_Engineering (1).pdf

Day 87: Day_87_MDL_Integrated_Shipbuilding_Quality_Engineering (1).pdf

Day 88: Day_88_IOCL_Refinery_Quality_Engineering (1).pdf

Day 89: Day_89_BPCL_Petroleum_Quality_Engineering-v2 (1).pdf

Day 90: Day_90_HPCL_Equipment_Reliability_Quality (1).pdf

Day 91: Day_91_AAI_Airport_Equipment_Quality_Management (1).pdf

Day 92: Day_92_India_Post_Service_Quality_Management (1) (1).pdf

Day 93: day_93_quality_engineering (1).pdf

Day 94: Day_94_Quality_Engineering_NABL_Case_Study-v3 (1).pdf

Day 95: Day_95_DVC_Case_Study (1).pdf

Day 96: Day 96 hydropower_case_study_ (1).pdf

Day 97: Day 97 nhai_quality_case_study_v6 (1).pdf

Day 98: Day 98 CPWD_Quality_Engineering_Case_Study-v2 (1).pdf

Day 99: Day 99 UPSC_Quality_Engineering_Case_Study_v3.pdf

Additional PDFs not confidently mapped to a day

About the Author

Jagriti Singh is a Quality Engineer and creator of the 100 Days of Quality Engineering Challenge. The initiative was driven by curiosity to understand how quality principles, engineering methodologies, reliability practices, process improvement strategies, and continuous improvement systems are applied across industries. Through this challenge, she aimed to connect theoretical knowledge with practical industrial applications by studying organisations, analysing quality practices, and understanding how engineering systems contribute to operational excellence. The project reflects an interest in learning beyond textbooks and exploring the practical role of Quality Engineering. The objective was not simply to study concepts but to understand how they work in industrial environments and how organisations approach quality, reliability, and continuous improvement. For her, quality is not merely a standard or final inspection activity; it is a continuous commitment to learning, improvement, accountability, reliability, and engineering excellence.

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